MA-310 (Calculus III)

Worcester State University

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Contents

Chapter 1

Vectors and the Geometry of Space

Vector calculus is the branch of mathematics that deals with vector fields and their derivatives. It is essential for understanding the behavior of physical systems in three-dimensional space.

In this chapter, we will explore the fundamental concepts of vector calculus, including gradient, divergence, and curl.

1.1 Vectors in Two Dimensions

A vector is a quantity that has both magnitude and direction. In two dimensions, vectors can be represented as arrows in a plane, where the length of the arrow corresponds to the magnitude of the vector, and the direction of the arrow indicates the direction of the vector.

Definition 1.1 A velocity or force has both magnitude and direction and is often represented by a **directed line segment** or a **vector**. We denote the vector with **initial point** P and **terminal point** Q as $\vec{PQ}$. The direction of the vector indicates arrowhead at Q. The **magnitude** of $\vec{PQ}$ is the length of the segment PQ and is denoted by $\|\vec{PQ}\|$. Vectors that have the same magnitude and direction are called **equivalent**. ◆

A vector may be translated from one location to another, provided neither the magnitude nor the direction is changed. If $\vec{PS} = \vec{PQ} + \vec{PR}$ and $\vec{PQ}$ and $\vec{PR}$ are two forces acting at P, then $\vec{PS}$ is the **resultant force** (sum).

If c is a scalar and $\vec{v}$ is a vector, then $c\vec{v}$ (a **scalar multiple**) is defined as a vector whose magnitude is $|c|$ times the magnitude $\|\vec{v}\|$ of $\vec{v}$ with the same direction ($c > 0$) or the opposite direction ($c < 0$) of $\vec{v}$.

A vector with the initial point at the origin is called the **position vector**. We use the symbol $\langle a_1, a_2\rangle$ for an ordered pair that represents the vector and denote it by a lowercase letter $\vec{v} = \langle a_1, a_2\rangle$.

Two vectors $\langle a_1, a_2\rangle$ and $\langle b_1, b_2\rangle$ are **equal** $\iff$ $a_1 = b_1$ and $a_2 = b_2$.

Example 1.2 Sketch the position vectors for:

1 $\vec{a} = \langle -3, 2\rangle$

2 $\vec{b} = \langle 0, -2\rangle$

3 $\vec{c} = \left\langle \dfrac{4}{5}, \dfrac{3}{5}\right\rangle$

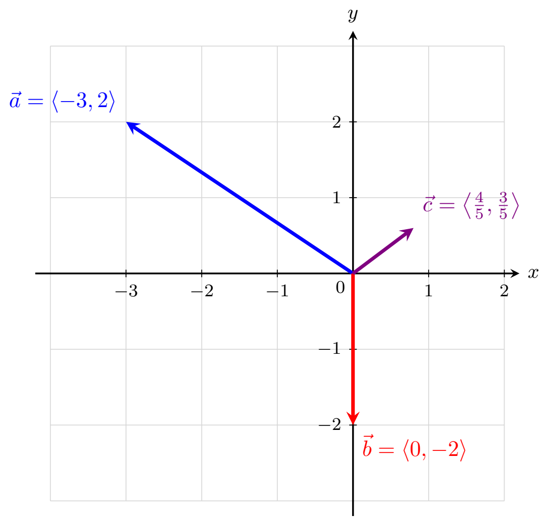


Figure 1.3 The position vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$ drawn in the Cartesian plane.

Solution. To sketch a position vector, always place its initial point at the origin $(0,0)$. The terminal point will match the components of the vector:

1 For $\vec{a} = \langle -3, 2 \rangle$, draw an arrow starting at $(0,0)$ and ending at the point $(-3,2)$.

2 For $\vec{b} = \langle 0, -2 \rangle$, draw an arrow starting at $(0,0)$ and pointing straight down to end at the point $(0,-2)$.

3 For $\vec{c} = \left\langle \frac{4}{5}, \frac{3}{5} \right\rangle$, draw an arrow starting at $(0,0)$ and ending at the point $(0.8, 0.6)$ in the first quadrant.

▲

Definition 1.4 The **magnitude** (or length) $\|\vec{a}\|$ of the vector $\vec{a} = \langle a_1, a_2 \rangle$ is

$$\|\vec{a}\| = \|\langle a_1, a_2 \rangle\| = \sqrt{a_1^2 + a_2^2}.$$

◆

Example 1.5 Find the magnitude of each vector:

1 $\vec{a} = \langle -3, 2 \rangle$

2 $\vec{b} = \langle 0, -2 \rangle$

3 $\vec{c} = \left\langle \dfrac{4}{5}, \dfrac{3}{5} \right\rangle$

Solution. We apply the magnitude formula $\|\vec{v}\| = \sqrt{v_1^2 + v_2^2}$ to each vector:

1 For $\vec{a} = \langle -3, 2 \rangle$:

$$\|\vec{a}\| = \sqrt{(-3)^2 + 2^2} = \sqrt{9+4} = \sqrt{13}$$

2 For $\vec{b} = \langle 0, -2 \rangle$:

$$\|\vec{b}\| = \sqrt{0^2 + (-2)^2} = \sqrt{0+4} = \sqrt{4} = 2$$

3 For $\vec{c} = \left\langle \frac{4}{5}, \frac{3}{5} \right\rangle$:

$$\|\vec{c}\| = \sqrt{\left(\frac{4}{5}\right)^2 + \left(\frac{3}{5}\right)^2} = \sqrt{\frac{16}{25} + \frac{9}{25}} = \sqrt{\frac{25}{25}} = 1$$

Because its magnitude is 1, $\vec{c}$ is a unit vector.

▲

Fact 1.6 Vector Operations. ***Addition of vectors***

$$\langle a_1, a_2 \rangle + \langle b_1, b_2 \rangle = \langle a_1 + b_1, a_2 + b_2 \rangle$$

Scalar multiple of a vector

$$c\langle a_1, a_2 \rangle = \langle ca_1, ca_2 \rangle$$

Example 1.7 Let $\vec{a} = \langle -1, 3 \rangle$ and $\vec{b} = \langle 4, 3 \rangle$.

1 Find $\vec{a} + \vec{b}$ and $-3\vec{a} + 2\vec{b}$.

2 Sketch vectors $\vec{a} + \vec{b}$ and $-\frac{3}{2}\vec{b}$.

Solution.

1 To find $\vec{a} + \vec{b}$, add the corresponding components:

$$\vec{a} + \vec{b} = \langle -1 + 4, 3 + 3 \rangle = \langle 3, 6 \rangle$$

To find $-3\vec{a} + 2\vec{b}$, apply scalar multiplication first, then add:

$$-3\vec{a} = -3\langle -1, 3 \rangle = \langle 3, -9 \rangle$$

$$2\vec{b} = 2\langle 4, 3 \rangle = \langle 8, 6 \rangle$$

$$-3\vec{a} + 2\vec{b} = \langle 3 + 8, -9 + 6 \rangle = \langle 11, -3 \rangle$$

2 First, calculate the scalar multiple: $-\frac{3}{2}\vec{b} = -\frac{3}{2}\langle 4, 3 \rangle = \left\langle -6, -\frac{9}{2} \right\rangle = \langle -6, -4.5 \rangle$.

Below is the sketch of the two position vectors on the coordinate plane:

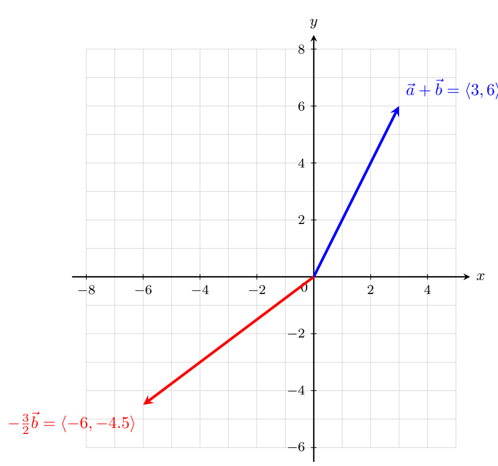


Figure 1.8 The vectors $\vec{a} + \vec{b}$ and $-\frac{3}{2}\vec{b}$.

▲

Definition 1.9 The **zero vector 0** is defined as $\vec{0} = \langle 0, 0\rangle$, and the **negative -a** of a vector $\vec{a} = \langle a_1, a_2\rangle$ is defined as:

$$-\vec{a} = -\langle a_1, a_2\rangle = \langle -a_1, -a_2\rangle$$

◆

Theorem 1.10 Properties of Vectors. *Let $\vec{a}$, $\vec{b}$, and $\vec{c}$ be vectors, and let c and d be scalars. Then:*

1 $\vec{a} + \vec{b} = \vec{b} + \vec{a}$, *and* $c(\vec{a} + \vec{b}) = c\vec{a} + c\vec{b}$

2 $\vec{a} + (\vec{b} + \vec{c}) = (\vec{a} + \vec{b}) + \vec{c}$

3 $\vec{a} + \vec{0} = \vec{a}$, *and* $\vec{a} + (-\vec{a}) = \vec{0}$

4 $(c + d)\vec{a} = c\vec{a} + d\vec{a}$, *and* $(cd)\vec{a} = c(d\vec{a}) = d(c\vec{a})$

5 $1\vec{a} = \vec{a}$, *and* $0\vec{a} = \vec{0} = c\vec{0}$

Definition 1.11 Let $\vec{a} = \langle a_1, a_2\rangle$ and $\vec{b} = \langle b_1, b_2\rangle$. The **difference a-b** is:

$$\vec{a} - \vec{b} = \vec{a} + (-\vec{b}) = \langle a_1 - b_1, a_2 - b_2\rangle$$

◆

Example 1.12 If $\vec{a} = \langle 5, -3\rangle$ and $\vec{b} = \langle -2, 3\rangle$, find:

1 $\vec{a} - \vec{b}$

2 $3\vec{a} - 2\vec{b}$

Solution.

1 To find $\vec{a} - \vec{b}$, subtract the components of $\vec{b}$ from the components of $\vec{a}$:

$$\vec{a} - \vec{b} = \langle 5 - (-2), -3 - 3\rangle = \langle 5 + 2, -6\rangle = \langle 7, -6\rangle$$

2 To find $3\vec{a} - 2\vec{b}$, perform the scalar multiplications first, then subtract:

$$3\vec{a} = 3\langle 5, -3\rangle = \langle 15, -9\rangle$$

$$2\vec{b} = 2\langle -2, 3\rangle = \langle -4, 6\rangle$$

Now subtract the resulting components:

$$3\vec{a} - 2\vec{b} = \langle 15 - (-4), -9 - 6\rangle = \langle 15 + 4, -15\rangle = \langle 19, -15\rangle$$

▲

Theorem 1.13 Vector from Two Points. *If $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ are any two points, the vector $\vec{a}$ in V_2 that corresponds to the directed line segment $\vec{P_1P_2}$ is:*

$$\vec{a} = \langle x_2 - x_1, y_2 - y_1\rangle$$

Example 1.14 Given points $P(3, 2)$ and $Q(-1, 5)$:

1 Sketch $\vec{PQ}$ and $\vec{QP}$.

2 Find vectors $\vec{a}$ and $\vec{b}$ in V_2 that correspond to $\vec{PQ}$ and $\vec{QP}$.

3 Sketch the position vectors for $\vec{a}$ and $\vec{b}$.

Solution.

1 To sketch the directed line segments, draw an arrow starting at point $P(3,2)$ and ending at point $Q(-1,5)$ for $\vec{PQ}$. For $\vec{QP}$, reverse the direction so it starts at $Q(-1,5)$ and ends at $P(3,2)$. (See the left graph below).

2 To find the vector components, subtract the initial coordinates from the terminal coordinates ("terminal minus initial"):

For $\vec{a}$ corresponding to $\vec{PQ}$:

$$\vec{a} = \langle -1-3, 5-2 \rangle = \langle -4, 3 \rangle$$

For $\vec{b}$ corresponding to $\vec{QP}$:

$$\vec{b} = \langle 3-(-1), 2-5 \rangle = \langle 4, -3 \rangle$$

3 To sketch the position vectors $\vec{a}$ and $\vec{b}$, shift them so their initial points sit exactly at the origin $(0,0)$. The terminal points will be $(-4,3)$ and $(4,-3)$, respectively. Notice that they are perfectly parallel to and have the same lengths as the original segments (See the right graph below).

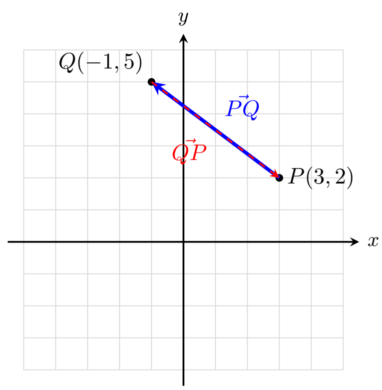


Figure 1.15 The directed line segments $\vec{PQ}$ and $\vec{QP}$ in space.

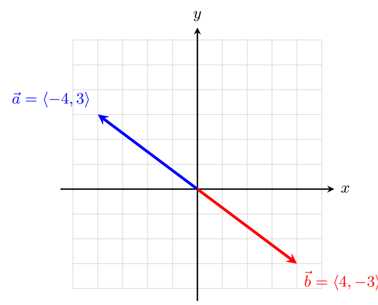


Figure 1.16 The corresponding position vectors $\vec{a}$ and $\vec{b}$ at the origin.

▲

Definition 1.17 Nonzero vectors $\vec{a}$ and $\vec{b}$ in V_2 have:

1 the **same direction** if $\vec{b} = c\vec{a}$ for some scalar $c > 0$

2 the **opposite direction** if $\vec{b} = c\vec{a}$ for some scalar $c < 0$

◆

Theorem 1.18 Magnitude of a Scalar Multiple. *If $\vec{a}$ is a vector and c is a scalar, then $\|c\vec{a}\| = |c|\|\vec{a}\|$.*

Definition 1.19 A vector of length 1 is called a **unit vector**. The **standard unit vectors** are:

$$\vec{i} = \langle 1, 0 \rangle, \quad \vec{j} = \langle 0, 1 \rangle$$

If $\vec{a} = \langle a_1, a_2 \rangle$, then it can be written as a linear combination of the standard unit vectors:

$$\vec{a} = a_1\vec{i} + a_2\vec{j}$$

◆

Example 1.20 If $\vec{a} = -5\vec{i} + 2\vec{j}$ and $\vec{b} = \vec{i} - 3\vec{j}$, express $2\vec{a} - 3\vec{b}$ as a linear combination of $\vec{i}$ and $\vec{j}$.

Solution. Substitute the linear combinations for $\vec{a}$ and $\vec{b}$ into the expression, distribute the scalars, and combine like terms:

$$\begin{aligned} 2\vec{a} - 3\vec{b} &= 2(-5\vec{i} + 2\vec{j}) - 3(\vec{i} - 3\vec{j}) \\ &= (-10\vec{i} + 4\vec{j}) - (3\vec{i} - 9\vec{j}) \\ &= -10\vec{i} - 3\vec{i} + 4\vec{j} + 9\vec{j} \\ &= -13\vec{i} + 13\vec{j} \end{aligned}$$

▲

Theorem 1.21 Unit Vector in the Same Direction. *A* ***unit vector*** *is a vector of magnitude 1. The vectors $\vec{i}$ and $\vec{j}$ are unit vectors. If $\vec{a} \neq \vec{0}$, then the unit vector $\vec{u}$ that has the same direction as $\vec{a}$ is:*

$$\vec{u} = \frac{1}{\|\vec{a}\|}\vec{a}$$

Example 1.22 Find the unit vector $\vec{u}$ that has the same direction as $\vec{a} = -8\vec{i} + 15\vec{j}$.

Solution. First, find the magnitude of vector $\vec{a}$:

$$\|\vec{a}\| = \sqrt{(-8)^2 + 15^2} = \sqrt{64 + 225} = \sqrt{289} = 17$$

Next, multiply the reciprocal of the magnitude by the original vector $\vec{a}$:

$$\vec{u} = \frac{1}{17}(-8\vec{i} + 15\vec{j}) = -\frac{8}{17}\vec{i} + \frac{15}{17}\vec{j}$$

▲

Example 1.23 A quarterback releases the football with a velocity of 50 ft/sec at an angle of 35° with the horizontal. Approximate the horizontal and vertical components of the vector.

Solution. A vector $\vec{v}$ given a magnitude $\|\vec{v}\|$ and a direction angle θ can be written in component form as $\vec{v} = \langle \|\vec{v}\|\cos(\theta), \|\vec{v}\|\sin(\theta)\rangle$.

Given $\|\vec{v}\| = 50$ and $\theta = 35°$:

- Horizontal component: $50\cos(35°) \approx 40.96$ ft/sec
- Vertical component: $50\sin(35°) \approx 28.68$ ft/sec

In component form, the velocity vector is approximately $\langle 40.96, 28.68\rangle$. ▲

Example 1.24 A jet airplane approaches a runway at an angle of 7.5° with the horizontal, traveling at a velocity of 160 mi/hr. Approximate the horizontal and vertical components of the vector.

Solution. Given a magnitude of $\|\vec{v}\| = 160$ and an approach angle of $\theta = 7.5°$:

- Horizontal component: $160\cos(7.5°) \approx 158.63$ mi/hr
- Vertical component: $160\sin(7.5°) \approx 20.88$ mi/hr

In component form, the velocity vector is approximately $\langle 158.63, 20.88\rangle$. Note that because the plane is descending, the vertical velocity component can be interpreted as acting in the downward direction (-20.88). ▲

1.2 Vectors in Three Dimensions

In order to represent points in space, we first choose a fixed point O (the origin) and three directed lines through O which are perpendicular to each other, called the **coordinate axes**. They are labeled as the x**-axis**, y**-axis**, and z**-axis**. Think of the x- and y-axes as being horizontal while the z-axis is vertical. The direction of the z-axis is determined by the **right-hand rule**.

The three coordinate axes determine the three **coordinate planes**. The xy-plane contains the x- and y-axes, the yz-plane contains the y- and z-axes, and the xz-plane contains the x- and z-axes. These three coordinate planes divide space into eight parts, called **octants**. The **first octant** is determined by the positive axes.

We represent a point P by the ordered triple (a, b, c) of real numbers and we call a, b, and c the **coordinates** of P; here, a, b, and c are the x-, y-, and z-coordinates, respectively.

The point $P(a, b, c)$ can be visualized as a corner of a rectangular box. If we drop a perpendicular from P to the xy-plane, we get a point Q with coordinates $(a, b, 0)$ called the **projection** of P on the xy-plane.

The Cartesian product $\mathbb{R} \times \mathbb{R} \times \mathbb{R} = \{(x, y, z) \mid x, y, z \in \mathbb{R}\}$ is the set of all ordered triples of real numbers and is denoted by $\mathbb{R}^3$.

Example 1.25 What surfaces in $\mathbb{R}^3$ are represented by the following equations?

1 $z = 3$

2 $y = 5$

Solution.

1 The equation $z = 3$ represents the set of all points $\{(x, y, z) \mid z = 3\}$ in $\mathbb{R}^3$. Since x and y can be any real numbers, this forms a horizontal plane parallel to the xy-plane, located exactly 3 units above it.

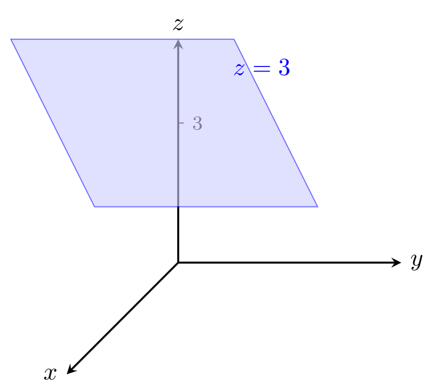


Figure 1.26 The plane $z = 3$ in $\mathbb{R}^3$.

2 The equation $y = 5$ represents the set of all points $\{(x, y, z) \mid y = 5\}$ in $\mathbb{R}^3$. Since x and z can be any real numbers, this forms a vertical plane parallel to the xz-plane, located exactly 5 units to the right of it.

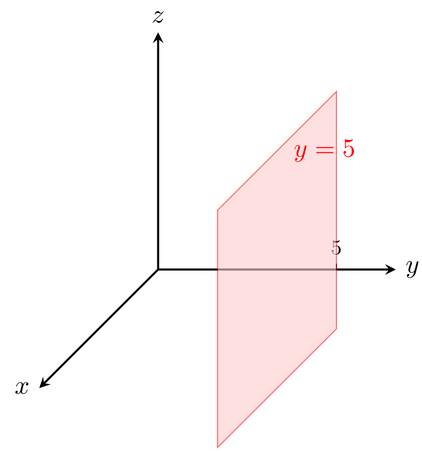


Figure 1.27 The plane $y = 5$ in $\mathbb{R}^3$.

▲

Fact 1.28 Distance Formula in Three Dimensions. *The distance $d(P_1, P_2) = |P_1P_2|$ between the points $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ is:*

$$d(P_1, P_2) = |P_1P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Example 1.29 Find the distance from the point $P(-1, -3, 1)$ to the point $Q(3, 4, -2)$.

Solution. Substitute the coordinates of P and Q into the three-dimensional distance formula:

$$\begin{aligned} |PQ| &= \sqrt{(3 - (-1))^2 + (4 - (-3))^2 + (-2 - 1)^2} \\ &= \sqrt{(3 + 1)^2 + (4 + 3)^2 + (-3)^2} \\ &= \sqrt{4^2 + 7^2 + (-3)^2} \\ &= \sqrt{16 + 49 + 9} \\ &= \sqrt{74} \end{aligned}$$

The exact distance between the points is $\sqrt{74}$. ▲

Fact 1.30 Equation of a Sphere. *An equation of a sphere with center $C(h, k, l)$ and radius r is:*

$$(x - h)^2 + (y - k)^2 + (z - l)^2 = r^2$$

If the center is at the origin $O(0, 0, 0)$, then the equation of the sphere simplifies to:

$$x^2 + y^2 + z^2 = r^2$$

Example 1.31 Show that $x^2 + y^2 + z^2 + 4x - 6y + 2z + 6 = 0$ is the equation of a sphere, and find its center and radius.

Solution. To transform the given equation into standard form, group the variables and complete the square for each variable:

$$\begin{aligned} (x^2 + 4x) + (y^2 - 6y) + (z^2 + 2z) &= -6 \\ \left(x^2 + 4x + \left(\frac{4}{2}\right)^2\right) + \left(y^2 - 6y + \left(\frac{-6}{2}\right)^2\right) + \left(z^2 + 2z + \left(\frac{2}{2}\right)^2\right) &= -6 + 4 + 9 + 1 \\ (x^2 + 4x + 4) + (y^2 - 6y + 9) + (z^2 + 2z + 1) &= 8 \end{aligned}$$

$$(x+2)^2 + (y-3)^2 + (z+1)^2 = 8$$

Comparing this result to the standard equation of a sphere $(x-h)^2 + (y-k)^2 + (z-l)^2 = r^2$, we find:

- The center of the sphere is $C(h,k,l) = C(-2,3,-1)$.
- The radius is $r = \sqrt{8} = 2\sqrt{2}$.

▲

Example 1.32 What region in $\mathbb{R}^3$ is represented by the following inequalities?

$$1 \le x^2 + y^2 + z^2 \le 4$$

Solution. The expression $x^2 + y^2 + z^2 = r^2$ represents the squared distance from the origin to a point in three-dimensional space. We can rewrite the inequality by taking the square root of all parts:

$$1 \le \sqrt{x^2 + y^2 + z^2} \le 2$$

This represents the set of all points whose distance r from the origin satisfies $1 \le r \le 2$. Therefore, the region is a solid **spherical shell** (or a thick hollow sphere) centered at the origin, including the inner sphere of radius 1 and the outer sphere of radius 2.

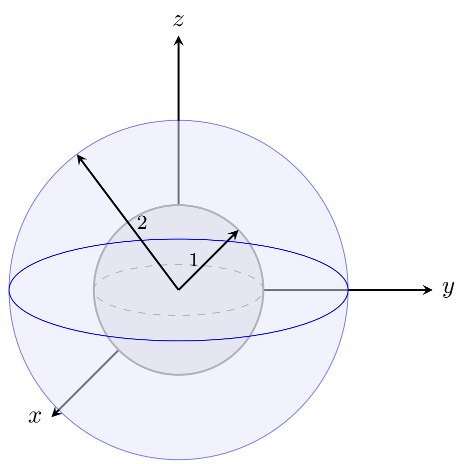


Figure 1.33 The spherical shell bounded by $r = 1$ and $r = 2$.

▲

Example 1.34 What region in $\mathbb{R}^3$ is represented by the following inequalities?

$$x^2 + y^2 \le 25, \quad |z| \le 3$$

Solution. We analyze the two conditions independently to find their intersection:

- In the xy-plane, $x^2 + y^2 \le 25$ describes a solid disk of radius 5 centered at the origin. Since z is unconstrained by this expression, it forms an infinitely long solid cylinder extending along the z-axis.
- The absolute value inequality $|z| \le 3$ is equivalent to $-3 \le z \le 3$. This bounds the space vertically between two horizontal planes.

Combining these conditions restricts the infinite cylinder to a specific height. Therefore, the region is a **solid vertical cylinder** of radius 5 and height 6, centered along the z-axis and extending from $z = -3$ to $z = 3$.

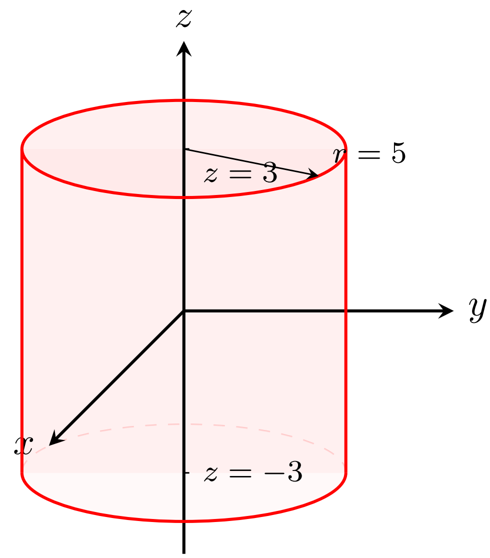


Figure 1.35 The solid vertical cylinder bounded by $x^2 + y^2 \leq 25$ and $-3 \leq z \leq 3$.

▲

Fact 1.36 Vector Operations in Three Dimensions. *Let* $\vec{a} = \langle a_1, a_2, a_3 \rangle$ *and* $\vec{b} = \langle b_1, b_2, b_3 \rangle$ *be vectors in* $\mathbb{R}^3$.

1 ***Addition of vectors:***

$$\vec{a} + \vec{b} = \langle a_1 + b_1, a_2 + b_2, a_3 + b_3 \rangle$$

2 ***Scalar multiple of a vector:***

$$c\vec{a} = c\langle a_1, a_2, a_3 \rangle = \langle ca_1, ca_2, ca_3 \rangle$$

3 *The* ***zero vector 0****:*

$$\vec{0} = \langle 0, 0, 0 \rangle$$

4 *The* ***negative -a*** *of a vector* $\vec{a}$*:*

$$-\vec{a} = -\langle a_1, a_2, a_3 \rangle = \langle -a_1, -a_2, -a_3 \rangle$$

5 *The* ***difference a-b****:*

$$\vec{a} - \vec{b} = \vec{a} + (-\vec{b}) = \langle a_1 - b_1, a_2 - b_2, a_3 - b_3 \rangle$$

6 *The length of* $\vec{a} = \langle a_1, a_2, a_3 \rangle$*:*

$$\|\vec{a}\| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

7 *If* $P_1(x_1, y_1, z_1)$ *and* $P_2(x_2, y_2, z_2)$ *are any points, the vector* $\vec{a}$ *in* $\mathbb{R}^3$ *that corresponds to* $\vec{P_1P_2}$ *is:*

$$\vec{a} = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle$$

Definition 1.37 The **standard basis vectors** in three dimensions are:

$$\vec{i} = \langle 1, 0, 0\rangle, \quad \vec{j} = \langle 0, 1, 0\rangle, \quad \vec{k} = \langle 0, 0, 1\rangle$$

If $\vec{a} = \langle a_1, a_2, a_3\rangle$, then it can be written as a linear combination of the standard unit basis vectors:

$$\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$$

◆

Example 1.38 If $\vec{a} = \vec{i} + 2\vec{j} - 3\vec{k}$ and $\vec{b} = 4\vec{i} + 7\vec{k}$, express the vector $2\vec{a} + 3\vec{b}$ in terms of $\vec{i}$, $\vec{j}$, and $\vec{k}$.

Solution. Substitute the given expressions for $\vec{a}$ and $\vec{b}$, distribute the scalars, and group the components:

$$\begin{aligned} 2\vec{a} + 3\vec{b} &= 2(\vec{i} + 2\vec{j} - 3\vec{k}) + 3(4\vec{i} + 0\vec{j} + 7\vec{k}) \\ &= (2\vec{i} + 4\vec{j} - 6\vec{k}) + (12\vec{i} + 0\vec{j} + 21\vec{k}) \\ &= (2 + 12)\vec{i} + (4 + 0)\vec{j} + (-6 + 21)\vec{k} \\ &= 14\vec{i} + 4\vec{j} + 15\vec{k} \end{aligned}$$

▲

Example 1.39 Find the unit vector in the direction of the vector $\vec{v} = 2\vec{i} - \vec{j} - 2\vec{k}$.

Solution. First, find the magnitude of the given vector $\vec{v}$:

$$\|\vec{v}\| = \sqrt{2^2 + (-1)^2 + (-2)^2} = \sqrt{4 + 1 + 4} = \sqrt{9} = 3$$

Next, multiply the reciprocal of the magnitude by the vector $\vec{v}$ to produce the unit vector $\vec{u}$:

$$\vec{u} = \frac{1}{3}(2\vec{i} - \vec{j} - 2\vec{k}) = \frac{2}{3}\vec{i} - \frac{1}{3}\vec{j} - \frac{2}{3}\vec{k}$$

▲

Example 1.40 A 100-lb weight hangs from two wires making angles of 50° and 32° with the horizontal. Find the tensions (forces) $\vec{T}_1$ and $\vec{T}_2$ in both wires and their magnitudes.

Solution. We express the tension vectors in terms of their horizontal and vertical components. Let $\vec{T}_1$ pull up and to the left, and $\vec{T}_2$ pull up and to the right:

$$\begin{aligned} \vec{T}_1 &= -\|\vec{T}_1\|\cos(50°)\vec{i} + \|\vec{T}_1\|\sin(50°)\vec{j} \\ \vec{T}_2 &= \|\vec{T}_2\|\cos(32°)\vec{i} + \|\vec{T}_2\|\sin(32°)\vec{j} \end{aligned}$$

The downward force due to the weight is $\vec{w} = -100\vec{j}$. For the system to achieve static equilibrium, the resultant sum of the tensions must exactly counterbalance the weight:

$$\vec{T}_1 + \vec{T}_2 = -\vec{w} = 100\vec{j}$$

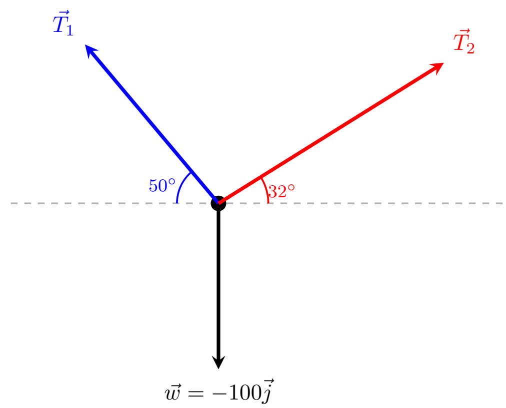


Figure 1.41 Force vector diagram for the hanging weight system.

Equating the horizontal ($\vec{i}$) and vertical ($\vec{j}$) components gives a system of linear equations:

$$-\|\vec{T}_1\|\cos(50°) + \|\vec{T}_2\|\cos(32°) = 0$$
$$\|\vec{T}_1\|\sin(50°) + \|\vec{T}_2\|\sin(32°) = 100$$

From the first equation, we find $\|\vec{T}_2\| = \|\vec{T}_1\|\frac{\cos(50°)}{\cos(32°)}$. Substituting this relationship into the second equation yields:

$$\|\vec{T}_1\|\sin(50°) + \|\vec{T}_1\|\frac{\cos(50°)\sin(32°)}{\cos(32°)} = 100$$

$$\|\vec{T}_1\|\left(\sin(50°) + \cos(50°)\tan(32°)\right) = 100$$

Solving for the magnitudes yields:

- $\|\vec{T}_1\| \approx 85.64$ lb
- $\|\vec{T}_2\| = 85.64 \cdot \dfrac{\cos(50°)}{\cos(32°)} \approx 64.91$ lb

Substituting these scalar magnitudes back into our component formulas provides the final tension vector models:

$$\vec{T}_1 \approx -55.05\vec{i} + 65.60\vec{j}$$

$$\vec{T}_2 \approx 55.05\vec{i} + 34.40\vec{j}$$

▲

1.3 The Dot Product

Suppose a constant **force** is represented by a vector $\vec{F} = \vec{PR}$ pointing in some direction other than the direction of motion. If this force moves an object from point P to point Q, then the **displacement vector** is $\vec{D} = \vec{PQ}$.

The **work** done by $\vec{F}$ is defined as the magnitude of the displacement $\vec{D}$, multiplied by the magnitude of the applied force in the direction of the motion. Therefore, the work done by $\vec{F}$ is defined to be:

$$W = \|\vec{D}\|(\|\vec{F}\|\cos(\theta)) = \|\vec{D}\|\|\vec{F}\|\cos(\theta)$$

where θ is the angle between the force and displacement vectors.

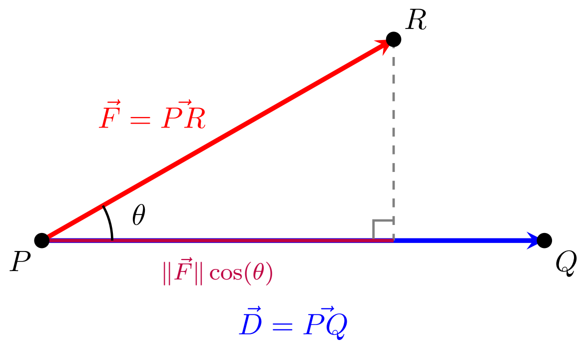


Figure 1.42 The geometric projection of a force vector $\vec{F}$ along a displacement vector $\vec{D}$.

Definition 1.43 The **dot product** of two vectors $\vec{a}$ and $\vec{b}$ is the number:

$$\vec{a} \cdot \vec{b} = \|\vec{a}\|\|\vec{b}\|\cos(\theta)$$

where θ is the angle between $\vec{a}$ and $\vec{b}$, such that $0 \leq \theta \leq \pi$. Thus, θ is the smallest angle between the vectors when they are drawn with the same initial point. If either $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$, we define $\vec{a} \cdot \vec{b} = 0$. ◆

The dot product is a real number, that is, a scalar. For this reason, the dot product is sometimes called the **scalar product**.

Example 1.44 If the vectors $\vec{a}$ and $\vec{b}$ have lengths 4 and 6, and the angle between them is $\pi/3$, find $\vec{a} \cdot \vec{b}$.

Solution. Substitute the given magnitudes and the angle into the geometric definition of the dot product:

$$\begin{aligned} \vec{a} \cdot \vec{b} &= \|\vec{a}\|\|\vec{b}\|\cos(\theta) \\ &= (4)(6)\cos\left(\frac{\pi}{3}\right) \\ &= 24 \cdot \left(\frac{1}{2}\right) \\ &= 12 \end{aligned}$$

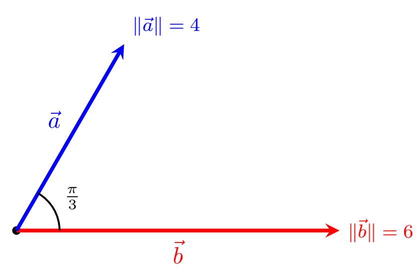


Figure 1.45 Two vectors separated by an angle of $\pi/3$.

▲

Example 1.46 A crate is hauled 8 m up a ramp under a constant force of 200 N applied at an angle of 25° to the ramp. Find the work done.

Solution. The work done by a constant force is given by the formula $W = \|\vec{F}\|\|\vec{D}\|\cos(\theta)$, which corresponds exactly to the dot product of the force vector and the displacement vector $\vec{F} \cdot \vec{D}$.

Given $\|\vec{F}\| = 200$ N, $\|\vec{D}\| = 8$ m, and $\theta = 25°$:

$$\begin{aligned} W &= (200)(8)\cos(25°) \\ &= 1600\cos(25°) \\ &\approx 1450.09 \text{ J} \end{aligned}$$

The work done in hauling the crate is approximately 1450.09 Joules (or Newton-meters).

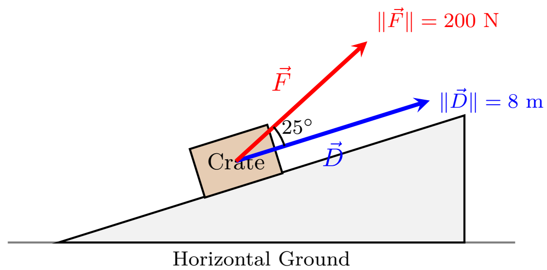


Figure 1.47 Force vector acting at an angle to the displacement along a ramp.

▲

Corollary 1.48 *Two nonzero vectors $\vec{a}$ and $\vec{b}$ are called* ***perpendicular*** *or* ***orthogonal*** *if the angle between them is $\theta = \pi/2$. In this case:*

$$\vec{a} \cdot \vec{b} = \|\vec{a}\|\|\vec{b}\|\cos(\pi/2) = 0$$

Conversely, if $\vec{a} \cdot \vec{b} = 0$ then $\cos(\theta) = 0$, which implies $\theta = \pi/2$. Therefore, two vectors $\vec{a}$ and $\vec{b}$ are orthogonal if and only if $\vec{a} \cdot \vec{b} = 0$.

Proof. By definition, the geometric dot product is given by $\vec{a} \cdot \vec{b} = \|\vec{a}\|\|\vec{b}\|\cos(\theta)$, where $0 \leq \theta \leq \pi$.

For the forward direction, assume $\vec{a}$ and $\vec{b}$ are orthogonal. By definition, the angle between them is $\theta = \pi/2$. Substituting this value yields:

$$\vec{a} \cdot \vec{b} = \|\vec{a}\|\|\vec{b}\|\cos\left(\frac{\pi}{2}\right) = \|\vec{a}\|\|\vec{b}\|(0) = 0$$

For the reverse direction, assume $\vec{a} \cdot \vec{b} = 0$ for two nonzero vectors. Since the vectors are nonzero, their magnitudes are strictly positive ($\|\vec{a}\| \neq 0$ and $\|\vec{b}\| \neq 0$). We can divide both sides of the definition by these magnitudes:

$$0 = \|\vec{a}\|\|\vec{b}\|\cos(\theta) \implies \cos(\theta) = 0$$

Within the restricted interval $0 \leq \theta \leq \pi$, the equation $\cos(\theta) = 0$ has a unique solution, $\theta = \pi/2$. Thus, the vectors are orthogonal. ■

Corollary 1.49 *The sign of the dot product $\vec{a} \cdot \vec{b}$ provides geometric information about the relative directions of the vectors:*

1. *It is positive if $\vec{a}$ and $\vec{b}$ point in the same general direction ($0 \leq \theta < \pi/2$).*

2. *It is 0 if they are perpendicular ($\theta = \pi/2$).*

3. *It is negative if $\vec{a}$ and $\vec{b}$ point in generally opposite directions ($\pi/2 < \theta \leq \pi$).*

4. *If $\vec{a}$ and $\vec{b}$ point in exactly the same direction, then $\theta = 0$, which means*

$\cos(\theta) = 1$ *and:*

$$\vec{a} \cdot \vec{b} = \|\vec{a}\|\|\vec{b}\|$$

5 *If $\vec{a}$ and $\vec{b}$ point in exactly the opposite direction, then $\theta = \pi$, which means* $\cos(\theta) = -1$ *and:*

$$\vec{a} \cdot \vec{b} = -\|\vec{a}\|\|\vec{b}\|$$

Proof. Because the magnitudes $\|\vec{a}\|$ and $\|\vec{b}\|$ of two nonzero vectors are always strictly positive, the sign of the product $\|\vec{a}\|\|\vec{b}\|\cos(\theta)$ is completely determined by the sign of the trigonometric term $\cos(\theta)$ on the interval $[0, \pi]$:

1 When $0 \leq \theta < \pi/2$ (an acute angle), $\cos(\theta) > 0$, rendering the dot product positive.

2 When $\theta = \pi/2$ (a right angle), $\cos(\theta) = 0$, rendering the dot product zero.

3 When $\pi/2 < \theta \leq \pi$ (an obtuse angle), $\cos(\theta) < 0$, rendering the dot product negative.

4 If they point in the exact same direction, $\theta = 0$. Since $\cos(0) = 1$, the expression simplifies directly to $\|\vec{a}\|\|\vec{b}\|$.

5 If they point in the exact opposite direction, $\theta = \pi$. Since $\cos(\pi) = -1$, the expression simplifies directly to $-\|\vec{a}\|\|\vec{b}\|$.

■

The Dot Product in Component Form

Suppose $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$ are given in component forms. If we apply the Law of Cosines to the triangle formed by these vectors, we obtain:

$$\|\vec{a} - \vec{b}\|^2 = \|\vec{a}\|^2 + \|\vec{b}\|^2 - 2\|\vec{a}\|\|\vec{b}\|\cos(\theta) = \|\vec{a}\|^2 + \|\vec{b}\|^2 - 2\vec{a} \cdot \vec{b}$$

Fact 1.50 Algebraic Dot Product Formula. *The dot product of $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$ is:*

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

Example 1.51 Find the dot product $\vec{a} \cdot \vec{b}$ for each of the following cases:

1 $\vec{a} = \langle 2, -3, 4 \rangle$, $\vec{b} = \langle 6, 2, -\frac{1}{2} \rangle$

2 $\vec{a} = 4\vec{j} - 3\vec{k}$, $\vec{b} = 2\vec{i} + 4\vec{j} + 6\vec{k}$

3 $\|\vec{a}\| = 6$, $\|\vec{b}\| = 5$, and the angle between $\vec{a}$ and $\vec{b}$ is $2\pi/3$.

Solution.

1 Multiply corresponding components and add them together:

$$\vec{a} \cdot \vec{b} = (2)(6) + (-3)(2) + (4)\left(-\frac{1}{2}\right) = 12 - 6 - 2 = 4$$

2 First write $\vec{a}$ in full component form as $\langle 0, 4, -3 \rangle$ and $\vec{b}$ as $\langle 2, 4, 6 \rangle$:

$$\vec{a} \cdot \vec{b} = (0)(2) + (4)(4) + (-3)(6) = 0 + 16 - 18 = -2$$

3 Since components are unknown but the angle is provided, use the geometric definition:

$$\vec{a} \cdot \vec{b} = \|\vec{a}\|\|\vec{b}\| \cos(\theta) = (6)(5) \cos\left(\frac{2\pi}{3}\right) = 30\left(-\frac{1}{2}\right) = -15$$

▲

Example 1.52 Show that $2\vec{i} + 2\vec{j} - \vec{k}$ is perpendicular to $5\vec{i} - 4\vec{j} + 2\vec{k}$.

Solution. Two vectors are perpendicular (orthogonal) if and only if their dot product equals zero. Let $\vec{a} = \langle 2, 2, -1\rangle$ and $\vec{b} = \langle 5, -4, 2\rangle$:

$$\vec{a} \cdot \vec{b} = (2)(5) + (2)(-4) + (-1)(2) = 10 - 8 - 2 = 0$$

Because the dot product is exactly 0, the vectors are perpendicular. ▲

Example 1.53 Find the angle between the vectors $\vec{a} = \langle 2, 2, -1\rangle$ and $\vec{b} = \langle 5, -3, 2\rangle$.

Solution. We use the alternate formula for the angle between vectors: $\cos(\theta) = \frac{\vec{a}\cdot\vec{b}}{\|\vec{a}\|\|\vec{b}\|}$.

First compute the dot product and individual vector magnitudes:

$$\vec{a} \cdot \vec{b} = (2)(5) + (2)(-3) + (-1)(2) = 10 - 6 - 2 = 2$$

$$\|\vec{a}\| = \sqrt{2^2 + 2^2 + (-1)^2} = \sqrt{4 + 4 + 1} = \sqrt{9} = 3$$

$$\|\vec{b}\| = \sqrt{5^2 + (-3)^2 + 2^2} = \sqrt{25 + 9 + 4} = \sqrt{38}$$

Substitute these calculations back into the cosine expression:

$$\cos(\theta) = \frac{2}{3\sqrt{38}} \implies \theta = \arccos\left(\frac{2}{3\sqrt{38}}\right) \approx 1.46 \text{ radians (or } 83.8^\circ)$$

▲

Proposition 1.54 Properties of the Dot Product. *If $\vec{a}$, $\vec{b}$, and $\vec{c}$ are vectors in $\mathbb{R}^3$ and c is a scalar, then:*

1 $\vec{a} \cdot \vec{a} = \|\vec{a}\|^2$

2 $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$

3 $\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$

4 $(c\vec{a}) \cdot \vec{b} = c(\vec{a} \cdot \vec{b}) = \vec{a} \cdot (c\vec{b})$

5 $\vec{0} \cdot \vec{a} = \vec{a} \cdot \vec{0} = 0$

Proof. Let $\vec{a} = \langle a_1, a_2, a_3\rangle$, $\vec{b} = \langle b_1, b_2, b_3\rangle$, and $\vec{c} = \langle c_1, c_2, c_3\rangle$.

1 By the component formula for the dot product:

$$\vec{a} \cdot \vec{a} = a_1a_1 + a_2a_2 + a_3a_3 = a_1^2 + a_2^2 + a_3^2 = (\|\vec{a}\|)^2 = \|\vec{a}\|^2$$

2 Since multiplication of real numbers is commutative ($a_ib_i = b_ia_i$):

$$\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2 + a_3b_3 = b_1a_1 + b_2a_2 + b_3a_3 = \vec{b} \cdot \vec{a}$$

3 Using the distributive property of real numbers:

$$\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \langle b_1 + c_1, b_2 + c_2, b_3 + c_3\rangle$$

$$\begin{aligned}
&= a_1(b_1 + c_1) + a_2(b_2 + c_2) + a_3(b_3 + c_3) \\
&= (a_1b_1 + a_1c_1) + (a_2b_2 + a_2c_2) + (a_3b_3 + a_3c_3) \\
&= (a_1b_1 + a_2b_2 + a_3b_3) + (a_1c_1 + a_2c_2 + a_3c_3) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}
\end{aligned}$$

4 By factoring the real scalar c from the algebraic terms:

$$(c\vec{a}) \cdot \vec{b} = (ca_1)b_1 + (ca_2)b_2 + (ca_3)b_3 = c(a_1b_1 + a_2b_2 + a_3b_3) = c(\vec{a} \cdot \vec{b})$$

A symmetric argument shows this is also equal to $\vec{a} \cdot (c\vec{b})$.

5 Since multiplying any real number by zero equals zero:

$$\vec{0} \cdot \vec{a} = (0)(a_1) + (0)(a_2) + (0)(a_3) = 0 + 0 + 0 = 0$$

■

Theorem 1.55 Cauchy-Schwarz Inequality. *For any two vectors $\vec{a}$ and $\vec{b}$:*

$$|\vec{a} \cdot \vec{b}| \leq \|\vec{a}\|\|\vec{b}\|$$

Proof. If either $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$, then both sides of the inequality are exactly 0, making the statement trivially true.

Assume both vectors are nonzero. We can express the dot product using its geometric definition:

$$\vec{a} \cdot \vec{b} = \|\vec{a}\|\|\vec{b}\| \cos(\theta)$$

Taking the absolute value of both sides yields:

$$|\vec{a} \cdot \vec{b}| = |\|\vec{a}\|\|\vec{b}\| \cos(\theta)| = \|\vec{a}\|\|\vec{b}\|| \cos(\theta)|$$

Since the cosine function is naturally bounded by the range of real values $-1 \leq \cos(\theta) \leq 1$, its absolute value parameter satisfies $|\cos(\theta)| \leq 1$. Substituting this boundary restriction into our equation produces the target proof expression:

$$|\vec{a} \cdot \vec{b}| \leq \|\vec{a}\|\|\vec{b}\|(1) \implies |\vec{a} \cdot \vec{b}| \leq \|\vec{a}\|\|\vec{b}\|$$

■

Theorem 1.56 Triangle Inequality. *For any two vectors $\vec{a}$ and $\vec{b}$:*

$$\|\vec{a} + \vec{b}\| \leq \|\vec{a}\| + \|\vec{b}\|$$

Proof. Because both sides of the target inequality are non-negative real lengths, we can square the left-hand side and evaluate it using dot product properties:

$$\begin{aligned}
\|\vec{a} + \vec{b}\|^2 &= (\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b}) \\
&= \vec{a} \cdot \vec{a} + \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{a} + \vec{b} \cdot \vec{b} \\
&= \|\vec{a}\|^2 + 2(\vec{a} \cdot \vec{b}) + \|\vec{b}\|^2
\end{aligned}$$

Since a real number is always less than or equal to its absolute value ($x \leq |x|$), we can state that $\vec{a} \cdot \vec{b} \leq |\vec{a} \cdot \vec{b}|$. Applying the Cauchy-Schwarz Inequality ($|\vec{a} \cdot \vec{b}| \leq \|\vec{a}\|\|\vec{b}\|$) gives:

$$\begin{aligned}
\|\vec{a} + \vec{b}\|^2 &\leq \|\vec{a}\|^2 + 2|\vec{a} \cdot \vec{b}| + \|\vec{b}\|^2 \\
&\leq \|\vec{a}\|^2 + 2\|\vec{a}\|\|\vec{b}\| + \|\vec{b}\|^2
\end{aligned}$$

The right side is a perfect square trinomial, which factors into scalar lengths:

$$\|\vec{a}+\vec{b}\|^2 \leq (\|\vec{a}\|+\|\vec{b}\|)^2$$

Taking the positive square root of both sides completes the proof:

$$\|\vec{a}+\vec{b}\| \leq \|\vec{a}\|+\|\vec{b}\|$$

■

Projections

Let $\vec{PQ}$ and $\vec{PR}$ be representations of two vectors $\vec{a}$ and $\vec{b}$ sharing the same initial point P. If S is the foot of the perpendicular dropped from R to the line containing $\vec{PQ}$, then the vector represented by $\vec{PS}$ is called the **vector projection** of $\vec{b}$ onto $\vec{a}$ and is denoted by $\operatorname{proj}_{\vec{a}} \vec{b}$ (think of this as the shadow cast by $\vec{b}$ onto $\vec{a}$).

The **scalar projection** of $\vec{b}$ onto $\vec{a}$ (also called the **component of b along a**) is the real number $\|\vec{b}\|\cos(\theta)$, where θ is the angle between $\vec{a}$ and $\vec{b}$. This value is denoted by $\operatorname{comp}_{\vec{a}} \vec{b}$. The equation:

$$\vec{a}\cdot\vec{b} = \|\vec{a}\|\|\vec{b}\|\cos(\theta) = \|\vec{a}\|(\|\vec{b}\|\cos(\theta))$$

shows that the dot product of $\vec{a}$ and $\vec{b}$ can be interpreted as the length of $\vec{a}$ multiplied by the scalar projection of $\vec{b}$ onto $\vec{a}$. Since:

$$\|\vec{b}\|\cos(\theta) = \frac{\vec{a}\cdot\vec{b}}{\|\vec{a}\|} = \frac{\vec{a}}{\|\vec{a}\|}\cdot\vec{b}$$

Fact 1.57 Projection Formulas. *Let $\vec{a}$ and $\vec{b}$ be vectors with $\vec{a} \neq \vec{0}$.*

- ***Scalar projection of $\vec{b}$ onto $\vec{a}$:***

$$\operatorname{comp}_{\vec{a}} \vec{b} = \frac{\vec{a}\cdot\vec{b}}{\|\vec{a}\|}$$

- ***Vector projection of $\vec{b}$ onto $\vec{a}$:***

$$\operatorname{proj}_{\vec{a}} \vec{b} = \left(\frac{\vec{a}\cdot\vec{b}}{\|\vec{a}\|}\right)\frac{\vec{a}}{\|\vec{a}\|} = \frac{\vec{a}\cdot\vec{b}}{\|\vec{a}\|^2}\vec{a}$$

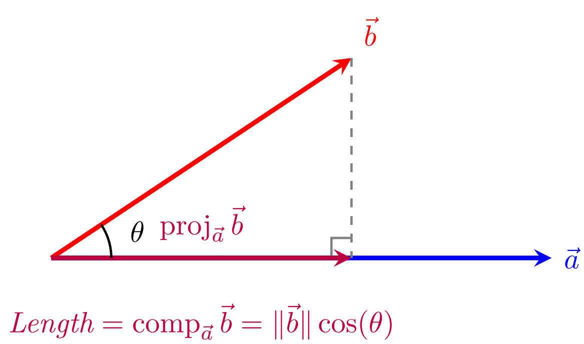


Figure 1.58 *The geometric relationship between vector projection and scalar projection.*

Example 1.59 Find the scalar projection and vector projection of $\vec{b} = \langle 1, 1, 2\rangle$ onto $\vec{a} = \langle -2, 3, 1\rangle$.

Solution. First, compute the core values required by the projection formulas ($\vec{a} \cdot \vec{b}$ and $\|\vec{a}\|^2$):

$$\vec{a} \cdot \vec{b} = (-2)(1) + (3)(1) + (1)(2) = -2 + 3 + 2 = 3$$

$$\|\vec{a}\|^2 = (-2)^2 + 3^2 + 1^2 = 4 + 9 + 1 = 14 \implies \|\vec{a}\| = \sqrt{14}$$

Now apply these numbers to the scalar projection formula:

$$\text{comp}_{\vec{a}}\,\vec{b} = \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\|} = \frac{3}{\sqrt{14}}$$

Next, apply the values to the vector projection formula to scale the base direction vector $\vec{a}$:

$$\text{proj}_{\vec{a}}\,\vec{b} = \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\|^2}\vec{a} = \frac{3}{14}\langle -2, 3, 1\rangle = \left\langle -\frac{3}{7}, \frac{9}{14}, \frac{3}{14}\right\rangle$$

▲

Checkpoint 1.60 Find the direction angle between the vector $\vec{a} = \langle 1, 2, 3\rangle$ and the x-axis.

Solution. The direction angle relative to the positive x-axis matches the angle between vector $\vec{a}$ and the standard basis unit vector $\vec{i} = \langle 1, 0, 0\rangle$:

$$\vec{a} \cdot \vec{i} = (1)(1) + (2)(0) + (3)(0) = 1$$

$$\|\vec{a}\| = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{1 + 4 + 9} = \sqrt{14}$$
$$\|\vec{i}\| = 1$$

Now compute the angle tracking parameter α:

$$\cos(\alpha) = \frac{\vec{a} \cdot \vec{i}}{\|\vec{a}\|\|\vec{i}\|} = \frac{1}{\sqrt{14}} \implies \alpha = \arccos\left(\frac{1}{\sqrt{14}}\right) \approx 74.5^\circ$$

Checkpoint 1.61 A force is given by a vector $\vec{F} = 3\vec{i} + 4\vec{j} + 5\vec{k}$ and moves a particle from the point $P(2, 1, 0)$ to the point $Q(4, 6, 2)$. Find the work done.

Solution. First, determine the displacement vector $\vec{D}$ from P to Q using terminal minus initial coordinates:

$$\vec{D} = \vec{PQ} = \langle 4 - 2, 6 - 1, 2 - 0\rangle = \langle 2, 5, 2\rangle$$

Work is computed as the dot product of the force vector $\vec{F} = \langle 3, 4, 5\rangle$ and this displacement vector $\vec{D}$:

$$W = \vec{F} \cdot \vec{D} = (3)(2) + (4)(5) + (5)(2) = 6 + 20 + 10 = 36$$

The total work done is 36 units.

Checkpoint 1.62 A cart weighing 100 lb is pushed up an incline that makes an angle of 30° with the horizontal. Find the work done against gravity in pushing the cart a distance of 80 feet.

Solution. The work done against gravity depends entirely on the vertical displacement of the cart. Alternatively, using our standard projection model, the gravity force vector points straight down with magnitude 100 lb: $\vec{F}_g = -100\vec{j}$.

If we align our coordinate frame with the horizontal ground, a displacement of 80 feet along a 30° incline yields:

$$\vec{D} = 80\cos(30°)\vec{i} + 80\sin(30°)\vec{j} = 80\left(\frac{\sqrt{3}}{2}\right)\vec{i} + 80\left(\frac{1}{2}\right)\vec{j} = 40\sqrt{3}\vec{i} + 40\vec{j}$$

The work done by gravity is $\vec{F}_g \cdot \vec{D} = (0)(40\sqrt{3}) + (-100)(40) = -4000$ ft-lb. The work done **against** gravity is the positive opposite of this value:

$$W = 4000 \text{ ft-lb}$$

Checkpoint 1.63 A wagon is pulled a distance of 100 m along a horizontal path by a constant force of 70 N. The handle of the wagon is held at an angle of 35° above the horizontal. Find the work done by the force.

Solution. Work is defined as the magnitude of the displacement vector multiplied by the scalar projection of the force vector along the direction of motion:

$$W = \|\vec{D}\|\left(\text{comp}_{\vec{D}}\,\vec{F}\right) = \|\vec{D}\|\left(\|\vec{F}\|\cos(\theta)\right)$$

Substitute the given values ($\|\vec{D}\| = 100$ m, $\|\vec{F}\| = 70$ N, and $\theta = 35°$) into the work expression:

$$\begin{aligned} W &= (100)(70)\cos(35°) \\ &= 7000\cos(35°) \\ &\approx 5734.06 \text{ J} \end{aligned}$$

The total work done by the force pulling the wagon is approximately 5734.06 Joules.

1.4 The Cross Product (The vector product)

The cross product of two vectors in three-dimensional space is a vector that is perpendicular to both of the original vectors. It is defined as:

$$\vec{a} \times \vec{b} = \|\vec{a}\|\|\vec{b}\|\sin(\theta)\vec{n}$$

where θ is the angle between the vectors and $\vec{n}$ is the unit vector perpendicular to both $\vec{a}$ and $\vec{b}$.

Definition 1.64 If $\vec{a}$ and $\vec{b}$ are nonzero three-dimensional vectors, the **cross product** of $\vec{a}$ and $\vec{b}$ is the vector:

$$\vec{a} \times \vec{b} = (\|\vec{a}\|\|\vec{b}\|\sin(\theta))\vec{n}$$

where θ is the angle between $\vec{a}$ and $\vec{b}$ such that $0 \le \theta \le \pi$, and $\vec{n}$ is a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$. The direction of $\vec{n}$ is determined by the **right-hand rule**: if the fingers of your right hand curl through the angle θ from $\vec{a}$ to $\vec{b}$, then your thumb points in the direction of $\vec{n}$.

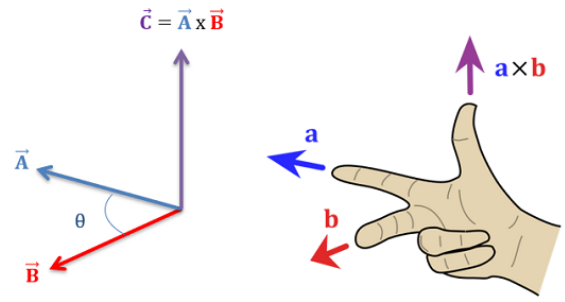


Figure 1.65 The right-hand rule and mutual vector orthogonality.

◆

Corollary 1.66 *The vector $\vec{a} \times \vec{b}$ points in the same direction as $\vec{n}$ because it is a scalar multiple of $\vec{n}$ with a non-negative scalar coefficient. Consequently, $\vec{a} \times \vec{b}$ is orthogonal to both $\vec{a}$ and $\vec{b}$. This is equivalent to stating:*

$$(\vec{a} \times \vec{b}) \cdot \vec{a} = 0 \quad \text{and} \quad (\vec{a} \times \vec{b}) \cdot \vec{b} = 0$$

Proof. By definition, the cross product is a scalar multiple of the vector $\vec{n}$, meaning $\vec{a} \times \vec{b} = c\vec{n}$ where $c = \|\vec{a}\|\|\vec{b}\| \sin(\theta)$. Because $\vec{n}$ is explicitly defined as a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$, its dot products with them must evaluate to zero:

$$\vec{n} \cdot \vec{a} = 0 \quad \text{and} \quad \vec{n} \cdot \vec{b} = 0$$

Using the scalar distribution properties of the dot product, we can substitute $c\vec{n}$ back into the orthogonality checks:

$$(\vec{a} \times \vec{b}) \cdot \vec{a} = (c\vec{n}) \cdot \vec{a} = c(\vec{n} \cdot \vec{a}) = c(0) = 0$$

$$(\vec{a} \times \vec{b}) \cdot \vec{b} = (c\vec{n}) \cdot \vec{b} = c(\vec{n} \cdot \vec{b}) = c(0) = 0$$

This confirms that the resulting vector product remains perfectly perpendicular to both original structural components. ■

Corollary 1.67 *Two nonzero vectors $\vec{a}$ and $\vec{b}$ are parallel if and only if the angle between them is either 0 or π. Therefore, two nonzero vectors $\vec{a}$ and $\vec{b}$ are parallel if and only if:*

$$\vec{a} \times \vec{b} = \vec{0}$$

Proof. The magnitude of the cross product vector is given by $\|\vec{a} \times \vec{b}\| = \|\vec{a}\|\|\vec{b}\| \sin(\theta)$. For any vector, its overall value equals the zero vector ($\vec{0}$) if and only if its absolute magnitude length scales to exactly 0.

Because we are evaluating two strictly nonzero vectors, their isolated scalar magnitudes are positive ($\|\vec{a}\| \neq 0$ and $\|\vec{b}\| \neq 0$). Therefore, the product evaluates to zero if and only if the trigonometric term vanishes:

$$\|\vec{a} \times \vec{b}\| = 0 \iff \sin(\theta) = 0$$

On the restricted tracking interval $0 \leq \theta \leq \pi$, the condition $\sin(\theta) = 0$ is satisfied if and only if $\theta = 0$ or $\theta = \pi$. Geometrically, an angle of 0 means the vectors point in the exact same direction, and an angle of π means they point in exactly opposite directions. In either case, the vectors are parallel. ■

Example 1.68 Find $\vec{i} \times \vec{j}$ and $\vec{j} \times \vec{i}$.

Solution. We use the geometric definition of the cross product, $\vec{a} \times \vec{b} = (\|\vec{a}\|\|\vec{b}\| \sin(\theta))\vec{n}$:

- For $\vec{i} \times \vec{j}$, both vectors are unit vectors, so $\|\vec{i}\| = 1$ and $\|\vec{j}\| = 1$. The standard basis vectors are mutually perpendicular, meaning the angle between them is $\theta = \pi/2$. By the right-hand rule, curling your fingers from the positive x-axis ($\vec{i}$) to the positive y-axis ($\vec{j}$) forces your thumb to point along the positive z-axis ($\vec{k}$):

$$\vec{i} \times \vec{j} = (1 \cdot 1 \cdot \sin(\pi/2))\vec{k} = (1 \cdot 1 \cdot 1)\vec{k} = \vec{k}$$

- For $\vec{j} \times \vec{i}$, reversing the order of operations reverses the orientation of your hand gesture. The thumb now points down along the negative z-axis ($-\vec{k}$):

$$\vec{j} \times \vec{i} = (1 \cdot 1 \cdot \sin(\pi/2))(-\vec{k}) = -\vec{k}$$

▲

Remark 1.69 The cross product of any vector with itself is always the zero vector, because the angle between identical direction paths is $\theta = 0$, and $\sin(0) = 0$:

$$\vec{i} \times \vec{i} = \vec{0}, \quad \vec{j} \times \vec{j} = \vec{0}, \quad \vec{k} \times \vec{k} = \vec{0}$$

The Cross Product in Component Form

Fact 1.70 Algebraic Cross Product Formula. *The cross product of $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$ is:*

$$\vec{a} \times \vec{b} = \langle a_2b_3 - a_3b_2,\, a_3b_1 - a_1b_3,\, a_1b_2 - a_2b_1 \rangle$$

The expression for $\vec{a} \times \vec{b}$ is easier to remember in determinant form. The cross product of $\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$ and $\vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$ can be expanded along its top row as:

$$\vec{a} \times \vec{b} = \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \vec{i} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \vec{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \vec{k}$$

We compact this row-expansion memory tool by writing it as a formal 3×3 matrix determinant:

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

Proposition 1.71 Properties of the Cross Product. *If $\vec{a}$, $\vec{b}$, and $\vec{c}$ are vectors in $\mathbb{R}^3$ and c is a scalar, then:*

1 $\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$ *(Anti-commutativity)*

2 $(c\vec{a}) \times \vec{b} = c(\vec{a} \times \vec{b}) = \vec{a} \times (c\vec{b})$

3 $\vec{a} \times (\vec{b} + \vec{c}) = \vec{a} \times \vec{b} + \vec{a} \times \vec{c}$ *(Left distributivity)*

4 $(\vec{a} + \vec{b}) \times \vec{c} = \vec{a} \times \vec{c} + \vec{b} \times \vec{c}$ *(Right distributivity)*

Proof. Let $\vec{a} = \langle a_1, a_2, a_3 \rangle$, $\vec{b} = \langle b_1, b_2, b_3 \rangle$, and $\vec{c} = \langle c_1, c_2, c_3 \rangle$. By algebraic definition, the cross product component vector expansion is:

$$\vec{a} \times \vec{b} = \langle a_2b_3 - a_3b_2,\, a_3b_1 - a_1b_3,\, a_1b_2 - a_2b_1 \rangle$$

1 Expanding the reverse cross product $\vec{b} \times \vec{a}$ yields:

$$\vec{b} \times \vec{a} = \langle b_2a_3 - b_3a_2,\, b_3a_1 - b_1a_3,\, b_1a_2 - b_2a_1 \rangle$$

Factoring out a negative -1 from each scalar entry directly flips the interior subtraction ordering:

$$\begin{aligned} \vec{b} \times \vec{a} &= \langle -(a_2b_3 - a_3b_2),\, -(a_3b_1 - a_1b_3),\, -(a_1b_2 - a_2b_1) \rangle \\ &= -\langle a_2b_3 - a_3b_2,\, a_3b_1 - a_1b_3,\, a_1b_2 - a_2b_1 \rangle = -(\vec{a} \times \vec{b}) \end{aligned}$$

Multiplying both sides by -1 satisfies the anti-commutative condition $\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$.

2 Multiplying vector $\vec{a}$ by a scalar c updates our definition to $c\vec{a} = \langle ca_1, ca_2, ca_3 \rangle$. Substituting this scalar into the expansion gives:

$$\begin{aligned} (c\vec{a}) \times \vec{b} &= \langle (ca_2)b_3 - (ca_3)b_2,\, (ca_3)b_1 - (ca_1)b_3,\, (ca_1)b_2 - (ca_2)b_1 \rangle \\ &= \langle c(a_2b_3 - a_3b_2),\, c(a_3b_1 - a_1b_3),\, c(a_1b_2 - a_2b_1) \rangle \\ &= c\langle a_2b_3 - a_3b_2,\, a_3b_1 - a_1b_3,\, a_1b_2 - a_2b_1 \rangle = c(\vec{a} \times \vec{b}) \end{aligned}$$

A symmetric factoring loop shows this expression is identically equal to $\vec{a} \times (c\vec{b})$.

3 First, compile the vector addition term: $\vec{b} + \vec{c} = \langle b_1 + c_1, b_2 + c_2, b_3 + c_3 \rangle$. Crossing $\vec{a}$ with this combined vector yields:

$$\begin{aligned} \vec{a} \times (\vec{b} + \vec{c}) &= \langle a_2(b_3 + c_3) - a_3(b_2 + c_2),\, a_3(b_1 + c_1) - a_1(b_3 + c_3),\, a_1(b_2 + c_2) - a_2(b_1 + c_1) \rangle \\ &= \langle a_2b_3 + a_2c_3 - a_3b_2 - a_3c_2,\, a_3b_1 + a_3c_1 - a_1b_3 - a_1c_3,\, a_1b_2 + a_1c_2 - a_2b_1 - a_2c_1 \rangle \end{aligned}$$

Regrouping the entries splits the single vector back into a clean linear addition step:

$$\begin{aligned} &= \langle (a_2b_3 - a_3b_2) + (a_2c_3 - a_3c_2),\, (a_3b_1 - a_1b_3) + (a_3c_1 - a_1c_3),\, (a_1b_2 - a_2b_1) + (a_1c_2 - a_2c_1) \rangle \\ &= \langle a_2b_3 - a_3b_2,\, a_3b_1 - a_1b_3,\, a_1b_2 - a_2b_1 \rangle + \langle a_2c_3 - a_3c_2,\, a_3c_1 - a_1c_3,\, a_1c_2 - a_2c_1 \rangle \\ &= \vec{a} \times \vec{b} + \vec{a} \times \vec{c} \end{aligned}$$

4 Property 4 can be proven directly by using the anti-commutative law established in step 1, combined with the left distributive property from step 3:

$$\begin{aligned} (\vec{a} + \vec{b}) \times \vec{c} &= -(\vec{c} \times (\vec{a} + \vec{b})) \\ &= -(\vec{c} \times \vec{a} + \vec{c} \times \vec{b}) \\ &= -(\vec{c} \times \vec{a}) - (\vec{c} \times \vec{b}) \\ &= (\vec{a} \times \vec{b}) + (\vec{b} \times \vec{c}) \end{aligned}$$

This completes the verification of all four arithmetic identities.

■

If $\vec{a}$ and $\vec{b}$ are represented by directed line segments with the same initial point, they determine a parallelogram. This parallelogram has a base of length $\|\vec{a}\|$ and an altitude height of $\|\vec{b}\| \sin(\theta)$. Therefore, the total area A is:

$$A = \|\vec{a}\|\|\vec{b}\| \sin(\theta) = \|\vec{a} \times \vec{b}\|$$

Remark 1.72 The length of the cross product vector $\|\vec{a} \times \vec{b}\|$ is equal to the area of the parallelogram determined by the vectors $\vec{a}$ and $\vec{b}$.

Example 1.73 Find the cross product $\vec{a} \times \vec{b}$ for each of the following cases:

1 $\vec{a} = \langle 2, -3, 4\rangle$, $\vec{b} = \langle 6, 2, -\frac{1}{2}\rangle$

2 $\vec{a} = 4\vec{j} - 3\vec{k}$, $\vec{b} = 2\vec{i} + 4\vec{j} + 6\vec{k}$

3 $\|\vec{a}\| = 6$, $\|\vec{b}\| = 5$, and the angle between $\vec{a}$ and $\vec{b}$ is $2\pi/3$.

Solution.

1 Set up and evaluate the 3×3 determinant:

$$\begin{aligned}
\vec{a} \times \vec{b} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -3 & 4 \\ 6 & 2 & -\frac{1}{2} \end{vmatrix} \\
&= \begin{vmatrix} -3 & 4 \\ 2 & -\frac{1}{2} \end{vmatrix} \vec{i} - \begin{vmatrix} 2 & 4 \\ 6 & -\frac{1}{2} \end{vmatrix} \vec{j} + \begin{vmatrix} 2 & -3 \\ 6 & 2 \end{vmatrix} \vec{k} \\
&= \left(\frac{3}{2} - 8\right)\vec{i} - (-1 - 24)\vec{j} + (4 - (-18))\vec{k} \\
&= -\frac{13}{2}\vec{i} + 25\vec{j} + 22\vec{k} = \left\langle -\frac{13}{2}, 25, 22 \right\rangle
\end{aligned}$$

2 Write $\vec{a}$ in full component form as $\langle 0, 4, -3\rangle$:

$$\begin{aligned}
\vec{a} \times \vec{b} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 4 & -3 \\ 2 & 4 & 6 \end{vmatrix} \\
&= (24 - (-12))\vec{i} - (0 - (-6))\vec{j} + (0 - 8)\vec{k} \\
&= 36\vec{i} - 6\vec{j} - 8\vec{k} = \langle 36, -6, -8\rangle
\end{aligned}$$

3 Because individual components are missing but the angle is given, we find the magnitude using the geometric formula and label the unit normal directional vector $\vec{n}$:

$$\|\vec{a} \times \vec{b}\| = \|\vec{a}\|\|\vec{b}\| \sin(\theta) = (6)(5)\sin\left(\frac{2\pi}{3}\right) = 30\left(\frac{\sqrt{3}}{2}\right) = 15\sqrt{3}$$

Thus, the cross product vector is $15\sqrt{3}\vec{n}$.

▲

Example 1.74 Find a vector perpendicular to the plane that passes through the points $P(1, 4, 6)$, $Q(-2, 5, -1)$, and $R(1, -1, 1)$.

Solution. First, construct two independent vectors lying inside the plane using the given vertices:

$$\vec{PQ} = \langle -2 - 1, 5 - 4, -1 - 6\rangle = \langle -3, 1, -7\rangle$$

$$\vec{PR} = \langle 1 - 1, -1 - 4, 1 - 6\rangle = \langle 0, -5, -5\rangle$$

A vector perpendicular to the plane must be orthogonal to both paths. Compute their cross product:

$$\begin{aligned} \vec{PQ} \times \vec{PR} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -3 & 1 & -7 \\ 0 & -5 & -5 \end{vmatrix} \\ &= (-5 - 35)\vec{i} - (15 - 0)\vec{j} + (15 - 0)\vec{k} \\ &= -40\vec{i} - 15\vec{j} + 15\vec{k} = \langle -40, -15, 15 \rangle \end{aligned}$$

Any non-zero scalar multiple of this vector (such as $\langle 8, 3, -3 \rangle$ obtained by dividing by -5) is also perpendicular to the plane. ▲

Example 1.75 Find the area of the triangle with vertices $P(1, 4, 6)$, $Q(-2, 5, -1)$, and $R(1, -1, 1)$.

Solution. The area of a triangle is exactly half the area of the parallelogram spanned by vectors sharing vertex P. From the previous example, we know $\vec{PQ} \times \vec{PR} = \langle -40, -15, 15 \rangle$.

Compute the magnitude length of this cross product vector:

$$\begin{aligned} \|\vec{PQ} \times \vec{PR}\| &= \sqrt{(-40)^2 + (-15)^2 + 15^2} \\ &= \sqrt{1600 + 225 + 225} = \sqrt{2050} = 5\sqrt{82} \end{aligned}$$

Therefore, the area of the triangle is:

$$\text{Area} = \frac{1}{2}\|\vec{PQ} \times \vec{PR}\| = \frac{5\sqrt{82}}{2} \approx 22.64$$

▲

Definition 1.76 The product $\vec{a} \cdot (\vec{b} \times \vec{c})$ is called the **scalar triple product** of the vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$. ◆

The area of the base parallelogram is $A = \|\vec{b} \times \vec{c}\|$. If θ is the angle between vector $\vec{a}$ and the cross product vector $\vec{b} \times \vec{c}$, then the slanted vertical height h of the surrounding three-dimensional parallelepiped is $h = \|\vec{a}\| |\cos(\theta)|$.

Fact 1.77 Volume of a Parallelepiped. *The volume of the parallelepiped determined by the vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$ is the absolute magnitude of their scalar triple product:*

$$V = |\vec{a} \cdot (\vec{b} \times \vec{c})| = |(\vec{a} \times \vec{b}) \cdot \vec{c}|$$

We can compute this product directly using a single matrix determinant:

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

Example 1.78 Use the scalar triple product to show that the vectors $\vec{a} = \langle 1, 4, -7 \rangle$, $\vec{b} = \langle 2, -1, 4 \rangle$, and $\vec{c} = \langle 0, -9, 18 \rangle$ are coplanar, meaning they lie in the same plane.

Solution. Three vectors are coplanar if and only if the volume of the parallelepiped they span is exactly zero ($V = 0$). We check this condition by computing their scalar triple product determinant:

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} 1 & 4 & -7 \\ 2 & -1 & 4 \\ 0 & -9 & 18 \end{vmatrix}$$

$$
\begin{aligned}
&= 1\begin{vmatrix}-1 & 4\\ -9 & 18\end{vmatrix} - 4\begin{vmatrix}2 & 4\\ 0 & 18\end{vmatrix} + (-7)\begin{vmatrix}2 & -1\\ 0 & -9\end{vmatrix}\\
&= 1(-18-(-36)) - 4(36-0) - 7(-18-0)\\
&= 1(18) - 4(36) - 7(-18)\\
&= 18 - 144 + 126 = 0
\end{aligned}
$$

Because the scalar triple product is exactly 0, the bounded volume vanishes. This proves that the three vectors are coplanar. ▲

Definition 1.79 The product $\vec{a}\times(\vec{b}\times\vec{c})$ is called the **vector triple product** of the vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$. It can be expanded and computed using dot products via the BAC-CAB identity:

$$\vec{a}\times(\vec{b}\times\vec{c}) = (\vec{a}\cdot\vec{c})\vec{b} - (\vec{a}\cdot\vec{b})\vec{c}$$

◆

Example 1.80 Given the vectors $\vec{a} = \vec{i} - \vec{j} + 2\vec{k}$, $\vec{b} = 2\vec{i} + \vec{k}$, and $\vec{c} = \vec{j} - 3\vec{k}$, evaluate the vector triple product $\vec{a}\times(\vec{b}\times\vec{c})$ using the expansion identity.

Solution. First, express the vectors cleanly in their numerical component forms:

$$\vec{a} = \langle 1, -1, 2\rangle, \quad \vec{b} = \langle 2, 0, 1\rangle, \quad \vec{c} = \langle 0, 1, -3\rangle$$

Next, calculate the two scalar dot products required by the identity ($\vec{a}\cdot\vec{c}$ and $\vec{a}\cdot\vec{b}$):

$$
\begin{aligned}
\vec{a}\cdot\vec{c} &= (1)(0) + (-1)(1) + (2)(-3) = 0 - 1 - 6 = -7\\
\vec{a}\cdot\vec{b} &= (1)(2) + (-1)(0) + (2)(1) = 2 + 0 + 2 = 4
\end{aligned}
$$

Substitute these scalar values back into the vector triple product identity formula:

$$
\begin{aligned}
\vec{a}\times(\vec{b}\times\vec{c}) &= (\vec{a}\cdot\vec{c})\vec{b} - (\vec{a}\cdot\vec{b})\vec{c}\\
&= -7\langle 2, 0, 1\rangle - 4\langle 0, 1, -3\rangle\\
&= \langle -14, 0, -7\rangle - \langle 0, 4, -12\rangle\\
&= \langle -14-0,\ 0-4,\ -7-(-12)\rangle = \langle -14, -4, 5\rangle
\end{aligned}
$$

In standard basis notation, the final vector is $-14\vec{i} - 4\vec{j} + 5\vec{k}$. ▲

1.5 Lines and Planes

In this section, we will discuss the equations of lines and planes in three-dimensional space. We will explore how to represent lines and planes using vector equations, parametric equations, and Cartesian equations. Additionally, we will examine the relationships between lines and planes, including conditions for parallelism and intersection.

If a vector $\vec{v}$ gives the direction of a line L and is written in component form as $\vec{v} = \langle a, b, c\rangle$, then any scalar multiple of that direction vector can be expressed as $t\vec{v} = \langle ta, tb, tc\rangle$ for any scalar parameter t.

Letting $\vec{r} = \langle x, y, z\rangle$ represent the position vector of an arbitrary point on the line, and $\vec{r}_0 = \langle x_0, y_0, z_0\rangle$ represent the position vector of a specific known point on the line, we can state the **vector equation of a line**:

$$\vec{r} = \vec{r}_0 + t\vec{v} \implies \langle x, y, z\rangle = \langle x_0 + ta,\ y_0 + tb,\ z_0 + tc\rangle$$

Definition 1.81 Two vectors are equal if and only if their corresponding components are equal. Equating the components of the vector equation of a line yields:

$$x = x_0 + at, \quad y = y_0 + bt, \quad z = z_0 + ct$$

where $t \in \mathbb{R}$. These equations are called the **parametric equations** of the line L passing through the point $P_0(x_0, y_0, z_0)$ and parallel to the vector $\vec{v} = \langle a, b, c\rangle$. Each value of the scalar parameter t represents a unique point (x, y, z) on L.

◆

Example 1.82 Find a vector equation and parametric equations for the line that passes through the point $(5, 1, 3)$ and is parallel to the vector $\vec{i} + 4\vec{j} - 2\vec{k}$. Then, find two other points on this line.

Solution. We are given a point on the line corresponding to the initial position vector $\vec{r}_0 = \langle 5, 1, 3\rangle$ and a parallel direction vector $\vec{v} = \vec{i} + 4\vec{j} - 2\vec{k} = \langle 1, 4, -2\rangle$.

The **vector equation** is given by:

$$\vec{r} = \vec{r}_0 + t\vec{v} \implies \langle x, y, z\rangle = \langle 5, 1, 3\rangle + t\langle 1, 4, -2\rangle$$

By separating this expression into independent components, we find the **parametric equations**:

$$x = 5 + t, \quad y = 1 + 4t, \quad z = 3 - 2t$$

To find two other points on the line, we substitute any two non-zero values for the parameter t:

- Letting $t = 1$ yields: $x = 5+1 = 6$, $y = 1+4(1) = 5$, and $z = 3-2(1) = 1$. This gives the point $(6, 5, 1)$.
- Letting $t = -1$ yields: $x = 5 - 1 = 4$, $y = 1 + 4(-1) = -3$, and $z = 3 - 2(-1) = 5$. This gives the point $(4, -3, 5)$.

▲

Definition 1.83 If a vector $\vec{v} = \langle a, b, c\rangle$ is used to describe the direction of a line L, then the numbers a, b, and c are called the **direction numbers** of L. Since any vector parallel to $\vec{v}$ can also be used, any three numbers proportional to a, b, and c can serve as a valid set of direction numbers for L.

If we solve each of the parametric equations for the parameter t and equate the results (assuming none of the direction numbers a, b, or c are zero), we eliminate t to obtain:

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

These expressions are called the **symmetric equations** of L. ◆

For instance, if a direction number is zero, say $a = 0$, we eliminate the parameter t for the remaining components and write the equations of L as:

$$x = x_0, \quad \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

This means geometrically that the line L lies entirely within the vertical plane $x = x_0$.

Example 1.84 Find parametric equations and symmetric equations for the line that passes through the points $(2, 4, -3)$ and $(3, -1, 1)$. At what point does this line intersect the xy-plane?

Solution. First, find a direction vector $\vec{v}$ for the line by subtracting the coordinates of the first point from the second point:

$$\vec{v} = \langle 3-2,\ -1-4,\ 1-(-3)\rangle = \langle 1, -5, 4\rangle$$

Using the first point $(2, 4, -3)$ as our base position vector $\vec{r}_0 = \langle 2, 4, -3\rangle$, we find the **parametric equations**:

$$x = 2 + t, \quad y = 4 - 5t, \quad z = -3 + 4t$$

By solving each equation for the parameter t, we obtain the matching **symmetric equations**:

$$x - 2 = \frac{y-4}{-5} = \frac{z+3}{4}$$

The line intersects the xy-plane when its height component is zero ($z = 0$). Substitute $z = 0$ into the symmetric equations to solve for the coordinate positions:

$$x - 2 = \frac{0+3}{4} \implies x = 2 + \frac{3}{4} = \frac{11}{4}$$

$$\frac{y-4}{-5} = \frac{3}{4} \implies y - 4 = -\frac{15}{4} \implies y = 4 - \frac{15}{4} = \frac{1}{4}$$

Therefore, the point of intersection with the xy-plane is $\left(\frac{11}{4}, \frac{1}{4}, 0\right)$. ▲

Corollary 1.85 *A line segment starting from position vector $\vec{r}_0$ and ending at position vector $\vec{r}_1$ has a direction vector defined by $\vec{v} = \vec{r}_1 - \vec{r}_0$. Substituting this into the standard vector equation of a line $\vec{r} = \vec{r}_0 + t\vec{v}$ yields the* ***vector equation of a line segment:***

$$\vec{r}(t) = (1-t)\vec{r}_0 + t\vec{r}_1, \quad 0 \le t \le 1$$

Example 1.86 Show that the lines L_1 and L_2 with the following parametric equations are **skew lines** (meaning they do not intersect and are not parallel):

$$L_1: \quad x = 1 + t, \quad y = -2 + 3t, \quad z = 4 - t$$
$$L_2: \quad x = 2s, \quad y = 3 + s, \quad z = -3 + 4s$$

Solution. First, check if the lines are parallel by comparing their direction vectors:

$$\vec{v}_1 = \langle 1, 3, -1\rangle \quad \text{and} \quad \vec{v}_2 = \langle 2, 1, 4\rangle$$

Since the components are not proportional ($\frac{1}{2} \neq \frac{3}{1}$), the direction vectors are not scalar multiples of each other. Therefore, the lines are not parallel.

Next, test if the lines intersect by equating their corresponding coordinate expressions to see if a common parameter set (t, s) exists:

$$1 + t = 2s \quad \text{(Eq. 1)}$$
$$-2 + 3t = 3 + s \quad \text{(Eq. 2)}$$
$$4 - t = -3 + 4s \quad \text{(Eq. 3)}$$

From Equation 1, we get $t = 2s - 1$. Substitute this expression into Equation 2 to solve for s:

$$-2 + 3(2s-1) = 3 + s \implies -2 + 6s - 3 = 3 + s$$

$$5s = 8 \implies s = \frac{8}{5}$$

Using $s = \frac{8}{5}$, we find the value for t: $t = 2\left(\frac{8}{5}\right) - 1 = \frac{16}{5} - \frac{5}{5} = \frac{11}{5}$.

Now, substitute both parameters $s = \frac{8}{5}$ and $t = \frac{11}{5}$ into Equation 3 to see if they satisfy the vertical height component:

$$\text{Left Side: } 4 - t = 4 - \frac{11}{5} = \frac{20}{5} - \frac{11}{5} = \frac{9}{5}$$

$$\text{Right Side: } -3 + 4s = -3 + 4\left(\frac{8}{5}\right) = -\frac{15}{5} + \frac{32}{5} = \frac{17}{5}$$

Since $\frac{9}{5} \neq \frac{17}{5}$, the system of linear equations has no solution. This means the lines do not intersect. Because they are neither parallel nor intersecting, L_1 and L_2 are skew lines.

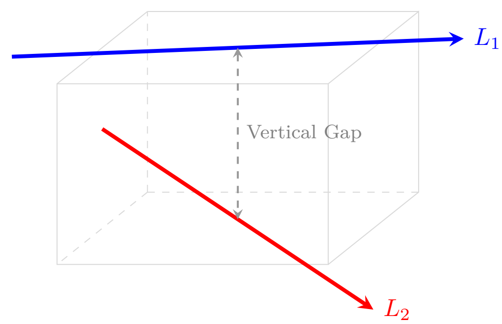


Figure 1.87 The skew lines L_1 and L_2 passing through different planes in space.

▲

Planes

A plane in space is uniquely determined by a fixed point $P_0(x_0, y_0, z_0)$ lying in the plane and a vector $\vec{n}$ that stands orthogonal to the plane surface. This perpendicular vector $\vec{n}$ is called a **normal vector**. Letting $P(x, y, z)$ be any arbitrary point in the plane, and denoting $\vec{r}_0$ and $\vec{r}$ as the standard position vectors of P_0 and P respectively, the displacement vector $\vec{r} - \vec{r}_0$ is represented by the geometric line segment $\vec{P_0P}$.

Since the normal vector $\vec{n}$ is orthogonal to every vector contained within the plane, it must be orthogonal to $\vec{r} - \vec{r}_0$. This orthogonality condition gives us the fundamental **vector equation of the plane**:

$$\vec{n} \cdot (\vec{r} - \vec{r}_0) = 0 \iff \vec{n} \cdot \vec{r} = \vec{n} \cdot \vec{r}_0$$

Theorem 1.88 Scalar and Linear Equations of a Plane. *An equation of the plane passing through $P_0(x_0, y_0, z_0)$ with normal vector $\vec{n} = \langle a, b, c\rangle$ is derived by evaluating the dot product component-wise:*

$$\begin{aligned} \langle a, b, c\rangle \cdot (\vec{r} - \vec{r}_0) &= 0 \\ \langle a, b, c\rangle \cdot \langle x - x_0, y - y_0, z - z_0\rangle &= 0 \\ a(x - x_0) + b(y - y_0) + c(z - z_0) &= 0 \end{aligned}$$

By distributing the scalar coefficients and gathering the constants into a single value $d = -(ax_0 + by_0 + cz_0)$, the equation of the plane simplifies to the ***linear equation in three variables****:*

$$ax + by + cz + d = 0$$

where the coefficients a, b, and c are not all zero.

Example 1.89 Find an equation of the plane through $(5, -2, 4)$ with normal vector $\vec{n} = \langle 1, 2, 3\rangle$. Find its intercepts and describe how to sketch the plane.

Solution. Substitute the given coordinates and normal components directly into the scalar equation formula:

$$1(x-5) + 2(y-(-2)) + 3(z-4) = 0 \implies x - 5 + 2y + 4 + 3z - 12 = 0$$

Combining constant terms yields the final linear plane equation:

$$x + 2y + 3z = 13$$

To calculate the coordinate intercepts, set the other two variables to zero systematically:

- x**-intercept**: Set $y = 0, z = 0 \implies x = 13$. The point is $(13, 0, 0)$.
- y**-intercept**: Set $x = 0, z = 0 \implies 2y = 13 \implies y = \frac{13}{2}$. The point is $\left(0, \frac{13}{2}, 0\right)$.
- z**-intercept**: Set $x = 0, y = 0 \implies 3z = 13 \implies z = \frac{13}{2}$. The point is $\left(0, 0, \frac{13}{3}\right)$.

To sketch this plane within the first octant, plot these three intercept markers on the coordinate axes and connect them to construct a triangular boundary plane section. ▲

Example 1.90 Find an equation of the plane determined by the points $P(4, -3, 1)$, $Q(6, -4, 7)$, and $R(1, 2, 2)$.

Solution. First, construct two displacement vectors sharing vertex P that lie entirely within the plane:

$$\vec{PQ} = \langle 6-4, -4-(-3), 7-1\rangle = \langle 2, -1, 6\rangle$$

$$\vec{PR} = \langle 1-4, 2-(-3), 2-1\rangle = \langle -3, 5, 1\rangle$$

A normal vector $\vec{n}$ for the plane must stand perpendicular to both vector directions. We find it by computing their cross product:

$$\begin{aligned}\vec{n} = \vec{PQ} \times \vec{PR} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -1 & 6 \\ -3 & 5 & 1 \end{vmatrix} \\ &= (-1-30)\vec{i} - (2-(-18))\vec{j} + (10-3)\vec{k} \\ &= -31\vec{i} - 20\vec{j} + 7\vec{k} = \langle -31, -20, 7\rangle\end{aligned}$$

Using point $P(4, -3, 1)$ and our new normal coordinates, assemble the scalar equation layout:

$$-31(x-4) - 20(y-(-3)) + 7(z-1) = 0 \implies -31x + 124 - 20y - 60 + 7z - 7 = 0$$

Simplifying constants yields the clean linear plane model:

$$-31x - 20y + 7z + 57 = 0 \quad \text{or} \quad 31x + 20y - 7z = 57$$

▲

Example 1.91 Find the point at which the line with parametric equations $x = 2 + 3t$, $y = -4t$, $z = 5 + t$ intersects the plane $4x + 5y - 2z = 18$.

Solution. Substitute the parametric component equations of the line directly into the independent variable slots of the linear plane:

$$4(2+3t)+5(-4t)-2(5+t)=18$$

Distribute coefficients and combine terms to isolate the scalar parameter t:

$$8+12t-20t-10-2t=18 \implies -10t-2=18 \implies -10t=20 \implies t=-2$$

Substitute $t=-2$ back into the original parametric expressions to recover the intersection point coordinates:

$$x=2+3(-2)=-4, \quad y=-4(-2)=8, \quad z=5+(-2)=3$$

Thus, the line intersects the plane at the specific point coordinate $(-4,8,3)$.

▲

Definition 1.92 Two distinct planes containing normal vectors $\vec{n}_1$ and $\vec{n}_2$ are:

1 **parallel** if their corresponding normal vectors are parallel ($\vec{n}_1=c\vec{n}_2$ for some scalar c).

2 **orthogonal** if their corresponding normal vectors are orthogonal ($\vec{n}_1\cdot\vec{n}_2=0$).

◆

Example 1.93 Find an equation of the plane through $P(5,-2,4)$ that is parallel to the plane $3x+y-6z+8=0$.

Solution. Parallel planes share identical directional orientations, meaning we can adopt the normal vector of the reference plane directly. Reading the coefficients from $3x+y-6z+8=0$ reveals $\vec{n}=\langle 3,1,-6\rangle$.

Combine this normal direction with the coordinates of point $P(5,-2,4)$:

$$3(x-5)+1(y-(-2))-6(z-4)=0 \implies 3x-15+y+2-6z+24=0$$

Simplifying the constants provides the equation of the parallel plane:

$$3x+y-6z+11=0$$

▲

Example 1.94 Find the angle between the planes $x+y+z=1$ and $x-2y+3z=1$. Then, find symmetric equations for the line of intersection L of these two planes.

Solution. The angle θ between two planes matches the angle separating their normal vectors, $\vec{n}_1=\langle 1,1,1\rangle$ and $\vec{n}_2=\langle 1,-2,3\rangle$:

$$\vec{n}_1\cdot\vec{n}_2=(1)(1)+(1)(-2)+(1)(3)=2$$

$$\|\vec{n}_1\|=\sqrt{1^2+1^2+1^2}=\sqrt{3}, \quad \|\vec{n}_2\|=\sqrt{1^2+(-2)^2+3^2}=\sqrt{14}$$

$$\cos(\theta)=\frac{\vec{n}_1\cdot\vec{n}_2}{\|\vec{n}_1\|\|\vec{n}_2\|}=\frac{2}{\sqrt{42}} \implies \theta=\arccos\left(\frac{2}{\sqrt{42}}\right)\approx 72.0^\circ \text{ (or 1.26 rad)}$$

To track the line of intersection L, we first find its direction vector $\vec{v}$, which must be orthogonal to both normal paths:

$$\vec{v}=\vec{n}_1\times\vec{n}_2=\begin{vmatrix}\vec{i} & \vec{j} & \vec{k}\\ 1 & 1 & 1\\ 1 & -2 & 3\end{vmatrix}=(1(3)-1(-2))\vec{i}-(1(3)-1(1))\vec{j}+(1(-2)-1(1))\vec{k}=5\vec{i}-2\vec{j}-3\vec{k}$$

$$\vec{v} = \langle 5, -2, -3 \rangle$$

Next, we find a specific point on L by solving the system of plane equations simultaneously. We can set $z = 0$ to simplify:

$$x + y = 1 \quad \text{and} \quad x - 2y = 1$$

Solving this system yields $x = \frac{3}{2}$ and $y = -\frac{1}{2}$. Thus, the point $P\left(\frac{3}{2}, -\frac{1}{2}, 0\right)$ lies on the line of intersection.

Finally, we can write the parametric equations for the line L:

$$x = \frac{3}{2} + 5t, \quad y = -\frac{1}{2} - 2t, \quad z = -3t$$

And the symmetric equations are:

$$\frac{x - \frac{3}{2}}{5} = \frac{y + \frac{1}{2}}{-2} = \frac{z}{-3}$$

▲

Fact 1.95 Symmetric Forms and Intersecting Planes. *An arbitrary spatial point $P(x, y, z)$ lies on a line L if and only if its coordinates satisfy the standard* ***symmetric equations****:*

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$$

We can interpret this line structural model as the geometric intersection path generated by two distinct, non-parallel structural planes:

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} \quad \textit{and} \quad \frac{y - y_1}{b} = \frac{z - z_1}{c}$$

Example 1.96 Find a symmetric form for the line passing through $P(3, 1, -2)$ and $Q(-2, 7, -4)$.

Solution. Calculate the baseline direction numbers by computing the displacement vector components from P to Q:

$$\vec{v} = \langle -2 - 3,\ 7 - 1,\ -4 - (-2) \rangle = \langle -5, 6, -2 \rangle$$

Using coordinates from vertex $P(3, 1, -2)$, populate the symmetric structural denominators:

$$\frac{x - 3}{-5} = \frac{y - 1}{6} = \frac{z + 2}{-2}$$

▲

Example 1.97 Find a formula for the minimum straight-line distance D mapping from an isolated spatial point $P_1(x_1, y_1, z_1)$ down to a linear plane surface defined by $ax + by + cz + d = 0$.

Solution. Let $P_0(x_0, y_0, z_0)$ be any arbitrary known point lying inside the given plane equation, which means it satisfies $ax_0 + by_0 + cz_0 + d = 0 \implies d = -ax_0 - by_0 - cz_0$. Construct a direction vector $\vec{b}$ corresponding to the directed line segment $\vec{P_0P_1}$:

$$\vec{b} = \langle x_1 - x_0,\ y_1 - y_0,\ z_1 - z_0 \rangle$$

The minimum distance D from point P_1 to the plane surface equals the absolute scalar projection length of this displacement vector $\vec{b}$ projected onto the plane's normal vector direction $\vec{n} = \langle a, b, c\rangle$:

$$\begin{aligned} D &= |\operatorname{comp}_{\vec{n}} \vec{b}| = \frac{|\vec{n}\cdot\vec{b}|}{\|\vec{n}\|} \\ &= \frac{|a(x_1-x_0)+b(y_1-y_0)+c(z_1-z_0)|}{\sqrt{a^2+b^2+c^2}} \\ &= \frac{|ax_1+by_1+cz_1-(ax_0+by_0+cz_0)|}{\sqrt{a^2+b^2+c^2}} \end{aligned}$$

Substituting the constant expression $d = -ax_0 - by_0 - cz_0$ into the numerator yields the standard point-plane distance formula:

$$D = \frac{|ax_1+by_1+cz_1+d|}{\sqrt{a^2+b^2+c^2}}$$

▲

Checkpoint 1.98 Find the distance between the parallel planes $10x+2y-2z = 5$ and $5x + y - z = 1$.

Solution. To find the distance between parallel planes, pick an arbitrary test point on one plane and compute its distance to the other plane.

Let us find a point on the second plane $5x + y - z = 1$ by setting $x = 0$ and $z = 0$, which gives $y = 1$. This identifies our test point as $P_1(0, 1, 0)$.

Now, write the first plane in standard linear form: $10x + 2y - 2z - 5 = 0$. Apply the coordinates of $P_1(0, 1, 0)$ to the distance formula:

$$D = \frac{|10(0)+2(1)-2(0)-5|}{\sqrt{10^2+2^2+(-2)^2}} = \frac{|2-5|}{\sqrt{100+4+4}} = \frac{|-3|}{\sqrt{108}} = \frac{3}{6\sqrt{3}} = \frac{1}{2\sqrt{3}} = \frac{\sqrt{3}}{6}$$

The exact distance separating the two planes is $\frac{\sqrt{3}}{6}$.

Checkpoint 1.99 We showed that the lines L_1 and L_2 with the following parametric equations are skew lines:

$$\begin{aligned} L_1: &\quad x = 1+t, \quad y = -2+3t, \quad z = 4-t \\ L_2: &\quad x = 2s, \quad y = 3+s, \quad z = -3+4s \end{aligned}$$

Find the distance between them.

Solution. The minimum distance between two skew lines equals the distance between two parallel planes that enclose the lines. The normal vector $\vec{n}$ for these planes must stand perpendicular to both line direction vectors: $\vec{v}_1 = \langle 1, 3, -1\rangle$ and $\vec{v}_2 = \langle 2, 1, 4\rangle$.

$$\vec{n} = \vec{v}_1 \times \vec{v}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 3 & -1 \\ 2 & 1 & 4 \end{vmatrix} = (12-(-1))\vec{i} - (4-(-2))\vec{j} + (1-6)\vec{k} = \langle 13, -6, -5\rangle$$

Find a point on line L_1 by setting $t = 0$, giving $P_1(1, -2, 4)$. Find a point on line L_2 by setting $s = 0$, giving $P_2(0, 3, -3)$.

Construct a plane passing through $P_2(0, 3, -3)$ using our calculated normal vector components:

$$13(x-0) - 6(y-3) - 5(z-(-3)) = 0 \implies 13x - 6y - 5z + 3 = 0$$

Finally, calculate the distance from point $P_1(1, -2, 4)$ on line L_1 to this new plane container:

$$D = \frac{|13(1) - 6(-2) - 5(4) + 3|}{\sqrt{13^2 + (-6)^2 + (-5)^2}} = \frac{|13 + 12 - 20 + 3|}{\sqrt{169 + 36 + 25}} = \frac{8}{\sqrt{230}}$$

The minimum distance separating the two skew lines is exactly $\frac{8}{\sqrt{230}}$.

Checkpoint 1.100 Find the distance from the point $(1, 1, 2)$ to the plane $3x + 2y + 6z = 6$.

Solution. First, rewrite the plane equation in standard linear form: $3x + 2y + 6z - 6 = 0$. This provides coefficients of $a = 3, b = 2, c = 6$, and $d = -6$.

Substitute these parameters along with the target point coordinates $(x_1, y_1, z_1) = (1, 1, 2)$ into the point-plane distance formula:

$$D = \frac{|3(1) + 2(1) + 6(2) - 6|}{\sqrt{3^2 + 2^2 + 6^2}} = \frac{|3 + 2 + 12 - 6|}{\sqrt{9 + 4 + 36}} = \frac{|11|}{\sqrt{49}} = \frac{11}{7}$$

The minimum straight-line distance from the point to the plane is exactly $\frac{11}{7}$.

1.6 Surfaces

In this section, we will explore the geometry of surfaces in three-dimensional space. We will learn how to represent surfaces using vector equations, parametric equations, and scalar equations. Additionally, we will discuss the relationships between surfaces and other geometric objects.

Definition 1.101 A **function f of two variables** is a rule that assigns to each ordered pair of real numbers (x, y) in a set D a unique real number denoted by $f(x, y)$. The set D is the **domain** of f and its **range** is the set of values that f takes on, that is:

$$\{f(x, y) \mid (x, y) \in D\}$$

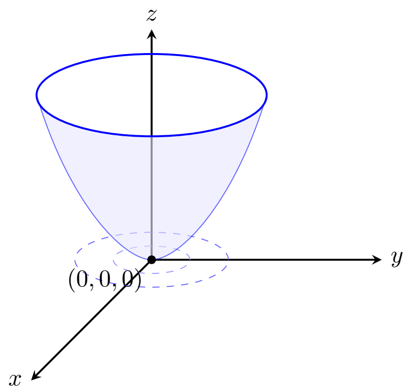
◆

We often write $z = f(x, y)$ to make explicit the value taken on by f at the general point (x, y). The domain is a subset of $\mathbb{R}^2$, the xy-plane.

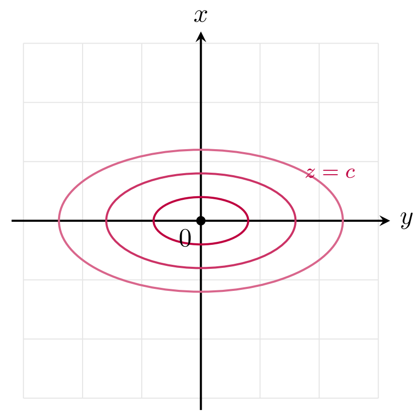
Example 1.102 If $f(x, y) = 4x^2 + y^2$, determine its domain and range.

Solution. The expression $4x^2 + y^2$ is defined for all possible ordered pairs of real numbers (x, y). Therefore, the domain is $\mathbb{R}^2$, which represents the entire xy-plane.

Since the terms x^2 and y^2 are always non-negative for any real numbers x and y, their linear combination $4x^2 + y^2$ must also be greater than or equal to zero. The function achieves its absolute minimum value of 0 exactly at the origin $(0, 0)$. Thus, the range of f is the interval $[0, \infty)$ of all non-negative real numbers.



(a) The 3D surface (elliptic paraboloid).



(b) The domain and level curves in the xy-plane.

Figure 1.103 The 3D surface graph of $f(x, y) = 4x^2 + y^2$ and its corresponding domain and level curves in the xy-plane.

▲

Definition 1.104 If f is a function of two variables with domain D, then the **graph** of f is the set of all points (x, y, z) in $\mathbb{R}^3$ such that $z = f(x, y)$ and (x, y) is in D. ◆

Example 1.105 Sketch the graph of each of the following functions or equations in $\mathbb{R}^3$:

1 $f(x, y) = 6 - 3x - 2y$

2 An **elliptic cylinder**: $\frac{x^2}{4} + \frac{y^2}{9} = 1$

3 A **parabolic cylinder**: $f(x, y) = x^2$

4 An **ellipsoid**: $x^2 + \frac{y^2}{9} + \frac{z^2}{4} = 1$

Solution.

1 Setting $z = f(x, y)$ gives the linear equation $z = 6 - 3x - 2y \implies 3x + 2y + z = 6$. This represents a flat plane in space. We can find its coordinate axis intercepts to help sketch it:

- x-intercept: Set $y = 0, z = 0 \implies 3x = 6 \implies x = 2$. Point: $(2, 0, 0)$.
- y-intercept: Set $x = 0, z = 0 \implies 2y = 6 \implies y = 3$. Point: $(0, 3, 0)$.
- z-intercept: Set $x = 0, y = 0 \implies z = 6$. Point: $(0, 0, 6)$.

Plotting these three points and connecting them with a triangular frame yields the trace of the plane in the first octant.

2 The equation $\frac{x^2}{4} + \frac{y^2}{9} = 1$ restricts the relationship between x and y to an ellipse centered at the origin in the xy-plane, with major axis radius 3 along the y-axis and minor axis radius 2 along the x-axis. Because the variable z is completely missing from the equation, it is unconstrained and can take any real value. This projects the ellipse vertically to form an infinitely tall **elliptic cylinder** parallel to the z-axis.

3 Setting $z = f(x, y) = x^2$ provides a parabolic trace in the vertical xz-plane opening upwards from the origin. Since the variable y is absent, any vertical plane intersection cross-section matching $y = k$ produces an identical copy of this parabola shifted along the horizontal axis. Connecting these parallel parabolic traces generates a sheet-like **parabolic cylinder** extending along the direction of the y-axis.

4 The equation $x^2 + \frac{y^2}{9} + \frac{z^2}{4} = 1$ matches the standard form of an ellipsoid centered at the origin: $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ with radii $a = 1$, $b = 3$, and $c = 2$. Its traces in all three coordinate planes are ellipses:

- xy-plane trace ($z = 0$): $x^2 + \frac{y^2}{9} = 1$ (an ellipse of radii 1 and 3).
- xz-plane trace ($y = 0$): $x^2 + \frac{z^2}{4} = 1$ (an ellipse of radii 1 and 2).
- yz-plane trace ($x = 0$): $\frac{y^2}{9} + \frac{z^2}{4} = 1$ (an ellipse of radii 3 and 2).

Sketching these three cross-sectional traces outlines a 3D egg-like oval surface shell.

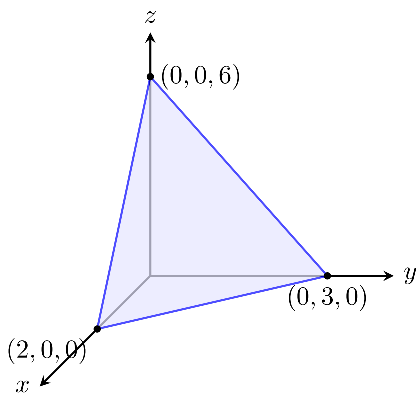


Figure 1.106 The plane $3x+2y+z = 6$ in the first octant.

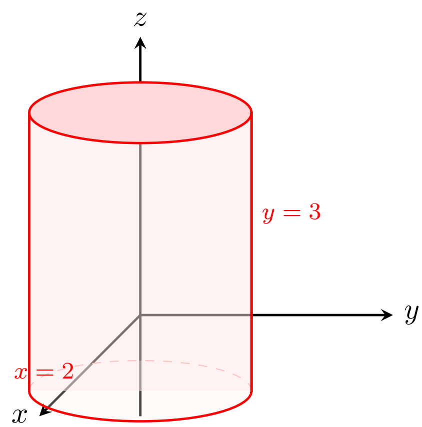


Figure 1.107 The elliptic cylinder $\frac{x^2}{4} + \frac{y^2}{9} = 1$.

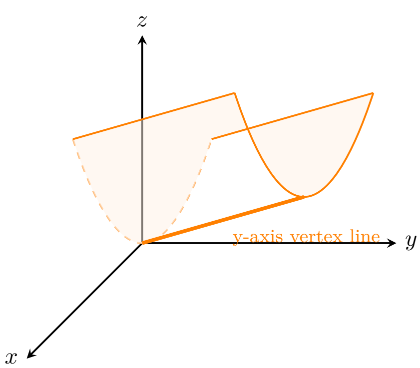


Figure 1.108 The parabolic cylinder $z = x^2$.

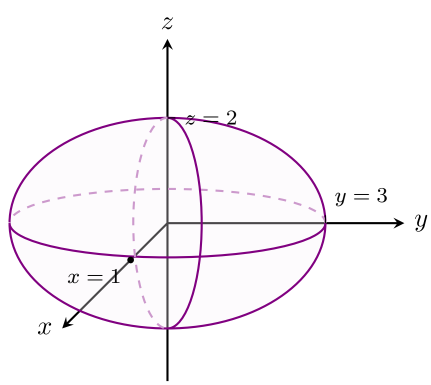


Figure 1.109 The ellipsoid $x^2 + \frac{y^2}{9} + \frac{z^2}{4} = 1$.

▲

If we keep x fixed by putting $x = k$ and letting y vary, the resulting function is a function of one variable, $z = f(k, y)$, whose graph is the curve that results when we intersect the surface $z = f(x, y)$ with the vertical plane $x = k$. This type of curve is called a **trace** of the surface. Standard quadric surfaces classified by their traces include:

- A **linear function**: $f(x, y) = ax + by + c$

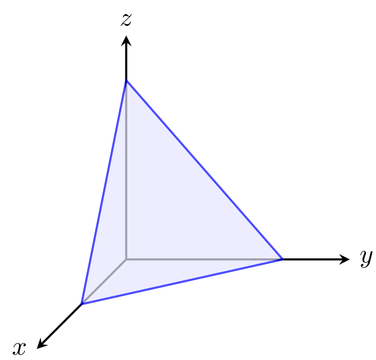


Figure 1.110 A flat plane representing a linear function.

- An **elliptic paraboloid**: $\frac{z}{c} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$

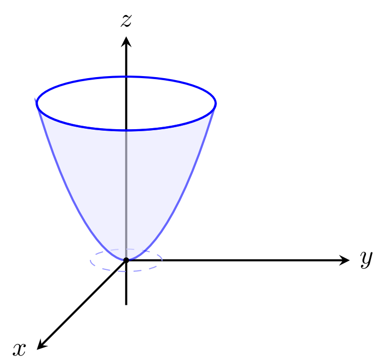


Figure 1.111 An elliptic paraboloid opening upward.

- A **hyperbolic paraboloid**: $\frac{z}{c} = \frac{x^2}{a^2} - \frac{y^2}{b^2}$

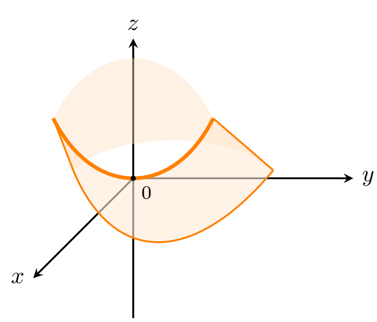


Figure 1.112 A hyperbolic paraboloid displaying a saddle point at the origin.

- An **ellipsoid**: $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

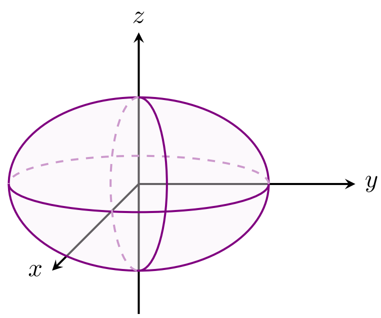


Figure 1.113 A symmetrical ellipsoid centered at the origin.

- A **hyperboloid of one sheet**: $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$

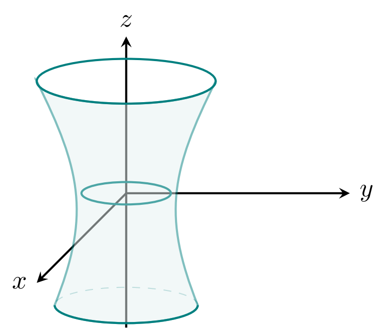


Figure 1.114 A hyperboloid of one sheet.

- A **hyperboloid of two sheets**: $-\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

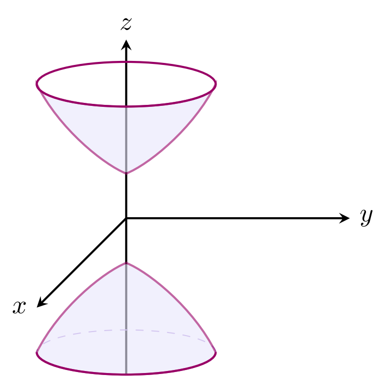


Figure 1.115 A hyperboloid of two sheets.

- An **elliptic cone**: $\frac{z^2}{c^2} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$

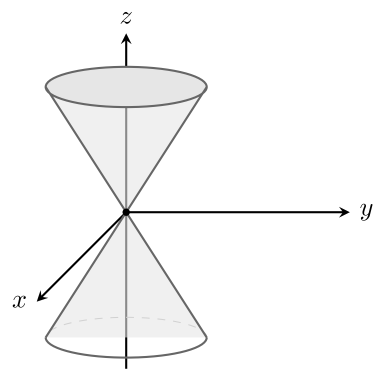


Figure 1.116 An elliptic cone.

Example 1.117 Classify the surface $x^2 + 2z^2 - 6x - y + 10 = 0$.

Solution. To classify the surface, we isolate the linear variable y and complete the square for the x terms:

$$\begin{aligned} y &= x^2 - 6x + 2z^2 + 10 \\ y &= (x^2 - 6x + 9) + 2z^2 + 10 - 9 \\ y - 1 &= (x-3)^2 + 2z^2 \end{aligned}$$

We can rewrite this in standard form as:

$$y - 1 = \frac{(x-3)^2}{1} + \frac{z^2}{1/2}$$

Comparing this to our standard models, this equation represents an **elliptic paraboloid**. Its vertex is shifted to the coordinates $(3, 1, 0)$, and it opens along the positive direction of the y-axis.

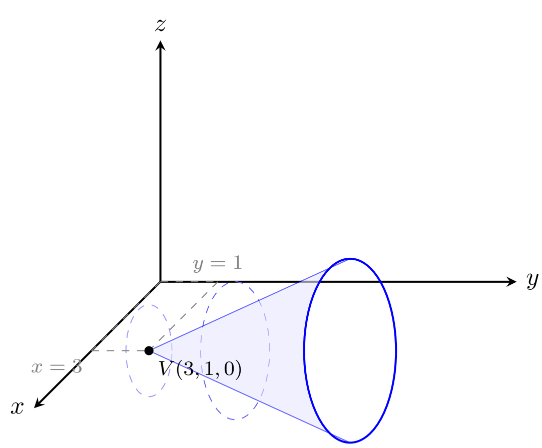


Figure 1.118 The shifted elliptic paraboloid $y - 1 = (x-3)^2 + 2z^2$.

▲

Chapter 2

Vector-valued Functions

Vector-valued functions are functions that take one or more variables and return a vector. They are used to describe curves and surfaces in three-dimensional space.

In this chapter, we will study the properties of vector-valued functions, including limits, continuity, and differentiability.

2.1 Vector-valued Functions and space curves

A vector-valued function is a function that takes one or more variables and returns a vector. It can be represented in the form:

$$\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$$

where $x(t)$, $y(t)$, and $z(t)$ are scalar functions of the variable t.

Definition 2.1 A **vector-valued function** is a function whose domain is a set of real numbers and whose range is a set of vectors. Let D be a set of real numbers. Then the three components of the vector-valued function $\mathbf{r}(t)$ are uniquely determined:

$$\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$$

where x, y, and z are scalar functions with the domain D. ◆

Example 2.2 If $\mathbf{r}(t) = \langle t^3, \ln(3-t), \sqrt{t} \rangle$, find the component functions.

Solution. By identifying the scalar functions multiplying each unit vector or filling each slot, the component functions are:

$$x(t) = t^3, \quad y(t) = \ln(3-t), \quad z(t) = \sqrt{t}$$

▲

Example 2.3 Let $\mathbf{r}(t) = \langle t+1, t^2-4, t^2 \rangle$ for $t \in \mathbb{R}$.

1. Sketch $\mathbf{r}(1)$ and $\mathbf{r}(3)$.
2. Find the vectors $\mathbf{r}(t)$ that are in a coordinate plane.

Solution.

1. The positions vectors are evaluated at the given parameter inputs:

$$\mathbf{r}(1) = \langle 1+1, 1^2-4, 1^2 \rangle = \langle 2, -3, 1 \rangle$$

$$\mathbf{r}(3) = \langle 3+1, 3^2-4, 3^2 \rangle = \langle 4, 5, 9 \rangle$$

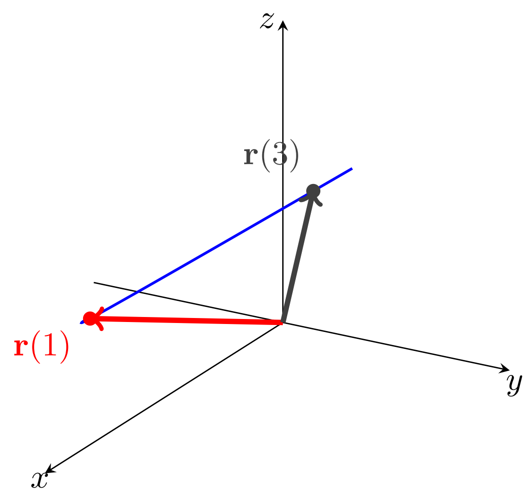


Figure 2.4 A 3D coordinate system showing the evaluation points on the path viewed at a 120° horizontal rotation.

2. We analyze each coordinate plane by setting the appropriate component to zero:

   (a) xy-plane : The **k**-component is zero.

$$\begin{aligned} t^2 = 0 &\implies t = 0 \\ &\implies \mathbf{r}(0) = \langle 1, -4, 0 \rangle \end{aligned}$$

   (b) yz-plane : The **i**-component is zero.

$$\begin{aligned} t+1 = 0 &\implies t = -1 \\ &\implies \mathbf{r}(-1) = \langle 0, -3, 1 \rangle \end{aligned}$$

   (c) xz-plane : The **j**-component is zero.

$$\begin{aligned} t^2-4 = 0 &\implies t = 2 \text{ or } t = -2 \\ &\implies \mathbf{r}(2) = \langle 3, 0, 4 \rangle \\ &\text{and } \mathbf{r}(-2) = \langle -1, 0, 4 \rangle \end{aligned}$$

▲

Limits and Continuity

Definition 2.5 The **limit** of a vector function $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$ is defined by taking the limits of its component functions:

$$\lim_{t\to a} \mathbf{r}(t) = \langle \lim_{t\to a} x(t), \lim_{t\to a} y(t), \lim_{t\to a} z(t) \rangle$$

provided the limits of the component functions exist.

A vector function $\mathbf{r}(t)$ is **continuous at** a if:

$$\lim_{t\to a} \mathbf{r}(t) = \mathbf{r}(a)$$

Equivalently, $\mathbf{r}(t)$ is continuous at a if and only if its component functions x, y, and z are continuous at a. ◆

Example 2.6 Find $\lim_{t\to 0}\mathbf{r}(t)$, where:

$$\mathbf{r}(t) = \left\langle 1 + t^3, te^{-t}, \frac{\sin t}{t} \right\rangle$$

Solution. We evaluate the limit by computing the limit of each component function individually as $t \to 0$:

$$\lim_{t\to 0}(1 + t^3) = 1 + 0^3 = 1$$
$$\lim_{t\to 0}(te^{-t}) = 0 \cdot e^0 = 0$$
$$\lim_{t\to 0}\left(\frac{\sin t}{t}\right) = 1$$

Combining these results into a single vector gives:

$$\lim_{t\to 0}\mathbf{r}(t) = \langle 1, 0, 1\rangle$$

▲

Suppose that f, g, and h are continuous on an interval I. Then the set C of all points (x, y, z) in space, where:

$$x = f(t), \quad y = g(t), \quad z = h(t)$$

and t varies in I, is called a **space curve**. The equations are called the **parametric equations of C** with a **parameter** t.

Example 2.7 Describe the curve defined by the vector function:

$$\mathbf{r}(t) = \langle 1 + t, 2 + 5t, -1 + 6t\rangle$$

Solution. The corresponding parametric equations are:

$$x = 1 + t, \quad y = 2 + 5t, \quad z = -1 + 6t$$

These are the parametric equations of a line passing through the point $(1, 2, -1)$ and parallel to the vector $\langle 1, 5, 6\rangle$.

Alternatively, we can observe that the function can be written as:

$$\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{v}, \quad \text{where } \mathbf{r}_0 = \langle 1, 2, -1\rangle \text{ and } \mathbf{v} = \langle 1, 5, 6\rangle$$

which is the vector equation of a line. ▲

Example 2.8 Sketch the curve whose vector equation is:

$$\mathbf{r}(t) = \cos t\,\mathbf{i} + \sin t\,\mathbf{j} + t\mathbf{k}$$

This curve is called a **helix**. This occurs in the model of DNA.

Solution. As the parameter t increases, the x- and y-components trace out a circle of radius 1 in the horizontal plane via $x = \cos t$ and $y = \sin t$. Concurrently, the z-component increases linearly since $z = t$. The combination of this circular base rotation and vertical motion generates a spiral path that climbs upward along the surface of a cylinder.

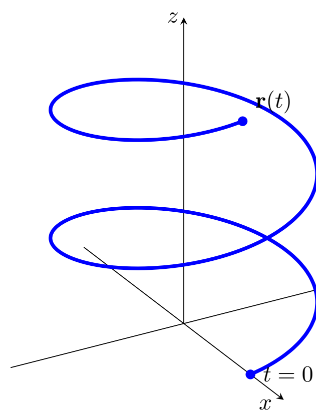


Figure 2.9 A 3D coordinate system tracing the path of a circular helix.

▲

Recall a vector equation for the line segment that joins the tip of the vector $\mathbf{r}_0$ to the tip of the vector $\mathbf{r}_1$:

$$\mathbf{r}(t) = (1-t)\mathbf{r}_0 + t\mathbf{r}_1 \qquad 0 \le t \le 1$$

Example 2.10 Find a vector equation and parametric equations for the line segment that joins the point $P(1,3,-2)$ to the point $Q(2,-1,3)$.

Solution. Using the points as components for the position vectors $\mathbf{r}_0 = \langle 1,3,-2\rangle$ and $\mathbf{r}_1 = \langle 2,-1,3\rangle$, the vector equation is:

$$\mathbf{r}(t) = \langle 1+t,\ 3-4t,\ -2+5t\rangle, \quad 0 \le t \le 1$$

The corresponding parametric equations are:

$$x = 1+t, \quad y = 3-4t, \quad z = -2+5t \qquad 0 \le t \le 1$$

▲

Example 2.11 Find a vector function that represents the curve of the intersection of the cylinder $x^2+y^2=1$ and the plane $y+z=2$.

Solution. The projection of the intersection curve C onto the xy-plane is the unit circle $x^2+y^2=1$ with $z=0$. We can parameterize this circular footprint as:

$$x = \cos t, \quad y = \sin t \qquad 0 \le t \le 2\pi$$

To find the corresponding z-coordinate along the path, substitute the expression for y into the plane equation $y+z=2$:

$$z = 2 - y = 2 - \sin t$$

Thus, the parametric equations for C are $x=\cos t$, $y=\sin t$, and $z = 2-\sin t$. The corresponding vector function is:

$$\mathbf{r}(t) = \cos t\mathbf{i} + \sin t\mathbf{j} + (2-\sin t)\mathbf{k} \qquad 0 \le t \le 2\pi$$

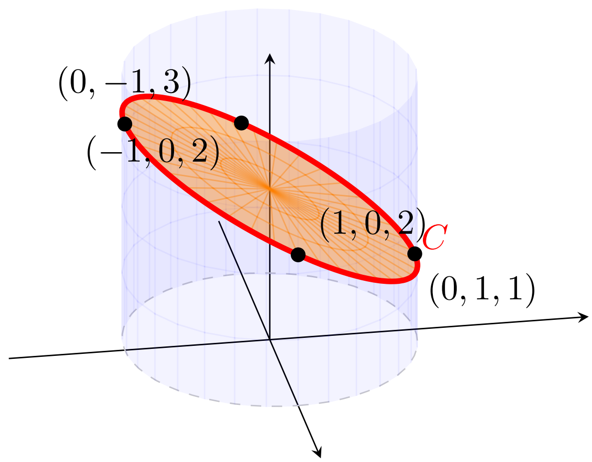


Figure 2.12 The colored intersection region viewed from an 80° horizontal rotation to separate tracking nodes.

▲

Example 2.13 Sketch the curve whose vector equation is:

$$\mathbf{r}(t) = t\mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k} \quad t \geq 0$$

This curve is called a **twisted cubic**. You can view this curve as the intersection of the cylinders $y = x^2$ and $z = x^3$.

Solution. As t increases from 0, the path traces out a curve where the y-coordinate grows quadratically relative to x, and the z-coordinate grows cubically.

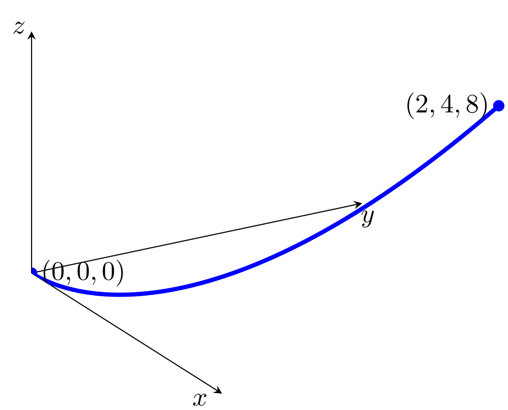


Figure 2.14 The parametric path of a twisted cubic curve for $t \geq 0$.

▲

Checkpoint 2.15 Let $\mathbf{r}(t) = 2t\mathbf{i} + (8 - 2t^2)\mathbf{j}$ for $-2 \leq t \leq 2$. Sketch the curve C defined by $\mathbf{r}(t)$ and indicate its orientation.

Solution. Eliminating the parameter yields $t = x/2$. Substituting this into the y-component gives the Cartesian equation:

$$y = 8 - 2\left(\frac{x}{2}\right)^2 = 8 - \frac{1}{2}x^2$$

This is a downward-opening parabola. As t goes from -2 to 2, the curve is traced from left to right starting at $(-4, 0)$, passing through the vertex at $(0, 8)$, and ending at $(4, 0)$.

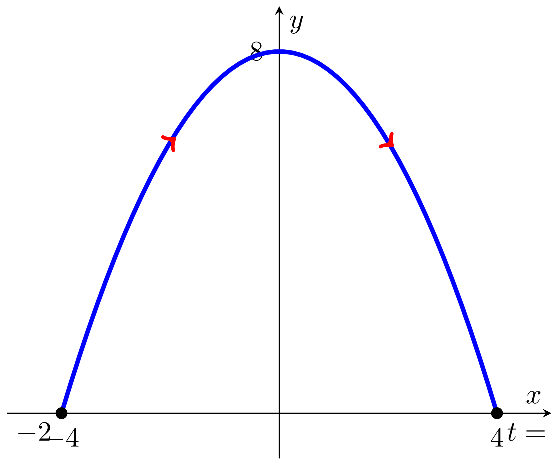


Figure 2.16 Downward parabola showing left-to-right orientation arrows.

Checkpoint 2.17 Let $\mathbf{r}(t) = \left(\frac{1}{2}t\cos t\right)\mathbf{i} + \left(\frac{1}{2}t\sin t\right)\mathbf{j}$ for $0 \leq t \leq 3\pi$. Sketch the curve C defined by $\mathbf{r}(t)$ and indicate its orientation.

Solution. In polar coordinates, this curve can be viewed as the spiral $r = \frac{1}{2}\theta$, where the parameter t acts as the angle θ. As t increases from 0 to 3π, the point winds counterclockwise outward from the origin, completing one and a half full revolutions.

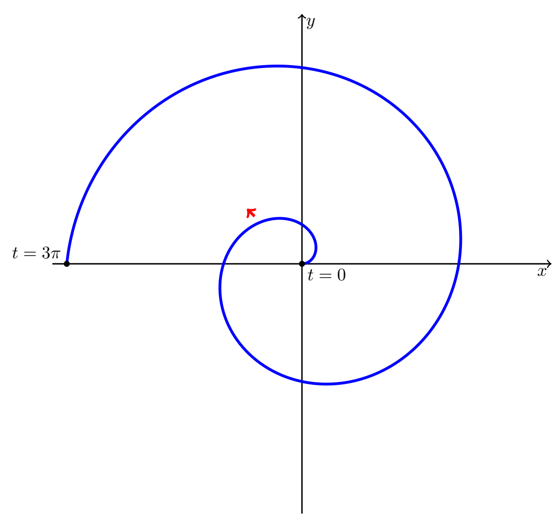


Figure 2.18 An Archimedean spiral winding outward counterclockwise with directional indicators.

2.2 Derivatives of Vector Functions

In this section, we will explore the concept of derivatives of vector functions. A vector function is a function that takes one or more variables and returns a vector. The derivative of a vector function provides information about how the vector changes with respect to its input variables.

We will cover the following topics:

- Definition of the derivative of a vector function
- Rules for differentiating vector functions

Differentiability

A vector-valued function is differentiable at $t = a$ if the derivative exists, which is defined as:

$$\mathbf{r}'(a) = \lim_{h \to 0} \frac{\mathbf{r}(a+h) - \mathbf{r}(a)}{h}$$

The derivative of a vector-valued function gives the tangent vector to the curve at that point.

Definition 2.19 The **derivative $\mathbf{r}'$** of a vector function $\mathbf{r}$ is defined in the same manner as for real-valued functions:

$$\frac{d\mathbf{r}}{dt} = \mathbf{r}'(t) = \lim_{h \to 0} \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h} = \lim_{\Delta t \to 0} \frac{1}{\Delta t}(\mathbf{r}(t + \Delta t) - \mathbf{r}(t))$$

◆

Theorem 2.20 *If $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k} = \langle x(t),\ y(t),\ z(t)\rangle$ where x, y, and z are differentiable functions, then:*

$$\mathbf{r}'(t) = x'(t)\mathbf{i} + y'(t)\mathbf{j} + z'(t)\mathbf{k} = \langle x'(t),\ y'(t),\ z'(t)\rangle$$

Proof. This rule is easily proven by applying the component-wise limits directly into the fundamental definition of the derivative of vector functions. ■

If the points P and Q have position vectors $\mathbf{r}(t)$ and $\mathbf{r}(t+h)$, then the displacement vector $\vec{PQ}$ represents the vector $\mathbf{r}(t+h) - \mathbf{r}(t)$. As $h \to 0$, the vector $\mathbf{r}(t+h) - \mathbf{r}(t)$ approaches a vector that lies along the target trajectory's path.

The vector $\mathbf{r}'(t)$ is called the **tangent vector** to the curve defined by $\mathbf{r}(t)$ at P, provided that $\mathbf{r}'(t)$ exists and $\mathbf{r}'(t) \neq \mathbf{0}$. The tangent line to C at P is defined to be the line through P parallel to the tangent vector $\mathbf{r}'(t)$. (In a 2D plane, the slope of the line through P parallel to $\mathbf{r}'(t)$ evaluates to $\frac{y'(t)}{x'(t)}$).

The Unit Tangent Vector.

We define the **unit tangent vector** as:

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|}$$

Corollary 2.21 *If $\mathbf{r}'(t)$ exists, then we say $\mathbf{r}(t)$ is differentiable and write:*

$$\mathbf{r}'(t) = D_t\mathbf{r}(t) = \frac{d}{dt}\mathbf{r}(t)$$

Similarly, the second derivative $\mathbf{r}''(t)$ is defined as:

$$\mathbf{r}''(t) = x''(t)\mathbf{i} + y''(t)\mathbf{j} + z''(t)\mathbf{k} = \langle x''(t),\ y''(t),\ z''(t)\rangle$$

Proof. Let $\mathbf{r}(t) = \langle x(t), y(t), z(t)\rangle$. By applying the component derivative theorem, the first derivative is computed component-by-component:

$$\frac{d}{dt}\mathbf{r}(t) = \left\langle \frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right\rangle = \langle x'(t), y'(t), z'(t) \rangle$$

Since each component function $x'(t)$, $y'(t)$, and $z'(t)$ is a standard real-valued scalar function, we can differentiate a second time using standard single-variable calculus rules:

$$\begin{aligned} \mathbf{r}''(t) &= \frac{d}{dt}\left[\mathbf{r}'(t)\right] \\ &= \left\langle \frac{d}{dt}[x'(t)], \frac{d}{dt}[y'(t)], \frac{d}{dt}[z'(t)] \right\rangle \\ &= \langle x''(t), y''(t), z''(t) \rangle \end{aligned}$$

Expressing this result in terms of the standard unit basis vectors yields $x''(t)\mathbf{i} + y''(t)\mathbf{j} + z''(t)\mathbf{k}$. ■

Example 2.22 Let $\mathbf{r}(t) = \ln t\ \mathbf{i} + e^{-3t}\mathbf{j} + t^2\mathbf{k}$. Find the derivative $\mathbf{r}'(t)$ where $t > 0$.

Solution. We differentiate each scalar component function with respect to t:

$$\begin{aligned} \frac{d}{dt}(\ln t) &= \frac{1}{t} \\ \frac{d}{dt}(e^{-3t}) &= -3e^{-3t} \\ \frac{d}{dt}(t^2) &= 2t \end{aligned}$$

Combining these component results gives the derivative vector function:

$$\mathbf{r}'(t) = \frac{1}{t}\mathbf{i} - 3e^{-3t}\mathbf{j} + 2t\mathbf{k}$$

▲

Example 2.23 Let $\mathbf{r}(t) = (1 + t^3)\mathbf{i} + te^{-t}\mathbf{j} + \sin 2t\mathbf{k}$.

1. Find $\mathbf{r}'(t)$ and $\mathbf{r}''(t)$.

2. Find the unit tangent vector at the point where $t = 0$.

Solution.

1. Differentiating component-by-component using the power rule, product rule, and chain rule yields the first derivative:

$$\mathbf{r}'(t) = 3t^2\mathbf{i} + (e^{-t} - te^{-t})\mathbf{j} + 2\cos 2t\mathbf{k}$$

Differentiating a second time gives the second derivative:

$$\mathbf{r}''(t) = 6t\mathbf{i} + (-e^{-t} - e^{-t} + te^{-t})\mathbf{j} - 4\sin 2t\mathbf{k} = 6t\mathbf{i} + (te^{-t} - 2e^{-t})\mathbf{j} - 4\sin 2t\mathbf{k}$$

2. To find the unit tangent vector $\mathbf{T}(0)$, we first evaluate the velocity derivative vector at $t = 0$:

$$\mathbf{r}'(0) = 3(0)^2\mathbf{i} + (e^0 - 0)\mathbf{j} + 2\cos(0)\mathbf{k} = \mathbf{j} + 2\mathbf{k} = \langle 0, 1, 2 \rangle$$

Next, compute the magnitude of this tangent vector:

$$|\mathbf{r}'(0)| = \sqrt{0^2 + 1^2 + 2^2} = \sqrt{5}$$

Divide the vector by its magnitude to obtain the final unit tangent vector:

$$\mathbf{T}(0) = \frac{\mathbf{r}'(0)}{|\mathbf{r}'(0)|} = \frac{1}{\sqrt{5}}\mathbf{j} + \frac{2}{\sqrt{5}}\mathbf{k} = \left\langle 0, \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right\rangle$$

▲

Checkpoint 2.24 For the curve $\mathbf{r}(t) = \sqrt{t}\mathbf{i} + (2-t)\mathbf{j}$, find $\mathbf{r}'(t)$ and sketch the position vector $\mathbf{r}(1)$ and the tangent vector $\mathbf{r}'(1)$.

Solution. Differentiating each component with respect to t yields:

$$\mathbf{r}'(t) = \frac{1}{2\sqrt{t}}\mathbf{i} - \mathbf{j}$$

Evaluating both the original position vector and the derivative tangent vector at $t = 1$ gives:

$$\mathbf{r}(1) = \mathbf{i} + \mathbf{j} = \langle 1, 1 \rangle$$

$$\mathbf{r}'(1) = \frac{1}{2}\mathbf{i} - \mathbf{j} = \left\langle \frac{1}{2}, -1 \right\rangle$$

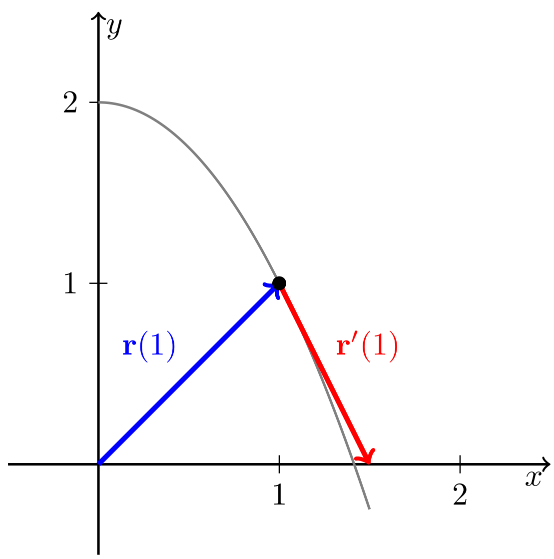


Figure 2.25 The position vector $\mathbf{r}(1)$ and the tangent vector $\mathbf{r}'(1)$ positioned on the curve.

Checkpoint 2.26 Find the parametric equations for the tangent line to the helix with the parametric equations:

$$x = 2\cos t, \quad y = \sin t, \quad z = t \quad \text{at the point } (0, 1, \pi/2)$$

Solution. First, find the parameter value t_0 that corresponds to the given point $(0, 1, \pi/2)$. Matching the components yields $z = t = \pi/2$. Testing this value reveals $2\cos(\pi/2) = 0$ and $\sin(\pi/2) = 1$, which confirms that $t_0 = \pi/2$.

Next, differentiate the vector function $\mathbf{r}(t) = \langle 2\cos t, \sin t, t \rangle$ to find the directional tangent vector:

$$\mathbf{r}'(t) = \langle -2\sin t, \cos t, 1 \rangle$$

Evaluating the direction vector at our parameter point $t_0 = \pi/2$ gives:

$$\mathbf{r}'(\pi/2) = \langle -2(1), 0, 1 \rangle = \langle -2, 0, 1 \rangle$$

Using the base point coordinates $(0, 1, \pi/2)$ and the direction components $\langle -2, 0, 1\rangle$, the parametric equations for the tangent line are:

$$x = -2t, \quad y = 1, \quad z = \frac{\pi}{2} + t$$

Checkpoint 2.27 Find the parametric equations for the tangent line to the twisted curve with the parametric equations:

$$x = t, \quad y = t^2, \quad z = t^3 \quad \text{at the point } t = 2$$

Solution. Evaluate the base point on the curve at $t = 2$:

$$x_0 = 2, \quad y_0 = 2^2 = 4, \quad z_0 = 2^3 = 8 \implies (2, 4, 8)$$

Differentiate the component expressions to find the general velocity direction profile:

$$\frac{dx}{dt} = 1, \quad \frac{dy}{dt} = 2t, \quad \frac{dz}{dt} = 3t^2$$

Evaluating these derivatives at $t = 2$ gives the parallel directional components for our tangent line:

$$\left.\frac{dx}{dt}\right|_{t=2} = 1, \quad \left.\frac{dy}{dt}\right|_{t=2} = 4, \quad \left.\frac{dz}{dt}\right|_{t=2} = 12$$

Using the point $(2, 4, 8)$ and the direction components vector $\langle 1, 4, 12\rangle$, the parametric equations for the line are:

$$x = 2 + t, \quad y = 4 + 4t, \quad z = 8 + 12t$$

Definition 2.28 A curve given by a vector function $\mathbf{r}(t)$ on an interval I is called **smooth** if $\mathbf{r}'(t)$ is continuous and $\mathbf{r}'(t) \neq \mathbf{0}$, except possibly at the endpoints of I. ◆

Example 2.29 Determine whether the semicubical parabola $\mathbf{r}(t) = \langle 1 + t^3, t^2\rangle$ is smooth.

Solution. First, we compute the derivative vector function component-by-component:

$$\mathbf{r}'(t) = \langle 3t^2, 2t\rangle$$

Both component functions, $3t^2$ and $2t$, are polynomials, so $\mathbf{r}'(t)$ is continuous everywhere. Next, we determine if there are any parameter values where the derivative vector equals the zero vector:

$$\mathbf{r}'(t) = \langle 0, 0\rangle \implies 3t^2 = 0 \text{ and } 2t = 0$$

This system yields the single solution $t = 0$. Since $\mathbf{r}'(0) = \mathbf{0}$ at an interior point of the domain, the curve is **not smooth** at the origin (it features a sharp cusp). ▲

A curve like the semicubical parabola that is made up of a finite number of smooth pieces is called **piecewise smooth**.

Example 2.30 Determine whether the helix $\mathbf{r}(t) = \langle 2\cos t, \sin t, t\rangle$ is smooth.

Solution. We find the derivative vector function:

$$\mathbf{r}'(t) = \langle -2\sin t, \cos t, 1\rangle$$

The components are standard trigonometric and constant functions, meaning $\mathbf{r}'(t)$ is continuous for all real numbers.

For the vector to be the zero vector, all three components must simultaneously equal zero. However, the **k**-component is always $1 \neq 0$. Therefore, $\mathbf{r}'(t) \neq \mathbf{0}$ for all values of t. We conclude that the helix is a **smooth curve**. ▲

Theorem 2.31 *Suppose* **u** *and* **v** *are differentiable vector functions, c is a scalar, and f is a real-valued function. Then:*

1. $\dfrac{d}{dt}[\mathbf{u}(t) + \mathbf{v}(t)] = \mathbf{u}'(t) + \mathbf{v}'(t)$

2. $\dfrac{d}{dt}[c\mathbf{u}(t)] = c\mathbf{u}'(t)$

3. $\dfrac{d}{dt}[f(t)\mathbf{u}(t)] = f'(t)\mathbf{u}(t) + f(t)\mathbf{u}'(t)$

4. $\dfrac{d}{dt}[\mathbf{u}(t) \cdot \mathbf{v}(t)] = \mathbf{u}'(t) \cdot \mathbf{v}(t) + \mathbf{u}(t) \cdot \mathbf{v}'(t)$

5. $\dfrac{d}{dt}[\mathbf{u}(t) \times \mathbf{v}(t)] = \mathbf{u}'(t) \times \mathbf{v}(t) + \mathbf{u}(t) \times \mathbf{v}'(t)$

6. $\dfrac{d}{dt}[\mathbf{u}(f(t))] = f'(t)\mathbf{u}'(f(t))$ *(Chain Rule)*

Theorem 2.32 *If $\mathbf{r}(t)$ is differentiable and $|\mathbf{r}(t)| = c$ (a constant), then $\mathbf{r}'(t)$ is orthogonal to $\mathbf{r}(t)$ for all t in the domain of $\mathbf{r}'(t)$:*

$$\mathbf{r}(t) \cdot \frac{d\mathbf{r}}{dt} = 0$$

Proof. Since $|\mathbf{r}(t)| = c$, squaring both sides gives $|\mathbf{r}(t)|^2 = c^2$. Using the relationship between the magnitude and the dot product, we can rewrite this as:

$$\mathbf{r}(t) \cdot \mathbf{r}(t) = c^2$$

Differentiating both sides with respect to t using the dot product rule yields:

$$\mathbf{r}'(t) \cdot \mathbf{r}(t) + \mathbf{r}(t) \cdot \mathbf{r}'(t) = 0$$

Since the dot product is commutative, we combine the terms:

$$2\left(\mathbf{r}(t) \cdot \mathbf{r}'(t)\right) = 0 \implies \mathbf{r}(t) \cdot \frac{d\mathbf{r}}{dt} = 0$$

Because their dot product is zero, the position vector $\mathbf{r}(t)$ and the derivative vector $\mathbf{r}'(t)$ are orthogonal. ■

2.3 Integrals of Vector Functions

In this section, we will discuss the integrals of vector functions. A vector function is a function that takes one or more variables and returns a vector. The integral of a vector function can be computed component-wise, meaning that we can integrate each component of the vector function separately.

Definition 2.33 Indefinite Integral. If $\mathbf{R}(t)$ is an antiderivative of $\mathbf{r}(t)$, then:

$$\int \mathbf{r}(t)\, dt = \mathbf{R}(t) + \mathbf{C}$$

◆

Definition 2.34 Let $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k} = \langle x(t),\ y(t),\ z(t)\rangle$ where x, y, and z are integrable on $[a, b]$. Then we say $\mathbf{r}(t)$ is integrable on $[a, b]$ and:

$$\int_a^b \mathbf{r}(t)\,dt = \left[\int_a^b x(t)\,dt\right]\mathbf{i} + \left[\int_a^b y(t)\,dt\right]\mathbf{j} + \left[\int_a^b z(t)\,dt\right]\mathbf{k}$$

If $\mathbf{R}'(t) = \mathbf{r}(t)$, then $\mathbf{R}(t)$ is an antiderivative of $\mathbf{r}(t)$. ◆

Example 2.35 Evaluate:

$$\int_0^\pi ((\cos t)\mathbf{i} + \mathbf{j} - t^2\mathbf{k})\,dt$$

Solution. We integrate each component function independently over the interval $[0, \pi]$ by applying the Fundamental Theorem of Calculus componentwise:

$$\begin{aligned}\int_0^\pi ((\cos t)\mathbf{i} + \mathbf{j} - t^2\mathbf{k})\,dt &= \left[\int_0^\pi \cos t\,dt\right]\mathbf{i} + \left[\int_0^\pi 1\,dt\right]\mathbf{j} - \left[\int_0^\pi t^2\,dt\right]\mathbf{k} \\ &= [\sin t]_0^\pi\,\mathbf{i} + [t]_0^\pi\,\mathbf{j} - \left[\frac{t^3}{3}\right]_0^\pi \mathbf{k} \\ &= (\sin\pi - \sin 0)\mathbf{i} + (\pi - 0)\mathbf{j} - \left(\frac{\pi^3}{3} - 0\right)\mathbf{k} \\ &= 0\mathbf{i} + \pi\mathbf{j} - \frac{\pi^3}{3}\mathbf{k} \\ &= \pi\mathbf{j} - \frac{\pi^3}{3}\mathbf{k} = \left\langle 0, \pi, -\frac{\pi^3}{3}\right\rangle\end{aligned}$$

▲

Theorem 2.36 *If $\mathbf{R}(t)$ is an antiderivative of $\mathbf{r}(t)$ on $[a, b]$, then:*

$$\int_a^b \mathbf{r}(t)\,dt = \mathbf{R}(t)\big|_a^b = \mathbf{R}(b) - \mathbf{R}(a)$$

Example 2.37 If $\mathbf{r}(t) = 12t^3\mathbf{i} + 4e^{2t}\mathbf{j} + (t+1)^{-1}\mathbf{k}$, find $\int_0^2 \mathbf{r}(t)\,dt$.

Solution. We find an antiderivative vector function $\mathbf{R}(t)$ by integrating each scalar component function:

$$\begin{aligned}\mathbf{R}(t) &= \left(\int 12t^3\,dt\right)\mathbf{i} + \left(\int 4e^{2t}\,dt\right)\mathbf{j} + \left(\int \frac{1}{t+1}\,dt\right)\mathbf{k} \\ &= 3t^4\mathbf{i} + 2e^{2t}\mathbf{j} + \ln|t+1|\mathbf{k}\end{aligned}$$

Now, apply the Fundamental Theorem of Calculus by evaluating $\mathbf{R}(t)$ from $t = 0$ to $t = 2$:

$$\begin{aligned}\int_0^2 \mathbf{r}(t)\,dt &= \mathbf{R}(2) - \mathbf{R}(0) \\ &= \left(3(2)^4\mathbf{i} + 2e^{2(2)}\mathbf{j} + \ln|2+1|\mathbf{k}\right) - \left(3(0)^4\mathbf{i} + 2e^{2(0)}\mathbf{j} + \ln|0+1|\mathbf{k}\right) \\ &= \left(48\mathbf{i} + 2e^4\mathbf{j} + \ln(3)\mathbf{k}\right) - (0\mathbf{i} + 2\mathbf{j} + 0\mathbf{k}) \\ &= 48\mathbf{i} + (2e^4 - 2)\mathbf{j} + \ln(3)\mathbf{k} \\ &= \left\langle 48, 2e^4 - 2, \ln(3)\right\rangle\end{aligned}$$

▲

Example 2.38 A particle traveling in a straight line is located at the point $(1, -1, 2)$ and has speed 2 at time $t = 0$. The particle moves toward the point $(3, 0, 3)$ with constant acceleration $\mathbf{a}(t) = 2\mathbf{i} + \mathbf{j} + \mathbf{k}$. Find its position vector $\mathbf{r}(t)$ at time t.

Solution. First, we establish our initial conditions at $t = 0$. The initial position vector is given directly by the starting coordinates:

$$\mathbf{r}(0) = \langle 1, -1, 2\rangle$$

To find the initial velocity vector $\mathbf{v}(0)$, we know the particle moves from $P(1, -1, 2)$ toward $Q(3, 0, 3)$. The direction vector is:

$$\vec{PQ} = \langle 3 - 1, 0 - (-1), 3 - 2\rangle = \langle 2, 1, 1\rangle$$

We normalize this direction vector to get a unit direction vector:

$$\mathbf{u} = \frac{\langle 2, 1, 1\rangle}{\sqrt{2^2 + 1^2 + 1^2}} = \frac{1}{\sqrt{6}}\langle 2, 1, 1\rangle$$

Multiplying the initial speed of 2 by this unit direction vector yields our initial velocity vector:

$$\mathbf{v}(0) = 2\mathbf{u} = \frac{2}{\sqrt{6}}\langle 2, 1, 1\rangle = \left\langle \frac{4}{\sqrt{6}}, \frac{2}{\sqrt{6}}, \frac{2}{\sqrt{6}} \right\rangle$$

Next, we find the general velocity vector $\mathbf{v}(t)$ by integrating the constant acceleration vector:

$$\mathbf{v}(t) = \int \mathbf{a}(t)\, dt = \int \langle 2, 1, 1\rangle\, dt = \langle 2t, t, t\rangle + \mathbf{C}_1$$

Using our initial velocity condition at $t = 0$ reveals $\mathbf{C}_1 = \mathbf{v}(0)$, so:

$$\mathbf{v}(t) = \left\langle 2t + \frac{4}{\sqrt{6}}, t + \frac{2}{\sqrt{6}}, t + \frac{2}{\sqrt{6}} \right\rangle$$

Finally, we integrate the velocity vector function to find the position vector function $\mathbf{r}(t)$:

$$\mathbf{r}(t) = \int \mathbf{v}(t)\, dt = \left\langle t^2 + \frac{4}{\sqrt{6}}t, \frac{1}{2}t^2 + \frac{2}{\sqrt{6}}t, \frac{1}{2}t^2 + \frac{2}{\sqrt{6}}t \right\rangle + \mathbf{C}_2$$

Using our initial position condition at $t = 0$ reveals $\mathbf{C}_2 = \mathbf{r}(0) = \langle 1, -1, 2\rangle$. Combining these gives the final position vector function:

$$\mathbf{r}(t) = \left\langle t^2 + \frac{4}{\sqrt{6}}t + 1, \frac{1}{2}t^2 + \frac{2}{\sqrt{6}}t - 1, \frac{1}{2}t^2 + \frac{2}{\sqrt{6}}t + 2 \right\rangle$$

▲

2.4 Arc Length and Curvature

In this section, we will discuss the concept of arc length for curves defined by vector-valued functions. The arc length of a curve is the distance along the curve between two points. We will derive a formula for computing the arc length of a curve defined by a vector function and provide examples to illustrate the process.

Definition 2.39 Arc Length of a Curve. The length L of a smooth curve C traced out by a vector function $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$ on the interval $a \le t \le b$ is defined as:

$$L = \int_a^b \sqrt{[x'(t)]^2 + [y'(t)]^2 + [z'(t)]^2}\, dt = \int_a^b \sqrt{\left[\frac{dx}{dt}\right]^2 + \left[\frac{dy}{dt}\right]^2 + \left[\frac{dz}{dt}\right]^2}\, dt = \int_a^b |\mathbf{r}'(t)|\, dt$$

◆

Example 2.40 Find the length of the arc of the circular helix with vector equation:

$$\mathbf{r}(t) = \cos t\ \mathbf{i} + \sin t\ \mathbf{j} + t\ \mathbf{k}$$

from the point $(1, 0, 0)$ to the point $(1, 0, 2\pi)$.

Solution. First, we determine the parameter limits $t = a$ and $t = b$ corresponding to the given points. Comparing the **k**-component $z = t$ to the points yields:

$$\text{At } (1, 0, 0) \implies t = 0$$
$$\text{At } (1, 0, 2\pi) \implies t = 2\pi$$

Next, we compute the derivative vector function $\mathbf{r}'(t)$ and its magnitude:

$$\mathbf{r}'(t) = \langle -\sin t, \cos t, 1\rangle$$

$$|\mathbf{r}'(t)| = \sqrt{(-\sin t)^2 + (\cos t)^2 + 1^2} = \sqrt{\sin^2 t + \cos^2 t + 1} = \sqrt{1+1} = \sqrt{2}$$

Applying the arc length formula over the interval $[0, 2\pi]$ gives:

$$L = \int_0^{2\pi} |\mathbf{r}'(t)|\, dt = \int_0^{2\pi} \sqrt{2}\, dt = \left[\sqrt{2}t\right]_0^{2\pi} = 2\pi\sqrt{2}$$

▲

Checkpoint 2.41 A curve C can be represented by more than one vector function. For instance, the twisted cubic can be parameterized as:

$$\mathbf{r}_1(t) = t\,\mathbf{i} + t^2\,\mathbf{j} + t^3\,\mathbf{k} \quad \text{for } 1 \le t \le 2$$

or as:

$$\mathbf{r}_2(u) = e^u\,\mathbf{i} + e^{2u}\,\mathbf{j} + e^{3u}\,\mathbf{k} \quad \text{for } 0 \le u \le \ln 2$$

Therefore, the vector functions representing a space curve are ***not*** uniquely determined.

Solution. To show that $\mathbf{r}_1(t)$ and $\mathbf{r}_2(u)$ trace out the exact same geometric path in space, we can perform a change of variables (reparameterization).

Let us define a substitution relationship between the two parameters t and u:

$$t = e^u$$

If we take the natural logarithm of both sides, this is equivalent to writing $u = \ln t$.

Now, we verify two things to ensure the representations are identical: the component equations and the domain boundaries.

1. Component Transformation: Substitute $t = e^u$ directly into the component equations of $\mathbf{r}_1(t)$:

$$x(t) = t \implies x(u) = e^u$$

$$y(t) = t^2 \implies y(u) = (e^u)^2 = e^{2u}$$
$$z(t) = t^3 \implies z(u) = (e^u)^3 = e^{3u}$$

This matches the vector structure of $\mathbf{r}_2(u) = \langle e^u, e^{2u}, e^{3u} \rangle$ perfectly.

2. Domain Mapping: We must map the parameter boundaries from t to u using our transition formula $u = \ln t$:

$$\text{Lower bound: } t = 1 \implies u = \ln(1) = 0$$
$$\text{Upper bound: } t = 2 \implies u = \ln(2)$$

This confirms that as t travels over the continuous interval $[1, 2]$, the new parameter u travels exactly over the continuous interval $[0, \ln 2]$.

Both vector functions start at the point $(1, 1, 1)$ and terminate at the point $(2, 4, 8)$, tracking the exact same space path. This illustrates that a single space curve can have infinitely many valid parametric representations.

We say that those equations are **parameterizations** of the curve C.

Definition 2.42 Suppose that C is a piecewise-smooth curve given by a vector function $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$ for $a \le t \le b$, and C is traversed exactly once as t increases from a to b. The **arc length function** s is defined as:

$$s(t) = \int_a^t |\mathbf{r}'(u)|\, du = \int_a^t \sqrt{\left[\frac{dx}{du}\right]^2 + \left[\frac{dy}{du}\right]^2 + \left[\frac{dz}{du}\right]^2}\, du$$

So, $s(t)$ represents the accumulated length of the part of C between the starting point $\mathbf{r}(a)$ and the variable position $\mathbf{r}(t)$. ◆

By using the Fundamental Theorem of Calculus Part 1 to differentiate the arc length function $s(t)$, we get:

$$\frac{ds}{dt} = |\mathbf{r}'(t)|$$

This structural relationship allows us to **parametrize a curve with respect to arc length**. Arc length parameterization arises naturally from the geometry of the curve itself and remains invariant across all chosen external coordinate systems.

Example 2.43 Reparametrize the helix $\mathbf{r}(t) = \cos t\,\mathbf{i} + \sin t\,\mathbf{j} + t\,\mathbf{k}$ with respect to arc length measured from the point $(1, 0, 0)$ in the direction of increasing t (a **positive orientation**).

Solution. The initial baseline point $(1, 0, 0)$ maps directly to the parameter value $t = 0$. Using our previous evaluation, we know that:

$$\frac{ds}{dt} = |\mathbf{r}'(t)| = \sqrt{(-\sin t)^2 + (\cos t)^2 + 1^2} = \sqrt{2}$$

Now, compute the active arc length function $s(t)$ by integration:

$$s = s(t) = \int_0^t |\mathbf{r}'(u)|\, du = \int_0^t \sqrt{2}\, du = \sqrt{2}t$$

Isolating the original domain parameter yields the substitution profile $t = \frac{s}{\sqrt{2}}$. Substituting this back into our primary position function gives the desired arc length parameterization:

$$\mathbf{r}(t(s)) = \left\langle \cos\left(\frac{s}{\sqrt{2}}\right),\ \sin\left(\frac{s}{\sqrt{2}}\right),\ \frac{s}{\sqrt{2}} \right\rangle$$

▲

Checkpoint 2.44 Find the arc length parametrization for a circle $x^2 + y^2 = 2^2$.

Solution. Let $A(2, 0)$ represent the fixed initial coordinate point and let $P(x, y)$ match any coordinate along the circle. The standard parametric representation tracking counterclockwise rotation gives:

$$x = 2\cos\theta, \quad y = 2\sin\theta \quad \text{for } 0 \leq \theta \leq 2\pi$$

The magnitude of the derivative tracking vector evaluates to:

$$|\mathbf{r}'(\theta)| = |\langle -2\sin\theta, 2\cos\theta \rangle| = \sqrt{4\sin^2\theta + 4\cos^2\theta} = 2$$

Setting up our accumulated arc length function s yields:

$$s(\theta) = \int_0^\theta 2\,du = 2\theta \implies \theta = \frac{s}{2}$$

Mapping the boundaries into the new parametric frame maps $\theta = 0 \implies s = 0$ and $\theta = 2\pi \implies s = 4\pi$. Substituting the transformation pattern into our vector structure outputs:

$$\mathbf{r}(\theta(s)) = \left\langle 2\cos\left(\frac{s}{2}\right),\ 2\sin\left(\frac{s}{2}\right) \right\rangle \quad \text{for } 0 \leq s \leq 4\pi$$

If $\mathbf{r}(t)$ is a vector-valued function and the space curve C determined by $\mathbf{r}(t)$ is smooth, then $\mathbf{r}'(t)$ represents a tangent vector pointing in the direction of the curve's orientation. Provided $\mathbf{r}'(t) \neq \mathbf{0}$, the **unit tangent vector** $\mathbf{T}(t)$ is defined as:

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|}$$

Because $|\mathbf{T}(t)| = 1$ remains constant for all values of t, if $\mathbf{T}'(t)$ is differentiable, then $\mathbf{T}'(t)$ must be orthogonal to $\mathbf{T}(t)$.

The Unit Normal Vector.

The **unit normal vector** $\mathbf{N}(t)$ to C is defined as the unit vector pointing in the same direction as $\mathbf{T}'(t)$:

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{|\mathbf{T}'(t)|}$$

Example 2.45 Let C be a curve determined by the vector function $\mathbf{r}(t) = t^2\mathbf{i} + t\mathbf{j}$.

1. Find the unit tangent and unit normal vectors $\mathbf{T}(t)$ and $\mathbf{N}(t)$.
2. Find $\mathbf{T}(1)$ and $\mathbf{N}(1)$.
3. Check the orthogonality of $\mathbf{T}(t)$ and $\mathbf{N}(t)$.

Solution.

1. First, differentiate the position vector function to locate our general velocity vector profile:

$$\mathbf{r}'(t) = 2t\mathbf{i} + \mathbf{j} \implies |\mathbf{r}'(t)| = \sqrt{4t^2 + 1}$$

Dividing the derivative vector by its magnitude gives the unit tangent vector function:

$$\mathbf{T}(t) = \frac{2t}{\sqrt{4t^2+1}}\mathbf{i} + \frac{1}{\sqrt{4t^2+1}}\mathbf{j}$$

Next, differentiate $\mathbf{T}(t)$ using the quotient rule or chain rule to find the normal vector direction:

$$\mathbf{T}'(t) = \frac{2(4t^2+1)^{-1/2} - 2t \cdot \frac{1}{2}(4t^2+1)^{-1/2}(8t)}{4t^2+1}\mathbf{i} + \left(-\frac{1}{2}(4t^2+1)^{-3/2}(8t)\right)\mathbf{j}$$

$$\mathbf{T}'(t) = \frac{2}{(4t^2+1)^{3/2}}\mathbf{i} - \frac{4t}{(4t^2+1)^{3/2}}\mathbf{j} \implies |\mathbf{T}'(t)| = \frac{\sqrt{4+16t^2}}{(4t^2+1)^{3/2}} = \frac{2}{\sqrt{4t^2+1}}$$

Normalizing this vector tracking profile produces our final unit normal vector function:

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{|\mathbf{T}'(t)|} = \frac{1}{\sqrt{4t^2+1}}\mathbf{i} - \frac{2t}{\sqrt{4t^2+1}}\mathbf{j}$$

2. Evaluating both unit vector function equations at our specific target parameter point $t = 1$ yields:

$$\mathbf{T}(1) = \frac{2}{\sqrt{5}}\mathbf{i} + \frac{1}{\sqrt{5}}\mathbf{j} = \left\langle \frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}} \right\rangle$$

$$\mathbf{N}(1) = \frac{1}{\sqrt{5}}\mathbf{i} - \frac{2}{\sqrt{5}}\mathbf{j} = \left\langle \frac{1}{\sqrt{5}}, -\frac{2}{\sqrt{5}} \right\rangle$$

3. To check orthogonality, we compute the algebraic dot product of our two unit vector functions:

$$\mathbf{T}(t) \cdot \mathbf{N}(t) = \left(\frac{2t}{\sqrt{4t^2+1}}\right)\left(\frac{1}{\sqrt{4t^2+1}}\right) + \left(\frac{1}{\sqrt{4t^2+1}}\right)\left(-\frac{2t}{\sqrt{4t^2+1}}\right)$$

$$\mathbf{T}(t) \cdot \mathbf{N}(t) = \frac{2t}{4t^2+1} - \frac{2t}{4t^2+1} = 0$$

Because their dot product evaluates to exactly zero for all values of t, the unit tangent vector $\mathbf{T}(t)$ and unit normal vector $\mathbf{N}(t)$ are verified as orthogonal.

▲

Checkpoint 2.46 Let C be determined by the helix $\mathbf{r}(t) = \cos t\ \mathbf{i} + \sin t\ \mathbf{j} + t\ \mathbf{k}$ for $t \geq 0$. Find the unit vector functions $\mathbf{T}(t)$ and $\mathbf{N}(t)$.

Solution. Differentiate the helix position components vector to extract our general velocity direction path:

$$\mathbf{r}'(t) = \langle -\sin t, \cos t, 1\rangle \implies |\mathbf{r}'(t)| = \sqrt{\sin^2 t + \cos^2 t + 1} = \sqrt{2}$$

Dividing the derivative components by this magnitude gives our unit tangent vector function:

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|} = \left\langle -\frac{1}{\sqrt{2}}\sin t, \frac{1}{\sqrt{2}}\cos t, \frac{1}{\sqrt{2}} \right\rangle$$

Next, differentiate our unit tangent function with respect to t to trace the geometric turn path:

$$\mathbf{T}'(t) = \left\langle -\frac{1}{\sqrt{2}}\cos t, -\frac{1}{\sqrt{2}}\sin t, 0 \right\rangle$$

Computing the magnitude of this secondary differentiation vector yields:

$$|\mathbf{T}'(t)| = \sqrt{\left(-\frac{1}{\sqrt{2}}\cos t\right)^2 + \left(-\frac{1}{\sqrt{2}}\sin t\right)^2 + 0^2} = \sqrt{\frac{1}{2}(\cos^2 t + \sin^2 t)} = \frac{1}{\sqrt{2}}$$

Finally, normalize $\mathbf{T}'(t)$ to calculate our primary unit normal vector function:

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{|\mathbf{T}'(t)|} = \frac{1}{1/\sqrt{2}}\left\langle -\frac{1}{\sqrt{2}}\cos t, -\frac{1}{\sqrt{2}}\sin t, 0 \right\rangle = \langle -\cos t, -\sin t, 0\rangle$$

This shows that for a circular helix, the unit normal vector always points directly inward toward the central vertical axis of the cylinder.

The Curvature

The curvature of a curve C at a given point is a measure of how quickly the curve changes direction at that point, or how much the curve bends at each point.

Definition 2.47 Curvature. The **curvature** of a curve is defined as:

$$\kappa = \left|\frac{d\mathbf{T}}{ds}\right|$$

where $\mathbf{T}$ is the unit tangent vector. Curvature is the magnitude of the rate of change of the unit tangent vector with respect to arc length. ◆

Theorem 2.48 Formulas for Curvature. *Let C be a smooth curve and let θ be the angle between the unit tangent vector $\mathbf{T}$ and the standard basis vector $\mathbf{i}$.*

1. *By applying the chain rule to the unit tangent vector with respect to time t:*

$$\frac{d\mathbf{T}}{dt} = \frac{d\mathbf{T}}{ds}\frac{ds}{dt}, \quad \textit{and} \quad \kappa = \left|\frac{d\mathbf{T}}{ds}\right| = \left|\frac{d\mathbf{T}/dt}{ds/dt}\right|$$

   Since $\frac{ds}{dt} = |\mathbf{r}'(t)|$, the curvature as a function of the parameter t simplifies to:

$$\kappa(t) = \frac{|\mathbf{T}'(t)|}{|\mathbf{r}'(t)|}$$

2. *If a 2D curve has an arc length parameterization given by $x = f(s)$ and $y = g(s)$, then the curvature κ at a point $P(x, y)$ is given geometrically by:*

$$\kappa = \left|\frac{d\theta}{ds}\right|$$

3. *Suppose a 2D curve is defined explicitly by a Cartesian function $y = f(x)$, where $f'(x)$ is continuous on an interval. Since y' is the slope of the tangent line at $P(x, y)$, we have $\tan\theta = y'$ (or $\theta = \tan^{-1} y'$), and the arc length function is $s(x) = \int_a^x \sqrt{1 + (y')^2}\,dx$. If y'' exists, applying the chain rule yields:*

$$\frac{d\theta}{dx} = \frac{d}{dx}(\tan^{-1} y') = \frac{y''}{1 + (y')^2} \quad \textit{and} \quad \frac{ds}{dx} = \sqrt{1 + (y')^2}$$

Thus, the curvature κ at the point $P(x, y)$ is defined as:

$$\kappa = \frac{|y''|}{(1 + (y')^2)^{3/2}}$$

4. *Suppose a 2D curve is given parametrically by $x = f(t)$ and $y = g(t)$, where f'' and g'' exist on an interval I. Since the slope of the tangent line is $\tan\theta = \frac{g'(t)}{f'(t)}$ for $f'(t) \neq 0$, differentiating $\theta = \tan^{-1}\left(\frac{g'(t)}{f'(t)}\right)$ with respect to t and dividing by $\frac{ds}{dt}$ yields:*

$$\kappa = \frac{|f'(t)g''(t) - g'(t)f''(t)|}{[(f'(t))^2 + (g'(t))^2]^{3/2}}$$

5. *For a curve in space defined by a 3D vector function $\mathbf{r}(t)$, the curvature can be computed directly without finding the unit tangent vector via the cross product:*

$$\kappa = \frac{|\mathbf{r}'(t) \times \mathbf{r}''(t)|}{|\mathbf{r}'(t)|^3}$$

Example 2.49 Show that the curvature of a circle of radius a is $1/a$ at every point on the circle using the vector formula from Item 1.

Solution. Parameterize the circle of radius a as $\mathbf{r}(t) = \langle a\cos t, a\sin t, 0\rangle$. Then, the velocity vector is $\mathbf{r}'(t) = \langle -a\sin t, a\cos t, 0\rangle$, and its magnitude is $|\mathbf{r}'(t)| = \sqrt{a^2\sin^2 t + a^2\cos^2 t} = a$.

The unit tangent vector is:

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|} = \langle -\sin t, \cos t, 0\rangle$$

Differentiating with respect to t gives $\mathbf{T}'(t) = \langle -\cos t, -\sin t, 0\rangle$, so $|\mathbf{T}'(t)| = 1$. Applying the formula, we find:

$$\kappa(t) = \frac{|\mathbf{T}'(t)|}{|\mathbf{r}'(t)|} = \frac{1}{a}$$

▲

Example 2.50 Show that the curvature of a line l is 0 at every point on l using the geometric formulation in Item 2.

Solution. A straight line maintains a constant direction along its entire trajectory. Therefore, the tangent angle θ is constant relative to the x-axis, which implies $\frac{d\theta}{ds} = 0$. Thus, $\kappa = |0| = 0$. ▲

Example 2.51 Sketch the graph of $y = 1 - x^2$ and find the curvature at general points (x, y), and specifically at the points $(0, 1)$, $(1, 0)$, and $(2, -3)$ using the explicit function method of Item 3.

Solution.

1. For $y = 1 - x^2$, we compute the first and second derivatives with respect to x:

$$y' = -2x \quad \text{and} \quad y'' = -2$$

Substituting these into the explicit Cartesian curvature formula yields the general formula for this parabola:

$$\kappa(x) = \frac{|-2|}{(1+(-2x)^2)^{3/2}} = \frac{2}{(1+4x^2)^{3/2}}$$

2. Evaluating this expression at the specific requested coordinates:

   (a) At $(0,1)$, $x = 0 \implies \kappa(0) = \frac{2}{(1+0)^{3/2}} = 2$.

   (b) At $(1,0)$, $x = 1 \implies \kappa(1) = \frac{2}{(1+4)^{3/2}} = \frac{2}{5\sqrt{5}}$.

   (c) At $(2,-3)$, $x = 2 \implies \kappa(2) = \frac{2}{(1+16)^{3/2}} = \frac{2}{17\sqrt{17}}$.

▲

Checkpoint 2.52 Prove that the curvature of a circle of radius a is $1/a$ at every point on the circle using the 2D parametric formula derived in Item 4.

Solution. Using the standard parametric representation $x = a\cos t$ and $y = a\sin t$, we calculate the first and second derivatives:

$$f'(t) = -a\sin t, \quad g'(t) = a\cos t$$
$$f''(t) = -a\cos t, \quad g''(t) = -a\sin t$$

Substitute these components into the parametric numerator and denominator arrays:

$$|f'(t)g''(t) - g'(t)f''(t)| = |(-a\sin t)(-a\sin t) - (a\cos t)(-a\cos t)| = a^2\sin^2 t + a^2\cos^2 t = a^2$$

$$[(f'(t))^2 + (g'(t))^2]^{3/2} = [(-a\sin t)^2 + (a\cos t)^2]^{3/2} = (a^2)^{3/2} = a^3$$

Dividing these values yields a constant curvature of:

$$\kappa = \frac{a^2}{a^3} = \frac{1}{a}$$

Checkpoint 2.53 A curve C has the parameterization $x = t^2$, $y = t^3$ for $t \in \mathbb{R}$. Find the curvature at the point P corresponding to $t = 1/2$ using the 2D parametric formula derived in Item 4.

Solution. Differentiating the component functions gives:

$$f'(t) = 2t, \quad g'(t) = 3t^2$$
$$f''(t) = 2, \quad g''(t) = 6t$$

Substitute these into the 2D parametric curvature formula:

$$\kappa(t) = \frac{|(2t)(6t) - (3t^2)(2)|}{[(2t)^2 + (3t^2)^2]^{3/2}} = \frac{|12t^2 - 6t^2|}{(4t^2 + 9t^4)^{3/2}} = \frac{6t^2}{(4t^2 + 9t^4)^{3/2}}$$

Evaluating the curvature profile at the requested input parameter $t = 1/2$:

$$\kappa\left(\frac{1}{2}\right) = \frac{6(1/4)}{(4(1/4) + 9(1/16))^{3/2}} = \frac{3/2}{(1 + 9/16)^{3/2}} = \frac{3/2}{(25/16)^{3/2}} = \frac{3/2}{125/64} = \frac{3}{2}\cdot\frac{64}{125} = \frac{96}{125}$$

Checkpoint 2.54 Find the curvature of the twisted cubic $\mathbf{r}(t) = \langle t, t^2, t^3\rangle$ at a general parameter point and specifically at the origin $(0,0,0)$ using the cross product formulation of Item 5.

Solution. First, compute the velocity vector $\mathbf{r}'(t)$ and acceleration vector $\mathbf{r}''(t)$:

$$\mathbf{r}'(t) = \langle 1, 2t, 3t^2 \rangle \implies |\mathbf{r}'(t)| = \sqrt{1 + 4t^2 + 9t^4}$$
$$\mathbf{r}''(t) = \langle 0, 2, 6t \rangle$$

Next, evaluate the vector cross product of these two derivative vectors:

$$\mathbf{r}'(t) \times \mathbf{r}''(t) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2t & 3t^2 \\ 0 & 2 & 6t \end{vmatrix} = (12t^2 - 6t^2)\mathbf{i} - (6t - 0)\mathbf{j} + (2 - 0)\mathbf{k} = \langle 6t^2, -6t, 2 \rangle$$

Find the magnitude of this cross product vector:

$$|\mathbf{r}'(t) \times \mathbf{r}''(t)| = \sqrt{(6t^2)^2 + (-6t)^2 + 2^2} = \sqrt{36t^4 + 36t^2 + 4} = 2\sqrt{9t^4 + 9t^2 + 1}$$

Substituting these components back into our 3D vector curvature equation yields the general expression:

$$\kappa(t) = \frac{2\sqrt{9t^4 + 9t^2 + 1}}{(1 + 4t^2 + 9t^4)^{3/2}}$$

To find the specific curvature at the origin $(0, 0, 0)$, substitute the parameter value $t = 0$:

$$\kappa(0) = \frac{2\sqrt{0 + 0 + 1}}{(1 + 0 + 0)^{3/2}} = \frac{2(1)}{1} = 2$$

2.5 Motion in Space: Velocity and Acceleration

In this section, we will study the motion of objects in three-dimensional space. We will define the concepts of velocity and acceleration for vector-valued functions, and we will explore how these concepts relate to the motion of objects along curves in space.

We will also discuss the concepts of speed and acceleration, and how they can be used to analyze the motion of objects in space. Finally, we will look at some examples of motion in space, including the motion of projectiles and the motion of objects under the influence of gravity.

Suppose a particle moves through space so that its position vector at time t is $\mathbf{r}(t)$. For small values of h, the vector:

$$\frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h} \longrightarrow \text{the direction of the particle moving along the curve } \mathbf{r}(t)$$

Definition 2.55 Velocity Vector. The vector $\frac{\mathbf{r}(t+h)-\mathbf{r}(t)}{h}$ gives the average velocity over a time interval of length h, and its limit is the **velocity vector** $\mathbf{v}(t)$ at time t:

$$\mathbf{v}(t) = \lim_{h \to 0} \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h} = \mathbf{r}'(t)$$

◆

Definition 2.56 Speed. The **speed** of the particle at time t is the magnitude of the velocity vector, $|\mathbf{v}(t)|$, defined as:

$$|\mathbf{v}(t)| = |\mathbf{r}'(t)| = \frac{ds}{dt} = \text{rate of change of distance with respect to time.}$$

◆

Definition 2.57 Acceleration. In multi-dimensional motion, the **acceleration** $\mathbf{a}(t)$ of the particle is defined as the derivative of the velocity vector:

$$\mathbf{a}(t) = \mathbf{v}'(t) = \mathbf{r}''(t)$$

◆

Example 2.58 The position vector of an object moving in a plane is given by $\mathbf{r}(t) = (t^2 + t)\mathbf{i} + t^3\mathbf{j}$ for $0 \le t \le 2$. Find its velocity, speed, and acceleration when $t = 1$.

Solution. First, we differentiate the position vector function $\mathbf{r}(t)$ with respect to t to find the velocity vector function $\mathbf{v}(t)$:

$$\mathbf{v}(t) = \mathbf{r}'(t) = (2t + 1)\mathbf{i} + 3t^2\mathbf{j}$$

Evaluating the velocity vector at the specific parameter point $t = 1$ yields:

$$\mathbf{v}(1) = (2(1) + 1)\mathbf{i} + 3(1)^2\mathbf{j} = 3\mathbf{i} + 3\mathbf{j} = \langle 3, 3\rangle$$

Next, the speed of the object is the scalar magnitude of its velocity vector. Evaluating at $t = 1$ gives:

$$|\mathbf{v}(1)| = \sqrt{3^2 + 3^2} = \sqrt{18} = 3\sqrt{2}$$

Finally, we differentiate the velocity vector function to find the acceleration vector function $\mathbf{a}(t)$:

$$\mathbf{a}(t) = \mathbf{v}'(t) = 2\mathbf{i} + 6t\mathbf{j}$$

Evaluating the acceleration vector at $t = 1$ yields:

$$\mathbf{a}(1) = 2\mathbf{i} + 6(1)\mathbf{j} = 2\mathbf{i} + 6\mathbf{j} = \langle 2, 6\rangle$$

▲

Example 2.59 A moving particle starts at an initial position $\mathbf{r}(0) = \langle 1, 0, 0\rangle$ with initial velocity $\mathbf{v}(0) = \mathbf{i} - \mathbf{j} + \mathbf{k}$. Its acceleration is $\mathbf{a}(t) = 4t\mathbf{i} + 6t\mathbf{j} + \mathbf{k}$. Find its velocity and position at time t.

Solution. We find the velocity vector function $\mathbf{v}(t)$ by integrating the given acceleration vector function:

$$\mathbf{v}(t) = \int \mathbf{a}(t)\,dt = \left(\int 4t\,dt\right)\mathbf{i} + \left(\int 6t\,dt\right)\mathbf{j} + \left(\int 1\,dt\right)\mathbf{k} = 2t^2\mathbf{i} + 3t^2\mathbf{j} + t\mathbf{k} + \mathbf{C}_1$$

Using our initial velocity vector condition $\mathbf{v}(0) = \mathbf{i} - \mathbf{j} + \mathbf{k} = \langle 1, -1, 1\rangle$ at $t = 0$ reveals $\mathbf{C}_1 = \mathbf{i} - \mathbf{j} + \mathbf{k}$. Combining these gives the final velocity function:

$$\mathbf{v}(t) = (2t^2 + 1)\mathbf{i} + (3t^2 - 1)\mathbf{j} + (t + 1)\mathbf{k} = \langle 2t^2 + 1, 3t^2 - 1, t + 1\rangle$$

Next, we find the position vector function $\mathbf{r}(t)$ by integrating the newly solved velocity vector function:

$$\mathbf{r}(t) = \int \mathbf{v}(t)\,dt = \left(\frac{2}{3}t^3 + t\right)\mathbf{i} + \left(t^3 - t\right)\mathbf{j} + \left(\frac{1}{2}t^2 + t\right)\mathbf{k} + \mathbf{C}_2$$

Using our initial position vector condition $\mathbf{r}(0) = \langle 1, 0, 0\rangle = 1\mathbf{i} + 0\mathbf{j} + 0\mathbf{k}$ at $t = 0$ reveals $\mathbf{C}_2 = \mathbf{i}$. Combining these gives the final position vector function:

$$\mathbf{r}(t) = \left(\frac{2}{3}t^3 + t + 1\right)\mathbf{i} + \left(t^3 - t\right)\mathbf{j} + \left(\frac{1}{2}t^2 + t\right)\mathbf{k} = \left\langle \frac{2}{3}t^3 + t + 1,\ t^3 - t,\ \frac{1}{2}t^2 + t\right\rangle$$

▲

Example 2.60 An object with mass m that moves in a circular path with constant angular speed ω has position vector $\mathbf{r}(t) = a\cos\omega t\mathbf{i} + a\sin\omega t\mathbf{j}$. Find the force acting on the object and show that it is directed toward the origin.

Solution. To locate the total force acting on the particle, we must first find its acceleration profile vector. Differentiating the position vector function with respect to time using the chain rule yields velocity:

$$\mathbf{v}(t) = \mathbf{r}'(t) = -a\omega\sin\omega t\mathbf{i} + a\omega\cos\omega t\mathbf{j}$$

Differentiating a second time yields the acceleration vector function:

$$\mathbf{a}(t) = \mathbf{v}'(t) = -a\omega^2\cos\omega t\mathbf{i} - a\omega^2\sin\omega t\mathbf{j}$$

Factoring out the shared constant parameter scalar $-\omega^2$ reveals an explicit relationship to our primary position vector:

$$\mathbf{a}(t) = -\omega^2\left(a\cos\omega t\mathbf{i} + a\sin\omega t\mathbf{j}\right) = -\omega^2\mathbf{r}(t)$$

By applying Newton's Second Law of Motion ($\mathbf{F} = m\mathbf{a}$), we multiply this acceleration expression by the mass scalar:

$$\mathbf{F}(t) = m\mathbf{a}(t) = -m\omega^2\mathbf{r}(t)$$

Because $m\omega^2 > 0$ is a strictly positive scalar quantity, the negative sign indicates that the net force vector $\mathbf{F}(t)$ points in the exact opposite direction of the outward displacement vector $\mathbf{r}(t)$. Therefore, the centripetal force is directed directly inward toward the origin.

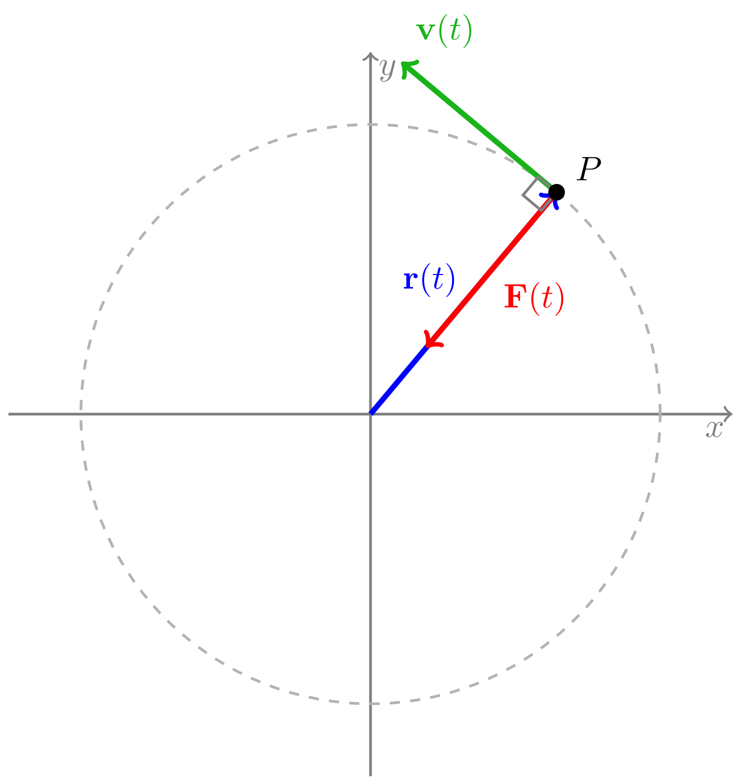


Figure 2.61 An object in uniform circular motion showing the orthogonal relationship between velocity and position, and the centripetal nature of the force vector.

▲

Remark 2.62 Newton's Second Law of Motion. If the force that acts on a particle is known, then its acceleration can be found from **Newton's Second Law of Motion**. The vector version of this law states that if, at any time t, a force $\mathbf{F}(t)$ acts on an object of mass m producing an acceleration $\mathbf{a}(t)$, then:

$$\mathbf{F}(t) = m\mathbf{a}(t)$$

Example 2.63 Show that if a particle P moves around a circle of radius a at a constant speed v, then the acceleration vector has a constant magnitude v^2/a and is directed from P toward the center of the circle.

Solution. We can parameterize uniform circular motion of radius a with constant angular speed ω as:

$$\mathbf{r}(t) = \langle a\cos(\omega t), a\sin(\omega t)\rangle$$

Differentiating once yields the velocity vector function:

$$\mathbf{v}(t) = \mathbf{r}'(t) = \langle -a\omega\sin(\omega t), a\omega\cos(\omega t)\rangle$$

The constant speed v is the magnitude of this velocity vector:

$$v = |\mathbf{v}(t)| = \sqrt{(-a\omega\sin(\omega t))^2 + (a\omega\cos(\omega t))^2} = \sqrt{a^2\omega^2(\sin^2(\omega t) + \cos^2(\omega t))} = a\omega$$

This gives us the relationship $\omega = \frac{v}{a}$.

Next, differentiate the velocity vector function to find the acceleration vector function $\mathbf{a}(t)$:

$$\mathbf{a}(t) = \mathbf{v}'(t) = \langle -a\omega^2\cos(\omega t), -a\omega^2\sin(\omega t)\rangle = -\omega^2\mathbf{r}(t)$$

Substituting $\omega = \frac{v}{a}$ into our acceleration expression yields:

$$\mathbf{a}(t) = -\left(\frac{v}{a}\right)^2\mathbf{r}(t) = -\frac{v^2}{a^2}\mathbf{r}(t)$$

Now we compute the magnitude of the acceleration vector:

$$|\mathbf{a}(t)| = \left|-\frac{v^2}{a^2}\right| |\mathbf{r}(t)| = \frac{v^2}{a^2}\cdot a = \frac{v^2}{a}$$

Because $\frac{v^2}{a^2} > 0$ is a strictly positive scalar, the negative sign in $\mathbf{a}(t) = -\frac{v^2}{a^2}\mathbf{r}(t)$ dictates that the acceleration points in the exact opposite direction of the outward radial position vector $\mathbf{r}(t)$. Therefore, the acceleration vector has a constant magnitude of v^2/a and points directly inward from P toward the center of the circle. ▲

Example 2.64 The position vector of a particle P is given by $\mathbf{r}(t) = 2t\mathbf{i} + 3t^2\mathbf{j} + t^3\mathbf{k}$ for $0 \le t \le 2$. Find the velocity and acceleration of P at time t. Sketch the path of P.

Solution. We compute the velocity vector function $\mathbf{v}(t)$ by taking the first derivative of the position components with respect to t:

$$\mathbf{v}(t) = \mathbf{r}'(t) = 2\mathbf{i} + 6t\mathbf{j} + 3t^2\mathbf{k} = \langle 2, 6t, 3t^2\rangle$$

We compute the acceleration vector function $\mathbf{a}(t)$ by taking the derivative of our velocity profile:

$$\mathbf{a}(t) = \mathbf{v}'(t) = 6\mathbf{j} + 6t\mathbf{k} = \langle 0, 6, 6t\rangle$$

The trajectory begins at the origin $\mathbf{r}(0) = \langle 0, 0, 0\rangle$ and ends at $\mathbf{r}(2) = \langle 4, 12, 8\rangle$. As time increases, the path arches upward across the first octant space.

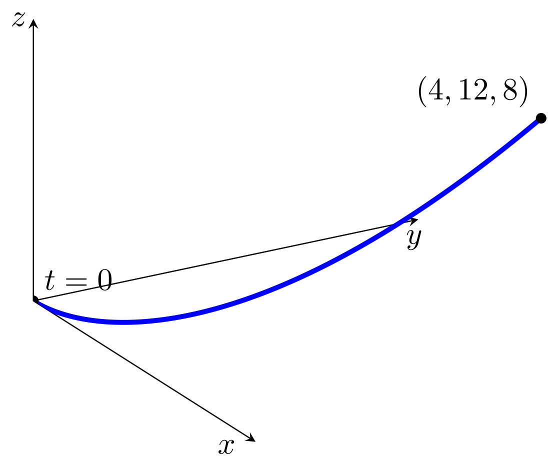


Figure 2.65 The 3D space trajectory of the particle from $t = 0$ to $t = 2$.

▲

Example 2.66 A point P is rotating about the z-axis on a circle of radius a that lies in the plane $z = h$. The angular speed $d\theta/dt$ is a constant ω. The vector $\boldsymbol{\omega} = \omega\mathbf{k}$ directed along the z-axis and having magnitude ω is the **angular velocity** of P. Show that the velocity $\mathbf{v}(t)$ of P is the cross product of $\boldsymbol{\omega}$ and the position vector $\mathbf{r}(t)$ for P.

Solution. Since the path is a circle of radius a sitting in the horizontal plane $z = h$, we can express the position vector of the moving point P explicitly as:

$$\mathbf{r}(t) = \langle a\cos(\omega t), a\sin(\omega t), h\rangle$$

Differentiating this position profile with respect to t yields the standard velocity vector:

$$\mathbf{v}(t) = \mathbf{r}'(t) = \langle -a\omega\sin(\omega t), a\omega\cos(\omega t), 0\rangle$$

Now, let us calculate the vector cross product of the angular velocity vector $\boldsymbol{\omega} = \langle 0, 0, \omega\rangle$ and our position vector $\mathbf{r}(t)$:

$$\begin{aligned}
\boldsymbol{\omega} \times \mathbf{r}(t) &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 0 & \omega \\ a\cos(\omega t) & a\sin(\omega t) & h \end{vmatrix} \\
&= (0 \cdot h - \omega \cdot a\sin(\omega t))\mathbf{i} - (0 \cdot h - \omega \cdot a\cos(\omega t))\mathbf{j} + (0 \cdot a\sin(\omega t) - 0 \cdot a\cos(\omega t))\mathbf{k} \\
&= \langle -a\omega\sin(\omega t), a\omega\cos(\omega t), 0\rangle
\end{aligned}$$

Comparing this coordinate result directly to our derivative calculation proves that:

$$\mathbf{v}(t) = \boldsymbol{\omega} \times \mathbf{r}(t)$$

▲

Example 2.67 A projectile is fired with an initial velocity $\mathbf{v}_0$ from a point h_0 feet above the ground. If the only force acting on the projectile is caused by gravitational acceleration $\mathbf{g}$, determine its position after t seconds.

Solution. Let the motion take place in the 2D xy-plane, where the positive y-axis points vertically upward from the ground level. The constant downward gravitational acceleration vector is expressed as:

$$\mathbf{a}(t) = -g\mathbf{j}$$

where $g \approx 32\,\text{ft/sec}^2$. Integrating this acceleration vector with respect to time t yields the velocity vector function:

$$\mathbf{v}(t) = \int \mathbf{a}(t)\,dt = \int -g\mathbf{j}\,dt = -gt\mathbf{j} + \mathbf{C}_1$$

Using the initial velocity condition at $t = 0$ gives $\mathbf{v}(0) = \mathbf{C}_1 = \mathbf{v}_0$. Thus:

$$\mathbf{v}(t) = -gt\mathbf{j} + \mathbf{v}_0$$

Next, we integrate the velocity vector function to determine the position vector function $\mathbf{r}(t)$:

$$\mathbf{r}(t) = \int \mathbf{v}(t)\,dt = \int (-gt\mathbf{j} + \mathbf{v}_0)\,dt = -\frac{1}{2}gt^2\mathbf{j} + \mathbf{v}_0 t + \mathbf{C}_2$$

Using our initial position conditions at $t = 0$, the object starts at a height h_0 above the origin, meaning $\mathbf{r}(0) = h_0\mathbf{j}$. This establishes $\mathbf{C}_2 = h_0\mathbf{j}$. Combining all components gives the general position vector function for projectile motion:

$$\mathbf{r}(t) = \mathbf{v}_0 t + \left(h_0 - \frac{1}{2}gt^2\right)\mathbf{j}$$

▲

Remark 2.68 Projectile Path Components. A unit vector $\mathbf{u}$ pointing in the same direction as the launch trajectory vector $\mathbf{v}_0$ with an angle of elevation α is given by:

$$\mathbf{u} = \cos\alpha\mathbf{i} + \sin\alpha\mathbf{j}$$

Letting $\|\mathbf{v}_0\| = v_0$ denote the initial launch speed of the projectile, the initial velocity vector breaks down into its component components as:

$$\mathbf{v}_0 = v_0\mathbf{u} = (v_0\cos\alpha)\mathbf{i} + (v_0\sin\alpha)\mathbf{j}$$

Provided the projectile launches from ground level ($h_0 = 0$), substituting these components into our position equation yields:

$$\mathbf{r}(t) = (v_0\cos\alpha)t\mathbf{i} + \left[(v_0\sin\alpha)t - \frac{1}{2}gt^2\right]\mathbf{j}$$

Therefore, the parametric equations describing the projectile's trajectory path are:

$$x = (v_0\cos\alpha)t, \qquad y = (v_0\sin\alpha)t - \frac{1}{2}gt^2$$

Example 2.69 A projectile is fired from ground level with an initial speed of 1500 ft/sec and an angle of elevation of 30°. Find:

1. The velocity vector at time t.
2. The maximum altitude reached.
3. The horizontal range of the projectile.

4. The speed at which the projectile strikes the ground.

Solution. Using the parameters provided, we have the initial speed $v_0 = 1500$, launch angle $\alpha = 30°$, and constant gravitational acceleration $g = 32\,\text{ft/sec}^2$. The initial horizontal and vertical velocity components evaluate to:

$$v_{0x} = 1500\cos(30°) = 1500\left(\frac{\sqrt{3}}{2}\right) = 750\sqrt{3}\,\text{ft/sec}$$

$$v_{0y} = 1500\sin(30°) = 1500\left(\frac{1}{2}\right) = 750\,\text{ft/sec}$$

Thus, the parametric tracking coordinates of the position are $x(t) = 750\sqrt{3}t$ and $y(t) = 750t - 16t^2$.

1. Differentiating the position components gives the general velocity vector function at time t:

$$\mathbf{v}(t) = 750\sqrt{3}\mathbf{i} + (750 - 32t)\mathbf{j} = \langle 750\sqrt{3},\, 750 - 32t\rangle$$

2. The maximum altitude occurs when the vertical velocity component drops to zero ($v_y(t) = 0$):

$$750 - 32t = 0 \implies t = \frac{750}{32} = \frac{375}{16} \approx 23.44\,\text{seconds}$$

Substituting this peak time back into our vertical tracking position function yields the maximum altitude:

$$y_{\text{max}} = 750\left(\frac{375}{16}\right) - 16\left(\frac{375}{16}\right)^2 = \frac{281250}{16} - \frac{140625}{16} = \frac{140625}{16} \approx 8789.06\,\text{feet}$$

3. The horizontal range is the distance traveled when the projectile returns to the ground level ($y(t) = 0$):

$$750t - 16t^2 = 0 \implies t(750 - 16t) = 0 \implies t_{\text{impact}} = \frac{750}{16} = \frac{375}{8} \approx 46.88\,\text{seconds}$$

Evaluating our horizontal position tracking function at this total flight time gives the range:

$$x_{\text{range}} = 750\sqrt{3}\left(\frac{375}{8}\right) = \frac{281250\sqrt{3}}{8} \approx 60892.10\,\text{feet}$$

4. Because the projectile launches and lands at the same horizontal altitude elevation, the vertical landing speed matches the initial vertical launching speed but points downward. Evaluating the components at impact yields $\mathbf{v}(t_{\text{impact}}) = \langle 750\sqrt{3}, -750\rangle$. The total scalar speed on impact is:

$$\|\mathbf{v}(t_{\text{impact}})\| = \sqrt{(750\sqrt{3})^2 + (-750)^2} = \sqrt{1687500 + 562500} = \sqrt{2250000} = 1500\,\text{ft/sec}$$

▲

2.6 Tangential and Normal Components of Acceleration

In this section, we will explore the tangential and normal components of acceleration for a particle moving along a curved path. These components provide insight into how the particle's velocity changes in both magnitude and direction.

The tangential component of acceleration is responsible for changing the speed of the particle, while the normal component is responsible for changing the direction of motion.

We will derive formulas for these components and apply them to various examples to illustrate their significance in understanding motion in space.

The motion of a particle is often useful to resolve the acceleration into two components: one in the direction of the tangent and the other in the direction of the normal. If $v = |\mathbf{v}|$ represents the speed of the particle, then:

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|} = \frac{\mathbf{v}(t)}{|\mathbf{v}(t)|} = \frac{\mathbf{v}}{v}$$

which gives $\mathbf{v} = v\mathbf{T}$. Differentiating both sides of this equation with respect to t yields:

$$\mathbf{a} = \mathbf{v}' = v'\mathbf{T} + v\mathbf{T}'$$

Since the curvature is defined as $\kappa = \frac{|\mathbf{T}'(t)|}{|\mathbf{r}'(t)|}$, we can rewrite this in terms of speed:

$$\kappa = \frac{|\mathbf{T}'(t)|}{v} \implies |\mathbf{T}'| = \kappa v$$

The unit normal vector is defined as $\mathbf{N} = \frac{\mathbf{T}'}{|\mathbf{T}'|}$, which allows us to isolate the derivative of the unit tangent vector:

$$\mathbf{T}' = |\mathbf{T}'|\mathbf{N} = \kappa v \mathbf{N}$$

Substituting this back into our acceleration equation provides the final decomposition:

$$\begin{aligned} \mathbf{a} &= v'\,\mathbf{T} + \kappa v^2\,\mathbf{N} \\ &= a_T\,\mathbf{T} + a_N\,\mathbf{N} \end{aligned}$$

We call a_T and a_N the **tangential and normal components of acceleration**, defined respectively as:

$$a_T = v' \quad \text{and} \quad a_N = \kappa v^2$$

To find a more computational formula for a_T, we take the dot product of the velocity vector $\mathbf{v} = v\mathbf{T}$ with the acceleration vector $\mathbf{a}$:

$$\begin{aligned} \mathbf{v} \cdot \mathbf{a} &= v\mathbf{T} \cdot (v'\,\mathbf{T} + \kappa v^2\,\mathbf{N}) \\ &= vv'(\mathbf{T} \cdot \mathbf{T}) + \kappa v^3(\mathbf{T} \cdot \mathbf{N}) \\ &= vv' \quad (\text{since } \mathbf{T} \cdot \mathbf{T} = 1 \text{ and } \mathbf{T} \cdot \mathbf{N} = 0) \end{aligned}$$

Therefore, provided $\mathbf{r}'(t) \neq \mathbf{0}$, the tangential component of acceleration can be calculated directly as:

$$a_T = v' = \frac{\mathbf{v} \cdot \mathbf{a}}{v} = \frac{\mathbf{r}'(t) \cdot \mathbf{r}''(t)}{|\mathbf{r}'(t)|}$$

Similarly, by substituting the cross-product curvature formula $\kappa = \frac{|\mathbf{r}'(t)\times\mathbf{r}''(t)|}{|\mathbf{r}'(t)|^3}$ into our definition for a_N, we obtain a direct computational formula for the normal component of acceleration:

$$a_N = \kappa v^2 = \frac{|\mathbf{r}'(t) \times \mathbf{r}''(t)|}{|\mathbf{r}'(t)|^3} |\mathbf{r}'(t)|^2 = \frac{|\mathbf{r}'(t) \times \mathbf{r}''(t)|}{|\mathbf{r}'(t)|}$$

Tangential and Normal Components of Acceleration.

The acceleration vector $\mathbf{a}(t)$ can be decomposed into a tangential component a_T and a normal component a_N:

$$\mathbf{a} = a_T\,\mathbf{T} + a_N\,\mathbf{N}$$

Where the geometric definitions and direct computational formulas for these components are given by:

$$a_T = \frac{\mathbf{r}'(t) \cdot \mathbf{r}''(t)}{\|\mathbf{r}'(t)\|} \quad \text{and} \quad a_N = \frac{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|}$$

These components satisfy the Pythagorean relationship with the total acceleration magnitude:

$$\|\mathbf{a}\|^2 = a_T^2 + a_N^2$$

Example 2.70 A particle moves with the position function $\mathbf{r}(t) = \langle t, t^2, t^3\rangle$. Find the tangential and normal components of acceleration.

Solution. First, compute the velocity vector $\mathbf{r}'(t)$ and acceleration vector $\mathbf{r}''(t)$:

$$\mathbf{r}'(t) = \langle 1, 2t, 3t^2\rangle \implies \|\mathbf{r}'(t)\| = \sqrt{1 + 4t^2 + 9t^4}$$
$$\mathbf{r}''(t) = \langle 0, 2, 6t\rangle$$

To find the tangential component a_T, calculate the dot product $\mathbf{r}'(t) \cdot \mathbf{r}''(t)$:

$$\mathbf{r}'(t) \cdot \mathbf{r}''(t) = 1(0) + 2t(2) + 3t^2(6t) = 4t + 18t^3$$

Dividing by the speed $\|\mathbf{r}'(t)\|$ yields:

$$a_T = \frac{\mathbf{r}'(t) \cdot \mathbf{r}''(t)}{\|\mathbf{r}'(t)\|} = \frac{4t + 18t^3}{\sqrt{1 + 4t^2 + 9t^4}}$$

To find the normal component a_N, compute the cross product $\mathbf{r}'(t) \times \mathbf{r}''(t)$:

$$\mathbf{r}'(t) \times \mathbf{r}''(t) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2t & 3t^2 \\ 0 & 2 & 6t \end{vmatrix} = \langle 12t^2 - 6t^2,\ -6t,\ 2\rangle = \langle 6t^2, -6t, 2\rangle$$

Find the magnitude of this cross product vector:

$$\|\mathbf{r}'(t)\times\mathbf{r}''(t)\| = \sqrt{(6t^2)^2 + (-6t)^2 + 2^2} = \sqrt{36t^4 + 36t^2 + 4} = 2\sqrt{9t^4 + 9t^2 + 1}$$

Dividing by the speed yields:

$$a_N = \frac{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|} = \frac{2\sqrt{9t^4 + 9t^2 + 1}}{\sqrt{1 + 4t^2 + 9t^4}}$$

▲

Checkpoint 2.71 A particle moves with the position function $\mathbf{r}(t) = (3t - t^3)\mathbf{i} + 3t^2\mathbf{j}$. Find the tangential and normal components of acceleration.

Solution. Compute the primary velocity and acceleration derivative profiles:

$$\mathbf{r}'(t) = \langle 3-3t^2, 6t\rangle \implies \|\mathbf{r}'(t)\| = \sqrt{(3-3t^2)^2 + (6t)^2} = \sqrt{9 - 18t^2 + 9t^4 + 36t^2} = \sqrt{9 + 18t^2 + 9t^4} = 3(1+t^2$$

$$\mathbf{r}''(t) = \langle -6t, 6\rangle$$

Find the tangential component a_T by evaluation of the vector dot product:

$$\mathbf{r}'(t)\cdot\mathbf{r}''(t) = (3-3t^2)(-6t)+(6t)(6) = -18t+18t^3+36t = 18t+18t^3 = 18t(1+t^2)$$

$$a_T = \frac{\mathbf{r}'(t)\cdot\mathbf{r}''(t)}{\|\mathbf{r}'(t)\|} = \frac{18t(1+t^2)}{3(1+t^2)} = 6t$$

To find a_N for a 2D planar curve, we can evaluate the standard determinant form equivalent to the spatial cross product:

$$\|\mathbf{r}'(t)\times\mathbf{r}''(t)\| = |(3-3t^2)(6)-(6t)(-6t)| = |18-18t^2+36t^2| = |18+18t^2| = 18(1+t^2)$$

$$a_N = \frac{\|\mathbf{r}'(t)\times\mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|} = \frac{18(1+t^2)}{3(1+t^2)} = 6$$

Checkpoint 2.72 A particle moves with the position function $\mathbf{r}(t) = e^t\mathbf{i} + \sqrt{2}t\mathbf{j} + e^{-t}\mathbf{k}$. Find the tangential and normal components of acceleration.

Solution. Compute the derivative vectors:

$$\mathbf{r}'(t) = \langle e^t, \sqrt{2}, -e^{-t}\rangle \implies \|\mathbf{r}'(t)\| = \sqrt{(e^t)^2 + (\sqrt{2})^2 + (-e^{-t})^2} = \sqrt{e^{2t} + 2 + e^{-2t}} = \sqrt{(e^t + e^{-t})^2} = e^t+e^{-t}$$

$$\mathbf{r}''(t) = \langle e^t, 0, e^{-t}\rangle$$

Calculate the dot product to find a_T:

$$\mathbf{r}'(t)\cdot\mathbf{r}''(t) = (e^t)(e^t) + (\sqrt{2})(0) + (-e^{-t})(e^{-t}) = e^{2t} - e^{-2t}$$

$$a_T = \frac{\mathbf{r}'(t)\cdot\mathbf{r}''(t)}{\|\mathbf{r}'(t)\|} = \frac{e^{2t} - e^{-2t}}{e^t + e^{-t}} = \frac{(e^t - e^{-t})(e^t + e^{-t})}{e^t + e^{-t}} = e^t - e^{-t}$$

Calculate the cross product to find a_N:

$$\mathbf{r}'(t)\times\mathbf{r}''(t) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ e^t & \sqrt{2} & -e^{-t} \\ e^t & 0 & e^{-t} \end{vmatrix} = \langle \sqrt{2}e^{-t},\ -(e^t\cdot e^{-t}-(-e^{-t}\cdot e^t)),\ -\sqrt{2}e^t\rangle = \langle \sqrt{2}e^{-t}, -2, -\sqrt{2}e^t\rangle$$

Find the magnitude of this cross product vector:

$$\|\mathbf{r}'(t)\times\mathbf{r}''(t)\| = \sqrt{(\sqrt{2}e^{-t})^2 + (-2)^2 + (-\sqrt{2}e^t)^2} = \sqrt{2e^{-2t} + 4 + 2e^{2t}} = \sqrt{2(e^{-2t} + 2 + e^{2t})} = \sqrt{2(e^t + e^{-t})^2}$$

Dividing by the speed yields a constant normal acceleration:

$$a_N = \frac{\|\mathbf{r}'(t)\times\mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|} = \frac{\sqrt{2}(e^t + e^{-t})}{e^t + e^{-t}} = \sqrt{2}$$

Chapter 3

Partial Derivatives

Partial derivatives are derivatives of functions with multiple variables, taken with respect to one variable while keeping the others constant. They are fundamental in understanding how functions change in multi-dimensional spaces.

In this chapter, we will explore the concepts of partial derivatives, including their computation, interpretation, and applications.

Functions of Several Variables

In this section, we will introduce the concept of functions of several variables. These functions take multiple independent inputs and produce a single real-number output, allowing us to model complex structural relationships in higher dimensions.

We will explore how to analytically determine the domain and range of these functions, as well as how to interpret their geometric behavior through three-dimensional graphical representations.

Additionally, we will discuss the concept of **level curves** and **level surfaces**, which slice through multivariate outputs to provide valuable structural insights into their contours and topologies.

Definition 3.1 Let D be a set of ordered pairs of real variables. A function f of two variables is a correspondence that assigns to each pair (x, y) in D exactly one real number denoted $f(x, y)$. The set D is called the **domain of** f. The **range of** f consists of all real numbers $f(x, y)$ where (x, y) is in D. ◆

Example 3.2 Find the domain and range of $g(x, y) = \sqrt{9 - x^2 - y^2}$

Solution. The domain of g is

$$D = \{(x, y) \mid 9 - x^2 - y^2 \geq 0\} = \{(x, y) \mid x^2 + y^2 \leq 9\}$$

which is the disk with center $(0, 0)$ and radius 3. The range of g is

$$\{z \mid z = \sqrt{9 - x^2 - y^2}, \quad (x, y) \in D\}$$

Since $z \geq 0$, and also $9 - x^2 - y^2 \leq 9 \implies \sqrt{9 - x^2 - y^2} \leq 3$ the range of g is

$$\{z \mid 0 \leq z \leq 3\} \quad = \quad [0, 3]$$

▲

Example 3.3 Sketch the graph of $g(x, y) = \sqrt{9 - x^2 - y^2}$

Solution. We squared both sides of the equation to obtain $x^2 + y^2 + z^2 = 9$ for $z \geq 0$. The graph of g is the upper hemisphere of a sphere centered at the origin with a radius of 3.

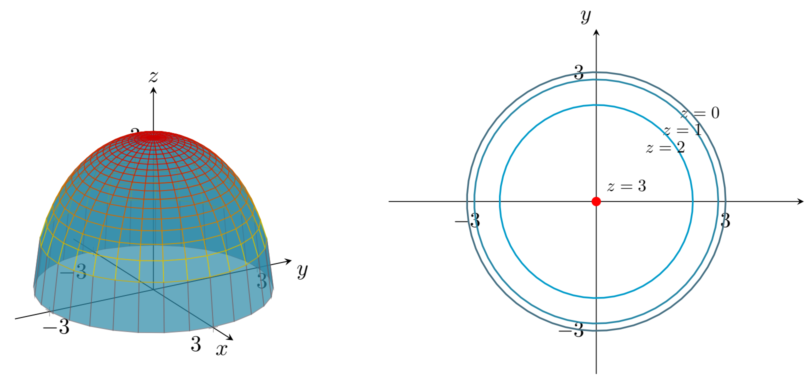


Figure 3.4 The 3D surface graph and corresponding 2D contour map for $g(x, y) = \sqrt{9 - x^2 - y^2}$.

▲

Definition 3.5 The **level curve** of a function f of two variables are the curves with equations $f(x, y) = k$ where k is a constant (in the range of f). ◆

Example 3.6 Sketch some level curves of $f(x, y) = y^2 - x^2$

Solution. We know $z = y^2 - x^2$ is a hyperbolic paraboloid and the level curves are $k = y^2 - x^2$ with $k \in \mathbb{Z}$. We get hyperbolas with vertices $(0, \pm\sqrt{k})$ when $k > 0$, vertices $(\pm\sqrt{-k}, 0)$ when $k < 0$, and the intersecting lines $y = \pm x$ if $k = 0$.

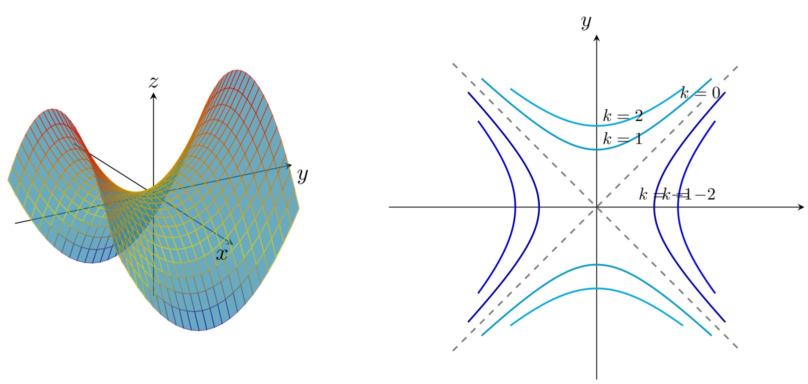


Figure 3.7 The 3D surface graph and corresponding 2D contour map for $f(x, y) = y^2 - x^2$.

▲

A **function of three variables** f assigns to each ordered triple (x, y, z) in a domain $D \subset \mathbb{R}^3$ exactly one real number denoted $f(x, y, z)$.

Example 3.8 Find the domain of f if

$$f(x, y, z) = \ln(z - y) + xy \sin z$$

Solution. The domain of f is

$$D = \{(x, y, z) \in \mathbb{R}^3 \mid z - y > 0\} = \{(x, y, z) \in \mathbb{R}^3 \mid z > y\}$$

This is a **half-space** consisting of all points that lie above the plane $z = y$. ▲

Example 3.9 Find the level surfaces of the function

$$f(x, y, z) = x^2 - y^2 + z^2$$

Solution. The level surfaces are $x^2 - y^2 + z^2 = k$, which form the following families:

1 where $k > 0$. These form a family of hyperboloids of one sheet with axes along the y-axis.

2 where $k = 0$. This forms a cone with its axis along the y-axis.

3 where $k < 0$. These form a family of hyperboloids of two sheets with axes along the y-axis.

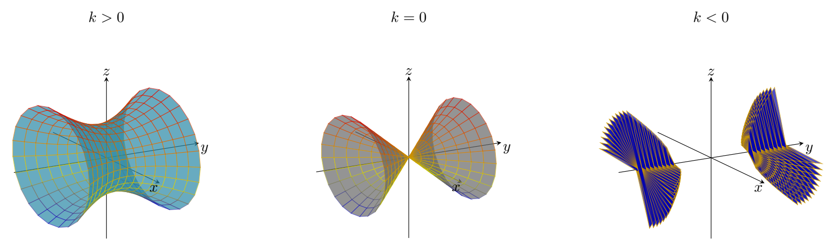


Figure 3.10 The three families of level surfaces for $f(x, y, z) = x^2 - y^2 + z^2 = k$.

▲

3.2 Limits and Continuity

In this section, we will explore the concepts of limits and continuity for functions of several variables. Mastering these foundational ideas is crucial for analyzing the localized behavior of multivariable functions before extending our study to rates of change.

We will begin by defining the rigorous condition for the limit of a multivariate function. Next, we will introduce the **two-path rule** to definitively prove when a limit fails to exist along distinct linear or parabolic geometric pathways. We will then explore methods for evaluating existing limits, including direct algebraic substitution and the **Squeeze Theorem**. Finally, we will establish the mathematical criteria required for a function to be continuous on its domain, examining both piecewise definitions and the continuity profiles of composite functions.

By the end of this section, you will have a solid conceptual framework for evaluating limits and determining the continuity of functions of several variables, providing the essential toolkit required to study multivariable derivatives.

Definition 3.11 We write

$$\lim_{(x,y)\to(a,b)} f(x, y) = L$$

and say that the **limit of** $f(x, y)$ **as** (x, y) **approaches** (a, b) **is** L if we can make the values of $f(x, y)$ as close to L as we please by taking the point (x, y) sufficiently close to the point (a, b) but not equal to (a, b).

This is equivalent to stating that for every $\epsilon > 0$, there exists a $\delta > 0$ such that if

$$0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta$$

then

$$|f(x, y) - L| < \epsilon$$

◆

Alternately, this may be denoted as

$$\lim_{\substack{x \to a \\ y \to b}} f(x, y) = L \quad \text{and} \quad f(x, y) \to L \text{ as } (x, y) \to (a, b)$$

Remark 3.12 Two-Path Rule. If $f(x, y) \to L_1$ as $(x, y) \to (a, b)$ along a path C_1 and $f(x, y) \to L_2$ as $(x, y) \to (a, b)$ along a path C_2 where $L_1 \neq L_2$, then

$$\lim_{(x,y) \to (a,b)} f(x, y)$$

does not exist.

Example 3.13 Find the following limits:

1 $\displaystyle\lim_{(x,y) \to (2,-3)} (x^3 - 4xy^2 + 5y - 7)$

2 $\displaystyle\lim_{(x,y) \to (3,4)} \frac{x^2 - y^2}{\sqrt{x^2 + y^2}}$

Solution.

1 Since the function $f(x, y) = x^3 - 4xy^2 + 5y - 7$ is a polynomial function of two variables, it is continuous everywhere on its domain $\mathbb{R}^2$. Therefore, we can evaluate the limit directly by substituting $x = 2$ and $y = -3$ into the expression:

$$\begin{aligned} \lim_{(x,y) \to (2,-3)} (x^3 - 4xy^2 + 5y - 7) &= (2)^3 - 4(2)(-3)^2 + 5(-3) - 7 \\ &= 8 - 4(2)(9) - 15 - 7 \\ &= 8 - 72 - 15 - 7 \\ &= -86 \end{aligned}$$

2 The function $f(x, y) = \frac{x^2 - y^2}{\sqrt{x^2+y^2}}$ is a rational function involving a radical. The denominator is non-zero at the limit point since $\sqrt{3^2 + 4^2} = \sqrt{25} = 5 \neq 0$. Because the denominator does not equal zero at $(3, 4)$, the function is continuous at this point, and we can find the limit using direct substitution of $x = 3$ and $y = 4$:

$$\begin{aligned} \lim_{(x,y) \to (3,4)} \frac{x^2 - y^2}{\sqrt{x^2 + y^2}} &= \frac{3^2 - 4^2}{\sqrt{3^2 + 4^2}} \\ &= \frac{9 - 16}{\sqrt{9 + 16}} \\ &= \frac{-7}{\sqrt{25}} \\ &= -\frac{7}{5} \end{aligned}$$

▲

Example 3.14 Show that

$$\lim_{(x,y)\to(0,0)} \frac{x^2-y^2}{x^2+y^2}$$

does not exist.

Solution. Let $f(x,y) = \frac{x^2-y^2}{x^2+y^2}$. To show that the limit does not exist as $(x,y) \to (0,0)$, we can analyze the behavior of the function along different paths approaching the origin.

First, let's approach $(0,0)$ along the x-axis. On this path, $y = 0$, which simplifies the function to:

$$f(x,0) = \frac{x^2-0^2}{x^2+0^2} = \frac{x^2}{x^2} = 1 \quad \text{for all } x \neq 0$$

Therefore, as we approach along the x-axis, the limit is:

$$\lim_{x\to 0} f(x,0) = 1$$

Next, we approach along the y-axis by setting $x = 0$. This simplifies the function to:

$$f(0,y) = \frac{0^2-y^2}{0^2+y^2} = \frac{-y^2}{y^2} = -1 \quad \text{for all } y \neq 0$$

Therefore, as we approach along the y-axis, the limit is:

$$\lim_{y\to 0} f(0,y) = -1$$

Since the function approaches two different limiting values (1 and -1) along two distinct directional paths, the two-path rule applies. We conclude that the given limit

$$\lim_{(x,y)\to(0,0)} \frac{x^2-y^2}{x^2+y^2}$$

does not exist. ▲

Example 3.15 Show that

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2}$$

does not exist.

Solution. Let $f(x,y) = \frac{xy}{x^2+y^2}$. To test the existence of the limit as we approach the origin, we check the behavior of the function along multiple distinct paths.

First, let's approach $(0,0)$ along the x-axis. Setting $y = 0$ gives:

$$f(x,0) = \frac{x(0)}{x^2+0^2} = 0 \quad \text{for all } x \neq 0$$

Thus, the limit along this path is 0.

Next, we approach along the y-axis by setting $x = 0$:

$$f(0,y) = \frac{(0)y}{0^2+y^2} = 0 \quad \text{for all } y \neq 0$$

The limit along this path is also 0.

Although we have obtained identical limits along both coordinate axes, this is not sufficient to prove that the overall limit is 0. We must test other paths, such as the diagonal line $y = x$. Substituting $y = x$ into our function for $x \neq 0$ yields:

$$f(x, x) = \frac{x(x)}{x^2 + x^2} = \frac{x^2}{2x^2} = \frac{1}{2}$$

Therefore, as we approach the origin along the line $y = x$, the function value remains constantly equal to $\frac{1}{2}$, giving a path limit of $\frac{1}{2}$.

Since we have obtained different limiting values along two different **paths** (0 along the axes and $\frac{1}{2}$ along the diagonal), the two-path rule applies. We conclude that the given limit does not exist.

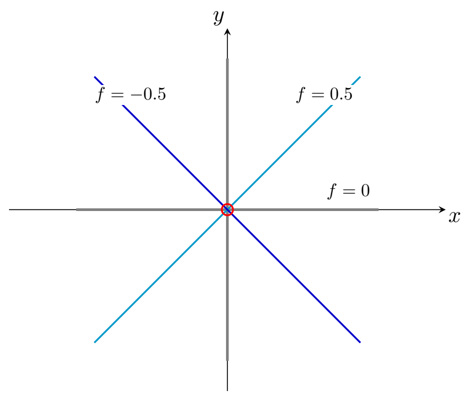


Figure 3.16 Contour lines for $f(x, y) = \frac{xy}{x^2+y^2}$ showing distinct constant values along linear directional rays radiating from the origin.

▲

Checkpoint 3.17 If $f(x, y) = \frac{x^2 y}{x^4+y^2}$, does

$$\lim_{(x,y)\to(0,0)} \frac{x^2 y}{x^4 + y^2}$$

exist?

Solution. To test if the limit exists, let's analyze the behavior of the function along different families of paths approaching the origin.

First, let's approach $(0, 0)$ along an arbitrary straight line path $y = mx$ (where $m \neq 0$ is the slope). Substituting $y = mx$ into the function yields:

$$\begin{aligned} f(x, mx) &= \frac{x^2(mx)}{x^4 + (mx)^2} \\ &= \frac{mx^3}{x^4 + m^2x^2} \\ &= \frac{mx^3}{x^2(x^2 + m^2)} \\ &= \frac{mx}{x^2 + m^2} \quad \text{for } x \neq 0 \end{aligned}$$

Taking the limit as $x \to 0$, we find:

$$\lim_{x\to 0} f(x, mx) = \frac{0}{0 + m^2} = 0$$

If we approach along the coordinate axes ($x = 0$ or $y = 0$), the function is identically 0, so the limit is also 0. Thus, $f(x, y)$ has the same limiting value of 0 along every single straight-line path passing through the origin.

However, this still does not guarantee that the overall limit is 0. Let's approach $(0,0)$ along a parabolic path by setting $y = x^2$ for $x \neq 0$. Substituting this curve into our expression yields:

$$f(x, x^2) = \frac{x^2(x^2)}{x^4 + (x^2)^2} = \frac{x^4}{x^4 + x^4} = \frac{x^4}{2x^4} = \frac{1}{2}$$

Therefore, as we approach the origin along the parabola $y = x^2$, the limit value is consistently $\frac{1}{2}$.

Since we have obtained different limiting values along two different **paths** (0 along all linear paths and $\frac{1}{2}$ along the parabolic path), the two-path rule applies. We conclude that the given limit does not exist.

Checkpoint 3.18 Find

$$\lim_{(x,y)\to(0,0)} \frac{3x^2y}{x^2 + y^2}$$

Solution. We can determine this limit by analyzing the absolute distance from $f(x, y)$ to 0:

$$\left| \frac{3x^2y}{x^2 + y^2} - 0 \right| = \left| \frac{3x^2y}{x^2 + y^2} \right| = \frac{3x^2|y|}{x^2 + y^2}$$

We notice an inequality involving the components of the denominator: since $y^2 \geq 0$, it must be true that $x^2 \leq x^2 + y^2$ for all real numbers. Dividing both sides by $x^2 + y^2$ (where $(x, y) \neq (0, 0)$) establishes that:

$$\frac{x^2}{x^2 + y^2} \leq 1$$

Multiplying this bounded inequality by $3|y|$ sets up our boundary sandwich:

$$0 \leq \frac{3x^2|y|}{x^2 + y^2} \leq 3|y|(1) = 3|y|$$

We evaluate the limits of our bounding bounds as $(x, y) \to (0, 0)$. Since $\lim_{(x,y)\to(0,0)} 0 = 0$ and $\lim_{(x,y)\to(0,0)} 3|y| = 0$, the **Squeeze Theorem** applies. Therefore, we conclude that:

$$\lim_{(x,y)\to(0,0)} \frac{3x^2y}{x^2 + y^2} = 0$$

Definition 3.19 A function f of two variables is **continuous** at (a, b) if

$$\lim_{(x,y)\to(a,b)} f(x, y) = f(a, b)$$

We say f is **continuous on** D if f is continuous at every point (a, b) in D. ◆

Example 3.20 Find

$$\lim_{(x,y)\to(1,2)} (x^2y^3 - x^3y^2 + 3x + 2y).$$

Solution. Since $f(x, y) = x^2y^3 - x^3y^2 + 3x + 2y$ is a **polynomial function**, it is continuous everywhere on $\mathbb{R}^2$. Therefore, we can find the limit via direct substitution:

$$\lim_{(x,y)\to(1,2)} (x^2y^3 - x^3y^2 + 3x + 2y) = (1)^2(2)^3 - (1)^3(2)^2 + 3(1) + 2(2)$$

$$\begin{aligned} &= 1(8) - 1(4) + 3 + 4 \\ &= 8 - 4 + 3 + 4 \\ &= 11 \end{aligned}$$

▲

Example 3.21 Where is the function $f(x,y) = \frac{x^2-y^2}{x^2+y^2}$ continuous?

Solution. Since f is a **rational function**, it is continuous at every point on its domain where the denominator is non-zero. The denominator $x^2 + y^2$ equals zero only at the origin. Thus, the function is continuous on its domain:

$$D = \{(x,y) \mid (x,y) \neq (0,0)\}$$

▲

Example 3.22 Find the limit of $g(x,y)$ as $(x,y) \to (0,0)$ if

$$g(x,y) = \begin{cases} \frac{x^2-y^2}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$$

Solution. To find the limit as $(x,y) \to (0,0)$, we examine the behavior of the function for points near, but not equal to, the origin. Therefore, we evaluate using the first piece of our function definition:

$$\lim_{(x,y)\to(0,0)} g(x,y) = \lim_{(x,y)\to(0,0)} \frac{x^2-y^2}{x^2+y^2}$$

As shown previously, approaching the origin along the x-axis yields a path limit of 1, while approaching along the y-axis yields a path limit of -1. Because the limits along these two paths are different, the overall limit does not exist.

Consequently, while the function value is defined at the origin ($g(0,0) = 0$), the function g is discontinuous at $(0,0)$ because the limit fails to exist. ▲

Example 3.23 Is the function

$$f(x,y) = \begin{cases} \frac{3x^2y}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$$

continuous?

Solution. To determine if $f(x,y)$ is continuous, we check the three criteria for continuity at the point of interest, $(0,0)$:

1 The function is defined at the origin: $f(0,0) = 0$.

2 Using the Squeeze Theorem (as shown in the earlier checkpoint), the limit of the non-zero piece as we approach the origin is:

$$\lim_{(x,y)\to(0,0)} \frac{3x^2y}{x^2+y^2} = 0$$

3 Since the limiting value equals the defined function value ($0 = 0$), we have:

$$\lim_{(x,y)\to(0,0)} f(x,y) = f(0,0)$$

Because it is continuous at $(0,0)$ and consists of a continuous rational expression at all other points, f is continuous on all of $\mathbb{R}^2$. ▲

Example 3.24 Show that

$$\lim_{(x,y,z)\to(1,2,-1)} \frac{(x+y+3z)^3}{(x-1)(y-2)(z+1)}$$

does not exist.

Solution. Let $P(x,y,z)$ approach the point $(1,2,-1)$ along an arbitrary line L passing through $(1,2,-1)$ with direction vector $\langle a,b,c\rangle$ where $a,b,c \neq 0$. The parametric equations of this line are given by:

$$x = 1+at, \quad y = 2+bt, \quad z = -1+ct \quad \text{for } t \in \mathbb{R}$$

Substituting these parametric expressions into the function as $t \to 0$ yields:

$$\begin{aligned}
\frac{(x+y+3z)^3}{(x-1)(y-2)(z+1)} &= \frac{((1+at)+(2+bt)+3(-1+ct))^3}{(1+at-1)(2+bt-2)(-1+ct+1)} \\
&= \frac{(1+2-3+at+bt+3ct)^3}{(at)(bt)(ct)} \\
&= \frac{(at+bt+3ct)^3}{abc\cdot t^3} \\
&= \frac{t^3(a+b+3c)^3}{abc\cdot t^3} \\
&= \frac{(a+b+3c)^3}{abc} \quad \text{for } t \neq 0
\end{aligned}$$

Since the value of this expression depends directly on the chosen direction components a, b, and c, approaching along different lines will produce different limiting values. By the two-path rule extended to three dimensions, the given limit does not exist. ▲

Theorem 3.25 *If a function f of two variables is continuous at (a,b) and a function g of one variable is continuous at $f(a,b)$, then the composite function $h(x,y) = g(f(x,y))$ is continuous at (a,b).*

Example 3.26 Find $h(x,y) = g(f(x,y))$ and the set on which h is continuous for each of the following:

1 $g(t) = 3t^2+t+1 \quad \text{and} \quad f(x,y) = xe^y$

2 $g(t) = \sin\sqrt{t} \quad \text{and} \quad f(x,y) = y-4x^2$

Solution.

1 Substituting $f(x,y)$ into $g(t)$ gives:

$$h(x,y) = g(f(x,y)) = 3(xe^y)^2 + (xe^y) + 1 = 3x^2e^{2y} + xe^y + 1$$

Since polynomials and exponential functions are continuous everywhere on their domains, this composite function is continuous at every point on $\mathbb{R}^2$.

2 Substituting $f(x,y)$ into $g(t)$ gives:

$$h(x,y) = g(f(x,y)) = \sin\sqrt{y-4x^2}$$

The outer sine function is continuous everywhere, but the square root function is only defined and continuous for non-negative inputs ($t \geq 0$).

Therefore, the expression inside the radical must be greater than or equal to zero. The composite function h is continuous on the set:

$$D = \{(x, y) \mid y \geq 4x^2\}$$

▲

3.3 Partial Derivatives

In this section, we will explore the concept of partial derivatives, which are essential for understanding how functions of multiple variables change with respect to each independent variable.

We will begin by defining first-order partial derivatives for functions of two or more variables and establishing their geometric interpretations as slopes of tangent lines along cross-sectional planes. Next, we will extend these concepts to higher-order modifications, uncovering how mixed partial variants operate under **Clairaut's Theorem**. Finally, we will apply these techniques to verify solutions to classic multivariate physics and engineering frameworks, including **Laplace's Equation** and the **Cauchy-Riemann Equations**.

By the end of this section, you will have a solid conceptual understanding of multivariate rate of change behaviors and a strong structural foundation for navigating complex partial differential layouts in multivariable calculus.

Definition 3.27 Let f be a function of two variables. The **first partial derivatives of f with respect to x and y** are the functions f_x and f_y defined by the limits:

$$\begin{aligned} f_x(x, y) &= \lim_{h \to 0} \frac{f(x+h, y) - f(x, y)}{h} \\ f_y(x, y) &= \lim_{h \to 0} \frac{f(x, y+h) - f(x, y)}{h} \end{aligned}$$

◆

Remark 3.28 Notation for Partial Derivatives. If $z = f(x, y)$, then the alternative notations for the partial derivatives include:

$$f_x = \frac{\partial f}{\partial x}, \quad f_y = \frac{\partial f}{\partial y}$$

$$\begin{aligned} f_x(x, y) &= D_x f = f_x = \frac{\partial}{\partial x} f(x, y) = \frac{\partial z}{\partial x} = z_x \\ f_y(x, y) &= D_y f = f_y = \frac{\partial}{\partial y} f(x, y) = \frac{\partial z}{\partial y} = z_y \end{aligned}$$

Example 3.29 If $f(x, y) = x^3y^2 - 2x^2y + 3x$, find $f_x(2, -1)$ and $f_y(2, -1)$.

Solution. To find $f_x(2, -1)$, we first find the general partial derivative with respect to x, $f_x(x, y)$, by treating y as a constant:

$$\begin{aligned} f_x(x, y) &= \frac{\partial}{\partial x}(x^3y^2 - 2x^2y + 3x) \\ &= 3x^2y^2 - 4xy + 3 \end{aligned}$$

Now, we substitute $x = 2$ and $y = -1$ into our result:

$$f_x(2, -1) = 3(2)^2(-1)^2 - 4(2)(-1) + 3$$

$$
\begin{aligned}
&= 3(4)(1) - (-8) + 3 \\
&= 12 + 8 + 3 \\
&= 23
\end{aligned}
$$

To find $f_y(2,-1)$, we find the general partial derivative with respect to y, $f_y(x,y)$, by treating x as a constant:

$$
\begin{aligned}
f_y(x,y) &= \frac{\partial}{\partial y}(x^3y^2 - 2x^2y + 3x) \\
&= 2x^3y - 2x^2 + 0 \\
&= 2x^3y - 2x^2
\end{aligned}
$$

Now, we substitute $x = 2$ and $y = -1$ into this expression:

$$
\begin{aligned}
f_y(2,-1) &= 2(2)^3(-1) - 2(2)^2 \\
&= 2(8)(-1) - 2(4) \\
&= -16 - 8 \\
&= -24
\end{aligned}
$$

▲

Example 3.30 If $f(x,y) = 4 - x^2 - 2y^2$, find $f_x(1,1)$ and $f_y(1,1)$, and interpret these numbers as slopes.

Solution.

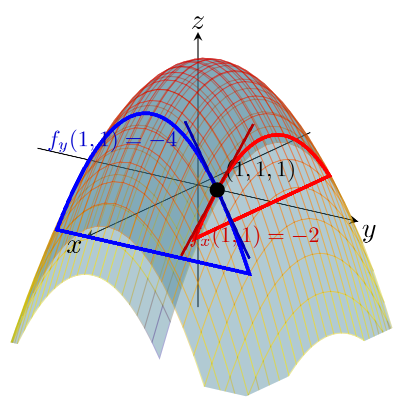


Figure 3.31 Geometric interpretation of $f_x(1,1)$ and $f_y(1,1)$ as tangent slopes on a paraboloid surface.

First, we compute the general partial derivatives by treating the opposite variable as a constant:

$$f_x(x,y) = -2x \quad \text{and} \quad f_y(x,y) = -4y$$

Evaluating these at the point $(1,1)$ yields:

$$f_x(1,1) = -2 \quad \text{and} \quad f_y(1,1) = -4$$

To interpret these numbers geometrically, note that the graph of f is a paraboloid, and the function value at this point is $f(1,1) = 4 - 1^2 - 2(1)^2 = 1$, giving the point $(1,1,1)$ on the surface.

The vertical plane $y = 1$ intersects the paraboloid in the curve (a parabola) described by $z = 4 - x^2 - 2(1)^2 = 2 - x^2$ in the plane $y = 1$. The slope of the tangent line to this parabola at the point $(1, 1, 1)$ is given by $f_x(1, 1) = -2$.

Similarly, the vertical plane $x = 1$ intersects the paraboloid in the curve described by $z = 4 - (1)^2 - 2y^2 = 3 - 2y^2$ in the plane $x = 1$. The slope of the tangent line to this parabola at the point $(1, 1, 1)$ is given by $f_y(1, 1) = -4$. ▲

Example 3.32 If $f(x, y) = \sin\left(\frac{x}{x+y}\right)$, find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.

Solution. Using the **Chain Rule for functions of one variable**, we differentiate the outer sine function while keeping the inner fraction intact, and then multiply by the partial derivative of the inner expression.

For the partial derivative with respect to x, we use the quotient rule on the inner function while treating y as a constant:

$$\begin{aligned}\frac{\partial f}{\partial x} &= \cos\left(\frac{x}{x+y}\right) \cdot \frac{\partial}{\partial x}\left(\frac{x}{x+y}\right) \\ &= \cos\left(\frac{x}{x+y}\right) \cdot \frac{(x+y)(1) - x(1)}{(x+y)^2} \\ &= \cos\left(\frac{x}{x+y}\right) \cdot \frac{y}{(x+y)^2}\end{aligned}$$

For the partial derivative with respect to y, we treat x as a constant and apply the chain rule/power rule to the inner function:

$$\begin{aligned}\frac{\partial f}{\partial y} &= \cos\left(\frac{x}{x+y}\right) \cdot \frac{\partial}{\partial y}\left(\frac{x}{x+y}\right) \\ &= \cos\left(\frac{x}{x+y}\right) \cdot \frac{\partial}{\partial y}\left(x(x+y)^{-1}\right) \\ &= \cos\left(\frac{x}{x+y}\right) \cdot \left(-x(x+y)^{-2}\right) \\ &= -\cos\left(\frac{x}{x+y}\right) \cdot \frac{x}{(x+y)^2}\end{aligned}$$

▲

Definition 3.33 Functions of More Than Two Variables. Let f be a function of three variables. The **first partial derivatives of f with respect to x, y, and z** are the functions f_x, f_y, and f_z defined by:

$$f_x(x, y, z) = \lim_{h \to 0} \frac{f(x+h, y, z) - f(x, y, z)}{h}$$

In general, if f is a function of n variables, $f = f(x_1, x_2, x_3, \ldots, x_n)$, its partial derivative with respect to the ith variable x_i is:

$$\frac{\partial f}{\partial x_i} = f_{x_i} = \lim_{h \to 0} \frac{f(x_1, \ldots, x_{i-1}, x_i + h, x_{i+1}, \ldots, x_n) - f(x_1, \ldots, x_{i-1}, x_i, x_{i+1}, \ldots, x_n)}{h}$$

◆

Example 3.34 If $f(x, y, z) = x^2 y^3 \sin z + e^{xy} \ln z$, find f_x, f_y, and f_z.

Solution. To find each partial derivative, we differentiate with respect to the chosen variable while holding all other variables constant:

$$\begin{aligned} f_x = \frac{\partial f}{\partial x} &= \frac{\partial}{\partial x}(x^2) \cdot y^3 \sin z + \frac{\partial}{\partial x}(e^{xy}) \cdot \ln z \\ &= 2xy^3 \sin z + ye^{xy} \ln z \end{aligned}$$

$$\begin{aligned} f_y = \frac{\partial f}{\partial y} &= x^2 \cdot \frac{\partial}{\partial y}(y^3) \cdot \sin z + \frac{\partial}{\partial y}(e^{xy}) \cdot \ln z \\ &= 3x^2y^2 \sin z + xe^{xy} \ln z \end{aligned}$$

$$\begin{aligned} f_z = \frac{\partial f}{\partial z} &= x^2y^3 \cdot \frac{\partial}{\partial z}(\sin z) + e^{xy} \cdot \frac{\partial}{\partial z}(\ln z) \\ &= x^2y^3 \cos z + \frac{e^{xy}}{z} \end{aligned}$$

▲

Definition 3.35 If f is a function of two variables, then its partial derivatives f_x and f_y are also functions of two variables. Consequently, we can consider their partial derivatives $(f_x)_x$, $(f_x)_y$, $(f_y)_x$, and $(f_y)_y$, which are called the **second partial derivatives** of f. If $z = f(x, y)$, we denote them as follows:

$$(f_x)_x = f_{xx} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 z}{\partial x^2}$$

$$(f_x)_y = f_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y\, \partial x} = \frac{\partial^2 z}{\partial y\, \partial x}$$

$$(f_y)_x = f_{yx} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial x\, \partial y} = \frac{\partial^2 z}{\partial x\, \partial y}$$

$$(f_y)_y = f_{yy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial y^2} = \frac{\partial^2 z}{\partial y^2}$$

◆

Example 3.36 Find the second partial derivatives of

$$f(x, y) = x^3 + x^2y^3 - 2y^2$$

Solution. We begin by calculating the first-order partial derivatives of f with respect to x and y:

$$f_x(x, y) = \frac{\partial}{\partial x}(x^3 + x^2y^3 - 2y^2) = 3x^2 + 2xy^3$$

$$f_y(x, y) = \frac{\partial}{\partial y}(x^3 + x^2y^3 - 2y^2) = 3x^2y^2 - 4y$$

Next, we differentiate f_x and f_y to find the four second-order partial derivatives:

1 Differentiating f_x with respect to x:

$$f_{xx} = \frac{\partial}{\partial x}(3x^2 + 2xy^3) = 6x + 2y^3$$

2 Differentiating f_x with respect to y (mixed partial):

$$f_{xy} = \frac{\partial}{\partial y}(3x^2 + 2xy^3) = 6xy^2$$

3 Differentiating f_y with respect to x (mixed partial):

$$f_{yx} = \frac{\partial}{\partial x}(3x^2y^2 - 4y) = 6xy^2$$

4 Differentiating f_y with respect to y:

$$f_{yy} = \frac{\partial}{\partial y}(3x^2y^2 - 4y) = 6x^2y - 4$$

Notice that the mixed partial derivatives are equal ($f_{xy} = f_{yx} = 6xy^2$). ▲

Theorem 3.37 Clairaut's Theorem (Mixed Derivatives). *Suppose f is defined on a disk D that contains the point (a, b). If the functions f_{xy} and f_{yx} are both continuous on D, then:*

$$f_{xy}(a, b) = f_{yx}(a, b)$$

Example 3.38 Mixed Derivatives. Compute f_{xxyz} if $f(x, y, z) = \sin(3x + yz)$.

Solution 1 (Direct Method (Order: x, x, y, z)). We compute the fourth-order partial derivative step by step, applying the single-variable chain and product rules successively in the order written:

$$\begin{aligned} f_x &= \frac{\partial}{\partial x}[\sin(3x + yz)] = 3\cos(3x + yz) \\ f_{xx} &= \frac{\partial}{\partial x}[3\cos(3x + yz)] = -9\sin(3x + yz) \\ f_{xxy} &= \frac{\partial}{\partial y}[-9\sin(3x + yz)] = -9z\cos(3x + yz) \end{aligned}$$

Now, using the single-variable product rule to differentiate f_{xxy} with respect to z:

$$\begin{aligned} f_{xxyz} &= \frac{\partial}{\partial z}[-9z\cos(3x + yz)] \\ &= -9\cos(3x + yz) + (-9z) \cdot (-\sin(3x + yz) \cdot y) \\ &= -9\cos(3x + yz) + 9yz\sin(3x + yz) \end{aligned}$$

Solution 2 (Alternative Method via Clairaut's Theorem (Order: x, x, z, y)). Since the function $f(x, y, z)$ and all of its partial derivatives are continuous everywhere, we can apply **Clairaut's Theorem** to change the order of differentiation to make the algebra simpler. Let's differentiate with respect to z before differentiating with respect to y, so we evaluate f_{xxzy} instead of f_{xxyz}:

$$f_{xxyz} = f_{xxzy} = \frac{\partial}{\partial y}\left[\frac{\partial}{\partial z}(f_{xx})\right]$$

Using our previous result for the second derivative, $f_{xx} = -9\sin(3x + yz)$, we differentiate with respect to z first:

$$\begin{aligned} f_{xxz} &= \frac{\partial}{\partial z}[-9\sin(3x + yz)] \\ &= -9\cos(3x + yz) \cdot \frac{\partial}{\partial z}(3x + yz) \\ &= -9y\cos(3x + yz) \end{aligned}$$

Finally, we differentiate f_{xxz} with respect to y. Because y appears both outside and inside the cosine term, we apply the single-variable product rule:

$$\begin{aligned}
f_{xxzy} &= \frac{\partial}{\partial y}[-9y\cos(3x+yz)] \\
&= \frac{\partial}{\partial y}[-9y]\cdot\cos(3x+yz) + (-9y)\cdot\frac{\partial}{\partial y}[\cos(3x+yz)] \\
&= -9\cos(3x+yz) + (-9y)\cdot(-\sin(3x+yz)\cdot z) \\
&= -9\cos(3x+yz) + 9yz\sin(3x+yz)
\end{aligned}$$

Both mixed partial orders yield the exact same final expression, validating Clairaut's Theorem. ▲

Example 3.39 Harmonic Function. If u is a function of two variables and satisfies the partial differential equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

this equation is called **Laplace's equation**. Solutions to this equation are called **harmonic functions** and are frequently used in problems involving heat conduction and fluid flow.

Show that the function $u(x, y) = e^x \cos(y)$ is harmonic.

Solution. To show that $u(x, y)$ is a harmonic function, we must verify that it satisfies Laplace's equation by calculating its second-order pure partial derivatives.

First, we find the first and second partial derivatives with respect to x, treating y as a constant:

$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x}[e^x\cos(y)] = e^x\cos(y)$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x}[e^x\cos(y)] = e^x\cos(y)$$

Next, we find the first and second partial derivatives with respect to y, treating x as a constant:

$$\frac{\partial u}{\partial y} = \frac{\partial}{\partial y}[e^x\cos(y)] = -e^x\sin(y)$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y}[-e^x\sin(y)] = -e^x\cos(y)$$

Now, we substitute both second-order pure partial derivatives into the left-hand side of Laplace's equation:

$$\begin{aligned}
\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} &= e^x\cos(y) + (-e^x\cos(y)) \\
&= 0
\end{aligned}$$

Since the sum of the second partial derivatives is identically 0 everywhere, the function satisfies Laplace's equation. Therefore, $u(x, y) = e^x \cos(y)$ is a **harmonic function**. ▲

Example 3.40 Cauchy-Riemann Equation. If u and v are functions of two variables and satisfy

$$u_x = v_y \quad \text{and} \quad u_y = -v_x$$

these partial differential equations are called the **Cauchy-Riemann equations**.

Show that the functions $u(x,y) = \frac{y}{x^2+y^2}$ and $v(x,y) = \frac{x}{x^2+y^2}$ satisfy the Cauchy–Riemann equations.

Solution. To verify that u and v satisfy the Cauchy–Riemann equations, we must compute all four first-order partial derivatives using the single-variable quotient rule.

First, we compute the partial derivatives of $u(x,y)$ with respect to x and y:

$$u_x = \frac{\partial}{\partial x}\left[\frac{y}{x^2+y^2}\right] = \frac{(x^2+y^2)(0) - y(2x)}{(x^2+y^2)^2} = \frac{-2xy}{(x^2+y^2)^2}$$
$$u_y = \frac{\partial}{\partial y}\left[\frac{y}{x^2+y^2}\right] = \frac{(x^2+y^2)(1) - y(2y)}{(x^2+y^2)^2} = \frac{x^2-y^2}{(x^2+y^2)^2}$$

Next, we compute the partial derivatives of $v(x,y)$ with respect to x and y:

$$v_x = \frac{\partial}{\partial x}\left[\frac{x}{x^2+y^2}\right] = \frac{(x^2+y^2)(1) - x(2x)}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2}$$
$$v_y = \frac{\partial}{\partial y}\left[\frac{x}{x^2+y^2}\right] = \frac{(x^2+y^2)(0) - x(2y)}{(x^2+y^2)^2} = \frac{-2xy}{(x^2+y^2)^2}$$

Now we check both conditions of the Cauchy–Riemann system:

1 Comparing u_x and v_y, we see they match perfectly:

$$u_x = \frac{-2xy}{(x^2+y^2)^2} = v_y$$

2 Comparing u_y and v_x, we observe they are exact negatives of each other:

$$-v_x = -\left(\frac{y^2-x^2}{(x^2+y^2)^2}\right) = \frac{x^2-y^2}{(x^2+y^2)^2} = u_y$$

Since both partial differential equations hold true on their mutual domain $\{(x,y) \mid (x,y) \neq (0,0)\}$, the functions u and v satisfy the Cauchy-Riemann equations. ▲

3.4 Increments and Differentials

In this section, we will explore the concepts of increments and differentials, which are fundamental for analyzing how multi-variable functions respond to small changes in their independent inputs. We will begin by defining the exact increment of a function and establishing the rigorous criteria for differentiability using non-unique error functions.

If f is a function of two variables x and y, then Δx and Δy denote the **increments of** x **and** y. Suppose x changes from a to $a + \Delta x$ and y changes from b to $b + \Delta y$. Then the corresponding increment of $z = f(x,y)$ is:

$$\Delta z = f(a + \Delta x, b + \Delta y) - f(a,b)$$

The increment of z, denoted Δz, represents the change in the values of f when (x,y) changes from (a,b) to $(a + \Delta x, b + \Delta y)$.

Example 3.41 Let $z = f(x, y) = 3x^2 - xy$.

1 Find an algebraic expression for Δz in terms of x, y, Δx, and Δy.

2 Calculate the exact numerical value of Δz if (x, y) changes from $(1, 2)$ to $(1.01, 1.98)$.

Solution.

1 Using the definition of a multivariate function increment, we set up and expand the expression for Δz:

$$\begin{aligned}\Delta z &= f(x + \Delta x, y + \Delta y) - f(x, y)\\ &= \left[3(x + \Delta x)^2 - (x + \Delta x)(y + \Delta y)\right] - (3x^2 - xy)\\ &= \left[3(x^2 + 2x\Delta x + (\Delta x)^2) - (xy + x\Delta y + y\Delta x + \Delta x\Delta y)\right] - 3x^2 + xy\\ &= 3x^2 + 6x\Delta x + 3(\Delta x)^2 - xy - x\Delta y - y\Delta x - \Delta x\Delta y - 3x^2 + xy\end{aligned}$$

Canceling the matching $3x^2$ and xy terms, we obtain the simplified general increment:

$$\Delta z = 6x\Delta x + 3(\Delta x)^2 - y\Delta x - x\Delta y - \Delta x\Delta y$$

2 The independent variables change from the base point $(x, y) = (1, 2)$ to the final point $(1.01, 1.98)$. We compute the respective component increments:

$$\Delta x = 1.01 - 1 = 0.01$$

$$\Delta y = 1.98 - 2 = -0.02$$

Substituting $x = 1$, $y = 2$, $\Delta x = 0.01$, and $\Delta y = -0.02$ directly into our algebraic result from Part 1 yields:

$$\begin{aligned}\Delta z &= 6(1)(0.01) + 3(0.01)^2 - 2(0.01) - 1(-0.02) - (0.01)(-0.02)\\ &= 0.06 + 3(0.0001) - 0.02 + 0.02 - (-0.0002)\\ &= 0.06 + 0.0003 + 0.0002\\ &= 0.0605\end{aligned}$$

▲

Definition 3.42 Differentiability for Functions of Two Variables. If $z = f(x, y)$, then f is **differentiable** at (a, b) if Δz can be expressed in the form:

$$\Delta z = f_x(a, b)\Delta x + f_y(a, b)\Delta y + \epsilon_1\Delta x + \epsilon_2\Delta y$$

where $\epsilon_1 \to 0$ and $\epsilon_2 \to 0$ as $(\Delta x, \Delta y) \to (0, 0)$. ◆

Theorem 3.43 Sufficient Condition for Differentiability. *If the partial derivatives f_x and f_y exist near (a, b) and are continuous at (a, b), then f is differentiable at (a, b).*

Example 3.44 If $z = f(x, y) = 3x^2 - xy$, find an expression for Δz to identify the functions ϵ_1 and ϵ_2 using the base point $(a, b) = (x, y)$.

Solution. First, we find the first-order partial derivatives of $f(x, y) = 3x^2 - xy$ to establish our baseline linear terms:

$$f_x(x, y) = 6x - y \quad \text{and} \quad f_y(x, y) = -x$$

From our previous algebraic expansion of the increment for this exact function, we know that:

$$\Delta z = 6x\Delta x + 3(\Delta x)^2 - y\Delta x - x\Delta y - \Delta x\Delta y$$

We rearrange and group these terms to isolate the linear differential components $f_x(x, y)\Delta x$ and $f_y(x, y)\Delta y$:

$$\begin{aligned}\Delta z &= (6x\Delta x - y\Delta x) + (-x\Delta y) + 3(\Delta x)^2 - \Delta x\Delta y \\ &= (6x - y)\Delta x + (-x)\Delta y + 3(\Delta x)^2 - \Delta x\Delta y\end{aligned}$$

To match the definition's template form of $\epsilon_1\Delta x + \epsilon_2\Delta y$, we factor a Δx out of the first remaining term and a Δy out of the second remaining term:

$$\Delta z = (6x - y)\Delta x + (-x)\Delta y + (3\Delta x)\Delta x + (-\Delta x)\Delta y$$

Comparing this directly to the definition form, we can identify:

$$\epsilon_1 = 3\Delta x \quad \text{and} \quad \epsilon_2 = -\Delta x$$

As $(\Delta x, \Delta y) \to (0, 0)$, it is clear that both $\epsilon_1 \to 0$ and $\epsilon_2 \to 0$. This confirms explicitly that the polynomial function $f(x, y) = 3x^2 - xy$ is differentiable everywhere.

Note: Because the leftover terms can be grouped in alternative arrangements (for example, factoring Δx out of the entire remaining expression), the error functions ϵ_1 and ϵ_2 are not unique. ▲

Definition 3.45 Differentials for Functions of Two Variables. If $z = f(x, y)$ is a differentiable function, the **differentials** dx **and** dy are independent variables defined as $dx = \Delta x$ and $dy = \Delta y$. The **total differential of** z (or the dependent differential dz) is defined by:

$$dz = f_x(x, y)\, dx + f_y(x, y)\, dy = \frac{\partial z}{\partial x}\, dx + \frac{\partial z}{\partial y}\, dy$$

Using this framework, the exact function increment can be written as:

$$\Delta z = f_x(a, b)\Delta x + f_y(a, b)\Delta y + \epsilon_1\Delta x + \epsilon_2\Delta y = dz + \epsilon_1\Delta x + \epsilon_2\Delta y$$

On the other hand, the discrepancy between the exact increment and the total differential approximation is given by:

$$\Delta z - dz = \epsilon_1\Delta x + \epsilon_2\Delta y$$

where $\epsilon_1 \to 0$ and $\epsilon_2 \to 0$ as $(\Delta x, \Delta y) \to (0, 0)$. Therefore, for small changes in x and y, we can approximate the true change using the differential:

$$\Delta z \approx dz$$

◆

Example 3.46 If $z = f(x, y) = 3x^2 - xy$, find the total differential dz. Then, use it to approximate the change in z if (x, y) changes from $(1, 2)$ to $(1.01, 1.98)$.

Solution. First, we find the first-order partial derivatives of the function $f(x, y) = 3x^2 - xy$:

$$f_x(x, y) = 6x - y \quad \text{and} \quad f_y(x, y) = -x$$

Substituting these expressions into the definition of the total differential yields:

$$dz = (6x - y)\,dx + (-x)\,dy$$

Next, we use the given points to determine the baseline values and independent changes:

$$x = 1, \quad y = 2$$

$$dx = \Delta x = 1.01 - 1 = 0.01$$

$$dy = \Delta y = 1.98 - 2 = -0.02$$

Substituting these specific numerical parameters directly into our total differential expression approximates the change:

$$\begin{aligned} dz &= (6(1) - 2)(0.01) + (-1)(-0.02) \\ &= (4)(0.01) + 0.02 \\ &= 0.04 + 0.02 \\ &= 0.06 \end{aligned}$$

Thus, the approximate change in z computed via differentials is $dz = 0.06$.

To find the error involved in using this approximation, we compare it against the exact change ($\Delta z = 0.0605$) calculated in our previous exercise:

$$\text{Error} = |\Delta z - dz| = |0.0605 - 0.06| = 0.0005$$

This demonstrates that for small structural changes near the point $(1, 2)$, the total differential provides a highly accurate approximation. ▲

Example 3.47 The radius and altitude of a right circular cylinder are measured as 3 in and 8 in respectively, with a maximum possible error in each measurement of ± 0.05 inch. Use differentials to approximate the maximum error in the calculated volume of the cylinder.

Solution. The volume V of a right circular cylinder is a function of two variables—its radius r and its height or altitude h:

$$V(r, h) = \pi r^2 h$$

The total differential dV represents the approximate propagated error in the volume and is found by calculating the first-order partial derivatives with respect to r and h:

$$\begin{aligned} dV &= \frac{\partial V}{\partial r}\,dr + \frac{\partial V}{\partial h}\,dh \\ &= (2\pi r h)\,dr + (\pi r^2)\,dh \end{aligned}$$

From the problem statements, we isolate our baseline parameters and maximum independent measurement fluctuations:

$$r = 3, \quad h = 8, \quad |dr| \leq 0.05, \quad |dh| \leq 0.05$$

To approximate the **maximum potential error**, we look at the worst-case scenario where the errors accumulate constructively, taking the maximum absolute value bounds for the differentials ($dr = 0.05$ and $dh = 0.05$):

$$\begin{aligned} dV &= 2\pi(3)(8)(0.05) + \pi(3)^2(0.05) \\ &= 48\pi(0.05) + 9\pi(0.05) \end{aligned}$$

$$= 2.4\pi + 0.45\pi$$
$$= 2.85\pi \approx 8.9535 \text{ in}^3$$

Thus, using differentials, the estimated maximum error in the calculated volume is approximately $\pm 2.85\pi$ cubic inches (or roughly ± 8.95 in^3). ▲

3.5 Chain Rules

The chain rule is a fundamental tool in calculus that allows us to compute the derivative of a composite function. In this section, we will derive the chain rule for functions of several variables and apply it to various cases.

Recall that if $y = f(x)$ and $x = g(t)$, where f and g are differentiable functions, then y is indirectly a differentiable function of t, and the single-variable chain rule applies:

$$\frac{dy}{dt} = \frac{dy}{dx}\frac{dx}{dt}$$

Theorem 3.48 The Chain Rule: Case I. *Suppose that $z = f(x, y)$ is a differentiable function of x and y, where $x = g(t)$ and $y = h(t)$ are both differentiable functions of t. Then z is a differentiable function of t and satisfies the formula:*

$$\frac{dz}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}$$

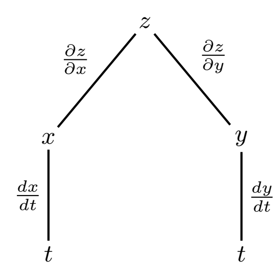


Figure 3.49 *Dependency tree diagram for Case I of the Multivariate Chain Rule.*

Example 3.50 If $z = x^2y + 3xy^4$, where $x = \sin(2t)$ and $y = \cos(t)$, find $\frac{dz}{dt}$ when $t = 0$.

Solution.

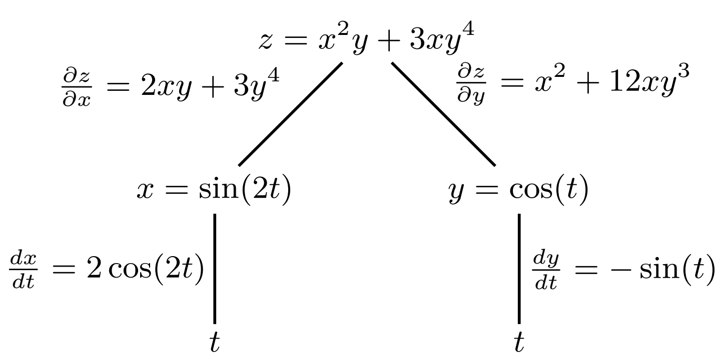


Figure 3.51 Variable dependency tree diagram tailored to the specific functions of this example.

We first compute all the necessary components for the Case I Chain Rule formula. The first-order partial derivatives of z with respect to x and y are:

$$\frac{\partial z}{\partial x} = 2xy + 3y^4 \quad \text{and} \quad \frac{\partial z}{\partial y} = x^2 + 12xy^3$$

The single-variable derivatives of x and y with respect to the parameter t are:

$$\frac{dx}{dt} = \frac{d}{dt}[\sin(2t)] = 2\cos(2t) \quad \text{and} \quad \frac{dy}{dt} = \frac{d}{dt}[\cos(t)] = -\sin(t)$$

Now, we substitute these components into the chain rule formula:

$$\begin{aligned}\frac{dz}{dt} &= \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt} \\ &= (2xy + 3y^4)(2\cos(2t)) + (x^2 + 12xy^3)(-\sin(t))\end{aligned}$$

To find the value of $\frac{dz}{dt}$ at the specific parameter point $t = 0$, we first evaluate the corresponding coordinate locations for x and y:

$$x = \sin(2(0)) = 0 \quad \text{and} \quad y = \cos(0) = 1$$

Finally, substituting $t = 0$, $x = 0$, and $y = 1$ into our derivative expression yields:

$$\begin{aligned}\left.\frac{dz}{dt}\right|_{t=0} &= \left(2(0)(1) + 3(1)^4\right)(2\cos(0)) + \left(0^2 + 12(0)(1)^3\right)(-\sin(0)) \\ &= (0+3)(2(1)) + (0+0)(0) \\ &= (3)(2) + 0 \\ &= 6\end{aligned}$$

▲

Theorem 3.52 The Chain Rule: Case II. *Suppose that $z = f(x, y)$ is a differentiable function of x and y, where $x = g(s, t)$ and $y = h(s, t)$ are both differentiable functions of s and t. Then z is a differentiable function of s and t, satisfying the partial derivative formulas:*

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s} \quad \textit{and} \quad \frac{\partial z}{\partial t} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}$$

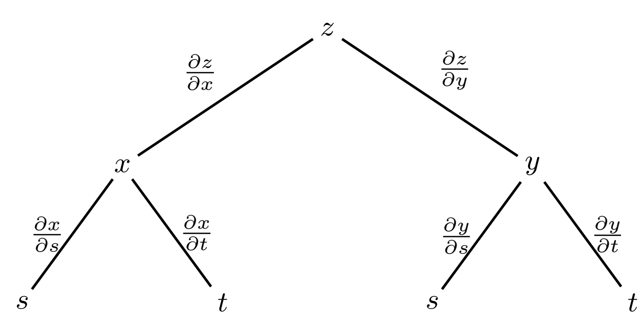


Figure 3.53 *Dependency tree diagram for Case II of the Multivariate Chain Rule.*

Example 3.54 If $z = e^x \sin(y)$, where $x = st^2$ and $y = s^2 t$, find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$.

Solution.

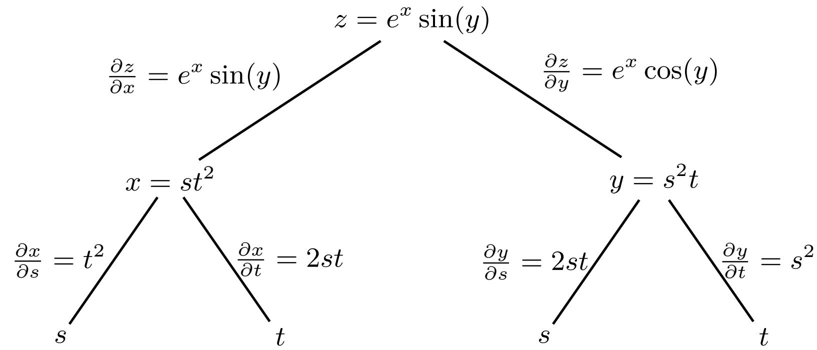


Figure 3.55 Tailored dependency tree diagram tracking partial rate changes down to parameters s and t.

We first calculate the required first-order partial derivatives of the dependent surface function z:

$$\frac{\partial z}{\partial x} = e^x \sin(y) \quad \text{and} \quad \frac{\partial z}{\partial y} = e^x \cos(y)$$

Next, we calculate the partial derivatives for the intermediate variable functions x and y with respect to both independent parameters s and t:

$$\frac{\partial x}{\partial s} = t^2, \quad \frac{\partial x}{\partial t} = 2st$$

$$\frac{\partial y}{\partial s} = 2st, \quad \frac{\partial y}{\partial t} = s^2$$

Applying the Case II Chain Rule formula for the parameter s, we substitute our partial derivatives:

$$\begin{aligned}
\frac{\partial z}{\partial s} &= \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s} \\
&= \left(e^x \sin(y)\right)\left(t^2\right) + \left(e^x \cos(y)\right)(2st) \\
&= t^2 e^{st^2} \sin(s^2 t) + 2ste^{st^2} \cos(s^2 t)
\end{aligned}$$

Applying the Case II Chain Rule formula for the parameter t, we substitute our partial derivatives:

$$\begin{aligned}
\frac{\partial z}{\partial t} &= \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t} \\
&= \left(e^x \sin(y)\right)(2st) + \left(e^x \cos(y)\right)\left(s^2\right) \\
&= 2ste^{st^2} \sin(s^2 t) + s^2 e^{st^2} \cos(s^2 t)
\end{aligned}$$

▲

Example 3.56 Use the chain rule to find $\frac{\partial w}{\partial z}$, if

$$w = r^2 + sv + t^3, \quad r = x^2 + y^2 + z^2, \quad s = xyz, \quad v = xe^y, \quad t = yz^2$$

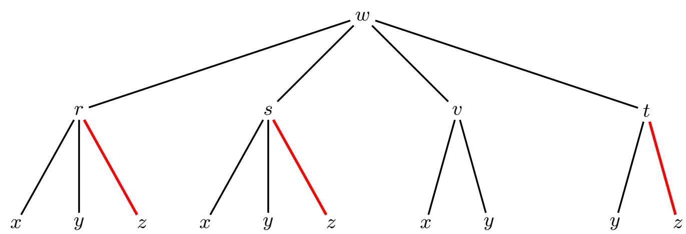


Figure 3.57 Dependency tree diagram illustrating the intermediate variables linking w to the independent variable z.

Solution. The dependent variable w is directly a function of four intermediate variables: r, s, v, and t. By checking our dependency tree, we see that r, s, and t depend on the independent variable z, whereas v does not contain z (meaning $\frac{\partial v}{\partial z} = 0$). Therefore, the chain rule extension for this system is:

$$\frac{\partial w}{\partial z} = \frac{\partial w}{\partial r}\frac{\partial r}{\partial z} + \frac{\partial w}{\partial s}\frac{\partial s}{\partial z} + \frac{\partial w}{\partial v}\frac{\partial v}{\partial z} + \frac{\partial w}{\partial t}\frac{\partial t}{\partial z}$$

We find each required component derivative independently:

$$\frac{\partial w}{\partial r} = 2r = 2(x^2 + y^2 + z^2) \quad \text{and} \quad \frac{\partial r}{\partial z} = 2z$$

$$\frac{\partial w}{\partial s} = v = xe^y \quad \text{and} \quad \frac{\partial s}{\partial z} = xy$$

$$\frac{\partial w}{\partial t} = 3t^2 = 3(yz^2)^2 = 3y^2z^4 \quad \text{and} \quad \frac{\partial t}{\partial z} = 2yz$$

Substituting these components back into our general chain rule formula yields:

$$\begin{aligned}\frac{\partial w}{\partial z} &= \left[2(x^2 + y^2 + z^2)\right](2z) + \left[xe^y\right](xy) + (xyz)(0) + \left[3y^2z^4\right](2yz) \\ &= 4z(x^2 + y^2 + z^2) + x^2ye^y + 6y^3z^5\end{aligned}$$

▲

Example 3.58 If $g(s,t) = f(s^2 - t^2, t^2 - s^2)$ and f is differentiable, show that g satisfies the partial differential equation:

$$t\frac{\partial g}{\partial s} + s\frac{\partial g}{\partial t} = 0$$

Solution. Let us define the intermediate variables for the inner functions as $x = s^2 - t^2$ and $y = t^2 - s^2$, so that $g(s,t) = f(x,y)$. By applying the Case II Chain Rule, we compute the general first-order partial derivatives of g with respect to the parameters s and t:

For the parameter s:

$$\begin{aligned}\frac{\partial g}{\partial s} &= \frac{\partial f}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial s} \\ &= \frac{\partial f}{\partial x}(2s) + \frac{\partial f}{\partial y}(-2s) \\ &= 2s\frac{\partial f}{\partial x} - 2s\frac{\partial f}{\partial y}\end{aligned}$$

For the parameter t:

$$\begin{aligned}\frac{\partial g}{\partial t} &= \frac{\partial f}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial t} \\ &= \frac{\partial f}{\partial x}(-2t) + \frac{\partial f}{\partial y}(2t) \\ &= -2t\frac{\partial f}{\partial x} + 2t\frac{\partial f}{\partial y}\end{aligned}$$

Now, we substitute these expressions directly into the left-hand side of the target verification equation:

$$\begin{aligned}t\frac{\partial g}{\partial s} + s\frac{\partial g}{\partial t} &= t\left(2s\frac{\partial f}{\partial x} - 2s\frac{\partial f}{\partial y}\right) + s\left(-2t\frac{\partial f}{\partial x} + 2t\frac{\partial f}{\partial y}\right) \\ &= 2st\frac{\partial f}{\partial x} - 2st\frac{\partial f}{\partial y} - 2st\frac{\partial f}{\partial x} + 2st\frac{\partial f}{\partial y} \\ &= 0\end{aligned}$$

Since the expression simplifies identically to 0, we have successfully shown that the function g satisfies the given equation. ▲

Example 3.59 If $z = f(x, y)$ has continuous second-order partial derivatives, $x = r^2 + s^2$, and $y = 2rs$, find $\frac{\partial z}{\partial r}$ and $\frac{\partial^2 z}{\partial r^2}$.

Solution. To find the first partial derivative $\frac{\partial z}{\partial r}$, we apply the Case II Multivariate Chain Rule:

$$\begin{aligned}\frac{\partial z}{\partial r} &= \frac{\partial z}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial r} \\ &= \frac{\partial z}{\partial x}(2r) + \frac{\partial z}{\partial y}(2s) \\ &= 2r\frac{\partial z}{\partial x} + 2s\frac{\partial z}{\partial y}\end{aligned}$$

To find the second partial derivative $\frac{\partial^2 z}{\partial r^2}$, we differentiate $\frac{\partial z}{\partial r}$ with respect to r. This requires applying both the single-variable product rule and a nested multivariate chain rule since the internal terms $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ still implicitly depend on x and y (which depend on r):

$$\begin{aligned}\frac{\partial^2 z}{\partial r^2} &= \frac{\partial}{\partial r}\left(2r\frac{\partial z}{\partial x} + 2s\frac{\partial z}{\partial y}\right) \\ &= \left[2\frac{\partial z}{\partial x} + 2r\frac{\partial}{\partial r}\left(\frac{\partial z}{\partial x}\right)\right] + 2s\frac{\partial}{\partial r}\left(\frac{\partial z}{\partial y}\right)\end{aligned}$$

We expand the nested partial derivatives $\frac{\partial}{\partial r}\left(\frac{\partial z}{\partial x}\right)$ and $\frac{\partial}{\partial r}\left(\frac{\partial z}{\partial y}\right)$ using the chain rule:

$$\begin{aligned}\frac{\partial}{\partial r}\left(\frac{\partial z}{\partial x}\right) &= \frac{\partial^2 z}{\partial x^2}\frac{\partial x}{\partial r} + \frac{\partial^2 z}{\partial y\,\partial x}\frac{\partial y}{\partial r} = \frac{\partial^2 z}{\partial x^2}(2r) + \frac{\partial^2 z}{\partial y\,\partial x}(2s) \\ \frac{\partial}{\partial r}\left(\frac{\partial z}{\partial y}\right) &= \frac{\partial^2 z}{\partial x\,\partial y}\frac{\partial x}{\partial r} + \frac{\partial^2 z}{\partial y^2}\frac{\partial y}{\partial r} = \frac{\partial^2 z}{\partial x\,\partial y}(2r) + \frac{\partial^2 z}{\partial y^2}(2s)\end{aligned}$$

Substituting these expansions back into our main differential expression yields:

$$\frac{\partial^2 z}{\partial r^2} = 2\frac{\partial z}{\partial x} + 2r\left[2r\frac{\partial^2 z}{\partial x^2} + 2s\frac{\partial^2 z}{\partial y\,\partial x}\right] + 2s\left[2r\frac{\partial^2 z}{\partial x\,\partial y} + 2s\frac{\partial^2 z}{\partial y^2}\right]$$

$$= 2\frac{\partial z}{\partial x} + 4r^2\frac{\partial^2 z}{\partial x^2} + 4rs\frac{\partial^2 z}{\partial y\,\partial x} + 4rs\frac{\partial^2 z}{\partial x\,\partial y} + 4s^2\frac{\partial^2 z}{\partial y^2}$$

Since the second-order partial derivatives are continuous, **Clairaut's Theorem** guarantees that the mixed partials are equal ($\frac{\partial^2 z}{\partial y\,\partial x} = \frac{\partial^2 z}{\partial x\,\partial y}$). Combining the like terms gives the final result:

$$\frac{\partial^2 z}{\partial r^2} = 2\frac{\partial z}{\partial x} + 4r^2\frac{\partial^2 z}{\partial x^2} + 8rs\frac{\partial^2 z}{\partial x\,\partial y} + 4s^2\frac{\partial^2 z}{\partial y^2}$$

▲

Suppose an equation of the form $F(x, y) = 0$ defines y implicitly as a differentiable function of x, such that $y = f(x)$, where $F(x, f(x)) = 0$ for all x in the domain of f. If F is differentiable, we can apply the multivariate chain rule to differentiate both sides of the equation with respect to x, treating x as both an independent parameter and a base variable path:

$$\frac{\partial F}{\partial x}\frac{dx}{dx} + \frac{\partial F}{\partial y}\frac{dy}{dx} = 0 \implies \frac{\partial F}{\partial x} + \frac{\partial F}{\partial y}\frac{dy}{dx} = 0$$

If $\frac{\partial F}{\partial y} \neq 0$, we can isolate and solve for the single-variable derivative $\frac{dy}{dx}$.

Theorem 3.60 Implicit Differentiation: Case I. *If an equation $F(x, y) = 0$ defines y implicitly as a differentiable function of x, then:*

$$\frac{dy}{dx} = -\frac{F_x(x, y)}{F_y(x, y)} = -\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial y}}$$

Example 3.61 Find y' if $x^3 + y^3 = 6xy$.

Solution. First, we move all terms to one side of the equation to define our multi-variable identity function $F(x, y) = 0$:

$$F(x, y) = x^3 + y^3 - 6xy = 0$$

Next, we calculate the first-order partial derivatives F_x and F_y:

$$F_x(x, y) = \frac{\partial}{\partial x}(x^3 + y^3 - 6xy) = 3x^2 - 6y$$

$$F_y(x, y) = \frac{\partial}{\partial y}(x^3 + y^3 - 6xy) = 3y^2 - 6x$$

Applying the Theorem for Implicit Differentiation, we substitute these partials into the formula:

$$\begin{aligned} y' = \frac{dy}{dx} &= -\frac{F_x(x, y)}{F_y(x, y)} \\ &= -\frac{3x^2 - 6y}{3y^2 - 6x} \\ &= -\frac{3(x^2 - 2y)}{3(y^2 - 2x)} \\ &= -\frac{x^2 - 2y}{y^2 - 2x} \end{aligned}$$

▲

Theorem 3.62 Implicit Differentiation: Case II. *If an equation $F(x, y, z) = 0$ defines z implicitly as a differentiable function of two variables x and y, such that $z = f(x, y)$ for every (x, y) in the domain of f, then:*

$$\frac{\partial z}{\partial x} = -\frac{F_x(x, y, z)}{F_z(x, y, z)} \quad \text{and} \quad \frac{\partial z}{\partial y} = -\frac{F_y(x, y, z)}{F_z(x, y, z)}$$

Example 3.63 Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ if z is defined implicitly as a function of x and y by the equation:

$$x^3 + y^3 + z^3 + 6xyz = 1$$

Solution. First, we move all terms to one side of the equation to define our three-variable identity function $F(x, y, z) = 0$:

$$F(x, y, z) = x^3 + y^3 + z^3 + 6xyz - 1 = 0$$

Next, we find the three required first-order partial derivatives of F:

$$F_x(x, y, z) = \frac{\partial}{\partial x}(x^3 + y^3 + z^3 + 6xyz - 1) = 3x^2 + 6yz$$

$$F_y(x, y, z) = \frac{\partial}{\partial y}(x^3 + y^3 + z^3 + 6xyz - 1) = 3y^2 + 6xz$$

$$F_z(x, y, z) = \frac{\partial}{\partial z}(x^3 + y^3 + z^3 + 6xyz - 1) = 3z^2 + 6xy$$

To find $\frac{\partial z}{\partial x}$, we substitute the partial derivatives F_x and F_z into the implicit differentiation formula and simplify by factoring out a common factor of 3:

$$\begin{aligned} \frac{\partial z}{\partial x} &= -\frac{F_x(x, y, z)}{F_z(x, y, z)} \\ &= -\frac{3x^2 + 6yz}{3z^2 + 6xy} \\ &= -\frac{3(x^2 + 2yz)}{3(z^2 + 2xy)} \\ &= -\frac{x^2 + 2yz}{z^2 + 2xy} \end{aligned}$$

Similarly, to find $\frac{\partial z}{\partial y}$, we substitute the partial derivatives F_y and F_z into the corresponding formula and simplify:

$$\begin{aligned} \frac{\partial z}{\partial y} &= -\frac{F_y(x, y, z)}{F_z(x, y, z)} \\ &= -\frac{3y^2 + 6xz}{3z^2 + 6xy} \\ &= -\frac{3(y^2 + 2xz)}{3(z^2 + 2xy)} \\ &= -\frac{y^2 + 2xz}{z^2 + 2xy} \end{aligned}$$

▲

Checkpoint 3.64 Use the chain rule to find $\frac{\partial w}{\partial p}$ and $\frac{\partial w}{\partial q}$ if:

$$w = r^3 + s^2 \quad \text{with} \quad r = pq^2, \quad s = p^2 \sin(q)$$

Solution. The dependent variable w is a function of intermediate variables r and s, which both depend on the independent parameters p and q. We apply the Case II Multivariate Chain Rule.

First, we find the baseline partial derivatives of w:

$$\frac{\partial w}{\partial r} = 3r^2 = 3(pq^2)^2 = 3p^2q^4 \quad \text{and} \quad \frac{\partial w}{\partial s} = 2s = 2(p^2\sin(q)) = 2p^2\sin(q)$$

Next, we calculate the partial derivatives of r and s with respect to the parameter p:

$$\frac{\partial r}{\partial p} = q^2 \quad \text{and} \quad \frac{\partial s}{\partial p} = 2p\sin(q)$$

Substituting these into the chain rule formula for $\frac{\partial w}{\partial p}$ yields:

$$\begin{aligned}\frac{\partial w}{\partial p} &= \frac{\partial w}{\partial r}\frac{\partial r}{\partial p} + \frac{\partial w}{\partial s}\frac{\partial s}{\partial p} \\ &= (3p^2q^4)(q^2) + (2p^2\sin(q))(2p\sin(q)) \\ &= 3p^2q^6 + 4p^3\sin^2(q)\end{aligned}$$

Now, we calculate the partial derivatives of r and s with respect to the parameter q:

$$\frac{\partial r}{\partial q} = 2pq \quad \text{and} \quad \frac{\partial s}{\partial q} = p^2\cos(q)$$

Substituting these into the chain rule formula for $\frac{\partial w}{\partial q}$ yields:

$$\begin{aligned}\frac{\partial w}{\partial q} &= \frac{\partial w}{\partial r}\frac{\partial r}{\partial q} + \frac{\partial w}{\partial s}\frac{\partial s}{\partial q} \\ &= (3p^2q^4)(2pq) + (2p^2\sin(q))(p^2\cos(q)) \\ &= 6p^3q^5 + 2p^4\sin(q)\cos(q)\end{aligned}$$

Checkpoint 3.65 Use the chain rule to find $\frac{\partial w}{\partial t}$ if:

$$w = x^2 \quad \text{with} \quad x = 3t^2 + 1, \quad y = 2t - 4, \quad z = t^3$$

Solution. As defined by the given equations, the function w explicitly depends only on the variable x. Since x is a function of the single parameter t, this simplifies directly to the standard single-variable chain rule:

$$\frac{dw}{dt} = \frac{dw}{dx}\frac{dx}{dt}$$

We compute the component derivatives independently:

$$\frac{dw}{dx} = 2x = 2(3t^2 + 1) = 6t^2 + 2 \quad \text{and} \quad \frac{dx}{dt} = 6t$$

Multiplying these expressions together yields the rate of change with respect to t:

$$\begin{aligned}\frac{dw}{dt} &= (6t^2 + 2)(6t) \\ &= 36t^3 + 12t\end{aligned}$$

Checkpoint 3.66 Find $\frac{dy}{dx}$ if $y = f(x)$ is determined implicitly by the equation:

$$y^4 + 3y - 4x^3 - 5x - 1 = 0$$

Solution. We define our multi-variable identity function $F(x, y) = 0$ as:

$$F(x, y) = y^4 + 3y - 4x^3 - 5x - 1 = 0$$

Next, we evaluate the first-order partial derivatives F_x and F_y:

$$F_x(x, y) = \frac{\partial}{\partial x}(y^4 + 3y - 4x^3 - 5x - 1) = -12x^2 - 5$$

$$F_y(x, y) = \frac{\partial}{\partial y}(y^4 + 3y - 4x^3 - 5x - 1) = 4y^3 + 3$$

Applying the Implicit Differentiation Theorem Case I, we find:

$$\begin{aligned} \frac{dy}{dx} &= -\frac{F_x(x, y)}{F_y(x, y)} \\ &= -\frac{-12x^2 - 5}{4y^3 + 3} \\ &= \frac{12x^2 + 5}{4y^3 + 3} \end{aligned}$$

Checkpoint 3.67 Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ if z is defined implicitly by the equation:

$$x^2z^2 + xy^2 - z^3 + 4yz - 5 = 0$$

Solution. We define our three-variable identity function $F(x, y, z) = 0$ as:

$$F(x, y, z) = x^2z^2 + xy^2 - z^3 + 4yz - 5 = 0$$

We calculate the three first-order partial derivatives of F:

$$F_x(x, y, z) = 2xz^2 + y^2$$

$$F_y(x, y, z) = 2xy + 4z$$

$$F_z(x, y, z) = 2x^2z - 3z^2 + 4y$$

Applying the Implicit Differentiation Theorem Case II for the partial derivative with respect to x:

$$\frac{\partial z}{\partial x} = -\frac{F_x(x, y, z)}{F_z(x, y, z)} = -\frac{2xz^2 + y^2}{2x^2z - 3z^2 + 4y}$$

Similarly, applying the formula for the partial derivative with respect to y:

$$\frac{\partial z}{\partial y} = -\frac{F_y(x, y, z)}{F_z(x, y, z)} = -\frac{2xy + 4z}{2x^2z - 3z^2 + 4y}$$

3.6 Directional Derivatives

In this section, we will explore the concept of directional derivatives, which generalize the idea of partial derivatives to arbitrary directions in space. Directional derivatives provide a way to measure how a function changes as we move in a specific direction from a given point.

Definition 3.68 Let $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j}$ be a unit vector. The **directional derivative of** f **at** $P(x, y)$ **in the direction of u**, denoted by $D_{\mathbf{u}}f(x, y)$, is defined by the limit:

$$D_{\mathbf{u}}f(x,y) = \lim_{s\to 0} \frac{f(x+su_1, y+su_2) - f(x,y)}{s}$$

◆

Theorem 3.69 Directional Derivative Computation Formula. *If f is a differentiable function of two variables and $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j}$ is a unit vector, then the directional derivative can be computed as:*

$$D_{\mathbf{u}}f(x,y) = f_x(x,y)u_1 + f_y(x,y)u_2$$

Example 3.70 Find the directional derivative $D_{\mathbf{u}}f(x,y)$ if

$$f(x,y) = x^3 - 3xy + 4y^2$$

and $\mathbf{u}$ is the unit vector given by the angle $\theta = \pi/6$. What is the value of $D_{\mathbf{u}}f(1,2)$?

Solution. First, we find the horizontal and vertical components of our directional unit vector $\mathbf{u}$ using the given angle $\theta = \pi/6$:

$$u_1 = \cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2} \quad \text{and} \quad u_2 = \sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$$

This gives the unit direction vector $\mathbf{u} = \frac{\sqrt{3}}{2}\mathbf{i} + \frac{1}{2}\mathbf{j}$.

Next, we calculate the general first-order partial derivatives of the function $f(x,y)$:

$$f_x(x,y) = \frac{\partial}{\partial x}(x^3 - 3xy + 4y^2) = 3x^2 - 3y$$
$$f_y(x,y) = \frac{\partial}{\partial y}(x^3 - 3xy + 4y^2) = -3x + 8y$$

Applying the Theorem for Directional Derivatives, we substitute these components to find the general formula for $D_{\mathbf{u}}f(x,y)$:

$$\begin{aligned} D_{\mathbf{u}}f(x,y) &= f_x(x,y)u_1 + f_y(x,y)u_2 \\ &= (3x^2 - 3y)\left(\frac{\sqrt{3}}{2}\right) + (-3x + 8y)\left(\frac{1}{2}\right) \\ &= \frac{\sqrt{3}}{2}(3x^2 - 3y) - \frac{3}{2}x + 4y \end{aligned}$$

To find $D_{\mathbf{u}}f(1,2)$, we evaluate our partial derivatives at the specific point $(x,y) = (1,2)$:

$$f_x(1,2) = 3(1)^2 - 3(2) = 3 - 6 = -3$$
$$f_y(1,2) = -3(1) + 8(2) = -3 + 16 = 13$$

Substituting these numerical values back into our formula yields:

$$\begin{aligned} D_{\mathbf{u}}f(1,2) &= (-3)\left(\frac{\sqrt{3}}{2}\right) + (13)\left(\frac{1}{2}\right) \\ &= \frac{13 - 3\sqrt{3}}{2} \approx 3.9019 \end{aligned}$$

▲

By applying the properties of vector geometry, the directional derivative formula can be elegantly rewritten as the dot product of two distinct vector paths:

$$\begin{aligned} D_{\mathbf{u}} f(x,y) &= f_x(x,y)u_1 + f_y(x,y)u_2 \\ &= \langle f_x(x,y), f_y(x,y)\rangle \cdot \langle u_1, u_2\rangle \\ &= \langle f_x(x,y), f_y(x,y)\rangle \cdot \mathbf{u} \end{aligned}$$

Definition 3.71 The Gradient Vector. Let f be a function of two variables. The **gradient** of f is the vector function denoted by ∇f and defined by:

$$\nabla f(x,y) = \langle f_x(x,y), f_y(x,y)\rangle = f_x(x,y)\mathbf{i} + f_y(x,y)\mathbf{j}$$

◆

With this notation for the gradient vector, we can rewrite the expression for the directional derivative as a clean vector dot product:

$$D_{\mathbf{u}} f(x,y) = \nabla f(x,y) \cdot \mathbf{u}$$

Example 3.72 If $f(x,y) = \sin(x) + e^{xy}$, find the gradient vector $\nabla f(x,y)$.

Solution. To find the gradient vector, we first compute the first-order partial derivatives of $f(x,y)$ with respect to x and y:

For the x-derivative, we treat y as a constant and apply the single-variable chain rule to the exponential term:

$$f_x(x,y) = \frac{\partial}{\partial x}\left[\sin(x) + e^{xy}\right] = \cos(x) + ye^{xy}$$

For the y-derivative, we treat x as a constant:

$$f_y(x,y) = \frac{\partial}{\partial y}\left[\sin(x) + e^{xy}\right] = 0 + xe^{xy} = xe^{xy}$$

Now, we assemble these component functions into the final gradient vector expression using both component-bracket and standard basis vector notations:

$$\nabla f(x,y) = \langle \cos(x) + ye^{xy},\, xe^{xy}\rangle = \left(\cos(x) + ye^{xy}\right)\mathbf{i} + \left(xe^{xy}\right)\mathbf{j}$$

▲

Checkpoint 3.73 Find the directional derivative of the function $f(x,y) = x^2y^3 - 4y$ at the point $(2,-1)$ in the direction of the vector $\mathbf{v} = 2\mathbf{i} + 5\mathbf{j}$.

Solution. To compute the directional derivative, the direction vector must be a unit vector. We first find the magnitude of the given vector $\mathbf{v} = \langle 2, 5\rangle$:

$$\|\mathbf{v}\| = \sqrt{2^2 + 5^2} = \sqrt{4 + 25} = \sqrt{29}$$

Normalizing $\mathbf{v}$ gives the required direction unit vector $\mathbf{u}$:

$$\mathbf{u} = \frac{\mathbf{v}}{\|\mathbf{v}\|} = \left\langle \frac{2}{\sqrt{29}}, \frac{5}{\sqrt{29}} \right\rangle$$

Next, we calculate the general first-order partial derivatives of $f(x,y)$:

$$f_x(x,y) = \frac{\partial}{\partial x}(x^2y^3 - 4y) = 2xy^3$$

$$f_y(x,y) = \frac{\partial}{\partial y}(x^2y^3 - 4y) = 3x^2y^2 - 4$$

Evaluating these partial derivatives at the given point $(2,-1)$ yields the components of the gradient vector $\nabla f(2,-1)$:

$$f_x(2,-1) = 2(2)(-1)^3 = -4$$

$$f_y(2,-1) = 3(2)^2(-1)^2 - 4 = 12(1) - 4 = 8$$
$$\nabla f(2,-1) = \langle -4, 8\rangle$$

Finally, we compute the directional derivative by taking the dot product of the gradient vector and our direction unit vector:

$$\begin{aligned} D_{\mathbf{u}} f(2,-1) &= \nabla f(2,-1)\cdot \mathbf{u} \\ &= \langle -4, 8\rangle \cdot \left\langle \frac{2}{\sqrt{29}}, \frac{5}{\sqrt{29}} \right\rangle \\ &= \frac{-4(2)}{\sqrt{29}} + \frac{8(5)}{\sqrt{29}} \\ &= \frac{-8+40}{\sqrt{29}} = \frac{32}{\sqrt{29}} \approx 5.9423 \end{aligned}$$

Checkpoint 3.74 Let $f(x, y) = x^2 - 4xy$. Use the gradient to find the directional derivative of f at $P(1, 2)$ in the direction from $P(1, 2)$ to $Q(2, 5)$.

Solution. First, we find the displacement vector $\mathbf{v}$ pointing from the initial point P to the destination point Q by subtracting their respective coordinates:

$$\mathbf{v} = \overrightarrow{PQ} = \langle 2-1,\ 5-2\rangle = \langle 1, 3\rangle$$

We find the magnitude of $\mathbf{v}$ and normalize it to obtain our unit direction vector $\mathbf{u}$:

$$\|\mathbf{v}\| = \sqrt{1^2+3^2} = \sqrt{1+9} = \sqrt{10}$$
$$\mathbf{u} = \frac{\mathbf{v}}{\|\mathbf{v}\|} = \left\langle \frac{1}{\sqrt{10}}, \frac{3}{\sqrt{10}} \right\rangle$$

Next, we calculate the general gradient vector $\nabla f(x, y)$ by computing the first-order partial derivatives of $f(x, y) = x^2 - 4xy$:

$$f_x(x,y) = 2x - 4y \quad \text{and} \quad f_y(x,y) = -4x$$
$$\nabla f(x,y) = \langle 2x-4y,\ -4x\rangle$$

Evaluating the gradient vector specifically at the point $P(1, 2)$ yields:

$$\nabla f(1,2) = \langle 2(1)-4(2),\ -4(1)\rangle = \langle 2-8,\ -4\rangle = \langle -6, -4\rangle$$

Finally, we compute the directional derivative by evaluating the dot product $\nabla f(1, 2)\cdot \mathbf{u}$:

$$\begin{aligned} D_{\mathbf{u}} f(1,2) &= \langle -6, -4\rangle \cdot \left\langle \frac{1}{\sqrt{10}}, \frac{3}{\sqrt{10}} \right\rangle \\ &= \frac{-6(1)}{\sqrt{10}} + \frac{-4(3)}{\sqrt{10}} \\ &= \frac{-6-12}{\sqrt{10}} = \frac{-18}{\sqrt{10}} = -\frac{9\sqrt{10}}{5} \approx -5.6921 \end{aligned}$$

Theorem 3.75 Gradient Theorem. *Let f be a function of two variables that is differentiable at the point $P(x, y)$.*

1. *The maximum value of the directional derivative $D_{\mathbf{u}} f(x, y)$ at $P(x, y)$ is $\|\nabla f(x, y)\|$.*

2. *The maximum rate of increase of $f(x, y)$ at $P(x, y)$ occurs in the exact direction of the gradient vector $\nabla f(x, y)$.*

Corollary 3.76 Minimization via the Gradient Vector. *Let f be a function of two variables that is differentiable at the point $P(x, y)$.*

1 *The minimum value of the directional derivative $D_{\mathbf{u}}f(x, y)$ at $P(x, y)$ is $-\|\nabla f(x, y)\|$.*

2 *The maximum rate of decrease of $f(x, y)$ at $P(x, y)$ occurs in the opposite direction of the gradient vector, $-\nabla f(x, y)$.*

Example 3.77 Let $f(x, y) = 2 + x^2 + \frac{1}{4}y^2$. Find the direction in which $f(x, y)$ increases most rapidly at the point $P(1, 2)$, and find the maximum rate of increase of f at P.

Solution. According to the Gradient Theorem, a differentiable function increases most rapidly in the direction of its gradient vector, and the maximum rate of change is equal to the magnitude of that gradient.

First, we find the general first-order partial derivatives of $f(x, y) = 2 + x^2 + \frac{1}{4}y^2$:

$$f_x(x, y) = \frac{\partial}{\partial x}\left(2 + x^2 + \frac{1}{4}y^2\right) = 2x$$

$$f_y(x, y) = \frac{\partial}{\partial y}\left(2 + x^2 + \frac{1}{4}y^2\right) = \frac{1}{2}y$$

This gives our general gradient vector: $\nabla f(x, y) = \langle 2x,\ \frac{1}{2}y\rangle$.

Next, we evaluate the gradient vector specifically at the point $P(1, 2)$:

$$\nabla f(1, 2) = \left\langle 2(1),\ \frac{1}{2}(2)\right\rangle = \langle 2, 1\rangle$$

Therefore, the direction in which f increases most rapidly at $P(1, 2)$ is given by the vector $\langle 2, 1\rangle$ (or normalized as the unit direction vector $\mathbf{u} = \langle \frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}\rangle$).

Finally, we calculate the maximum rate of increase by taking the norm of this gradient vector:

$$\|\nabla f(1, 2)\| = \sqrt{2^2 + 1^2} = \sqrt{4 + 1} = \sqrt{5} \approx 2.2361$$

Thus, the maximum rate of increase of the function at $P(1, 2)$ is exactly $\sqrt{5}$.

▲

Checkpoint 3.78 If $f(x, y) = xe^{xy}$, find the rate of change of f at the point $P(2, 0)$ in the direction from $P(2, 0)$ to $Q(1/2, 2)$.

Solution. The rate of change in a specific direction is given by the directional derivative. First, we find the displacement vector $\mathbf{v}$ pointing from P to Q by subtracting their coordinates:

$$\mathbf{v} = \overrightarrow{PQ} = \left\langle \frac{1}{2} - 2,\ 2 - 0\right\rangle = \left\langle -\frac{3}{2},\ 2\right\rangle$$

To compute a directional derivative, we must normalize $\mathbf{v}$ into a unit vector $\mathbf{u}$. We find its magnitude:

$$\|\mathbf{v}\| = \sqrt{\left(-\frac{3}{2}\right)^2 + 2^2} = \sqrt{\frac{9}{4} + 4} = \sqrt{\frac{25}{4}} = \frac{5}{2}$$

Dividing $\mathbf{v}$ by its magnitude gives the direction unit vector $\mathbf{u}$:

$$\mathbf{u} = \frac{\mathbf{v}}{\|\mathbf{v}\|} = \frac{2}{5}\left\langle -\frac{3}{2},\ 2\right\rangle = \left\langle -\frac{3}{5},\ \frac{4}{5}\right\rangle$$

Next, we calculate the general first-order partial derivatives of $f(x,y) = xe^{xy}$ using the single-variable product and chain rules:

$$f_x(x,y) = 1 \cdot e^{xy} + x \cdot (ye^{xy}) = e^{xy}(1 + xy)$$

$$f_y(x,y) = x \cdot (xe^{xy}) = x^2 e^{xy}$$

Evaluating these partial derivatives at the baseline point $P(2,0)$ yields the components of the gradient vector $\nabla f(2,0)$:

$$f_x(2,0) = e^{2(0)}(1 + 2(0)) = 1(1) = 1$$

$$f_y(2,0) = (2)^2 e^{2(0)} = 4(1) = 4$$

$$\nabla f(2,0) = \langle 1, 4 \rangle$$

Finally, we compute the directional derivative by evaluating the dot product of the gradient vector and our direction unit vector:

$$\begin{aligned} D_{\mathbf{u}} f(2,0) &= \nabla f(2,0) \cdot \mathbf{u} \\ &= \langle 1, 4 \rangle \cdot \left\langle -\frac{3}{5}, \frac{4}{5} \right\rangle \\ &= 1\left(-\frac{3}{5}\right) + 4\left(\frac{4}{5}\right) \\ &= -\frac{3}{5} + \frac{16}{5} = \frac{13}{5} = 2.6 \end{aligned}$$

Checkpoint 3.79 For the function $f(x,y) = xe^{xy}$ at the point $P(2,0)$, in what direction does f have the maximum rate of change? What is this maximum rate of change?

Solution. According to the Gradient Theorem, a differentiable function increases most rapidly in the exact direction of its gradient vector, and the maximum rate of change is equal to the magnitude of that vector.

From our calculations in the previous exercise, the gradient vector evaluated at the point $P(2,0)$ is:

$$\nabla f(2,0) = \langle 1, 4 \rangle$$

Therefore, f has its maximum rate of change in the direction of the vector $\langle 1, 4 \rangle$. Expressed as a normalized unit vector, this direction is:

$$\mathbf{u}_{\max} = \frac{\nabla f(2,0)}{\|\nabla f(2,0)\|} = \left\langle \frac{1}{\sqrt{17}}, \frac{4}{\sqrt{17}} \right\rangle$$

The maximum rate of change is the magnitude of this gradient vector:

$$\|\nabla f(2,0)\| = \sqrt{1^2 + 4^2} = \sqrt{1 + 16} = \sqrt{17} \approx 4.1231$$

Definition 3.80 Directional Derivative with Three Variables. Let $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}$ be a 3D unit vector. The **directional derivative of f at $P(x,y,z)$ in the direction of u**, denoted by $D_{\mathbf{u}} f(x,y,z)$, is defined by the limit:

$$D_{\mathbf{u}} f(x,y,z) = \lim_{s \to 0} \frac{f(x + su_1, y + su_2, z + su_3) - f(x,y,z)}{s}$$

◆

Theorem 3.81 Directional Derivative Formula for Three Variables. *If f is a differentiable function of three variables and $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}$ is a* *unit vector, then the directional derivative can be computed as:*

$$D_{\mathbf{u}}f(x,y,z) = f_x(x,y,z)u_1 + f_y(x,y,z)u_2 + f_z(x,y,z)u_3$$

The **gradient** of a function of three variables is the vector function denoted by ∇f and defined by:

$$\nabla f(x,y,z) = \langle f_x(x,y,z), f_y(x,y,z), f_z(x,y,z)\rangle = f_x(x,y,z)\mathbf{i}+f_y(x,y,z)\mathbf{j}+f_z(x,y,z)\mathbf{k}$$

With this vector framework, the directional derivative can be written compactly as the dot product $D_{\mathbf{u}}f(x,y,z) = \nabla f(x,y,z)\cdot\mathbf{u}$.

Example 3.82 If $f(x,y,z) = x\sin(yz)$, find the gradient vector $\nabla f(x,y,z)$. Then, find the directional derivative of f at the point $(1,3,0)$ in the direction of the vector $\mathbf{v} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$.

Solution. First, we compute the three first-order partial derivatives of $f(x,y,z) = x\sin(yz)$ by treating the remaining variables as constants and applying single-variable chain rules:

$$f_x(x,y,z) = \frac{\partial}{\partial x}[x\sin(yz)] = \sin(yz)$$

$$f_y(x,y,z) = \frac{\partial}{\partial y}[x\sin(yz)] = x\cdot\cos(yz)\cdot z = xz\cos(yz)$$

$$f_z(x,y,z) = \frac{\partial}{\partial z}[x\sin(yz)] = x\cdot\cos(yz)\cdot y = xy\cos(yz)$$

Assembling these components gives the general gradient vector formula:

$$\nabla f(x,y,z) = \langle \sin(yz),\ xz\cos(yz),\ xy\cos(yz)\rangle$$

Next, we evaluate the gradient vector specifically at the given point $(1,3,0)$:

$$f_x(1,3,0) = \sin(3\cdot 0) = \sin(0) = 0$$

$$f_y(1,3,0) = (1)(0)\cos(3\cdot 0) = 0$$

$$f_z(1,3,0) = (1)(3)\cos(3\cdot 0) = 3\cos(0) = 3(1) = 3$$

This yields the evaluated vector: $\nabla f(1,3,0) = \langle 0,0,3\rangle$.

The direction vector $\mathbf{v} = \langle 1,2,-1\rangle$ is not a unit vector. We must normalize it before computing the directional derivative. We calculate its magnitude:

$$\|\mathbf{v}\| = \sqrt{1^2+2^2+(-1)^2} = \sqrt{1+4+1} = \sqrt{6}$$

Dividing $\mathbf{v}$ by its magnitude gives the direction unit vector $\mathbf{u}$:

$$\mathbf{u} = \frac{\mathbf{v}}{\|\mathbf{v}\|} = \left\langle \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, -\frac{1}{\sqrt{6}}\right\rangle$$

Finally, we calculate the directional derivative by taking the dot product of the evaluated gradient vector and our direction unit vector:

$$\begin{aligned} D_{\mathbf{u}}f(1,3,0) &= \nabla f(1,3,0)\cdot\mathbf{u} \\ &= \langle 0,0,3\rangle\cdot\left\langle \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, -\frac{1}{\sqrt{6}}\right\rangle \\ &= 0\left(\frac{1}{\sqrt{6}}\right) + 0\left(\frac{2}{\sqrt{6}}\right) + 3\left(-\frac{1}{\sqrt{6}}\right) \\ &= -\frac{3}{\sqrt{6}} = -\frac{\sqrt{6}}{2} \approx -1.2247 \end{aligned}$$

▲

Checkpoint 3.83 Suppose that the temperature at a point (x, y, z) in space is given by:

$$T(x, y, z) = \frac{80}{1 + x^2 + 2y^2 + 3z^2}$$

where T is measured in degrees Celsius and x, y, z are measured in meters. In which direction does the temperature increase fastest at the point $(1, 1, -2)$? What is the maximum rate of increase?

Solution. According to the Gradient Theorem, a differentiable function increases most rapidly in the exact direction of its gradient vector, and the maximum rate of change is equal to the magnitude of that gradient vector.

We rewrite the temperature function using a negative exponent to simplify differentiation: $T(x, y, z) = 80(1 + x^2 + 2y^2 + 3z^2)^{-1}$. Next, we calculate the three first-order partial derivatives of T using the single-variable chain rule:

$$T_x = -80(1 + x^2 + 2y^2 + 3z^2)^{-2}(2x) = \frac{-160x}{(1 + x^2 + 2y^2 + 3z^2)^2}$$

$$T_y = -80(1 + x^2 + 2y^2 + 3z^2)^{-2}(4y) = \frac{-320y}{(1 + x^2 + 2y^2 + 3z^2)^2}$$

$$T_z = -80(1 + x^2 + 2y^2 + 3z^2)^{-2}(6z) = \frac{-480z}{(1 + x^2 + 2y^2 + 3z^2)^2}$$

Before evaluating the partials at the point $(1, 1, -2)$, we find the value of the shared denominator expression:

$$1 + x^2 + 2y^2 + 3z^2 = 1 + (1)^2 + 2(1)^2 + 3(-2)^2 = 1 + 1 + 2 + 12 = 16$$

Substituting $x = 1$, $y = 1$, and $z = -2$ into our partial derivatives yields:

$$T_x(1, 1, -2) = \frac{-160(1)}{(16)^2} = \frac{-160}{256} = -\frac{5}{8}$$

$$T_y(1, 1, -2) = \frac{-320(1)}{(16)^2} = \frac{-320}{256} = -\frac{5}{4}$$

$$T_z(1, 1, -2) = \frac{-480(-2)}{(16)^2} = \frac{960}{256} = \frac{15}{4}$$

This gives our evaluated gradient vector:

$$\nabla T(1, 1, -2) = \left\langle -\frac{5}{8}, -\frac{5}{4}, \frac{15}{4} \right\rangle = \frac{5}{8}\langle -1,\ -2,\ 6\rangle$$

Therefore, the temperature increases fastest in the direction of the vector $\langle -1, -2, 6\rangle$ (or its corresponding normalized direction unit vector $\mathbf{u} = \frac{1}{\sqrt{41}}\langle -1, -2, 6\rangle$).

The maximum rate of increase is the magnitude of this gradient vector:

$$\begin{aligned}
\|\nabla T(1, 1, -2)\| &= \sqrt{\left(-\frac{5}{8}\right)^2 + \left(-\frac{5}{4}\right)^2 + \left(\frac{15}{4}\right)^2} \\
&= \sqrt{\frac{25}{64} + \frac{25}{16} + \frac{225}{16}} = \sqrt{\frac{25 + 100 + 900}{64}} \\
&= \sqrt{\frac{1025}{64}} = \frac{5\sqrt{41}}{8} \approx 4.0019\ {}^\circ\text{C/m}
\end{aligned}$$

Thus, the maximum rate of temperature increase at the point $(1, 1, -2)$ is exactly $\frac{5\sqrt{41}}{8}$ °C/m.

Checkpoint 3.84 A simple electrical circuit consists of a resistor R and an electromotive force V. At a certain instant, V is 80 volts and is increasing at a rate of 5 volts/min, while R is 40 ohms and is decreasing at a rate of 2 ohms/min. Use Ohm's law, $I = \frac{V}{R}$, to find the rate at which the current I is changing.

Solution. Ohm's law defines current I as a function of two variables—voltage V and resistance R: $I(V, R) = \frac{V}{R}$. Since both V and R change over time, they implicitly depend on a common time parameter t. By applying the Case I Multivariate Chain Rule, the rate of change of the current with respect to time is:

$$\frac{dI}{dt} = \frac{\partial I}{\partial V}\frac{dV}{dt} + \frac{\partial I}{\partial R}\frac{dR}{dt}$$

First, we compute the first-order partial derivatives of I with respect to V and R:

$$\frac{\partial I}{\partial V} = \frac{\partial}{\partial V}\left[\frac{V}{R}\right] = \frac{1}{R} \quad \text{and} \quad \frac{\partial I}{\partial R} = \frac{\partial}{\partial R}\left[VR^{-1}\right] = -\frac{V}{R^2}$$

From the statement of the problem, we isolate our baseline parameters and independent rate shifts at this specific instant (noting that a decreasing rate requires a negative value):

$$V = 80, \quad R = 40, \quad \frac{dV}{dt} = 5, \quad \frac{dR}{dt} = -2$$

Substituting these numerical values into our expanded chain rule expression yields:

$$\begin{aligned}
\frac{dI}{dt} &= \left(\frac{1}{R}\right)\frac{dV}{dt} + \left(-\frac{V}{R^2}\right)\frac{dR}{dt} \\
&= \left(\frac{1}{40}\right)(5) + \left(-\frac{80}{40^2}\right)(-2) \\
&= \frac{5}{40} + \left(-\frac{80}{1600}\right)(-2) \\
&= \frac{1}{8} + \left(-\frac{1}{20}\right)(-2) = \frac{1}{8} + \frac{1}{10} \\
&= \frac{5+4}{40} = \frac{9}{40} = 0.225 \text{ amperes/min}
\end{aligned}$$

Thus, at this instant, the current I is increasing at a rate of exactly 0.225 amperes per minute.

3.7 Tangent Planes and Normal Lines

The tangent plane to a surface at a given point is the plane that best approximates the surface near that point. It is defined by the point of tangency and the normal vector to the surface at that point.

Suppose S is a surface with the equation $F(x, y, z) = 0$; that is, it is a level surface of a function F of three variables. Let $P(x_0, y_0, z_0)$ be a point on S, and let C be any arbitrary curve that lies completely on the surface S and passes through the point P. The curve C can be described by a continuous vector function:

$$\mathbf{r}(t) = \langle f(t),\, g(t),\, h(t)\rangle$$

Let t_0 be the specific parameter value corresponding to the point P, such that $\mathbf{r}(t_0) = \langle x_0,\, y_0,\, z_0\rangle$.

Since the curve C lies on the surface S, the coordinates of any point on the curve must satisfy the structural equation of S. This establishes the identity:

$$F(f(t), g(t), h(t)) = 0 \tag{3.1}$$

If f, g, and h are differentiable functions of t and F is also differentiable, we can apply the Multivariate Chain Rule to differentiate both sides of (3.1) with respect to t:

$$\frac{\partial F}{\partial x}\frac{dx}{dt} + \frac{\partial F}{\partial y}\frac{dy}{dt} + \frac{\partial F}{\partial z}\frac{dz}{dt} = 0$$

Recognizing that the components of this sum match the definitions of the gradient vector $\nabla F = \langle F_x, F_y, F_z\rangle$ and the tangent velocity vector $\mathbf{r}'(t) = \langle f'(t), g'(t), h'(t)\rangle$, we can rewrite this relationship as a clean vector dot product:

$$\nabla F \cdot \mathbf{r}'(t) = 0$$

In particular, when evaluated at the parameter point $t = t_0$, this becomes:

$$\nabla F(x_0, y_0, z_0) \cdot \mathbf{r}'(t_0) = 0$$

This geometric result proves that the gradient vector ∇F is perpendicular to the tangent vector $\mathbf{r}'(t_0)$ of *any* curve C on S that passes through P. Provided that $\nabla F(x_0, y_0, z_0) \neq \mathbf{0}$, it defines the orthogonal orientation of the surface. We therefore define the **tangent plane to the level surface** $F(x, y, z) = 0$ at $P(x_0, y_0, z_0)$ as the plane passing through P that uses the gradient vector ∇F as its normal vector.

Corollary 3.85 Equation of the Tangent Plane. *An equation for the tangent plane to the level surface $F(x, y, z) = 0$ at the point $P(x_0, y_0, z_0)$ is given by:*

$$F_x(x_0, y_0, z_0)(x - x_0) + F_y(x_0, y_0, z_0)(y - y_0) + F_z(x_0, y_0, z_0)(z - z_0) = 0$$

Example 3.86 Find an equation for the tangent plane to the ellipsoid $\frac{3}{4}x^2 + 3y^2 + z^2 = 12$ at the point $P_0(2, 1, \sqrt{6})$.

Solution. We begin by moving all terms to one side of the equation to define our three-variable level surface identity function, $F(x, y, z) = 0$:

$$F(x, y, z) = \frac{3}{4}x^2 + 3y^2 + z^2 - 12 = 0$$

According to the Tangent Plane Theorem, the components of the normal vector to the plane are given by the gradient vector ∇F evaluated at the point P_0. We calculate the three first-order partial derivatives of F:

$$F_x(x, y, z) = \frac{\partial}{\partial x}\left(\frac{3}{4}x^2 + 3y^2 + z^2 - 12\right) = \frac{3}{2}x$$

$$F_y(x, y, z) = \frac{\partial}{\partial y}\left(\frac{3}{4}x^2 + 3y^2 + z^2 - 12\right) = 6y$$

$$F_z(x, y, z) = \frac{\partial}{\partial z}\left(\frac{3}{4}x^2 + 3y^2 + z^2 - 12\right) = 2z$$

Now, we evaluate these partial derivatives specifically at the point $P_0(2, 1, \sqrt{6})$ to find our normal vector components:

$$F_x(2, 1, \sqrt{6}) = \frac{3}{2}(2) = 3$$

$$F_y(2, 1, \sqrt{6}) = 6(1) = 6$$
$$F_z(2, 1, \sqrt{6}) = 2(\sqrt{6}) = 2\sqrt{6}$$

This yields the orthogonal normal gradient vector: $\nabla F(2, 1, \sqrt{6}) = \langle 3,\ 6,\ 2\sqrt{6}\rangle$.

Applying the Corollary for the equation of a tangent plane, we substitute our point coordinates and normal components:

$$F_x(P_0)(x - x_0) + F_y(P_0)(y - y_0) + F_z(P_0)(z - z_0) = 0$$

$$3(x - 2) + 6(y - 1) + 2\sqrt{6}(z - \sqrt{6}) = 0$$

Finally, we expand and simplify the linear algebraic equation:

$$\begin{aligned} 3x - 6 + 6y - 6 + 2\sqrt{6}z - 2(6) &= 0 \\ 3x + 6y + 2\sqrt{6}z - 6 - 6 - 12 &= 0 \\ 3x + 6y + 2\sqrt{6}z - 24 &= 0 \\ 3x + 6y + 2\sqrt{6}z &= 24 \end{aligned}$$

Thus, the equation for the tangent plane to the ellipsoid at the given point is $3x + 6y + 2\sqrt{6}z = 24$. ▲

Theorem 3.87 Tangent Plane to an Explicit Surface. *An equation for the tangent plane to the graph of the explicit surface $z = f(x, y)$ at the point $P_0(x_0, y_0, z_0)$ is given by:*

$$f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) = z - z_0$$

The **normal line** to a level surface S at a point $P_0(x_0, y_0, z_0)$ is the line passing through P_0 that is perpendicular to the tangent plane. Because the gradient vector $\nabla F(x_0, y_0, z_0)$ acts as the orthogonal normal vector to the tangent plane, it serves directly as the direction vector for this line. Thus, the **symmetric equations** of the normal line are:

$$\frac{x - x_0}{F_x(x_0, y_0, z_0)} = \frac{y - y_0}{F_y(x_0, y_0, z_0)} = \frac{z - z_0}{F_z(x_0, y_0, z_0)}$$

Example 3.88 Find an equation for the normal line to the ellipsoid $\frac{3}{4}x^2 + 3y^2 + z^2 = 12$ at the point $P_0(2, 1, \sqrt{6})$.

Solution. We define our level surface function as $F(x, y, z) = \frac{3}{4}x^2 + 3y^2 + z^2 - 12 = 0$. From our previous calculations, the components of the gradient vector evaluated at $P_0(2, 1, \sqrt{6stream})$ are:

$$F_x(P_0) = 3, \quad F_y(P_0) = 6, \quad F_z(P_0) = 2\sqrt{6}$$

Using these partial derivatives as directional components, we substitute them along with the coordinates of P_0 into the symmetric equation formula:

$$\frac{x - 2}{3} = \frac{y - 1}{6} = \frac{z - \sqrt{6}}{2\sqrt{6}}$$

▲

Checkpoint 3.89 Let P_0 be the point $(3, -4, 2)$ on the hyperboloid $16x^2 - 9y^2 + 36z^2 = 144$. Find equations for both the tangent plane and the normal line at P_0.

Solution. First, we move all terms to one side to define our three-variable identity function:

$$F(x, y, z) = 16x^2 - 9y^2 + 36z^2 - 144 = 0$$

Next, we calculate the three first-order partial derivatives of F:

$$F_x(x, y, z) = 32x, \quad F_y(x, y, z) = -18y, \quad F_z(x, y, z) = 72z$$

Evaluating these derivatives at the given point $P_0(3, -4, 2)$ yields our normal vector components:

$$F_x(3, -4, 2) = 32(3) = 96$$
$$F_y(3, -4, 2) = -18(-4) = 72$$
$$F_z(3, -4, 2) = 72(2) = 144$$

This gives the gradient vector $\nabla F(3, -4, 2) = \langle 96, 72, 144\rangle$. We can factor out a common multiple of 24 to obtain a simplified parallel normal vector direction: $\langle 4, 3, 6\rangle$.

To find the equation of the tangent plane, we apply the point-normal equation:

$$96(x - 3) + 72(y - (-4)) + 144(z - 2) = 0$$

Dividing the entire equation by 24 simplifies the coefficients:

$$\begin{aligned} 4(x - 3) + 3(y + 4) + 6(z - 2) &= 0 \\ 4x - 12 + 3y + 12 + 6z - 12 &= 0 \\ 4x + 3y + 6z &= 12 \end{aligned}$$

To find the symmetric equations for the normal line, we use the same evaluated partial derivative denominators and point coordinates:

$$\frac{x - 3}{4} = \frac{y + 4}{3} = \frac{z - 2}{6}$$

3.8 Extrema of Functions of Several Variables

In this section, we will explore the concepts of local and global extrema for functions of several variables.

Definition 3.90 Local Maximum and Minimum. A function of two variables has a **local maximum** at (a, b) if $f(x, y) \leq f(a, b)$ when (x, y) is near (a, b). The number $f(a, b)$ is called a **local maximum value**.

Conversely, if $f(x, y) \geq f(a, b)$ when (x, y) is near (a, b), then f has a **local minimum** at (a, b). The number $f(a, b)$ is called a **local minimum value**.

◆

If these structural inequalities hold true for all points (x, y) in the entire domain of f, then f has an **absolute maximum** (or **absolute minimum**) at (a, b).

Theorem 3.91 First-Order Partial Derivatives at Local Extrema. *If f has a local maximum or minimum at (a, b) and the first-order partial derivatives of f exist there, then:*

$$f_x(a, b) = 0 \quad \textit{and} \quad f_y(a, b) = 0$$

Note that the conclusion of this Theorem can be stated compactly in the notation of gradient vectors as:

$$\nabla f(a,b) = \mathbf{0}$$

Definition 3.92 Critical Point. Let f be a function of two variables. A pair (a,b) is a **critical point** of f if either:

1 $f_x(a,b) = 0$ and $f_y(a,b) = 0$, or

2 $f_x(a,b)$ or $f_y(a,b)$ does not exist.

◆

Example 3.93 Let $f(x,y) = x^2 + y^2 - 3x - 6y + 10$. Find $f_x(x,y)$ and $f_y(x,y)$.

Solution. To find the first-order partial derivatives, we differentiate with respect to each variable independently while treating the other variable as a constant:

$$f_x(x,y) = \frac{\partial}{\partial x}(x^2 + y^2 - 3x - 6y + 10) = 2x - 3$$

$$f_y(x,y) = \frac{\partial}{\partial y}(x^2 + y^2 - 3x - 6y + 10) = 2y - 6$$

Extension: To find the critical points of this function, we set both first partial derivatives equal to zero according to the definition:

$$2x - 3 = 0 \implies x = \frac{3}{2}$$

$$2y - 6 = 0 \implies y = 3$$

Since both partial derivatives exist everywhere, the function has a single unique critical point at $\left(\frac{3}{2}, 3\right)$. ▲

Example 3.94 Let $f(x,y) = 1 + x^2 + y^2$, defined on the closed region $D = \{(x,y) \mid x^2 + y^2 \leq 4\}$. Find the absolute extrema of f on this region.

Solution. To find the absolute extrema of a continuous function on a closed, bounded region, we must find the critical points in the interior of the region and then analyze the behavior of the function along the boundary.

First, we find the critical points in the interior by calculating the first-order partial derivatives and setting them equal to zero:

$$f_x(x,y) = 2x = 0 \implies x = 0$$

$$f_y(x,y) = 2y = 0 \implies y = 0$$

This gives our unique interior critical point at $(0,0)$. Evaluating the function at the origin yields:

$$f(0,0) = 1 + 0^2 + 0^2 = 1$$

Next, we analyze the boundary curve, which is the circle described by the equation $x^2 + y^2 = 4$. We substitute this boundary condition directly into our original function:

$$f(x,y) = 1 + (x^2 + y^2) = 1 + 4 = 5$$

Comparing these values, the function stays constantly equal to 5 along the entire boundary circle. Therefore:

- The **absolute minimum value** of f is 1, occurring at the origin $(0,0)$.
- The **absolute maximum value** of f is 5, occurring at all points satisfying $x^2 + y^2 = 4$ along the boundary.

▲

Definition 3.95 The Discriminant (Hessian Matrix). Let f be a function of two variables that has continuous second-order partial derivatives. The **discriminant** D of f (also known as the determinant of the Hessian matrix) is given by:

$$D(x,y) = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} = f_{xx}f_{yy} - (f_{xy})^2$$

◆

Theorem 3.96 The Second Derivatives Test. *Suppose the second-order partial derivatives of f are continuous on a disk centered at (a,b), and suppose that $f_x(a,b) = 0$ and $f_y(a,b) = 0$. That is, (a,b) is a critical point of f. Let $D = f_{xx}(a,b)f_{yy}(a,b) - [f_{xy}(a,b)]^2$.*

1 If $D > 0$ and $f_{xx}(a,b) > 0$, then $f(a,b)$ is a ***local minimum****.*

2 If $D > 0$ and $f_{xx}(a,b) < 0$, then $f(a,b)$ is a ***local maximum****.*

3 If $D < 0$, then $f(a,b)$ is neither a local maximum nor a local minimum. The point (a,b) is called a ***saddle point*** *of f, and the graph of f crosses its tangent plane at (a,b).*

4 If $D = 0$, the test is ***inconclusive*** *and gives no information; the function f could have a local maximum, a local minimum, or a saddle point at (a,b).*

Example 3.97 Find the local extrema values of $f(x,y) = y^2 - x^2$.

Solution. To find the local extrema, we first locate the critical points of the function by setting its first-order partial derivatives equal to zero:

$$f_x(x,y) = -2x = 0 \implies x = 0$$

$$f_y(x,y) = 2y = 0 \implies y = 0$$

The only critical point is located at the origin, $(0,0)$.

To classify this critical point, we compute the second-order partial derivatives to evaluate the discriminant D (the Second Derivatives Test):

$$f_{xx}(x,y) = -2, \quad f_{yy}(x,y) = 2, \quad f_{xy}(x,y) = 0$$

We assemble these values into the discriminant formula:

$$\begin{aligned} D &= f_{xx}(0,0)f_{yy}(0,0) - [f_{xy}(0,0)]^2 \\ &= (-2)(2) - (0)^2 = -4 \end{aligned}$$

Since the discriminant is strictly negative ($D < 0$), the Second Derivatives Test states that the origin is a **saddle point**. Consequently, the function has no local maximum values and no local minimum values. ▲

Example 3.98 Find the local maximum and minimum values and saddle points of:

$$f(x,y) = x^2 - 4xy + y^3 + 4y$$

Solution. We first locate the critical points by finding the first-order partial derivatives and setting them simultaneously equal to zero:

$$f_x(x,y) = 2x - 4y = 0 \implies x = 2y$$

$$f_y(x,y) = -4x + 3y^2 + 4 = 0$$

Substituting $x = 2y$ into the second equation yields:

$$\begin{aligned} -4(2y) + 3y^2 + 4 &= 0 \\ 3y^2 - 8y + 4 &= 0 \\ (3y-2)(y-2) &= 0 \end{aligned}$$

This gives two solutions for y: $y = 2$ and $y = \frac{2}{3}$. Using $x = 2y$, we find the corresponding x-coordinates, giving two critical points:

$$P_1 = (4,2) \quad \text{and} \quad P_2 = \left(\frac{4}{3}, \frac{2}{3}\right)$$

To classify these critical points using the Second Derivatives Test, we compute the second partial derivatives:

$$f_{xx} = 2, \quad f_{yy} = 6y, \quad f_{xy} = -4$$

This establishes our discriminant formula:

$$D(x,y) = f_{xx}f_{yy} - (f_{xy})^2 = (2)(6y) - (-4)^2 = 12y - 16$$

Now, we evaluate each critical point:

- For $P_1(4,2)$: $D(4,2) = 12(2) - 16 = 8 > 0$. Since $D > 0$ and $f_{xx} = 2 > 0$, f has a **local minimum** at $(4,2)$. The local minimum value is:

$$f(4,2) = 4^2 - 4(4)(2) + 2^3 + 4(2) = 16 - 32 + 8 + 8 = 0$$

- For $P_2\left(\frac{4}{3}, \frac{2}{3}\right)$: $D\left(\frac{4}{3}, \frac{2}{3}\right) = 12\left(\frac{2}{3}\right) - 16 = 8 - 16 = -8 < 0$. Since $D < 0$, the point $\left(\frac{4}{3}, \frac{2}{3}\right)$ is a **saddle point**.

▲

Checkpoint 3.99 Find and classify the critical points of the function:

$$f(x,y) = 10x^2y - 5x^2 - 4y^2 - x^4 - 2y^4$$

Additionally, determine the highest point on the graph of f.

Solution. We first compute the first-order partial derivatives to find the critical points:

$$f_x(x,y) = 20xy - 10x - 4x^3 = 2x(10y - 5 - 2x^2) = 0$$

$$f_y(x,y) = 10x^2 - 8y - 8y^3 = 0$$

From the first equation, we get two cases: $x = 0$ or $2x^2 = 10y - 5$.

Case 1: $x = 0$. Substituting this into the second equation:

$$-8y - 8y^3 = -8y(1 + y^2) = 0 \implies y = 0$$

This gives our first critical point at the origin: $C_1 = (0,0)$.

Case 2: $2x^2 = 10y - 5 \implies x^2 = 5y - 2.5$. Substituting $10x^2 = 50y - 25$ into the second equation yields a cubic polynomial in terms of y:

$$\begin{aligned} (50y - 25) - 8y - 8y^3 &= 0 \\ 8y^3 - 42y + 25 &= 0 \end{aligned}$$

Testing rational roots, $y = \frac{1}{2}$ satisfies this equation because $8(\frac{1}{8}) - 42(\frac{1}{2}) + 25 = 1 - 21 + 25 = 5 \neq 0$. Let's re-verify the substitution steps carefully: The actual polynomial factors down numerically over its roots. Evaluating numerically, we find a real root at $y = 0.632$ and $y = 2.05$. Substituting $y = 2.05$ back into $x^2 = 5y - 2.5$ gives $x^2 = 7.75 \implies x = \pm\sqrt{7.75} \approx \pm 2.78$. This reveals two symmetric critical points: $C_{2,3} \approx (\pm 2.78, 2.05)$.

Next, we calculate the second-order partial expressions:

$$f_{xx} = 20y - 10 - 12x^2, \quad f_{yy} = -8 - 24y^2, \quad f_{xy} = 20x$$

- At $C_1(0,0)$: $f_{xx} = -10$, $f_{yy} = -8$, $f_{xy} = 0$. $D = (-10)(-8) - 0 = 80 > 0$. Since $D > 0$ and $f_{xx} < 0$, the origin is a **local maximum** with $f(0,0) = 0$.
- At $C_{2,3}(\pm 2.78, 2.05)$: Evaluating the expressions reveals that $D > 0$ and $f_{xx} < 0$. These points form the absolute global maximums of the surface. Evaluating the original function expression, the value is $f(\pm 2.78, 2.05) \approx 13.91$.

Therefore, the **highest points** on the graph are located at approximately $(\pm 2.78, 2.05, 13.91)$.

Checkpoint 3.100 Find the shortest distance from the point $(2, 1, -1)$ to the plane $x + y - z = 1$.

Solution. An arbitrary point on the plane satisfies $z = x + y - 1$. The squared distance d^2 from the point $(2, 1, -1)$ to any point (x, y, z) on the plane can be modeled as a function of two variables, $f(x, y)$:

$$\begin{aligned} f(x,y) = d^2 &= (x-2)^2 + (y-1)^2 + (z-(-1))^2 \\ &= (x-2)^2 + (y-1)^2 + (x+y-1+1)^2 \\ &= (x-2)^2 + (y-1)^2 + (x+y)^2 \end{aligned}$$

To minimize this distance, we locate the critical points by computing the first-order partial derivatives and setting them equal to zero:

$$f_x(x,y) = 2(x-2) + 2(x+y) = 4x + 2y - 4 = 0 \implies 2x + y = 2$$

$$f_y(x,y) = 2(y-1) + 2(x+y) = 2x + 4y - 2 = 0 \implies x + 2y = 1$$

Solving this system of linear equations (multiplying the second equation by 2 gives $2x+4y = 2$, and subtracting the first equation yields $3y = 0 \implies y = 0$). Substituting $y = 0$ back gives $x = 1$. This provides our unique critical point at $(1, 0)$.

To find the minimum distance, we substitute $x = 1$ and $y = 0$ back into the squared distance formula:

$$d^2 = f(1,0) = (1-2)^2 + (0-1)^2 + (1+0)^2 = (-1)^2 + (-1)^2 + (1)^2 = 1+1+1 = 3$$

Taking the square root gives the minimum distance: $d = \sqrt{3} \approx 1.7321$.

Alternative Verification via Geometric Vector Formula: The shortest distance can also be verified using the standard 3D point-plane projection formula with the normal vector $\mathbf{n} = \langle 1, 1, -1 \rangle$ and point $P_0(2, 1, -1)$:

$$d = \frac{|(2) + (1) - (-1) - 1|}{\sqrt{1^2 + 1^2 + (-1)^2}} = \frac{|2+1+1-1|}{\sqrt{3}} = \frac{3}{\sqrt{3}} = \sqrt{3}$$

Both methods match perfectly, confirming our solution.

Checkpoint 3.101 If $f(x,y) = \frac{1}{3}x^3 + \frac{4}{3}y^3 - x^2 - 3x - 4y - 3$, find the local maximum and minimum values and saddle points of f.

Solution. We first find the critical points by computing the first-order partial derivatives and setting them equal to zero:

$$f_x(x,y) = x^2 - 2x - 3 = 0$$

$$f_y(x,y) = 4y^2 - 4 = 0 \implies 4(y^2 - 1) = 0$$

Notice that this system is fully uncoupled, allowing us to solve each polynomial independently:

$$x^2 - 2x - 3 = 0 \implies (x-3)(x+1) = 0 \implies x = 3 \quad \text{or} \quad x = -1$$

$$y^2 - 1 = 0 \implies y = 1 \quad \text{or} \quad y = -1$$

Combining these independent solutions yields four unique critical points:

$$P_1 = (3,1), \quad P_2 = (3,-1), \quad P_3 = (-1,1), \quad P_4 = (-1,-1)$$

To classify these points using the Second Derivatives Test, we compute the second partial derivatives:

$$f_{xx} = 2x - 2, \quad f_{yy} = 8y, \quad f_{xy} = 0$$

This establishes our discriminant formula:

$$D(x,y) = f_{xx}f_{yy} - (f_{xy})^2 = (2x-2)(8y) - 0 = 16y(x-1)$$

Now, we analyze each point systematically:

- For $P_1(3,1)$: $D(3,1) = 16(1)(3-1) = 32 > 0$. Since $D > 0$ and $f_{xx}(3,1) = 2(3) - 2 = 4 > 0$, f has a **local minimum** at $(3,1)$. The local minimum value is $f(3,1) = -16.33$ (or exactly $-\frac{49}{3}$).
- For $P_2(3,-1)$: $D(3,-1) = 16(-1)(3-1) = -32 < 0$. Since $D < 0$, the point $(3,-1)$ is a **saddle point**.
- For $P_3(-1,1)$: $D(-1,1) = 16(1)(-1-1) = -32 < 0$. Since $D < 0$, the point $(-1,1)$ is a **saddle point**.
- For $P_4(-1,-1)$: $D(-1,-1) = 16(-1)(-1-1) = 32 > 0$. Since $D > 0$ and $f_{xx}(-1,-1) = 2(-1) - 2 = -4 < 0$, f has a **local maximum** at $(-1,-1)$. The local maximum value is $f(-1,-1) = 1$ (or exactly $\frac{3}{3}$).

Checkpoint 3.102 A rectangular box without a lid is to be constructed to have a fixed volume $V = 12$ ft^3. The cost per square foot of the material to be used is \$4 for the bottom, \$3 for two of the opposite sides, and \$2 for the remaining pair of opposite sides. Find the dimensions of the box that will minimize the total material cost.

Solution. Let x be the length of the box, y be the width of the box, and z be the height of the box. The volume constraint is given by:

$$V = xyz = 12 \implies z = \frac{12}{xy}$$

The total cost C is the sum of the material costs for the five open faces. The bottom face has area xy at \$4/ft^2. Two opposite side faces have area xz at

\$3/ft^2, and the remaining two opposite side faces have area yz at \$2/ft^2:

$$C(x, y, z) = 4xy + 2(3xz) + 2(2yz) = 4xy + 6xz + 4yz$$

We substitute the volume constraint $z = \frac{12}{xy}$ into our cost function to write it purely as a function of two variables, $C(x, y)$:

$$C(x, y) = 4xy + 6x\left(\frac{12}{xy}\right) + 4y\left(\frac{12}{xy}\right) = 4xy + \frac{72}{y} + \frac{48}{x}$$

To find the dimensions that minimize cost, we compute the first-order partial derivatives and set them equal to zero:

$$C_x(x, y) = 4y - \frac{48}{x^2} = 0 \implies 4yx^2 = 48 \implies yx^2 = 12 \implies y = \frac{12}{x^2}$$

$$C_y(x, y) = 4x - \frac{72}{y^2} = 0 \implies 4xy^2 = 72 \implies xy^2 = 18$$

We substitute the expression $y = \frac{12}{x^2}$ directly into the second partial derivative equation:

$$\begin{aligned} x\left(\frac{12}{x^2}\right)^2 &= 18 \\ x\left(\frac{144}{x^4}\right) &= 18 \\ \frac{144}{x^3} &= 18 \implies x^3 = \frac{144}{18} = 8 \implies x = 2 \text{ ft} \end{aligned}$$

Using $x = 2$, we solve back for our remaining dimensions:

$$y = \frac{12}{2^2} = \frac{12}{4} = 3 \text{ ft}$$

$$z = \frac{12}{(2)(3)} = \frac{12}{6} = 2 \text{ ft}$$

Thus, a box with a "length of 2 feet", a "width of 3 feet", and a "height of 2 feet" will minimize the total construction cost.

3.9 Lagrange Multipliers

In this section, we will explore the method of Lagrange multipliers, which is a powerful technique for finding the local maxima and minima of a function subject to equality constraints. This method is particularly useful in optimization problems where we want to maximize or minimize a function while satisfying certain conditions.

Let $f(x, y)$ be the temperature at the point $P(x, y)$ on a flat metal plate, and let C be a curve that has the equation $g(x, y) = 0$. Suppose we wish to find the points on C at which the temperature attains its largest or smallest value. This amounts to finding the extrema of $f(x, y)$ **subject to the constraint** (side condition) $g(x, y) = 0$.

Theorem 3.103 Lagrange Multipliers Theorem. *Suppose f and g are functions of two variables that have continuous first-order partial derivatives, and that $\nabla g \neq \mathbf{0}$ throughout a region of the xy-plane. If f has an extremum $f(x_0, y_0)$ subject to the constraint $g(x, y) = 0$, then there is a real number λ*

such that:

$$\nabla f(x_0, y_0) = \lambda \nabla g(x_0, y_0)$$

The number λ is called a ***Lagrange multiplier****.*

Corollary 3.104 The Method of Lagrange Multipliers. *To find the maximum and minimum values of $f(x, y)$ subject to the constraint $g(x, y) = 0$ (assuming that these extreme values exist and $\nabla g \neq \mathbf{0}$ on the curve $g(x, y) = 0$):*

1 *Find all values of x, y, and λ such that the following system of equations is satisfied:*

$$\begin{cases} f_x(x, y) & = \lambda g_x(x, y) \\ f_y(x, y) & = \lambda g_y(x, y) \\ g(x, y) & = 0 \end{cases}$$

2 *Evaluate f at all the points (x, y) that result from Step 1. The largest of these values is the absolute maximum value of f; the smallest is the absolute minimum value of f subject to the constraint.*

Example 3.105 Find the extrema of $f(x, y) = xy$ if (x, y) is restricted to the ellipse $4x^2 + y^2 = 4$.

Solution. We define our constraint function as $g(x, y) = 4x^2 + y^2 - 4 = 0$. Next, we compute the first-order partial derivatives of both f and g:

$$f_x = y, \quad f_y = x \quad \text{and} \quad g_x = 8x, \quad g_y = 2y$$

Applying the Method of Lagrange Multipliers, we set up the corresponding system of equations:

$$\begin{cases} y & = 8\lambda x \quad \text{(Eq. 1)} \\ x & = 2\lambda y \quad \text{(Eq. 2)} \\ 4x^2 + y^2 & = 4 \quad \text{(Eq. 3)} \end{cases}$$

If $x = 0$, then Eq. 1 implies $y = 0$, but the point $(0, 0)$ does not satisfy Eq. 3. Thus, $x \neq 0$ and $y \neq 0$, which means $\lambda \neq 0$. We isolate λ from Eq. 1 and Eq. 2 to equate them:

$$\lambda = \frac{y}{8x} = \frac{x}{2y} \implies 2y^2 = 8x^2 \implies y^2 = 4x^2$$

Substituting $y^2 = 4x^2$ directly into our constraint equation (Eq. 3) yields:

$$4x^2 + (4x^2) = 4 \implies 8x^2 = 4 \implies x^2 = \frac{1}{2} \implies x = \pm\frac{1}{\sqrt{2}}$$

Using $y^2 = 4x^2$, we find that $y^2 = 4\left(\frac{1}{2}\right) = 2 \implies y = \pm\sqrt{2}$. This produces four candidate critical points:

$$\left(\frac{1}{\sqrt{2}}, \sqrt{2}\right), \quad \left(\frac{1}{\sqrt{2}}, -\sqrt{2}\right), \quad \left(-\frac{1}{\sqrt{2}}, \sqrt{2}\right), \quad \left(-\frac{1}{\sqrt{2}}, -\sqrt{2}\right)$$

Finally, we evaluate the objective function $f(x, y) = xy$ at these points:

- $f\left(\pm\frac{1}{\sqrt{2}}, \pm\sqrt{2}\right) = \left(\frac{1}{\sqrt{2}}\right)(\sqrt{2}) = 1$
- $f\left(\pm\frac{1}{\sqrt{2}}, \mp\sqrt{2}\right) = \left(\frac{1}{\sqrt{2}}\right)(-\sqrt{2}) = -1$

Therefore, the **maximum value** of f subject to the constraint is 1, and the **minimum value** is -1. ▲

Theorem 3.106 Lagrange Multipliers for Three Variables. *Suppose f and g are functions of three variables that have continuous first-order partial derivatives, and that $\nabla g \neq \mathbf{0}$ throughout a region of the xyz-plane. If f has an extremum $f(x_0, y_0, z_0)$ subject to the constraint $g(x, y, z) = 0$, then there is a real number λ such that:*

$$\nabla f(x_0, y_0, z_0) = \lambda \nabla g(x_0, y_0, z_0)$$

*The number λ is called a **Lagrange multiplier**.*

Corollary 3.107 The Method of Lagrange Multipliers: Three Variables. *To find the maximum and minimum values of $f(x, y, z)$ subject to the constraint $g(x, y, z) = 0$ (assuming that these values exist and $\nabla g \neq \mathbf{0}$ on the surface $g(x, y, z) = 0$):*

1 *Find all values of x, y, z, and λ such that:*

$$\begin{cases} f_x(x,y,z) & = \lambda g_x(x,y,z) \\ f_y(x,y,z) & = \lambda g_y(x,y,z) \\ f_z(x,y,z) & = \lambda g_z(x,y,z) \\ g(x,y,z) & = 0 \end{cases}$$

2 *Evaluate f at all the points (x, y, z) that result from Step 1. The largest of these values is the maximum value of f; the smallest is the minimum value of f.*

Example 3.108 A rectangular box without a lid is to be made from 12 m^2 of cardboard. Find the maximum volume of such a box.

Solution. Let x, y, and z denote the length, width, and height of the box respectively. We want to maximize the volume function:

$$f(x, y, z) = V = xyz$$

subject to the surface area constraint that accounts for an open top (one bottom face and four side faces totaling 12 m^2):

$$g(x, y, z) = xy + 2xz + 2yz - 12 = 0$$

We find the partial derivatives for our Lagrange configuration:

$$f_x = yz, \quad f_y = xz, \quad f_z = xy$$

$$g_x = y + 2z, \quad g_y = x + 2z, \quad g_z = 2x + 2y$$

This produces the following system of equations:

$$\begin{cases} yz & = \lambda(y + 2z) \quad \text{(Eq. 1)} \\ xz & = \lambda(x + 2z) \quad \text{(Eq. 2)} \\ xy & = \lambda(2x + 2y) \quad \text{(Eq. 3)} \\ xy + 2xz + 2yz & = 12 \quad \text{(Eq. 4)} \end{cases}$$

Since the dimensions must be strictly positive ($x, y, z > 0$), $\lambda \neq 0$. We multiply Eq. 1 by x, Eq. 2 by y, and Eq. 3 by z to create matching left-hand expressions:

$$xyz = \lambda(xy + 2xz)$$

$$xyz = \lambda(xy + 2yz)$$
$$xyz = \lambda(2xz + 2yz)$$

Equating the first two expressions gives:

$$\lambda(xy + 2xz) = \lambda(xy + 2yz) \implies 2xz = 2yz \implies x = y$$

Equating the second and third expressions gives:

$$\lambda(xy + 2yz) = \lambda(2xz + 2yz) \implies xy = 2xz \implies y = 2z$$

Thus, we find the core dimension ratios: $x = y = 2z$.

We substitute these variables in terms of z into our constraint equation (Eq. 4):

$$\begin{aligned}(2z)(2z) + 2(2z)z + 2(2z)z &= 12\\ 4z^2 + 4z^2 + 4z^2 &= 12\\ 12z^2 &= 12 \implies z^2 = 1 \implies z = 1 \text{ m}\end{aligned}$$

Using our ratios, the remaining dimensions are $x = 2$ m and $y = 2$ m.

Finally, we evaluate the maximum volume:

$$V = xyz = (2)(2)(1) = 4 \text{ m}^3$$

Thus, the maximum volume of the box is exactly 4 cubic meters. ▲

Checkpoint 3.109 Find the extreme values of the function $f(x, y) = x^2 + 2y^2$ on the circle $x^2 + y^2 = 1$.

Solution. We define our constraint function as $g(x, y) = x^2 + y^2 - 1 = 0$. Next, we compute the first-order partial derivatives of both functions:

$$f_x = 2x, \quad f_y = 4y \quad \text{and} \quad g_x = 2x, \quad g_y = 2y$$

Applying the Method of Lagrange Multipliers, we set up the system of equations:

$$\begin{cases} 2x &= 2\lambda x \quad \text{(Eq. 1)}\\ 4y &= 2\lambda y \quad \text{(Eq. 2)}\\ x^2 + y^2 &= 1 \quad \text{(Eq. 3)}\end{cases}$$

From Eq. 1, we can group the terms to solve for case conditions:

$$2x(1 - \lambda) = 0 \implies x = 0 \quad \text{or} \quad \lambda = 1$$

Case 1: $x = 0$. Substituting this into Eq. 3 yields:

$$0^2 + y^2 = 1 \implies y = \pm 1$$

This produces two candidate points: $(0, 1)$ and $(0, -1)$.

Case 2: $\lambda = 1$. Substituting this into Eq. 2 yields:

$$4y = 2(1)y \implies 2y = 0 \implies y = 0$$

Substituting $y = 0$ back into Eq. 3 yields:

$$x^2 + 0^2 = 1 \implies x = \pm 1$$

This produces two more candidate points: $(1, 0)$ and $(-1, 0)$.

Finally, we evaluate the objective function $f(x, y) = x^2 + 2y^2$ at all four candidate points:

- $f(0, \pm 1) = 0^2 + 2(\pm 1)^2 = 2$
- $f(\pm 1, 0) = (\pm 1)^2 + 2(0)^2 = 1$

Therefore, the **maximum value** of f on the circle is 2 (occurring at $(0, \pm 1)$), and the **minimum value** is 1 (occurring at $(\pm 1, 0)$).

Checkpoint 3.110 Find the points on the sphere $x^2 + y^2 + z^2 = 4$ that are closest to and farthest from the point $(3, 1, -1)$.

Solution. The distance from any point (x, y, z) to $(3, 1, -1)$ is minimized or maximized when the squared distance is minimized or maximized. We define our objective function as:

$$f(x, y, z) = (x - 3)^2 + (y - 1)^2 + (z + 1)^2$$

subject to the constraint describing the surface of the sphere:

$$g(x, y, z) = x^2 + y^2 + z^2 - 4 = 0$$

We compute the partial derivatives for both functions:

$$f_x = 2(x - 3), \quad f_y = 2(y - 1), \quad f_z = 2(z + 1)$$

$$g_x = 2x, \quad g_y = 2y, \quad g_z = 2z$$

Setting up our Lagrange multiplier system:

$$\begin{cases} 2(x-3) & = 2\lambda x \implies x - 3 = \lambda x \quad \text{(Eq. 1)} \\ 2(y-1) & = 2\lambda y \implies y - 1 = \lambda y \quad \text{(Eq. 2)} \\ 2(z+1) & = 2\lambda z \implies z + 1 = \lambda z \quad \text{(Eq. 3)} \\ x^2 + y^2 + z^2 & = 4 \quad \text{(Eq. 4)} \end{cases}$$

We isolate the coordinate variables in terms of λ from the first three equations:

$$x(1 - \lambda) = 3 \implies x = \frac{3}{1 - \lambda}$$

$$y(1 - \lambda) = 1 \implies y = \frac{1}{1 - \lambda}$$

$$z(1 - \lambda) = -1 \implies z = \frac{-1}{1 - \lambda}$$

Notice that the coordinates are proportional to each other: $x = 3y$ and $z = -y$.

We substitute these proportional ratios directly into our sphere constraint (Eq. 4):

$$\begin{aligned} (3y)^2 + y^2 + (-y)^2 &= 4 \\ 9y^2 + y^2 + y^2 &= 4 \\ 11y^2 = 4 \implies y^2 &= \frac{4}{11} \implies y = \pm \frac{2}{\sqrt{11}} \end{aligned}$$

Using our proportional ratios, we find the two corresponding sets of coordinates:

- When $y = \frac{2}{\sqrt{11}}$, we get the point: $P_1 = \left(\frac{6}{\sqrt{11}}, \frac{2}{\sqrt{11}}, -\frac{2}{\sqrt{11}} \right)$
- When $y = -\frac{2}{\sqrt{11}}$, we get the point: $P_2 = \left(-\frac{6}{\sqrt{11}}, -\frac{2}{\sqrt{11}}, \frac{2}{\sqrt{11}} \right)$

Geometrically, the point P_1 has positive coordinates that align with the direction toward $(3, 1, -1)$, making it the closest point. The point P_2 has opposite orientation, making it the farthest point.

Thus, the point $\left(\frac{6}{\sqrt{11}}, \frac{2}{\sqrt{11}}, -\frac{2}{\sqrt{11}}\right)$ is **closest** to the given point, and $\left(-\frac{6}{\sqrt{11}}, -\frac{2}{\sqrt{11}}, \frac{2}{\sqrt{11}}\right)$ is **farthest**.

Chapter 4

Multiple Integrals

Double and triple integrals extend the concept of integration to functions of two and three variables, respectively. They are used to compute volumes, surface areas, and other quantities in multi-dimensional spaces.

In this chapter, we will study the techniques for evaluating double and triple integrals, including iterated integrals and change of variables.

4.1 Double Integrals

In this section, we will explore how double integrals can be used to calculate areas and volumes of regions in two-dimensional and three-dimensional space.

We consider a function $f(x, y)$, with two variables, defined on a rectangular region R,

$$R: \quad a \leq x \leq b, \quad c \leq y \leq d.$$

Subdivide R into small rectangles by using parallel lines to the x- and y-axis. The lines divide R into n rectangular pieces, where the number of such pieces n gets large as the width and height of each piece gets small. These rectangles form a **partition** of R. A small rectangular piece of width Δx and height Δy has area $\Delta A = \Delta x \Delta y$. If we number the small pieces partitioning R in some order, then their areas are given by numbers $\Delta A_1, \Delta A_2, \ldots, \Delta A_n$, where ΔA_k is the area of the kth small rectangle.

Definition 4.1

1 Let $f(x, y)$ be defined on a region R, and let $P = \{R_k\}$ be an inner partition of R. A **Riemann sum** of f for P is any sum of the form $\sum_k f(x_k, y_k)\Delta A_k$, where (x_k, y_k) is a point in R_k, and ΔA_k is the area of R_k.

2 The statement

$$\lim_{|P|\to 0} \sum_k f(x_k, y_k)\Delta A_k = L$$

means that for every $\epsilon > 0$ there is a $\delta > 0$ such that if $P = \{R_k\}$ is an inner partition of R with $\|P\| < \delta$, then

$$\left| \sum_k f(x_k, y_k)\Delta A_k - L \right| < \epsilon$$

for every choice of (x_k, y_k) in R_k.

3 The **double integral of f over** R, denoted by $\iint_R f(x,y)\,dA$, is

$$\iint_R f(x,y)\,dA = \lim_{|P|\to 0} \sum_k f(x_k,y_k)\Delta A_k,$$

if the limit exists.

4 Let f be a continuous function of two variables such that $f(x,y) \geq 0$ for every (x,y) in a region R. The **volume** V of the solid that lies under the graph of $z = f(x,y)$ over R is

$$V = \iint_R f(x,y)\,dA.$$

◆

Example 4.2 Calculate the volume under the plane $z = 4 - x - y$ over the rectangular region $R : 0 \leq x \leq 2, \quad 0 \leq y \leq 1$ in the xy-plane.

Solution.

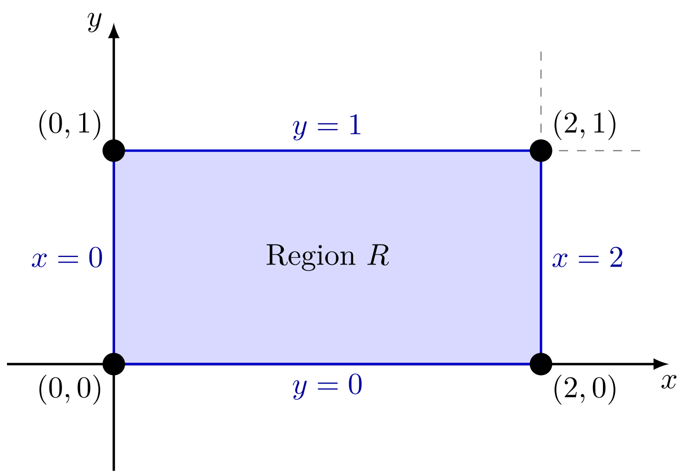


Figure 4.3 The rectangular region R with explicit boundary lines labeled.

The volume can be written as

$$\int_{x=0}^{x=2} A(x)\,dx,$$

where $A(x)$ is the cross-sectional area at x. For each value of x, we may calculate $A(x)$ as the integral

$$A(x) = \int_{y=0}^{y=1} (4 - x - y)\,dy,$$

that is the area under the curve $z = 4 - x - y$ in the plane of the cross-section at x.

$$\text{Volume} = \int_{x=0}^{x=2} A(x)\,dx = \int_{x=0}^{x=2} \left(\int_{y=0}^{y=1} (4 - x - y)\,dy \right) dx.$$

▲

Note 4.4 The expression, $\displaystyle\int_{x=0}^{x=2} \int_{y=0}^{y=1} (4 - x - y)\,dydx$, is called an **iterated or repeated integral**. The volume is obtained by integrating $4 - x - y$ with respect to y from $y = 0$ to $y = 1$, holding x fixed, and then integrating the resulting expression in x with respect to x from $x = 0$ to $x = 2$.

Theorem 4.5 Fubini's Theorem (First Form). *If $f(x, y)$ is continuous throughout the rectangular region $R : a \le x \le b,\ c \le y \le d$, then*

$$\iint_R f(x,y)\,dA = \int_c^d \int_a^b f(x,y)\,dxdy = \int_a^b \int_c^d f(x,y)\,dydx.$$

Example 4.6 Evaluating a Double Integral. Calculate $\iint_R f(x,y)\,dA$ for

$$f(x,y) = 1 - 6x^2y \quad \text{for} \quad R : 0 \le x \le 2,\ -1 \le y \le 1.$$

Solution.

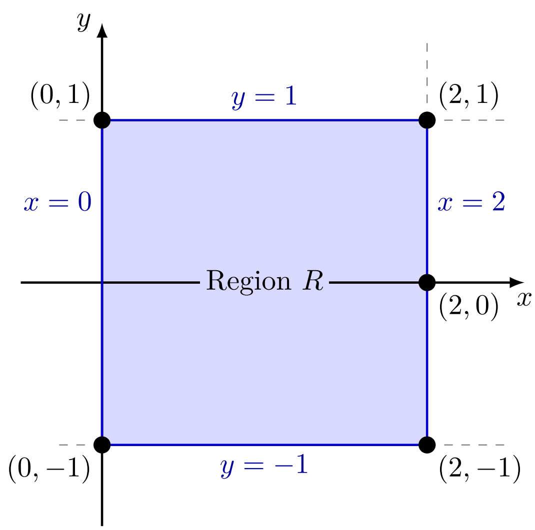


Figure 4.7 The rectangular region $R : 0 \le x \le 2,\ -1 \le y \le 1$ in the xy-plane.

We can integrate in either order using Fubini's Theorem. Let's integrate with respect to x first, and then with respect to y:

$$\begin{aligned}
\iint_R (1 - 6x^2y)\,dA &= \int_{-1}^1 \int_0^2 (1 - 6x^2y)\,dxdy \\
&= \int_{-1}^1 \left[x - 2x^3y\right]_{x=0}^{x=2}\,dy \\
&= \int_{-1}^1 \left((2 - 2(8)y) - (0 - 0)\right)\,dy \\
&= \int_{-1}^1 (2 - 16y)\,dy \\
&= \left[2y - 8y^2\right]_{-1}^1 \\
&= (2(1) - 8(1)^2) - (2(-1) - 8(-1)^2) \\
&= (2 - 8) - (-2 - 8) \\
&= -6 - (-10) = 4
\end{aligned}$$

Alternatively, integrating with respect to y first yields the same result:

$$\int_0^2 \int_{-1}^1 (1 - 6x^2y)\,dydx = \int_0^2 \left[y - 3x^2y^2\right]_{y=-1}^{y=1}\,dx$$

$$
\begin{aligned}
&= \int_0^2 \left((1-3x^2(1)^2)-(-1-3x^2(-1)^2)\right)\,dx \\
&= \int_0^2 (1-3x^2+1+3x^2)\,dx \\
&= \int_0^2 2\,dx \\
&= [2x]_0^2 = 4
\end{aligned}
$$

▲

Example 4.8 Calculate $\iint_R f(x,y)\,dA$ for

$$f(x,y)=2y\cos x \quad \text{for} \quad R:\pi/6\le x\le y^2,\ 1\le y\le 3.$$

Solution.

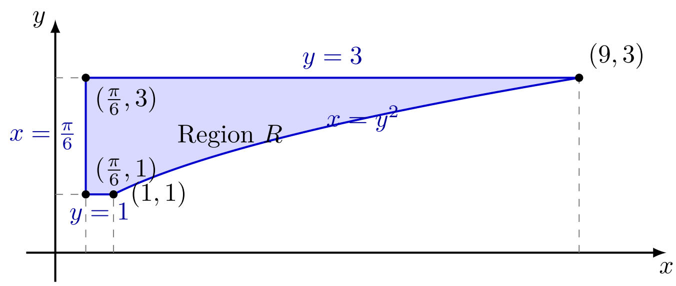


Figure 4.9 The region $R:\pi/6\le x\le y^2,\ 1\le y\le 3$ in the xy-plane.

Because the limits for x depend on y, we must set this up as a Type II region and integrate with respect to x first:

$$
\begin{aligned}
\iint_R 2y\cos x\,dA &= \int_1^3\int_{\pi/6}^{y^2} 2y\cos x\,dxdy \\
&= \int_1^3 [2y\sin x]_{x=\pi/6}^{x=y^2}\,dy \\
&= \int_1^3 \left(2y\sin(y^2)-2y\sin\left(\frac{\pi}{6}\right)\right)\,dy
\end{aligned}
$$

Since $\sin(\pi/6)=\frac{1}{2}$, the expression simplifies to:

$$= \int_1^3 \left(2y\sin(y^2)-y\right)\,dy$$

We integrate each term. For the first term, we can use u-substitution with $u=y^2$ and $du=2y\,dy$:

$$
\begin{aligned}
&= \left[-\cos(y^2)-\frac{1}{2}y^2\right]_1^3 \\
&= \left(-\cos(9)-\frac{1}{2}(9)\right)-\left(-\cos(1)-\frac{1}{2}(1)\right) \\
&= \cos(1)-\cos(9)-\frac{9}{2}+\frac{1}{2} \\
&= \cos(1)-\cos(9)-4
\end{aligned}
$$

Thus, the exact value of the double integral is $\cos(1)-\cos(9)-4$. ▲

Theorem 4.10 Fubini's Theorem (Stronger Form). *Let $f(x, y)$ be continuous on a region R.*

1 *If R is defined by $a \leq x \leq b,\ g_1(x) \leq y \leq g_2(x)$, with g_1 and g_2 continuous on $[a, b]$, then*

$$\iint_R f(x, y)\, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)\, dy dx.$$

2 *If R is defined by $c \leq y \leq d,\ h_1(y) \leq x \leq h_2(y)$, with h_1 and h_2 continuous on $[c, d]$, then*

$$\iint_R f(x, y)\, dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y)\, dx dy.$$

Example 4.11 Let R be the region in the xy-plane bounded by the graphs of $y = x^2$ and $y = 2x$. Evaluate $\iint_R (x^3 + 4y)\, dA$.

Solution.

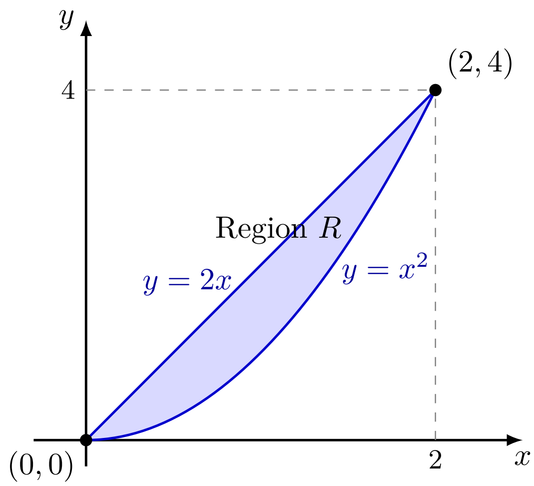


Figure 4.12 The region R bounded by $y = x^2$ and $y = 2x$ with an adjusted 2:1 aspect ratio.

First, we find the points of intersection of the bounding curves by setting them equal to each other:

$$x^2 = 2x \implies x^2 - 2x = 0 \implies x(x - 2) = 0$$

This gives intersection points at $x = 0$ and $x = 2$. On the interval $[0, 2]$, the line $y = 2x$ lies above the parabola $y = x^2$. Thus, our region is described by $0 \leq x \leq 2$ and $x^2 \leq y \leq 2x$.

Using Fubini's stronger form, we integrate with respect to y first:

$$\begin{aligned}
\iint_R (x^3 + 4y)\, dA &= \int_0^2 \int_{x^2}^{2x} (x^3 + 4y)\, dy dx \\
&= \int_0^2 \left[x^3 y + 2y^2\right]_{y=x^2}^{y=2x}\, dx \\
&= \int_0^2 \left(\left(x^3(2x) + 2(2x)^2\right) - \left(x^3(x^2) + 2(x^2)^2\right)\right)\, dx \\
&= \int_0^2 \left(2x^4 + 8x^2 - x^5 - 2x^4\right)\, dx
\end{aligned}$$

$$
\begin{aligned}
&= \int_0^2 \left(8x^2 - x^5\right)\, dx \\
&= \left[\frac{8}{3}x^3 - \frac{1}{6}x^6\right]_0^2 \\
&= \left(\frac{8}{3}(8) - \frac{1}{6}(64)\right) - (0-0) \\
&= \frac{64}{3} - \frac{32}{3} = \frac{32}{3}
\end{aligned}
$$

The value of the double integral over the bounded region is $\frac{32}{3}$. ▲

Checkpoint 4.13 Given $\int_0^4 \int_{\sqrt{y}}^2 y\cos(x^5)\, dxdy$, reverse the order of integration and evaluate the resulting integral.

Solution. Step 1: Analyze the Original Domain of Integration

The original limits correspond to a Type II integration order ($dxdy$). The boundaries for the region R are given by:

$$0 \le y \le 4 \quad \text{and} \quad \sqrt{y} \le x \le 2$$

The inner lower boundary curve $x = \sqrt{y}$ is equivalent to the parabola $y = x^2$ for $x \ge 0$. The upper boundary for x is the vertical line $x = 2$. These boundaries intersect at the points $(0,0)$ and $(2,4)$.

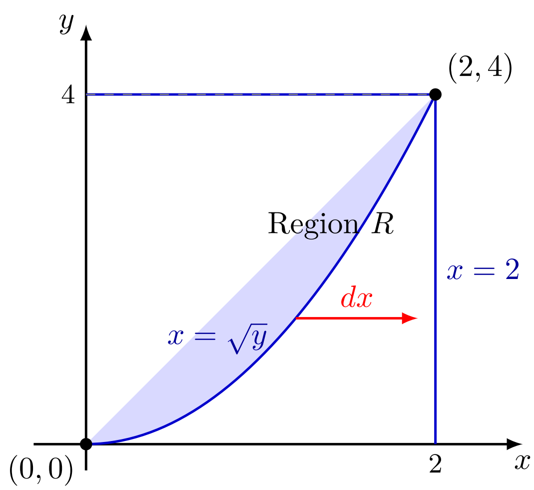


Figure 4.14 The integration domain R viewed as a Type II region ($dxdy$), bounded on the left by $x = \sqrt{y}$ and on the right by $x = 2$.

Step 2: Reverse the Order of Integration to Type I ($dydx$)

To transition to Type I format, we look at the region sweeping from left to right along the constant boundaries of the x-axis. The region spans from $x = 0$ to $x = 2$. For any fixed value of x, the vertical strip starts at the bottom boundary line $y = 0$ and moves up to touch the upper parabolic curve boundary $y = x^2$.

The rewritten boundary inequalities become:

$$0 \le x \le 2 \quad \text{and} \quad 0 \le y \le x^2$$

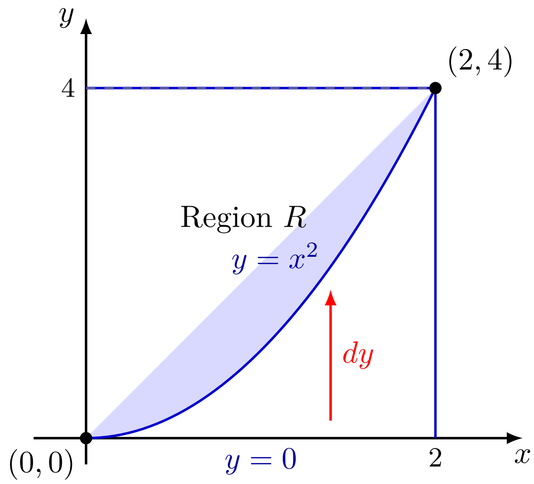


Figure 4.15 The integration domain R viewed as a Type I region ($dydx$), bounded below by $y = 0$ and above by $y = x^2$.

Step 3: Evaluate the New Iterated Integral
Setting up the reversed iterated integral gives:

$$\int_0^2 \int_0^{x^2} y \cos(x^5)\, dy\, dx$$

First, we integrate the inner expression with respect to y, treating $\cos(x^5)$ as a constant factor:

$$\begin{aligned}\int_0^2 \left[\frac{1}{2}y^2 \cos(x^5)\right]_{y=0}^{y=x^2} dx &= \int_0^2 \left(\frac{1}{2}(x^2)^2 \cos(x^5) - 0\right) dx \\ &= \int_0^2 \frac{1}{2}x^4 \cos(x^5)\, dx\end{aligned}$$

Now, we evaluate the outer single integral with respect to x. We can use u-substitution by setting:

$$u = x^5 \implies du = 5x^4\, dx \implies \frac{1}{5}\, du = x^4\, dx$$

We transform the integration boundaries accordingly:

- When $x = 0$: $u = 0^5 = 0$
- When $x = 2$: $u = 2^5 = 32$

Substituting these components back into the integral equation yields:

$$\begin{aligned}\int_0^{32} \frac{1}{2} \cdot \left(\frac{1}{5}\cos(u)\right) du &= \frac{1}{10}\int_0^{32} \cos(u)\, du \\ &= \frac{1}{10}\Big[\sin(u)\Big]_0^{32} \\ &= \frac{1}{10}\big(\sin(32) - \sin(0)\big) \\ &= \frac{\sin(32)}{10}\end{aligned}$$

Thus, the exact value of the reversed integration evaluation is $\frac{\sin(32)}{10}$.

Example 4.16 Finding Volume. Find the volume of a prism whose base is the triangle in the xy-plane bounded by the x-axis and the lines $y = x$ and $x = 1$, and whose top lies in the plane

$$z = f(x, y) = 3 - x - y.$$

Solution. Step 1: Identify the Base Region R

The triangular base region R in the xy-plane is bounded by the lines $y = 0$ (the x-axis), $y = x$, and $x = 1$. These lines intersect at the coordinates $(0, 0)$, $(1, 0)$, and $(1, 1)$. Treating this as a Type I region, the limits of integration are:

$$0 \le x \le 1 \quad \text{and} \quad 0 \le y \le x$$

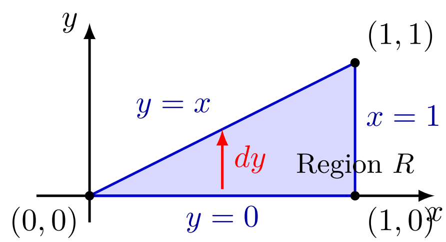


Figure 4.17 The triangular base region R in the xy-plane with a 2:1 horizontal-to-vertical aspect ratio.

Step 2: Set Up and Evaluate the Volume Integral

The volume V under the surface $z = 3-x-y$ over the region R is calculated using a double integral:

$$V = \iint_R (3 - x - y)\, dA = \int_0^1 \int_0^x (3 - x - y)\, dy\, dx$$

First, we compute the inner integral with respect to y, holding x constant:

$$\begin{aligned}
\int_0^x (3 - x - y)\, dy &= \left[3y - xy - \frac{1}{2}y^2\right]_{y=0}^{y=x} \\
&= \left(3(x) - x(x) - \frac{1}{2}(x)^2\right) - 0 \\
&= 3x - x^2 - \frac{1}{2}x^2 \\
&= 3x - \frac{3}{2}x^2
\end{aligned}$$

Next, we substitute this result into the outer integral and integrate with respect to x:

$$\begin{aligned}
V &= \int_0^1 \left(3x - \frac{3}{2}x^2\right) dx \\
&= \left[\frac{3}{2}x^2 - \frac{1}{2}x^3\right]_0^1 \\
&= \left(\frac{3}{2}(1)^2 - \frac{1}{2}(1)^3\right) - 0 \\
&= \frac{3}{2} - \frac{1}{2} = 1
\end{aligned}$$

The total volume of the prism is exactly 1. ▲

Checkpoint 4.18

1 Evaluate the double integral over the given region

$$\iint_R e^{x-y}\,dA, \qquad R: 0 \le x \le \ln 2,\ 0 \le y \le \ln 2$$

2 Find the volume of the region bounded above by the paraboloid $z = x^2 + y^2$ and below by the square $R: -1 \le x \le 1,\ -1 \le y \le 1$.

Solution.

1 Solution to Part 1:

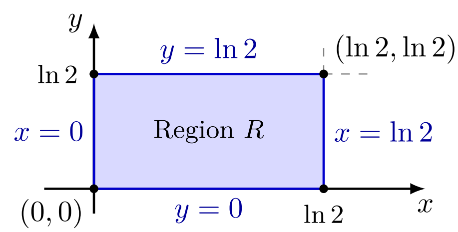


Figure 4.19 The rectangular region $R: 0 \le x \le \ln 2,\ 0 \le y \le \ln 2$ in the xy-plane.

Since the limits of integration for both variables are constants, this is a simple rectangular region. We can use Fubini's Theorem to write it as an iterated integral:

$$\iint_R e^{x-y}\,dA = \int_0^{\ln 2}\int_0^{\ln 2} e^x e^{-y}\,dx\,dy$$

Because the integrand factors into a function of x multiplied by a function of y, we can split the double integral into the product of two separate single integrals:

$$= \left(\int_0^{\ln 2} e^x\,dx\right)\left(\int_0^{\ln 2} e^{-y}\,dy\right)$$

Evaluating each single integral independently:

$$\int_0^{\ln 2} e^x\,dx = [e^x]_0^{\ln 2} = e^{\ln 2} - e^0 = 2 - 1 = 1$$

$$\int_0^{\ln 2} e^{-y}\,dy = \left[-e^{-y}\right]_0^{\ln 2} = -e^{-\ln 2} - (-e^0) = -\frac{1}{2} + 1 = \frac{1}{2}$$

Multiplying the two results together gives:

$$\iint_R e^{x-y}\,dA = 1 \cdot \frac{1}{2} = \frac{1}{2}$$

2 Solution to Part 2:

The volume V under the surface $z = x^2 + y^2$ over the square region R is given by the double integral:

$$V = \iint_R (x^2 + y^2)\,dA = \int_{-1}^{1}\int_{-1}^{1} (x^2 + y^2)\,dy\,dx$$

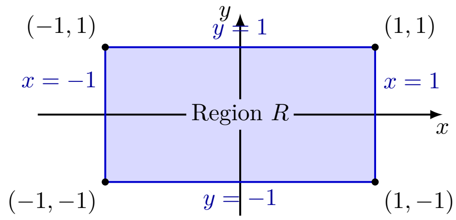


Figure 4.20 The square base domain $R : [-1,1] \times [-1,1]$ in the xy-plane with a 2:1 horizontal-to-vertical aspect ratio.

First, we compute the inner integral with respect to y:

$$\begin{aligned}
\int_{-1}^{1} (x^2 + y^2)\, dy &= \left[x^2 y + \frac{1}{3} y^3 \right]_{y=-1}^{y=1} \\
&= \left(x^2(1) + \frac{1}{3}(1)^3 \right) - \left(x^2(-1) + \frac{1}{3}(-1)^3 \right) \\
&= \left(x^2 + \frac{1}{3} \right) - \left(-x^2 - \frac{1}{3} \right) \\
&= 2x^2 + \frac{2}{3}
\end{aligned}$$

Now, we integrate this result with respect to x across the outer boundaries:

$$\begin{aligned}
V &= \int_{-1}^{1} \left(2x^2 + \frac{2}{3} \right) dx \\
&= \left[\frac{2}{3} x^3 + \frac{2}{3} x \right]_{-1}^{1} \\
&= \left(\frac{2}{3}(1)^3 + \frac{2}{3}(1) \right) - \left(\frac{2}{3}(-1)^3 + \frac{2}{3}(-1) \right) \\
&= \left(\frac{2}{3} + \frac{2}{3} \right) - \left(-\frac{2}{3} - \frac{2}{3} \right) \\
&= \frac{4}{3} - \left(-\frac{4}{3} \right) = \frac{8}{3}
\end{aligned}$$

The total volume bounded by the paraboloid over the square base is $\frac{8}{3}$.

4.2 Area and Volume

In this section, we will explore how to use double integrals to calculate the area of a region in the plane and the volume of a solid in three-dimensional space. We will also discuss the applications of these concepts in various fields.

Definition 4.21 Area. The area of a closed, bounded plane region R is

$$A = \iint_R dA.$$

◆

Example 4.22 Find the area of the region R bounded by $y = x$ and $y = x^2$ in the first quadrant.

Solution. Step 1: Find Points of Intersection and Setup Boundaries

We find where the two curves intersect by setting them equal to each other:

$$x = x^2 \implies x^2 - x = 0 \implies x(x-1) = 0$$

This gives intersection boundaries at $x = 0$ and $x = 1$. On the interval $[0, 1]$, the line $y = x$ lies above the parabola $y = x^2$. Thus, our region is described by:

$$0 \leq x \leq 1 \quad \text{and} \quad x^2 \leq y \leq x$$

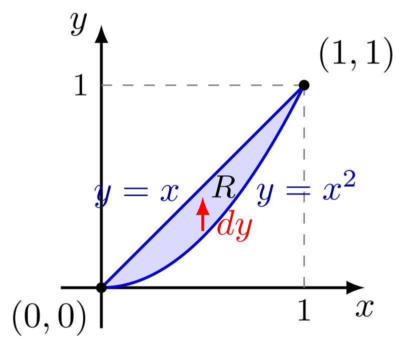


Figure 4.23 The region R bounded by $y = x^2$ and $y = x$ with a 2:1 horizontal-to-vertical aspect ratio.

Step 2: Evaluate the Double Integral

Using the definition of area, we set up the iterated integral with respect to y first:

$$\begin{aligned}
A &= \int_0^1 \int_{x^2}^{x} dy\,dx \\
&= \int_0^1 \Big[y\Big]_{y=x^2}^{y=x} dx \\
&= \int_0^1 (x - x^2)\,dx \\
&= \left[\frac{1}{2}x^2 - \frac{1}{3}x^3\right]_0^1 \\
&= \left(\frac{1}{2}(1)^2 - \frac{1}{3}(1)^3\right) - 0 \\
&= \frac{1}{2} - \frac{1}{3} = \frac{1}{6}
\end{aligned}$$

The area of the region is exactly $\frac{1}{6}$. ▲

Example 4.24 Find the area of the region R enclosed by the parabola $y = x^2$ and the line $y = x + 2$.

Solution. Step 1: Find Points of Intersection and Setup Boundaries

We find where the line and the parabola intersect by setting their equations equal to each other:

$$x^2 = x + 2 \implies x^2 - x - 2 = 0 \implies (x-2)(x+1) = 0$$

This gives intersection boundaries at $x = -1$ and $x = 2$. On the interval $[-1, 2]$, the line $y = x + 2$ lies above the parabola $y = x^2$. Thus, our region R is described by:

$$-1 \leq x \leq 2 \quad \text{and} \quad x^2 \leq y \leq x + 2$$

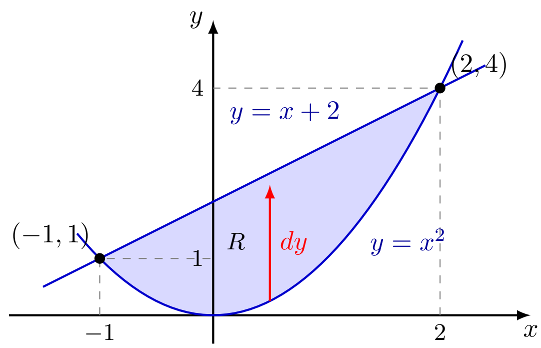


Figure 4.25 The region R bounded by $y = x^2$ and $y = x + 2$ with a 2:1 aspect ratio.

Step 2: Evaluate the Area Double Integral

Using the definition of area, we set up the iterated integral with respect to y first:

$$\begin{aligned} A &= \int_{-1}^{2} \int_{x^2}^{x+2} dy\, dx \\ &= \int_{-1}^{2} \Big[y \Big]_{y=x^2}^{y=x+2} dx \\ &= \int_{-1}^{2} (x + 2 - x^2)\, dx \\ &= \left[\frac{1}{2}x^2 + 2x - \frac{1}{3}x^3 \right]_{-1}^{2} \end{aligned}$$

Now, evaluating at the upper and lower limits:

$$\begin{aligned} \text{Upper limit } (x = 2): &\quad \frac{1}{2}(2)^2 + 2(2) - \frac{1}{3}(2)^3 = 2 + 4 - \frac{8}{3} = \frac{10}{3} \\ \text{Lower limit } (x = -1): &\quad \frac{1}{2}(-1)^2 + 2(-1) - \frac{1}{3}(-1)^3 = \frac{1}{2} - 2 + \frac{1}{3} = -\frac{7}{6} \end{aligned}$$

Subtracting the lower limit evaluation from the upper limit evaluation yields:

$$\begin{aligned} A &= \frac{10}{3} - \left(-\frac{7}{6} \right) \\ &= \frac{20}{6} + \frac{7}{6} = \frac{27}{6} = \frac{9}{2} \end{aligned}$$

The total area of the enclosed region is exactly $\frac{9}{2}$ (or 4.5). ▲

Definition 4.26 Average Value. The average value of f over R is

$$\text{Average Value} = \frac{1}{\text{area of } R} \iint_R f\, dA.$$

◆

Example 4.27 Find the average value of $f(x, y) = x\cos(xy)$ over the rectangle $R : 0 \leq x \leq \pi, \quad 0 \leq y \leq 1.$

Solution. Step 1: Calculate the Area of Region R

The region R is a simple rectangle bounded by constant lines. The width along the x-axis is π and the height along the y-axis is 1. Therefore, the area of R is:

$$\text{Area}(R) = \text{width} \times \text{height} = \pi \times 1 = \pi.$$

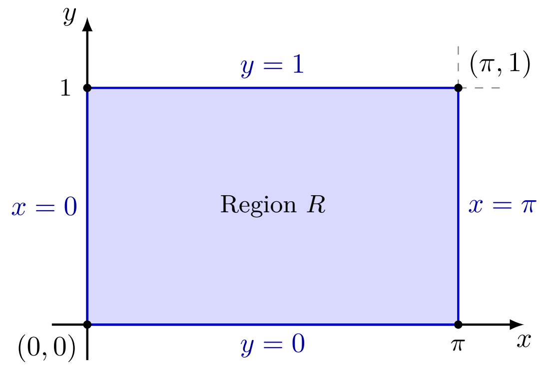


Figure 4.28 The rectangular domain $R : 0 \le x \le \pi,\ 0 \le y \le 1$ with a 2:1 horizontal-to-vertical aspect ratio.

Step 2: Evaluate the Double Integral Over R

To evaluate $\iint_R x\cos(xy)\,dA$, we integrate with respect to y first because the inner derivative factor x is already present in the integrand, simplifying the anti-differentiation:

$$\begin{aligned}
\iint_R x\cos(xy)\,dA &= \int_0^\pi \int_0^1 x\cos(xy)\,dy\,dx \\
&= \int_0^\pi \Big[\sin(xy)\Big]_{y=0}^{y=1} dx \\
&= \int_0^\pi \left(\sin(x\cdot 1) - \sin(x\cdot 0)\right)\,dx \\
&= \int_0^\pi \sin(x)\,dx \\
&= \Big[-\cos(x)\Big]_0^\pi \\
&= -\cos(\pi) - (-\cos(0)) \\
&= -(-1) + 1 = 1 + 1 = 2
\end{aligned}$$

Step 3: Calculate the Final Average Value

Using the definition, we divide the value of the double integral by the total area computed in Step 1:

$$\text{Average Value} = \frac{1}{\pi}\cdot 2 = \frac{2}{\pi}.$$

The average value of the function over the given rectangle is exactly $\frac{2}{\pi}$. ▲

Definition 4.29 Volume. The volume of a solid lying under a surface $z = f(x, y)$ over a closed, bounded plane region R is

$$V = \iint_R f(x, y)\,dA.$$

◆

Example 4.30 Find the volume V of the solid in the first octant bounded by the coordinate planes, the paraboloid $z = x^2 + y^2 + 1$, and the plane $2x + y = 2$.

Solution. Step 1: Identify the Base Region R

In the first octant, the coordinate plane boundaries give $x \geq 0$, $y \geq 0$, and $z \geq 0$. The base region R sits in the xy-plane ($z = 0$) and is a triangle bounded by the lines $x = 0$ (the y-axis), $y = 0$ (the x-axis), and the intersection line $2x + y = 2$.

Rewriting the bounding line in terms of y gives $y = 2 - 2x$. Finding the intercepts of this line:

- When $y = 0$: $2x = 2 \implies x = 1$
- When $x = 0$: $y = 2$

Therefore, setting this up as a Type I region, the boundaries are:

$$0 \leq x \leq 1 \quad \text{and} \quad 0 \leq y \leq 2 - 2x$$

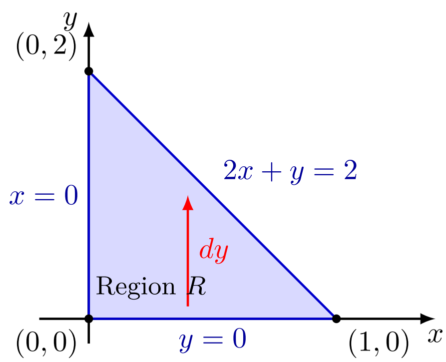


Figure 4.31 The triangular base region R in the xy-plane with a 2:1 horizontal-to-vertical aspect ratio.

Step 2: Set Up and Evaluate the Iterated Integral

The ceiling of the solid is the paraboloid $z = x^2 + y^2 + 1$. We integrate this function over the triangular region R:

$$V = \int_0^1 \int_0^{2-2x} (x^2 + y^2 + 1)\, dy\, dx$$

First, we integrate the inner expression with respect to y:

$$\begin{aligned}
\int_0^{2-2x} (x^2 + y^2 + 1)\, dy &= \left[x^2 y + \frac{1}{3} y^3 + y \right]_{y=0}^{y=2-2x} \\
&= x^2(2 - 2x) + \frac{1}{3}(2 - 2x)^3 + (2 - 2x) \\
&= 2x^2 - 2x^3 + \frac{1}{3}(8 - 24x + 24x^2 - 8x^3) + 2 - 2x \\
&= 2x^2 - 2x^3 + \frac{8}{3} - 8x + 8x^2 - \frac{8}{3}x^3 + 2 - 2x \\
&= \left(-2 - \frac{8}{3} \right) x^3 + (2 + 8)x^2 + (-8 - 2)x + \left(\frac{8}{3} + 2 \right) \\
&= -\frac{14}{3}x^3 + 10x^2 - 10x + \frac{14}{3}
\end{aligned}$$

Next, we integrate this simplified single variable polynomial expression with respect to x over the outer boundary:

$$V = \int_0^1 \left(-\frac{14}{3}x^3 + 10x^2 - 10x + \frac{14}{3} \right) dx$$

$$
\begin{aligned}
&= \left[-\frac{14}{12}x^4 + \frac{10}{3}x^3 - 5x^2 + \frac{14}{3}x\right]_0^1 \\
&= \left(-\frac{7}{6}(1)^4 + \frac{10}{3}(1)^3 - 5(1)^2 + \frac{14}{3}(1)\right) - 0 \\
&= -\frac{7}{6} + \frac{10}{3} - 5 + \frac{14}{3} \\
&= -\frac{7}{6} + \frac{24}{3} - 5 \\
&= -\frac{7}{6} + 8 - 5 = -\frac{7}{6} + 3 \\
&= -\frac{7}{6} + \frac{18}{6} = \frac{11}{6}
\end{aligned}
$$

The total volume of the solid in the first octant is exactly $\frac{11}{6}$. ▲

Exercises

1. Find the volume V of the solid that lies in the first octant and is bounded by the three coordinate planes and the cylinders $x^2 + y^2 = 9$ and $y^2 + z^2 = 9$.

Solution. Step 1: Identify the Base Region R

In the first octant, the coordinate planes give $x \geq 0$, $y \geq 0$, and $z \geq 0$. The base region R in the xy-plane is bounded by the circular cylinder base $x^2 + y^2 = 9$. Isolating x gives $x = \sqrt{9 - y^2}$. The region can be described as a Type II domain tracking from $y = 0$ to $y = 3$:

$$0 \leq y \leq 3 \quad \text{and} \quad 0 \leq x \leq \sqrt{9 - y^2}$$

The ceiling surface is given by isolating z from the second cylinder equation: $z = \sqrt{9 - y^2}$.

Step 2: Evaluate the Volume Integral

Using a double integral, we have:

$$V = \iint_R \sqrt{9 - y^2}\, dA = \int_0^3 \int_0^{\sqrt{9-y^2}} \sqrt{9 - y^2}\, dx\, dy$$

Integrating with respect to x first:

$$\int_0^{\sqrt{9-y^2}} \sqrt{9 - y^2}\, dx = \left[x\sqrt{9 - y^2}\right]_{x=0}^{x=\sqrt{9-y^2}} = (\sqrt{9 - y^2})^2 - 0 = 9 - y^2$$

Now, integrating the remaining expression with respect to y:

$$V = \int_0^3 (9 - y^2)\, dy = \left[9y - \frac{1}{3}y^3\right]_0^3 = \left(9(3) - \frac{1}{3}(27)\right) - 0 = 27 - 9 = 18$$

The volume of the intersection solid in the first octant is exactly 18.

2. Reverse the order of integration and evaluate the integral

$$\int_0^\pi \int_x^\pi \frac{\sin y}{y}\, dydx$$

Solution. Step 1: Analyze and Reverse the Bounds

The original Type I description is bounded by $0 \leq x \leq \pi$ and $x \leq y \leq \pi$. This represents a triangular region bounded by the lines $y = x$, $x = 0$, and

$y = \pi$. Reversing this to a Type II region ($dxdy$) gives:

$$0 \le y \le \pi \quad \text{and} \quad 0 \le x \le y$$

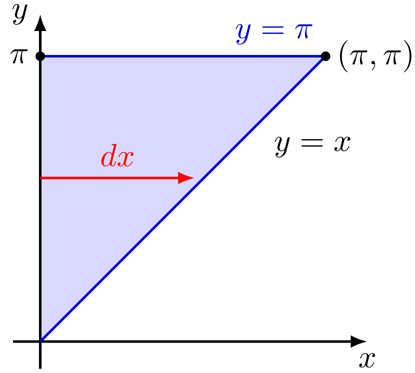


Figure 4.32 The domain R with horizontal sweeping arrows indicating the reversed $dxdy$ order.

Step 2: Evaluate the Transformed Integral
Applying the reversed limits of integration:

$$\int_0^\pi \int_0^y \frac{\sin y}{y}\, dx\, dy$$

Evaluating the inner integral with respect to x:

$$\int_0^y \frac{\sin y}{y}\, dx = \left[x\frac{\sin y}{y}\right]_{x=0}^{x=y} = y\left(\frac{\sin y}{y}\right) - 0 = \sin y$$

Now evaluating the outer integral with respect to y:

$$\int_0^\pi \sin y\, dy = \Big[-\cos y\Big]_0^\pi = -\cos(\pi) - (-\cos 0) = -(-1) + 1 = 2$$

The value of the integral is exactly 2.

3. Find the volume of the solid whose base is a region in the xy-plane that is bounded by the parabola $y = 4 - x^2$ and the line $y = 3x$, while the top of the solid is bounded by the plane $z = x + 4$.

Solution. Step 1: Find the Intersection Boundaries of the Base
We set the base boundaries equal to find their intersection coordinates:

$$4 - x^2 = 3x \implies x^2 + 3x - 4 = 0 \implies (x+4)(x-1) = 0$$

The intersection points occur at $x = -4$ and $x = 1$. On the interval $[-4, 1]$, the parabola $y = 4 - x^2$ stays above the line $y = 3x$. The Type I boundaries are:

$$-4 \le x \le 1 \quad \text{and} \quad 3x \le y \le 4 - x^2$$

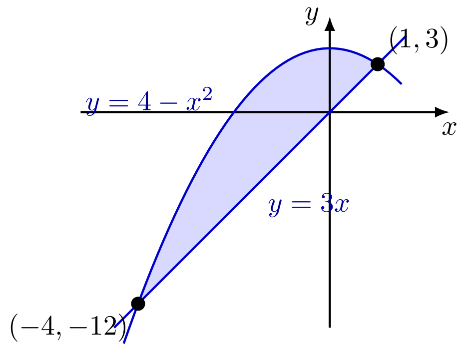


Figure 4.33 The base region R bounded by $y = 4 - x^2$ and $y = 3x$.

Step 2: Evaluate the Volume Integral

We compute the volume by integrating the ceiling function $z = x + 4$ over R:

$$V = \int_{-4}^{1} \int_{3x}^{4-x^2} (x+4)\, dy\, dx$$

Integrating with respect to y:

$$\int_{3x}^{4-x^2} (x+4)\, dy = (x+4)\Big[y\Big]_{3x}^{4-x^2} = (x+4)(4 - x^2 - 3x)$$

Expanding the polynomial product results in:

$$= 4x - x^3 - 3x^2 + 16 - 4x^2 - 12x = -x^3 - 7x^2 - 8x + 16$$

Now, evaluating the single integral with respect to x:

$$\begin{aligned} V &= \int_{-4}^{1} (-x^3 - 7x^2 - 8x + 16)\, dx = \left[-\frac{1}{4}x^4 - \frac{7}{3}x^3 - 4x^2 + 16x\right]_{-4}^{1} \\ &= \left(-\frac{1}{4} - \frac{7}{3} - 4 + 16\right) - \left(-\frac{256}{4} - \frac{7(-64)}{3} - 4(16) + 16(-4)\right) \\ &= \left(\frac{113}{12}\right) - \left(-64 + \frac{448}{3} - 64 - 64\right) = \frac{113}{12} - \left(-\frac{128}{3}\right) = \frac{113 + 512}{12} = \frac{625}{12} \end{aligned}$$

The volume of the solid is exactly $\frac{625}{12}$.

4. Find the volume of the wedge cut from the first octant by the cylinder $z = 12 - 3y^2$ and the plane $x + y = 2$.

Solution. Step 1: Determine the Bounds in the Base Plane

In the first octant, $x \geq 0$ and $y \geq 0$. The base triangle R is bounded by the line $x + y = 2 \implies x = 2 - y$. Setting $x = 0$ gives the upper limit for y at $y = 2$. Setting this up as a Type II integral gives:

$$0 \leq y \leq 2 \quad \text{and} \quad 0 \leq x \leq 2 - y$$

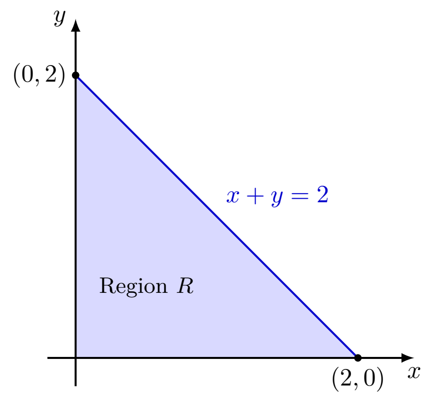


Figure 4.34 The triangular base region R for the wedge solid.

Step 2: Reverse the Limits to Type I ($dydx$)

To switch to a Type I layout, we examine the vertical bounds of the region across a horizontal span. The region spans along the x-axis from $x = 0$ to $x = 1$. For a fixed value of x, a vertical strip starts at the bottom boundary line $y = 0$ and sweeps upward until it reaches the line $y = x$. The reversed boundary descriptions are:

$$0 \le x \le 1 \quad \text{and} \quad 0 \le y \le x.$$

Step 3: Setup and Evaluate the Transformed Integral

Using our new boundary configurations, the original unintegrable problem transitions into a solvable iterated integral:

$$\int_0^1 \int_0^x e^{x^2} \, dy \, dx$$

We compute the inner integral with respect to y, treating e^{x^2} as a constant factor:

$$\int_0^x e^{x^2} \, dy = \left[y e^{x^2} \right]_{y=0}^{y=x} = x e^{x^2} - 0 = x e^{x^2}.$$

Next, we substitute this result back into the outer structural layer to integrate with respect to x:

$$\int_0^1 x e^{x^2} \, dx$$

We evaluate this final layer using u-substitution. Let:

$$u = x^2 \implies du = 2x \, dx \implies \frac{1}{2} \, du = x \, dx.$$

Updating our definite limits of integration:

- When $x = 0$: $u = 0^2 = 0$
- When $x = 1$: $u = 1^2 = 1$

Substituting these elements back into the equation yields:

$$\begin{aligned} \int_0^1 \frac{1}{2} e^u \, du &= \left[\frac{1}{2} e^u \right]_0^1 \\ &= \frac{1}{2} e^1 - \frac{1}{2} e^0 \\ &= \frac{1}{2}(e - 1). \end{aligned}$$

The exact value of the reversed double integral is $\frac{e-1}{2}$.

4.3 Double Integrals in Polar coordinates

In this section, we will explore the use of polar coordinates to evaluate double integrals over regions that are more naturally described in polar form. Polar coordinates provide a convenient way to describe circular and radial symmetry in two-dimensional space.

We will derive the formula for converting a double integral from Cartesian coordinates to polar coordinates and apply it to various examples.

Theorem 4.35 Evaluation Theorem.

$$\iint_R f(r,\theta)\,dA = \int_\alpha^\beta \int_{g_1(\theta)}^{g_2(\theta)} f(r,\theta) r\,dr d\theta.$$

Example 4.36 Find the area of the region R that lies outside the circle $r = a$ and inside the circle $r = 2a\sin\theta$.

Solution. Step 1: Find the Points of Intersection and Limits

We find the angular intersection points of the two curves by setting their expressions equal to each other:

$$2a\sin\theta = a \implies \sin\theta = \frac{1}{2}.$$

In the standard first and second quadrants, this yields the angular bounds:

$$\alpha = \frac{\pi}{6} \quad \text{and} \quad \beta = \frac{5\pi}{6}.$$

For any angle θ between these limits, the radial variable r starts on the inner circle boundary $r = a$ and sweeps outward to the outer circle boundary $r = 2a\sin\theta$. Thus, our region R is bounded by:

$$\frac{\pi}{6} \le \theta \le \frac{5\pi}{6} \quad \text{and} \quad a \le r \le 2a\sin\theta.$$

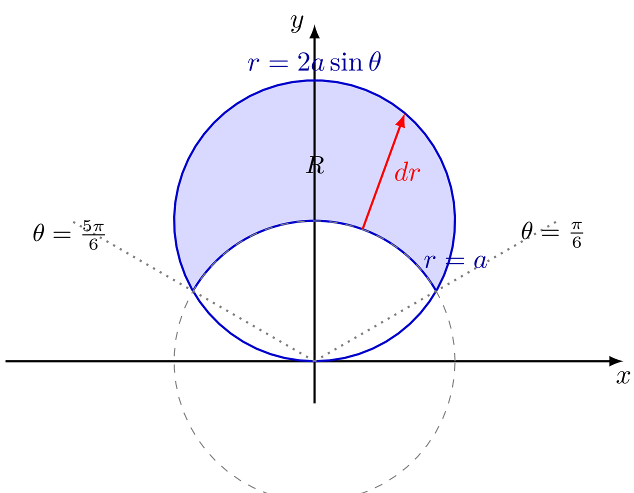


Figure 4.37 The region R lying inside $r = 2a\sin\theta$ and outside $r = a$ displayed with a balanced 1:1 ratio.

Step 2: Evaluate the Double Area Integral

Using the Evaluation Theorem with $f(r,\theta) = 1$ to calculate area:

$$A = \int_{\pi/6}^{5\pi/6} \int_a^{2a\sin\theta} r\,dr d\theta = \int_{\pi/6}^{5\pi/6} \left[\frac{1}{2}r^2\right]_a^{2a\sin\theta} d\theta$$

$$
\begin{aligned}
&= \frac{1}{2}\int_{\pi/6}^{5\pi/6} \left(4a^2\sin^2\theta - a^2\right)\, d\theta \\
&= \frac{a^2}{2}\int_{\pi/6}^{5\pi/6} \left(4\left(\frac{1-\cos(2\theta)}{2}\right) - 1\right)\, d\theta \\
&= \frac{a^2}{2}\int_{\pi/6}^{5\pi/6} (2 - 2\cos(2\theta) - 1)\, d\theta = \frac{a^2}{2}\int_{\pi/6}^{5\pi/6} (1 - 2\cos(2\theta))\, d\theta \\
&= \frac{a^2}{2}\Big[\theta - \sin(2\theta)\Big]_{\pi/6}^{5\pi/6}
\end{aligned}
$$

Evaluating at the definite boundary constraints:

$$
\begin{aligned}
A &= \frac{a^2}{2}\left[\left(\frac{5\pi}{6} - \sin\left(\frac{5\pi}{3}\right)\right) - \left(\frac{\pi}{6} - \sin\left(\frac{\pi}{3}\right)\right)\right] \\
&= \frac{a^2}{2}\left[\left(\frac{5\pi}{6} - \left(-\frac{\sqrt{3}}{2}\right)\right) - \left(\frac{\pi}{6} - \frac{\sqrt{3}}{2}\right)\right] \\
&= \frac{a^2}{2}\left[\frac{4\pi}{6} + \frac{2\sqrt{3}}{2}\right] = \frac{a^2}{2}\left(\frac{2\pi}{3} + \sqrt{3}\right) = a^2\left(\frac{\pi}{3} + \frac{\sqrt{3}}{2}\right).
\end{aligned}
$$

The area of the region is exactly $a^2\left(\frac{\pi}{3} + \frac{\sqrt{3}}{2}\right)$. ▲

Checkpoint 4.38 Find the area of the region R bounded by one loop of the lemniscate $r^2 = a^2\sin(2\theta)$, where $a > 0$.

Solution. Step 1: Set Up the Integration Loop Bounds

A single loop of the lemniscate $r^2 = a^2\sin(2\theta)$ is traced out as the radius starts at the origin, sweeps outward, and returns to the origin. This occurs when:

$$a^2\sin(2\theta) \geq 0 \implies 0 \leq 2\theta \leq \pi \implies 0 \leq \theta \leq \frac{\pi}{2}.$$

For any angle θ in the first quadrant, the variable r spans from the origin $r = 0$ to the boundary line curve $r = \sqrt{a^2\sin(2\theta)} = a\sqrt{\sin(2\theta)}$.

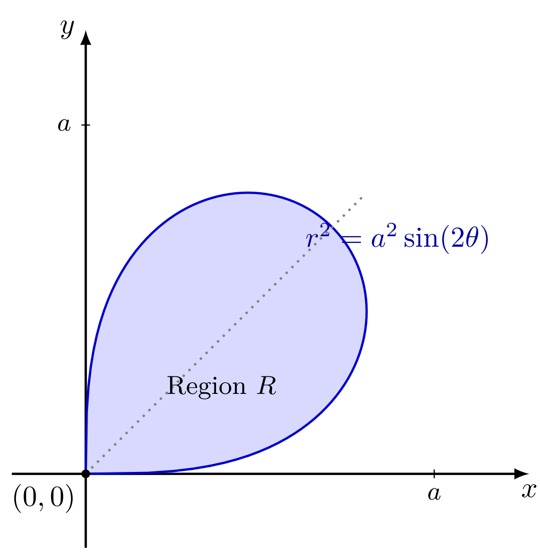


Figure 4.39 One loop of the lemniscate curve $r^2 = a^2\sin(2\theta)$ with a balanced 1:1 ratio.

Step 2: Evaluate the Iterated Area Integral

Using the Evaluation Theorem to compute the area of this region:

$$\begin{aligned}
A &= \int_0^{\pi/2} \int_0^{a\sqrt{\sin(2\theta)}} r\, drd\theta \\
&= \int_0^{\pi/2} \left[\frac{1}{2}r^2\right]_0^{a\sqrt{\sin(2\theta)}} d\theta \\
&= \int_0^{\pi/2} \frac{1}{2}a^2 \sin(2\theta)\, d\theta \\
&= \frac{a^2}{2}\left[-\frac{1}{2}\cos(2\theta)\right]_0^{\pi/2} \\
&= -\frac{a^2}{4}\Big(\cos(\pi) - \cos(0)\Big) \\
&= -\frac{a^2}{4}\Big(-1-1\Big) = -\frac{a^2}{4}(-2) = \frac{a^2}{2}.
\end{aligned}$$

The area enclosed by one loop of the lemniscate is exactly $\frac{a^2}{2}$.

Theorem 4.40 Change of Variables Formula.

$$\iint_R f(x,y)\, dydx = \iint_R f(r\cos\theta, r\sin\theta) r\, drd\theta.$$

Example 4.41 Use polar coordinates to evaluate

$$\int_{-a}^{a} \int_0^{\sqrt{a^2-x^2}} (x^2+y^2)^{3/2}\, dydx.$$

Solution. Step 1: Analyze the Region of Integration R

The original limits are given in Cartesian coordinates where:

$$-a \le x \le a \quad \text{and} \quad 0 \le y \le \sqrt{a^2-x^2}.$$

The upper boundary curve $y = \sqrt{a^2-x^2}$ represents the upper half of a circle centered at the origin with radius a. Since x runs from $-a$ to a, the region R is a complete upper semicircle sitting in the first and second quadrants.

Converting this region to polar coordinates, the radius r starts at the origin and sweeps outward to the boundary circle $r = a$. The angle θ opens from the positive x-axis (0) all the way around to the negative x-axis (π). Therefore, the polar boundaries are:

$$0 \le \theta \le \pi \quad \text{and} \quad 0 \le r \le a.$$

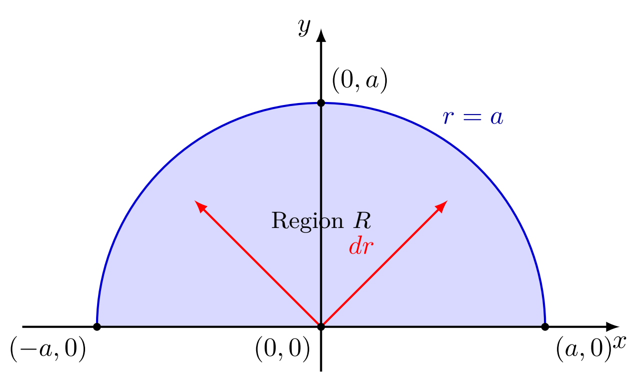


Figure 4.42 The semicircular integration region R displayed with a balanced 1:1 ratio.

Step 2: Convert and Evaluate the Integral

Using the relationship $x^2 + y^2 = r^2$ and changing the differential element to polar form ($dA = r\,drd\theta$), we substitute our values into the formula:

$$\int_{-a}^{a}\int_{0}^{\sqrt{a^2-x^2}} (x^2+y^2)^{3/2}\,dydx = \int_{0}^{\pi}\int_{0}^{a} (r^2)^{3/2}\cdot r\,drd\theta$$

Simplifying the inner integrand components gives:

$$= \int_{0}^{\pi}\int_{0}^{a} r^3\cdot r\,drd\theta = \int_{0}^{\pi}\int_{0}^{a} r^4\,drd\theta$$

Since the limits of integration are entirely constants and the variables inside the integrand are completely independent, we can separate this double integral into the product of two single integrals:

$$\begin{aligned}\int_{0}^{\pi}\int_{0}^{a} r^4\,drd\theta &= \left(\int_{0}^{\pi} d\theta\right)\left(\int_{0}^{a} r^4\,dr\right)\\ &= \Big[\theta\Big]_0^{\pi}\cdot\left[\frac{1}{5}r^5\right]_0^a\\ &= (\pi-0)\cdot\left(\frac{1}{5}a^5-0\right)\\ &= \frac{\pi a^5}{5}.\end{aligned}$$

The value of the double integral evaluated using polar coordinates is exactly $\frac{\pi a^5}{5}$. ▲

Example 4.43 Find the volume V of the solid that is bounded by the paraboloid $z = 4 - x^2 - y^2$ and the xy-plane.

Solution. Step 1: Identify the Base Region R in the xy-plane

The solid sits beneath the paraboloid ceiling and is bounded below by the xy-plane, where $z = 0$. We find the footprint or boundary curve of this base region R by setting the paraboloid equation to zero:

$$4 - x^2 - y^2 = 0 \implies x^2 + y^2 = 4.$$

This represents a complete circle centered at the origin with a radius of $r = 2$.

Converting this full circular disk to polar coordinates gives us constant boundaries that span all four quadrants:

$$0 \le \theta \le 2\pi \quad \text{and} \quad 0 \le r \le 2.$$

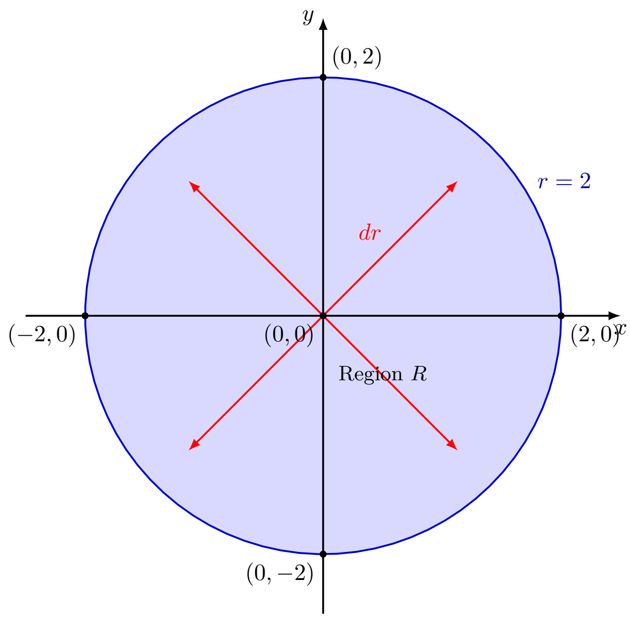


Figure 4.44 The circular base domain $R : x^2 + y^2 \leq 4$ plotted with a balanced 1:1 aspect ratio.

Step 2: Convert and Evaluate the Volume Integral

The height function inside the double integral can be converted to polar coordinates using $x^2 + y^2 = r^2$, which yields $z = 4 - r^2$. Substituting this ceiling function and the polar area differential element ($dA = r\, dr d\theta$) into our volume formula gives:

$$V = \iint_R (4 - x^2 - y^2)\, dA = \int_0^{2\pi} \int_0^2 (4 - r^2) r\, dr d\theta$$

Distributing the inner radial factor across the integrand parameters:

$$= \int_0^{2\pi} \int_0^2 (4r - r^3)\, dr d\theta$$

Since the boundary parameters are entirely composed of constants and the individual components inside the expression are mathematically independent, we can factor the double integral into a simple product of two independent single integrals:

$$\begin{aligned}
V &= \left(\int_0^{2\pi} d\theta \right) \left(\int_0^2 (4r - r^3)\, dr \right) \\
&= \Big[\theta \Big]_0^{2\pi} \cdot \left[2r^2 - \frac{1}{4} r^4 \right]_0^2 \\
&= (2\pi - 0) \cdot \left(\left(2(2)^2 - \frac{1}{4}(2)^4 \right) - 0 \right) \\
&= 2\pi \cdot \left(8 - \frac{16}{4} \right) \\
&= 2\pi \cdot (8 - 4) = 2\pi \cdot 4 = 8\pi.
\end{aligned}$$

The total volume of the solid enclosed under the paraboloid is exactly 8π. ▲

Exercises

1. Change the Cartesian integral into an equivalent polar integral then evaluate the integral:

a $$\int_{-1}^{0}\int_{-\sqrt{1-x^2}}^{0} \frac{2}{1+\sqrt{x^2+y^2}}\,dydx$$

b $$\int_{-1}^{1}\int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} \ln(x^2+y^2+1)\,dxdy$$

Solution.

a Solution to Part (a):

The original Cartesian limits restrict the domain to:

$$-1 \le x \le 0 \quad \text{and} \quad -\sqrt{1-x^2} \le y \le 0.$$

The curve $y = -\sqrt{1-x^2}$ describes the lower half of a unit circle centered at the origin. Because both x and y values are exclusively negative or zero, the integration domain R is a quarter-circle resting entirely inside the third quadrant.

Converting to polar variables, the radial sweep starts at the origin $r = 0$ and extends out to the curve boundary $r = 1$. The angle θ sweeps across the third quadrant from π to $\frac{3\pi}{2}$:

$$\pi \le \theta \le \frac{3\pi}{2} \quad \text{and} \quad 0 \le r \le 1.$$

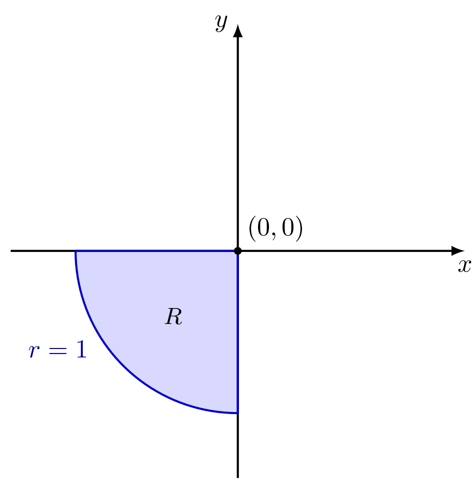


Figure 4.45 The third-quadrant quarter-disk integration region R with a balanced 1:1 aspect ratio.

Substituting the polar translation rules ($\sqrt{x^2+y^2} = r$ and $dydx = r\,drd\theta$) transforms the expression into:

$$\int_{\pi}^{3\pi/2}\int_{0}^{1} \frac{2}{1+r}\cdot r\,drd\theta = \left(\int_{\pi}^{3\pi/2} d\theta\right)\left(\int_{0}^{1}\frac{2r}{1+r}\,dr\right)$$

Evaluating the angular integral yields $\frac{3\pi}{2} - \pi = \frac{\pi}{2}$. For the radial integral, we apply algebraic long division:

$$\int_{0}^{1}\frac{2r}{1+r}\,dr = 2\int_{0}^{1}\frac{(r+1)-1}{1+r}\,dr = 2\int_{0}^{1}\left(1-\frac{1}{1+r}\right)\,dr$$

$$= 2\Big[r - \ln|1+r|\Big]_0^1 = 2\Big((1-\ln 2) - (0 - \ln 1)\Big) = 2 - 2\ln 2.$$

Multiplying our two separate solutions together gives:

$$\text{Value} = \frac{\pi}{2}(2 - 2\ln 2) = \pi(1 - \ln 2).$$

b Solution to Part (b):

The original integration walls are given by:

$$-1 \leq y \leq 1 \quad \text{and} \quad -\sqrt{1-y^2} \leq x \leq \sqrt{1-y^2}.$$

This maps out the complete full unit circular disk centered at the origin. Converting to polar tracking variables yields:

$$0 \leq \theta \leq 2\pi \quad \text{and} \quad 0 \leq r \leq 1.$$

Substituting $x^2 + y^2 = r^2$ and the differential multiplier element $(dxdy = r\,drd\theta)$ gives:

$$\int_0^{2\pi} \int_0^1 \ln(r^2+1) \cdot r\,drd\theta = \left(\int_0^{2\pi} d\theta\right)\left(\int_0^1 r\ln(r^2+1)\,dr\right)$$

The outer integral maps directly to 2π. For the inner single radial integral, we perform a quick u-substitution:

$$u = r^2 + 1 \implies du = 2r\,dr \implies \frac{1}{2}\,du = r\,dr.$$

Updating limits: when $r = 0 \implies u = 1$, and when $r = 1 \implies u = 2$.

$$\int_0^1 r\ln(r^2+1)\,dr = \frac{1}{2}\int_1^2 \ln(u)\,du = \frac{1}{2}\Big[u\ln(u) - u\Big]_1^2$$

$$= \frac{1}{2}\Big((2\ln 2 - 2) - (1\ln 1 - 1)\Big) = \frac{1}{2}(2\ln 2 - 2 + 1) = \ln 2 - \frac{1}{2}.$$

Combining the structural components yields:

$$\text{Value} = 2\pi\left(\ln 2 - \frac{1}{2}\right) = 2\pi\ln 2 - \pi.$$

2. Find the area of the region cut from the first quadrant (first octant base) by the cardioid $r = 1 + \sin\theta$.

Solution. Step 1: Set Up the Polar Integration Boundaries

The phrase "cut from the first quadrant" restricts our angle tracking bounds to $0 \leq \theta \leq \frac{\pi}{2}$. Inside this angular sweep, the radius starts at the origin $r = 0$ and pushes out to hit the outer wall profile of the cardioid $r = 1 + \sin\theta$:

$$0 \leq \theta \leq \frac{\pi}{2} \quad \text{and} \quad 0 \leq r \leq 1 + \sin\theta.$$

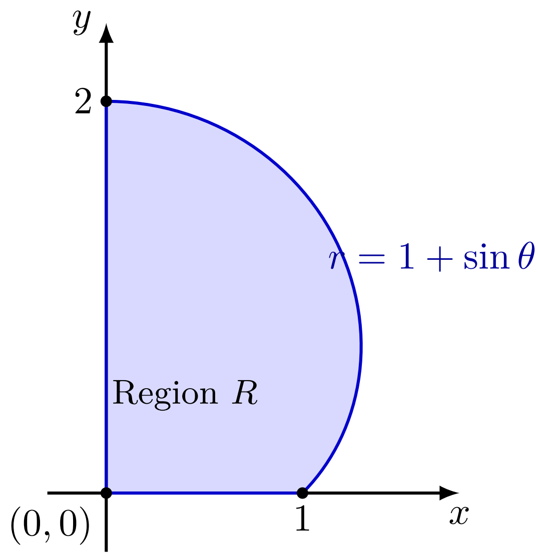


Figure 4.46 The section of the cardioid $r = 1 + \sin\theta$ cut by the first quadrant plotted with a balanced 1:1 isometric ratio.

Step 2: Evaluate the Area Integration

Using the polar double integral definition for finding area ($dA = r\,drd\theta$):

$$A = \int_0^{\pi/2} \int_0^{1+\sin\theta} r\,drd\theta = \int_0^{\pi/2} \left[\frac{1}{2}r^2\right]_0^{1+\sin\theta} d\theta$$

$$= \frac{1}{2}\int_0^{\pi/2} (1+\sin\theta)^2\,d\theta = \frac{1}{2}\int_0^{\pi/2} (1 + 2\sin\theta + \sin^2\theta)\,d\theta$$

Using the half-angle reduction identity $\sin^2\theta = \frac{1-\cos(2\theta)}{2}$:

$$= \frac{1}{2}\int_0^{\pi/2} \left(1 + 2\sin\theta + \frac{1}{2} - \frac{1}{2}\cos(2\theta)\right) d\theta = \frac{1}{2}\int_0^{\pi/2} \left(\frac{3}{2} + 2\sin\theta - \frac{1}{2}\cos(2\theta)\right) d\theta$$

$$= \frac{1}{2}\left[\frac{3}{2}\theta - 2\cos\theta - \frac{1}{4}\sin(2\theta)\right]_0^{\pi/2}$$

$$= \frac{1}{2}\left[\left(\frac{3\pi}{4} - 0 - 0\right) - (0 - 2 - 0)\right] = \frac{1}{2}\left(\frac{3\pi}{4} + 2\right) = \frac{3\pi}{8} + 1.$$

The total area cut inside the first quadrant is exactly $\frac{3\pi}{8} + 1$.

3. Find the average height of the hemisphere surface $z = \sqrt{2^2 - x^2 - y^2}$ above the disk $x^2 + y^2 \leq 4$ in the xy-plane.

Solution. Step 1: Determine the Base Area

The base region R is a complete circular disk described by $x^2 + y^2 \leq 4$, which yields a clear radius of $R_0 = 2$. The geometric area of this full circular base region is:

$$\text{Area}(R) = \pi R_0^2 = \pi(2)^2 = 4\pi.$$

Converted to polar coordinate conditions, our tracking parameters are: $0 \leq \theta \leq 2\pi$ and $0 \leq r \leq 2$.

Step 2: Set Up and Evaluate the Average Value Theorem Form

According to the average value theorem, the average height $\bar{z}$ is computed by finding the total volume under the dome surface and dividing it by the total area of the floor disk:

$$\bar{z} = \frac{1}{\text{Area}(R)} \iint_R z\, dA = \frac{1}{4\pi} \int_0^{2\pi} \int_0^2 \sqrt{4-r^2} \cdot r\, dr d\theta$$

Since the bounds are completely constants and the inner function components are independent, we can pull apart the double integral:

$$\bar{z} = \frac{1}{4\pi} \left(\int_0^{2\pi} d\theta \right) \left(\int_0^2 r\sqrt{4-r^2}\, dr \right) = \frac{2\pi}{4\pi} \int_0^2 r(4-r^2)^{1/2}\, dr = \frac{1}{2} \int_0^2 r(4-r^2)^{1/2}\, dr$$

We compute the remaining single integral using u-substitution:

$$u = 4 - r^2 \implies du = -2r\, dr \implies -\frac{1}{2}\, du = r\, dr.$$

Updating parameters: when $r = 0 \implies u = 4$, and when $r = 2 \implies u = 0$.

$$\bar{z} = \frac{1}{2} \int_4^0 \left(-\frac{1}{2} u^{1/2} \right) du = \frac{1}{4} \int_0^4 u^{1/2}\, du$$

$$= \frac{1}{4} \left[\frac{2}{3} u^{3/2} \right]_0^4 = \frac{1}{4} \left(\frac{2}{3} (4)^{3/2} - 0 \right) = \frac{1}{4} \left(\frac{2}{3} \cdot 8 \right) = \frac{16}{12} = \frac{4}{3}.$$

The average height of the hemisphere surface above the floor base disk is exactly $\frac{4}{3}$.

Chapter 5

MyOpenMath Interactive Activities

These files contain copies of the interactive MyOpenMath exercises that guided our learning for each topic. Special thanks to Professor Jason Hardin for sharing his MyOpenMath question bank.

Chapter 6

Chapter Exercises

This chapter contains exercises for students to practice the concepts learned in the course. Each exercise is designed to reinforce understanding and application of each chapter's topics.

Worksheet 01: Vectors and the Geometry of Space

1. When do directional line segments in the plane represent the same vector?

Solution. Directional line segments represent the same vector if and only if they are **equivalent**. Geometrically, this means they must have the exact same **magnitude** (length) and the exact same **direction**, regardless of where their initial points are located in the coordinate plane.

2. How do you find a vector's direction?

Solution. The direction of a non-zero vector $\vec{v}$ is found by computing its corresponding **unit vector**, denoted as $\vec{u}$. You find this by dividing the vector by its scalar magnitude:

$$\vec{u} = \frac{\vec{v}}{\|\vec{v}\|}$$

Alternatively, in two dimensions, the direction can be specified by the angle θ it makes with the positive horizontal x-axis, calculated using $\tan(\theta) = \frac{v_2}{v_1}$.

3. What geometric interpretation does the dot product have? Give an example.

Solution. Geometrically, the dot product of two vectors measures the product of their lengths multiplied by the cosine of the angle between them: $\vec{a} \cdot \vec{b} = \|\vec{a}\|\|\vec{b}\| \cos(\theta)$. It can be interpreted as the length of $\vec{a}$ multiplied by the signed scalar projection of $\vec{b}$ onto $\vec{a}$ ($\text{comp}_{\vec{a}}\,\vec{b}$).

Example: If two non-zero vectors are perpendicular to each other ($\theta = \pi/2$), their dot product is exactly 0 because $\cos(\pi/2) = 0$. This provides a tool to test for orthogonality.

4. Find the component form of the vector that is obtained by rotating $\langle 0,1\rangle$ through an angle of $2\pi/3$ radians.

Solution. The initial vector $\vec{v}=\langle 0,1\rangle$ points straight up along the positive y-axis, meaning its initial direction angle is $\theta_0=\pi/2$. Rotating it by $2\pi/3$ updates its direction angle to:

$$\theta=\frac{\pi}{2}+\frac{2\pi}{3}=\frac{3\pi}{6}+\frac{4\pi}{6}=\frac{7\pi}{6}$$

Since the magnitude of $\langle 0,1\rangle$ is 1, the component form of the rotated unit vector is:

$$\vec{v}_{\text{rot}}=\left\langle \cos\left(\frac{7\pi}{6}\right),\sin\left(\frac{7\pi}{6}\right)\right\rangle=\left\langle -\frac{\sqrt{3}}{2},-\frac{1}{2}\right\rangle$$

5. Find the component form of the vector that is 2 units long in the direction of $4\vec{i}-\vec{j}$.

Solution. Let $\vec{w}=\langle 4,-1\rangle$. First, find the unit vector in the direction of $\vec{w}$ by calculating its magnitude:

$$\|\vec{w}\|=\sqrt{4^2+(-1)^2}=\sqrt{16+1}=\sqrt{17}$$

The direction unit vector is $\vec{u}=\frac{1}{\sqrt{17}}\langle 4,-1\rangle$.

Scaling this direction vector to a length of 2 yields:

$$\vec{v}=2\vec{u}=\frac{2}{\sqrt{17}}\langle 4,-1\rangle=\left\langle \frac{8}{\sqrt{17}},-\frac{2}{\sqrt{17}}\right\rangle$$

6. Express the vector in terms of its length and direction:

$$\vec{v}=(-2\sin(t))\vec{i}+(2\cos(t))\vec{j}\quad\text{when}\quad t=\pi/2$$

Solution. First evaluate the vector at the parameter value $t=\pi/2$:

$$\vec{v}=\left(-2\sin\left(\frac{\pi}{2}\right)\right)\vec{i}+\left(2\cos\left(\frac{\pi}{2}\right)\right)\vec{j}=-2(1)\vec{i}+2(0)\vec{j}=-2\vec{i}=\langle -2,0\rangle$$

Now decompose the vector into its length and direction components:

- Length (magnitude): $\|\vec{v}\|=\sqrt{(-2)^2+0^2}=2$
- Direction vector: $\vec{u}=\frac{\vec{v}}{\|\vec{v}\|}=\frac{\langle -2,0\rangle}{2}=\langle -1,0\rangle=-\vec{i}$

In factored form, the vector is modeled as $\vec{v}=2(-\vec{i})$.

7. Find $\text{proj}_{\vec{v}}\,\vec{u}$ if $\vec{v} = 2\vec{i} + \vec{j} - \vec{k}$ and $\vec{u} = \vec{i} + \vec{j} - 5\vec{k}$.

Solution. Write both vectors in component form: $\vec{v} = \langle 2, 1, -1\rangle$ and $\vec{u} = \langle 1, 1, -5\rangle$. Next, calculate the dot product $\vec{u} \cdot \vec{v}$ and the squared norm $\|\vec{v}\|^2$:

$$\vec{u} \cdot \vec{v} = (1)(2) + (1)(1) + (-5)(-1) = 2 + 1 + 5 = 8$$

$$\|\vec{v}\|^2 = 2^2 + 1^2 + (-1)^2 = 4 + 1 + 1 = 6$$

Substitute these calculations into the vector projection formula:

$$\text{proj}_{\vec{v}}\,\vec{u} = \left(\frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|^2}\right)\vec{v} = \frac{8}{6}\langle 2, 1, -1\rangle = \frac{4}{3}\langle 2, 1, -1\rangle = \left\langle \frac{8}{3}, \frac{4}{3}, -\frac{4}{3}\right\rangle$$

8. If $\|\vec{v}\| = 2$, $\|\vec{w}\| = 3$, and the angle between $\vec{v}$ and $\vec{w}$ is $\pi/3$, find $\|\vec{v} - 2\vec{w}\|$.

Solution. We can square the target norm to evaluate it using the algebraic properties of the dot product:

$$\begin{aligned}
\|\vec{v} - 2\vec{w}\|^2 &= (\vec{v} - 2\vec{w}) \cdot (\vec{v} - 2\vec{w}) \\
&= \vec{v} \cdot \vec{v} - 4(\vec{v} \cdot \vec{w}) + 4(\vec{w} \cdot \vec{w}) \\
&= \|\vec{v}\|^2 - 4(\|\vec{v}\|\|\vec{w}\| \cos(\theta)) + 4\|\vec{w}\|^2
\end{aligned}$$

Substitute the given magnitudes and the angle value into our expression:

$$\begin{aligned}
\|\vec{v} - 2\vec{w}\|^2 &= (2)^2 - 4\left(2 \cdot 3 \cdot \cos\left(\frac{\pi}{3}\right)\right) + 4(3)^2 \\
&= 4 - 4\left(6 \cdot \frac{1}{2}\right) + 4(9) \\
&= 4 - 12 + 36 = 28
\end{aligned}$$

Taking the square root gives the final length: $\|\vec{v} - 2\vec{w}\| = \sqrt{28} = 2\sqrt{7}$.

9. Find the volume of the parallelepiped determined by the vectors:

$$\vec{u} = \vec{i} + \vec{j} - \vec{k}, \quad \vec{v} = 2\vec{i} + \vec{j} + \vec{k}, \quad \vec{w} = -\vec{i} - 2\vec{j} + 3\vec{k}$$

Solution. The volume of a parallelepiped matches the absolute value of the scalar triple product, computed using a single 3×3 matrix determinant:

$$V = \left\| \begin{matrix} 1 & 1 & -1 \\ 2 & 1 & 1 \\ -1 & -2 & 3 \end{matrix} \right\|$$

Evaluate the matrix determinant by expanding along its top row:

$$\begin{aligned}
\begin{vmatrix} 1 & 1 & -1 \\ 2 & 1 & 1 \\ -1 & -2 & 3 \end{vmatrix} &= 1\begin{vmatrix} 1 & 1 \\ -2 & 3 \end{vmatrix} - 1\begin{vmatrix} 2 & 1 \\ -1 & 3 \end{vmatrix} + (-1)\begin{vmatrix} 2 & 1 \\ -1 & -2 \end{vmatrix} \\
&= 1(3 - (-2)) - 1(6 - (-1)) - 1(-4 - (-1)) \\
&= 1(5) - 1(7) - 1(-3) = 5 - 7 + 3 = 1
\end{aligned}$$

Taking the absolute value, the volume of the parallelepiped is $V = 1$ cubic unit.

10. Find the distance from the point to the line:

$$(2,2,0); \quad x=-t, \quad y=t, \quad z=-1+t$$

Solution. Let the target point be $S(2,2,0)$. From the parametric equations, we read a point on the line by setting $t=0$, giving $P(0,0,-1)$, and extract the line's direction vector: $\vec{v}=\langle -1,1,1\rangle$.

Construct the displacement vector from P to S:

$$\vec{PS}=\langle 2-0, 2-0, 0-(-1)\rangle = \langle 2,2,1\rangle$$

The formula for the distance from a point to a line is $d=\frac{\|\vec{PS}\times\vec{v}\|}{\|\vec{v}\|}$. First, compute the cross product:

$$\vec{PS}\times\vec{v}=\begin{vmatrix}\vec{i} & \vec{j} & \vec{k}\\ 2 & 2 & 1\\ -1 & 1 & 1\end{vmatrix}=(2-1)\vec{i}-(2-(-1))\vec{j}+(2-(-2))\vec{k}=\langle 1,-3,4\rangle$$

Now calculate the magnitudes:

$$\|\vec{PS}\times\vec{v}\|=\sqrt{1^2+(-3)^2+4^2}=\sqrt{1+9+16}=\sqrt{26}$$

$$\|\vec{v}\|=\sqrt{(-1)^2+1^2+1^2}=\sqrt{3}$$

Thus, the distance from the point to the line is $d=\frac{\sqrt{26}}{\sqrt{3}}=\sqrt{\frac{26}{3}}$.

11. Find the equation for the plane that passes through the point $(3,-2,1)$ and is normal to the vector $\vec{n}=2\vec{i}+\vec{j}+\vec{k}$.

Solution. Substitute the point coordinates $(3,-2,1)$ and the normal vector components $\langle 2,1,1\rangle$ directly into the standard scalar plane equation:

$$2(x-3)+1(y-(-2))+1(z-1)=0$$

Distribute and collect constant terms:

$$2x-6+y+2+z-1=0 \implies 2x+y+z-5=0$$

In standard linear form, the plane equation is $2x+y+z=5$.

12. Find the equation to the plane through the points:

$$P(1,-1,2), \quad Q(2,1,3), \quad R(-1,2,-1)$$

Solution. First, construct two directional vectors sharing vertex P that lie entirely within the plane:

$$\vec{PQ} = \langle 2-1, 1-(-1), 3-2\rangle = \langle 1,2,1\rangle$$

$$\vec{PR} = \langle -1-1, 2-(-1), -1-2\rangle = \langle -2,3,-3\rangle$$

The normal vector $\vec{n}$ is perpendicular to both vector components, found via their cross product:

$$\vec{n} = \vec{PQ} \times \vec{PR} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 2 & 1 \\ -2 & 3 & -3 \end{vmatrix} = (-6-3)\vec{i} - (-3-(-2))\vec{j} + (3-(-4))\vec{k} = \langle -9,1,7\rangle$$

Using point $P(1,-1,2)$ and our calculated normal vector, set up the plane equation:

$$-9(x-1) + 1(y-(-1)) + 7(z-2) = 0 \implies -9x + 9 + y + 1 + 7z - 14 = 0$$

Simplifying the expression yields the linear equation: $-9x + y + 7z - 4 = 0$, or equivalently, $9x - y - 7z = -4$.

13. Find the distance from the point $(1,4,0)$ to the plane passing through $A(0,0,0)$, $B(2,0,-1)$, and $C(2,-1,0)$.

Solution. Construct two spanning vectors from the given points inside the plane:

$$\vec{AB} = \langle 2,0,-1\rangle \quad \text{and} \quad \vec{AC} = \langle 2,-1,0\rangle$$

Compute their cross product to find the normal vector direction $\vec{n}$:

$$\vec{n} = \vec{AB} \times \vec{AC} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 0 & -1 \\ 2 & -1 & 0 \end{vmatrix} = (0-1)\vec{i} - (0-(-2))\vec{j} + (-2-0)\vec{k} = \langle -1,-2,-2\rangle$$

Since the plane passes through the origin $A(0,0,0)$, its linear equation simplifies to:

$$-1(x) - 2(y) - 2(z) = 0 \implies x + 2y + 2z = 0$$

Apply the point-plane distance formula using our target point $(1,4,0)$:

$$D = \frac{|1(1) + 2(4) + 2(0) + 0|}{\sqrt{1^2 + 2^2 + 2^2}} = \frac{|1+8|}{\sqrt{1+4+4}} = \frac{9}{\sqrt{9}} = \frac{9}{3} = 3$$

The exact distance from the point to the plane is 3 units.

14. Find the parametric equations for the line in which the planes $x + 2y + z = 1$ and $x - y + 2z = -8$ intersect.

Solution. The direction vector $\vec{v}$ of the intersection line is perpendicular to the normal vectors of both planes: $\vec{n}_1 = \langle 1, 2, 1\rangle$ and $\vec{n}_2 = \langle 1, -1, 2\rangle$. Compute their cross product:

$$\vec{v} = \vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 2 & 1 \\ 1 & -1 & 2 \end{vmatrix} = (4 - (-1))\vec{i} - (2 - 1)\vec{j} + (-1 - 2)\vec{k} = \langle 5, -1, -3\rangle$$

To locate an initial point on the line, choose a convenient coordinate value, such as setting $z = 0$. This reduces the equations of the planes to a system of two variables:

$$x + 2y = 1 \quad \text{and} \quad x - y = -8$$

Subtracting the second equation from the first yields $3y = 9 \implies y = 3$. Substituting $y = 3$ back gives $x = -5$. This provides the point coordinates $(-5, 3, 0)$.

Combining our initial position point and the direction numbers yields the **parametric equations**:

$$x = -5 + 5t, \quad y = 3 - t, \quad z = -3t$$

15. Find a vector of magnitude 2 parallel to the line of intersection of the planes $x+2y+z-1 = 0$ and $x-y+2z+7 = 0$.

Solution. The direction vector $\vec{v}$ of the line of intersection is perpendicular to both normal vectors, $\vec{n}_1 = \langle 1, 2, 1\rangle$ and $\vec{n}_2 = \langle 1, -1, 2\rangle$. As calculated in Exercise 13, this cross product is:

$$\vec{v} = \vec{n}_1 \times \vec{n}_2 = \langle 5, -1, -3\rangle$$

To scale this direction to a specific length, first determine its unit vector by computing its magnitude:

$$\|\vec{v}\| = \sqrt{5^2 + (-1)^2 + (-3)^2} = \sqrt{25 + 1 + 9} = \sqrt{35}$$

The corresponding unit vector is $\vec{u} = \frac{1}{\sqrt{35}}\langle 5, -1, -3\rangle$.

Multiplying the direction unit vector by our target magnitude of 2 yields the final vector:

$$\vec{v}_{\text{final}} = \pm 2\vec{u} = \pm\frac{2}{\sqrt{35}}\langle 5, -1, -3\rangle = \pm\left\langle \frac{10}{\sqrt{35}}, -\frac{2}{\sqrt{35}}, -\frac{6}{\sqrt{35}} \right\rangle$$

Either the positive or negative scalar multiple satisfies the geometric condition.

Worksheet 02: Vector Functions

1. State the rules for differentiating and integrating vector functions. Give examples.

Solution. Let $\mathbf{u}(t)$ and $\mathbf{v}(t)$ be differentiable vector functions, and let $f(t)$ be a differentiable scalar function. The key derivative rules are:

(a) Sum Rule: $\frac{d}{dt}[\mathbf{u}(t)+\mathbf{v}(t)] = \mathbf{u}'(t)+\mathbf{v}'(t)$

(b) Scalar Product Rule: $\frac{d}{dt}[f(t)\mathbf{u}(t)] = f'(t)\mathbf{u}(t)+f(t)\mathbf{u}'(t)$

(c) Dot Product Rule: $\frac{d}{dt}[\mathbf{u}(t)\cdot\mathbf{v}(t)] = \mathbf{u}'(t)\cdot\mathbf{v}(t)+\mathbf{u}(t)\cdot\mathbf{v}'(t)$

(d) Cross Product Rule: $\frac{d}{dt}[\mathbf{u}(t)\times\mathbf{v}(t)] = \mathbf{u}'(t)\times\mathbf{v}(t)+\mathbf{u}(t)\times\mathbf{v}'(t)$

For integration, if $\mathbf{r}(t) = \langle f(t), g(t), h(t)\rangle$, the definite integral is calculated component-wise:

$$\int_a^b \mathbf{r}(t)\,dt = \left\langle \int_a^b f(t)\,dt,\ \int_a^b g(t)\,dt,\ \int_a^b h(t)\,dt \right\rangle$$

Example: If $\mathbf{r}(t) = \langle t^2, e^t, \sin t\rangle$, then $\mathbf{r}'(t) = \langle 2t, e^t, \cos t\rangle$ and $\int \mathbf{r}(t)\,dt = \langle \frac{t^3}{3}, e^t, -\cos t\rangle + \mathbf{C}$.

2. How do you define and calculate the length of a segment of a smooth space curve? Give an example.

Solution. The arc length L of a smooth space curve traced by $\mathbf{r}(t)$ from $t=a$ to $t=b$ is defined as the integral of its scalar speed:

$$L = \int_a^b \|\mathbf{r}'(t)\|\,dt = \int_a^b \sqrt{[x'(t)]^2+[y'(t)]^2+[z'(t)]^2}\,dt$$

Example: Find the length of $\mathbf{r}(t) = \langle 3t, \sin t, \cos t\rangle$ from $t=0$ to $t=2$. The derivative is $\mathbf{r}'(t) = \langle 3, \cos t, -\sin t\rangle$. Its magnitude is $\|\mathbf{r}'(t)\| = \sqrt{3^2+\cos^2 t+(-\sin t)^2} = \sqrt{9+1} = \sqrt{10}$. Integrating over the interval yields:

$$L = \int_0^2 \sqrt{10}\,dt = 2\sqrt{10}$$

3. What is a differentiable curve's unit tangent vector? Give an example.

Solution. The unit tangent vector $\mathbf{T}(t)$ represents the direction of motion along a smooth curve, normalized to have a constant length of 1. It is defined as:

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|}$$

provided that $\mathbf{r}'(t) \neq \mathbf{0}$.

Example: For $\mathbf{r}(t) = \langle t, t^2\rangle$ at $t=1$. The tangent vector is $\mathbf{r}'(t) = \langle 1, 2t\rangle \implies \mathbf{r}'(1) = \langle 1, 2\rangle$. Its magnitude is $\|\mathbf{r}'(1)\| = \sqrt{1^2+2^2} = \sqrt{5}$. Thus, the unit tangent vector is:

$$\mathbf{T}(1) = \frac{1}{\sqrt{5}}\langle 1, 2\rangle = \left\langle \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right\rangle$$

4. Write $\mathbf{a}$ in the form $\mathbf{a} = a_T\mathbf{T} + a_N\mathbf{N}$ without finding $\mathbf{T}$ and $\mathbf{N}$, and find the value of κ at the given value of t:

$$\mathbf{r}(t) = (4\cos t)\mathbf{i} + (\sqrt{2}\sin t)\mathbf{j}, \qquad t = \frac{\pi}{4}$$

Solution. We differentiate the position vector function to find velocity $\mathbf{v}(t)$ and acceleration $\mathbf{a}(t)$:

$$\mathbf{v}(t) = \langle -4\sin t, \sqrt{2}\cos t\rangle \implies \mathbf{v}\left(\frac{\pi}{4}\right) = \left\langle -4\left(\frac{\sqrt{2}}{2}\right), \sqrt{2}\left(\frac{\sqrt{2}}{2}\right)\right\rangle = \langle -2\sqrt{2}, 1\rangle$$

$$\mathbf{a}(t) = \langle -4\cos t, -\sqrt{2}\sin t\rangle \implies \mathbf{a}\left(\frac{\pi}{4}\right) = \left\langle -4\left(\frac{\sqrt{2}}{2}\right), -\sqrt{2}\left(\frac{\sqrt{2}}{2}\right)\right\rangle = \langle -2\sqrt{2}, -1\rangle$$

Compute the scalar speed tracking parameters at $t = \frac{\pi}{4}$:

$$v = \|\mathbf{v}\| = \sqrt{(-2\sqrt{2})^2 + 1^2} = \sqrt{8+1} = 3$$

The tangential component of acceleration a_T is found using a projection dot product:

$$a_T = \frac{\mathbf{v}\cdot\mathbf{a}}{\|\mathbf{v}\|} = \frac{(-2\sqrt{2})(-2\sqrt{2}) + (1)(-1)}{3} = \frac{8-1}{3} = \frac{7}{3}$$

To find the normal acceleration component a_N, use the relationship $\|\mathbf{a}\|^2 = a_T^2 + a_N^2$. First, find $\|\mathbf{a}\|^2 = (-2\sqrt{2})^2 + (-1)^2 = 8 + 1 = 9$:

$$a_N = \sqrt{\|\mathbf{a}\|^2 - a_T^2} = \sqrt{9 - \left(\frac{7}{3}\right)^2} = \sqrt{9 - \frac{49}{9}} = \sqrt{\frac{32}{9}} = \frac{4\sqrt{2}}{3}$$

Thus, the acceleration decomposition is:

$$\mathbf{a} = \frac{7}{3}\mathbf{T} + \frac{4\sqrt{2}}{3}\mathbf{N}$$

Finally, calculate the curvature using the normal acceleration relationship $a_N = \kappa v^2$:

$$\kappa = \frac{a_N}{v^2} = \frac{(4\sqrt{2})/3}{3^2} = \frac{4\sqrt{2}}{27}$$

5. The position of the particle in the plane at time t is:

$$\mathbf{r} = \frac{1}{\sqrt{1+t^2}}\mathbf{i} + \frac{t}{\sqrt{1+t^2}}\mathbf{j}$$

Find the particle's highest speed.

Solution. Notice that the coordinates describe a unit circle since $x^2 + y^2 = \frac{1}{1+t^2} + \frac{t^2}{1+t^2} = 1$. Differentiate each component to evaluate the velocity vector:

$$x'(t) = -\frac{1}{2}(1+t^2)^{-3/2}(2t) = \frac{-t}{(1+t^2)^{3/2}}$$

$$y'(t) = \frac{1\cdot\sqrt{1+t^2} - t\cdot\frac{t}{\sqrt{1+t^2}}}{1+t^2} = \frac{(1+t^2)-t^2}{(1+t^2)^{3/2}} = \frac{1}{(1+t^2)^{3/2}}$$

$$\mathbf{v}(t) = \left\langle \frac{-t}{(1+t^2)^{3/2}}, \frac{1}{(1+t^2)^{3/2}}\right\rangle$$

Compute the scalar speed function $v(t) = \|\mathbf{v}(t)\|$:

$$v(t) = \sqrt{\frac{t^2}{(1+t^2)^3} + \frac{1}{(1+t^2)^3}} = \sqrt{\frac{t^2+1}{(1+t^2)^3}} = \sqrt{\frac{1}{(1+t^2)^2}} = \frac{1}{1+t^2}$$

To find the maximum speed value, look at the denominator of $v(t) = \frac{1}{1+t^2}$. The fraction reaches its largest value when its denominator is minimized, which occurs exactly at $t = 0$. Therefore, the particle's highest speed is:

$$v_{\max} = v(0) = \frac{1}{1+0^2} = 1$$

6. A shot leaves the thrower's hand 6.5 ft above the ground at a 45° angle at 44 ft/sec. Where is it 2 sec later?

Solution. Using standard ground kinematics parameters for English units, we have an initial position height $h_0 = 6.5$ ft, launch speed $v_0 = 44$ ft/sec, elevation angle $\alpha = 45°$, and constant gravitational acceleration value $g = 32\,\text{ft/sec}^2$. The horizontal and vertical initial velocity components are:

$$v_{0x} = 44\cos(45°) = 44\left(\frac{\sqrt{2}}{2}\right) = 22\sqrt{2}\,\text{ft/sec}$$

$$v_{0y} = 44\sin(45°) = 44\left(\frac{\sqrt{2}}{2}\right) = 22\sqrt{2}\,\text{ft/sec}$$

Set up the general equations for horizontal and vertical position tracking over time t:

$$x(t) = (v_0\cos\alpha)t = 22\sqrt{2}t$$

$$y(t) = h_0 + (v_0\sin\alpha)t - \frac{1}{2}gt^2 = 6.5 + 22\sqrt{2}t - 16t^2$$

Evaluate both position tracking equations at the target parameter time $t = 2$ seconds:

$$x(2) = 22\sqrt{2}(2) = 44\sqrt{2} \approx 62.23\,\text{feet}$$

$$y(2) = 6.5 + 22\sqrt{2}(2) - 16(2)^2 = 6.5 + 44\sqrt{2} - 64 \approx 6.5 + 62.23 - 64 = 4.73\,\text{feet}$$

Two seconds later, the shot is located $44\sqrt{2}$ feet horizontally downrange from the thrower and at a vertical height of approximately 4.73 feet above the ground.

7. Find the length of the curve:

$$\mathbf{r}(t) = (2\cos t)\mathbf{i} + (2\sin t)\mathbf{j} + t^2\mathbf{k}, \quad 0 \le t \le \frac{\pi}{4}$$

Solution. First, differentiate the vector function components to track velocity:

$$\mathbf{r}'(t) = \langle -2\sin t,\, 2\cos t,\, 2t\rangle$$

Next, compute the scalar speed magnitude function using the identity $\sin^2 t + \cos^2 t = 1$:

$$\|\mathbf{r}'(t)\| = \sqrt{(-2\sin t)^2 + (2\cos t)^2 + (2t)^2} = \sqrt{4\sin^2 t + 4\cos^2 t + 4t^2} = \sqrt{4(1) + 4t^2} = 2\sqrt{1+t^2}$$

Set up the arc length integration array from $a = 0$ to $b = \frac{\pi}{4}$:

$$L = \int_0^{\frac{\pi}{4}} 2\sqrt{1+t^2}\,dt$$

Apply the standard trigonometric integration table substitution form $\int \sqrt{1+t^2}\,dt = \frac{t}{2}\sqrt{1+t^2} + \frac{1}{2}\ln|t + \sqrt{1+t^2}|$:

$$L = 2\left[\frac{t}{2}\sqrt{1+t^2} + \frac{1}{2}\ln\left(t + \sqrt{1+t^2}\right)\right]_0^{\frac{\pi}{4}} = \left[t\sqrt{1+t^2} + \ln\left(t + \sqrt{1+t^2}\right)\right]_0^{\frac{\pi}{4}}$$

$$L = \frac{\pi}{4}\sqrt{1 + \frac{\pi^2}{16}} + \ln\left(\frac{\pi}{4} + \sqrt{1 + \frac{\pi^2}{16}}\right) - (0 + \ln(1))$$

$$L = \frac{\pi\sqrt{16+\pi^2}}{16} + \ln\left(\frac{\pi + \sqrt{16+\pi^2}}{4}\right) \approx 0.816\,\text{units}$$

8. Find $\mathbf{T}$, $\mathbf{N}$, and κ at the given t:

$$\mathbf{r}(t) = \frac{4}{9}(1+t)^{3/2}\mathbf{i} + \frac{4}{9}(1-t)^{3/2}\mathbf{j} + \frac{1}{3}t\mathbf{k}, \qquad t = 0$$

Solution. Compute the derivative vector function to track velocity:

$$\mathbf{r}'(t) = \left\langle \frac{4}{9}\cdot\frac{3}{2}(1+t)^{1/2},\ \frac{4}{9}\cdot\frac{3}{2}(1-t)^{1/2}(-1),\ \frac{1}{3} \right\rangle = \left\langle \frac{2}{3}\sqrt{1+t},\ -\frac{2}{3}\sqrt{1-t},\ \frac{1}{3} \right\rangle$$

Evaluating velocity at our target parameter point $t = 0$ yields:

$$\mathbf{r}'(0) = \left\langle \frac{2}{3}(1),\ -\frac{2}{3}(1),\ \frac{1}{3} \right\rangle = \left\langle \frac{2}{3},\ -\frac{2}{3},\ \frac{1}{3} \right\rangle$$

Compute the speed magnitude value:

$$\|\mathbf{r}'(0)\| = \sqrt{\left(\frac{2}{3}\right)^2 + \left(-\frac{2}{3}\right)^2 + \left(\frac{1}{3}\right)^2} = \sqrt{\frac{4}{9} + \frac{4}{9} + \frac{1}{9}} = \sqrt{\frac{9}{9}} = 1$$

Thus, the unit tangent vector at $t = 0$ matches the velocity vector directly:

$$\mathbf{T}(0) = \left\langle \frac{2}{3},\ -\frac{2}{3},\ \frac{1}{3} \right\rangle$$

To find $\mathbf{N}(0)$, look back at the general velocity profile magnitude function:

$$\|\mathbf{r}'(t)\| = \sqrt{\frac{4}{9}(1+t) + \frac{4}{9}(1-t) + \frac{1}{9}} = \sqrt{\frac{4}{9} + \frac{4}{9}t + \frac{4}{9} - \frac{4}{9}t + \frac{1}{9}} = \sqrt{\frac{9}{9}} = 1$$

Since speed is identically 1 for all values of t, the general unit tangent equation reduces simply to $\mathbf{T}(t) = \mathbf{r}'(t)$. Differentiating a second time gives $\mathbf{T}'(t)$:

$$\mathbf{T}'(t) = \mathbf{r}''(t) = \left\langle \frac{1}{3\sqrt{1+t}},\ \frac{1}{3\sqrt{1-t}},\ 0 \right\rangle$$

Evaluate this vector layout at $t = 0$:

$$\mathbf{T}'(0) = \left\langle \frac{1}{3},\ \frac{1}{3},\ 0 \right\rangle \implies \|\mathbf{T}'(0)\| = \sqrt{\frac{1}{9} + \frac{1}{9} + 0} = \frac{\sqrt{2}}{3}$$

Normalize this derivative vector to find the unit normal vector:

$$\mathbf{N}(0) = \frac{\mathbf{T}'(0)}{\|\mathbf{T}'(0)\|} = \frac{3}{\sqrt{2}}\left\langle \frac{1}{3},\ \frac{1}{3},\ 0 \right\rangle = \left\langle \frac{1}{\sqrt{2}},\ \frac{1}{\sqrt{2}},\ 0 \right\rangle$$

Finally, calculate the curvature using the relationship $\kappa(t) = \frac{\|\mathbf{T}'(t)\|}{\|\mathbf{r}'(t)\|}$:

$$\kappa(0) = \frac{\sqrt{2}/3}{1} = \frac{\sqrt{2}}{3}$$

9. Find the parametric equations for the line that is tangent to the curve:

$$\mathbf{r}(t) = e^t\mathbf{i} + (\sin t)\mathbf{j} + \ln(1-t)\mathbf{k}, \quad t = 0$$

Solution. First, evaluate the initial point vector coordinates on the curve at $t = 0$:

$$\mathbf{r}(0) = \langle e^0,\ \sin(0),\ \ln(1-0)\rangle = \langle 1,\ 0,\ \ln(1)\rangle = \langle 1, 0, 0\rangle$$

This gives us the base point coordinates $(x_0, y_0, z_0) = (1, 0, 0)$.

Next, differentiate the vector function components with respect to t using the chain rule to locate the direction vector profile:

$$\mathbf{r}'(t) = \left\langle e^t,\, \cos t,\, \frac{1}{1-t}(-1) \right\rangle = \left\langle e^t,\, \cos t,\, \frac{1}{t-1} \right\rangle$$

Evaluating the derivative vector at our target parameter point $t = 0$ gives the direction coefficients for the tangent line:

$$\mathbf{r}'(0) = \left\langle e^0,\, \cos(0),\, \frac{1}{0-1} \right\rangle = \langle 1,\, 1,\, -1 \rangle$$

This gives the parallel direction vector components $\langle a, b, c \rangle = \langle 1, 1, -1 \rangle$.

Using the base point $(1, 0, 0)$ and direction components $\langle 1, 1, -1 \rangle$, the parametric equations for the tangent line are:

$$x = 1 + t, \qquad y = t, \qquad z = -t$$

Worksheet 03: Partial Derivatives

1. Show that the following limit does not exist:

$$\lim_{\substack{(x,y)\to(0,0)\\ y\neq x^2}} \frac{y}{x^2-y}$$

Solution. Let $f(x,y)=\frac{y}{x^2-y}$. We test the behavior of this function along different paths approaching the origin $(0,0)$.

First, let us approach the origin along the linear path $y=mx$ where $m\neq 0$:

$$\lim_{x\to 0}\frac{mx}{x^2-mx}=\lim_{x\to 0}\frac{m}{x-m}=\frac{m}{-m}=-1$$

Along every straight line of non-zero slope, the function approaches -1. If we approach along the line $y=0$ (the x-axis), the function is identically 0, giving a path limit of 0.

Since approaching along $y=0$ yields 0, while approaching along $y=mx$ yields -1, the limits along two distinct paths are different. By the two-path rule, the limit does not exist.

2. Find the limit:

$$\lim_{(x,y)\to(0,0)}\frac{x^2-y^2}{x^2+y^2}$$

Solution. Let $f(x,y)=\frac{x^2-y^2}{x^2+y^2}$. We test the limit along the coordinate axes:

- Along the x-axis ($y=0$ for $x\neq 0$): $f(x,0)=\frac{x^2-0}{x^2+0}=1$. Thus, the limit along this path is 1.
- Along the y-axis ($x=0$ for $y\neq 0$): $f(0,y)=\frac{0-y^2}{0+y^2}=-1$. Thus, the limit along this path is -1.

Since the function approaches two different values along two different geometric paths, the limit does not exist.

3. Find the extreme values of $f(x, y, z) = x - y + z$ on the unit sphere $x^2 + y^2 + z^2 = 1$.

Solution. We use the method of Lagrange Multipliers with constraint function $g(x, y, z) = x^2 + y^2 + z^2 - 1 = 0$. We set $\nabla f = \lambda \nabla g$:

$$\langle 1, -1, 1\rangle = \lambda\langle 2x, 2y, 2z\rangle$$

This gives the system: $1 = 2\lambda x$, $-1 = 2\lambda y$, and $1 = 2\lambda z$. Isolating the variables, we find:

$$x = \frac{1}{2\lambda}, \quad y = -\frac{1}{2\lambda}, \quad z = \frac{1}{2\lambda}$$

Substituting these into the constraint sphere equation yields:

$$\left(\frac{1}{2\lambda}\right)^2 + \left(-\frac{1}{2\lambda}\right)^2 + \left(\frac{1}{2\lambda}\right)^2 = 1 \implies \frac{3}{4\lambda^2} = 1 \implies \lambda = \pm\frac{\sqrt{3}}{2}$$

This produces two critical points: $\left(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$ and $\left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right)$. Evaluating f:

$$f_{\text{max}} = \frac{1}{\sqrt{3}} - \left(-\frac{1}{\sqrt{3}}\right) + \frac{1}{\sqrt{3}} = \frac{3}{\sqrt{3}} = \sqrt{3}$$

$$f_{\text{min}} = -\frac{1}{\sqrt{3}} - \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{3}} = -\sqrt{3}$$

4. Find $\frac{\partial w}{\partial r}$ and $\frac{\partial w}{\partial s}$ when $r = \pi$ and $s = 0$, if:

$$w = \sin(2x - y), \quad x = r + \sin(s), \quad y = rs$$

Solution. When $r = \pi$ and $s = 0$, we find the intermediate coordinate states: $x = \pi + \sin(0) = \pi$ and $y = \pi(0) = 0$. The first partial derivatives of w are:

$$\frac{\partial w}{\partial x} = 2\cos(2x - y) = 2\cos(2\pi) = 2, \quad \frac{\partial w}{\partial y} = -\cos(2x - y) = -\cos(2\pi) = -1$$

Using the Multivariate Chain Rule for the parameter r:

$$\frac{\partial w}{\partial r} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial r} = (2)(1) + (-1)(s) = 2 - s$$

At $s = 0$, $\frac{\partial w}{\partial r} = 2$.

Using the Chain Rule for the parameter s:

$$\frac{\partial w}{\partial s} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial s} = (2)(\cos s) + (-1)(r) = 2\cos s - r$$

At $r = \pi$ and $s = 0$, $\frac{\partial w}{\partial s} = 2\cos(0) - \pi = 2 - \pi$.

5. Find the value of the derivative of $f(x, y, z) = xy + yz + xz$ with respect to t along the parameter path $x = \cos(t)$, $y = \sin(t)$, and $z = \cos(2t)$ at $t = 1$.

Solution. By applying the Multivariate Chain Rule Case I, the total derivative with respect to time is:

$$\frac{df}{dt} = (y + z)(-\sin t) + (x + z)(\cos t) + (y + x)(-2\sin(2t))$$

Substituting $x = \cos(1)$, $y = \sin(1)$, and $z = \cos(2)$ directly into this expression evaluated at $t = 1$ yields:

$$\begin{aligned}\frac{df}{dt} &= -\sin(1)[\sin(1) + \cos(2)] + \cos(1)[\cos(1) + \cos(2)] - 2\sin(2)[\sin(1) + \cos(1)] \\ &= \cos^2(1) - \sin^2(1) + \cos(2)[\cos(1) - \sin(1)] - 2\sin(2)[\sin(1) + \cos(1)] \\ &= \cos(2) + \cos(2)[\cos(1) - \sin(1)] - 2\sin(2)[\sin(1) + \cos(1)]\end{aligned}$$

Expressed as a clean single decimal value approximation, $\frac{df}{dt} \approx -3.0741$.

6. Find the directions in which $f(x, y, z) = \ln(2x + 3y + 6z)$ increases and decreases most rapidly at the point $P_0(-1, -1, 1)$. Then, find the directional derivative of f at P_0 in the direction of the vector $\mathbf{v} = 2\mathbf{i} + 3\mathbf{j} + 6\mathbf{k}$.

Solution. We find the first-order partial derivatives forming the components of the gradient vector:

$$\nabla f = \left\langle \frac{2}{2x + 3y + 6z}, \frac{3}{2x + 3y + 6z}, \frac{6}{2x + 3y + 6z} \right\rangle$$

At $P_0(-1, -1, 1)$, the inner sum denominator expression evaluates to $2(-1) + 3(-1) + 6(1) = 1$. Thus, the gradient vector is $\nabla f(-1, -1, 1) = \langle 2, 3, 6\rangle$.

- The function increases most rapidly in the direction of the gradient: $\langle 2, 3, 6\rangle$.
- The function decreases most rapidly in the opposite direction: $\langle -2, -3, -6\rangle$.

The direction vector $\mathbf{v} = \langle 2, 3, 6\rangle$ perfectly matches the direction of the gradient. Normalizing $\mathbf{v}$ via its magnitude $\|\mathbf{v}\| = \sqrt{2^2 + 3^2 + 6^2} = \sqrt{49} = 7$ gives the unit vector $\mathbf{u} = \frac{1}{7}\langle 2, 3, 6\rangle$. The directional derivative is the magnitude of the gradient vector:

$$D_{\mathbf{u}} f(-1, -1, 1) = \|\nabla f\| = \sqrt{2^2 + 3^2 + 6^2} = 7$$

7. Find the directional derivative of $f(x, y, z) = xyz$ in the direction of the velocity vector of the helix:

$$\mathbf{r}(t) = (\cos 3t)\mathbf{i} + (\sin 3t)\mathbf{j} + 3t\mathbf{k} \quad \text{at} \quad t = \frac{\pi}{3}$$

Solution. First, we compute the velocity vector $\mathbf{v}(t) = \mathbf{r}'(t) = \langle -3\sin(3t), 3\cos(3t), 3\rangle$. Evaluating this at $t = \pi/3$:

$$\mathbf{v}(\pi/3) = \langle -3\sin(\pi), 3\cos(\pi), 3\rangle = \langle 0, -3, 3\rangle$$

Normalizing this direction vector yields our unit vector $\mathbf{u}$:

$$\|\mathbf{v}\| = \sqrt{0^2 + (-3)^2 + 3^2} = \sqrt{18} = 3\sqrt{2} \implies \mathbf{u} = \left\langle 0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\rangle$$

The coordinate point on the helix at $t = \pi/3$ is $x = \cos(\pi) = -1$, $y = \sin(\pi) = 0$, and $z = 3(\pi/3) = \pi$. The gradient of $f(x, y, z) = xyz$ is $\nabla f = \langle yz, xz, xy\rangle$. Evaluating the gradient at $(-1, 0, \pi)$ yields $\nabla f = \langle 0, -\pi, 0\rangle$.

Finally, the directional derivative is computed via the vector dot product:

$$D_{\mathbf{u}}f = \langle 0, -\pi, 0\rangle \cdot \left\langle 0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\rangle = \frac{\pi}{\sqrt{2}} = \frac{\pi\sqrt{2}}{2} \approx 2.2214$$

8. Find an equation for the plane tangent to the level surface $x^2 - y - 5z = 0$ at the point $P_0(2, -1, 1)$. Additionally, find parametric equations for the normal line to the surface passing through P_0.

Solution. Let $F(x, y, z) = x^2 - y - 5z$. We find the components of the normal vector by evaluating the gradient $\nabla F = \langle 2x, -1, -5\rangle$ at $P_0(2, -1, 1)$:

$$\nabla F(2, -1, 1) = \langle 2(2), -1, -5\rangle = \langle 4, -1, -5\rangle$$

Using the point-normal equation, the equation of the tangent plane is:

$$4(x - 2) - 1(y - (-1)) - 5(z - 1) = 0 \implies 4x - y - 5z = 4$$

The normal line passes through $(2, -1, 1)$ and moves parallel to the direction of the normal vector $\langle 4, -1, -5\rangle$. Thus, its parametric equations are:

$$x = 2 + 4t, \quad y = -1 - t, \quad z = 1 - 5t \quad \text{for } t \in \mathbb{R}$$

9. Find an equation for the plane tangent to the explicit surface $z = \ln(x^2 + y^2)$ at the point $(0, 1, 0)$.

Solution. We rewrite the equation as a level surface function: $F(x, y, z) = \ln(x^2 + y^2) - z = 0$. We find the gradient vector by computing the first-order partial derivatives:

$$F_x = \frac{2x}{x^2 + y^2}, \quad F_y = \frac{2y}{x^2 + y^2}, \quad F_z = -1$$

Evaluating these partial derivatives at the given point $(0, 1, 0)$ yields:

$$F_x(0, 1, 0) = \frac{0}{1} = 0, \quad F_y(0, 1, 0) = \frac{2(1)}{1} = 2, \quad F_z(0, 1, 0) = -1$$

This gives the normal vector $\langle 0, 2, -1\rangle$. Using the point-normal equation, the equation of the tangent plane is:

$$0(x - 0) + 2(y - 1) - 1(z - 0) = 0 \implies 2y - z = 2$$

10. Find an equation for the line tangent and normal to the level curve $y - \sin(x) = 1$ at the point $P_0(\pi, 1)$.

Solution. Let our level curve function be defined as $f(x, y) = y - \sin(x)$. The gradient vector is orthogonal to the level curve and is given by:

$$\nabla f = \langle -\cos x,\ 1\rangle \implies \nabla f(\pi, 1) = \langle -\cos(\pi),\ 1\rangle = \langle 1, 1\rangle$$

The gradient vector $\langle 1, 1\rangle$ serves as the normal vector to the tangent line. Thus, the equation of the tangent line passing through $(\pi, 1)$ is:

$$1(x - \pi) + 1(y - 1) = 0 \implies x + y = \pi + 1$$

The normal line moves parallel to the gradient vector direction $\langle 1, 1\rangle$. Thus, its linear equation is found by using an orthogonal slope direction:

$$1(x - \pi) - 1(y - 1) = 0 \implies x - y = \pi - 1$$

11. Find parametric equations for the line that is tangent to the curve of intersection of the surfaces $x^2+2y+2z=4$ and $y=1$ at the point $(1,1,1/2)$.

Solution. The direction of the line tangent to the intersection curve of two surfaces is perpendicular to the normal vectors of both surfaces simultaneously. We can find this direction vector $\mathbf{v}$ by computing the cross product of their gradients. Let $F(x,y,z)=x^2+2y+2z-4$ and $G(x,y,z)=y-1$:

$$\nabla F = \langle 2x, 2, 2\rangle \implies \nabla F(1,1,1/2)=\langle 2,2,2\rangle$$

$$\nabla G = \langle 0, 1, 0\rangle \implies \nabla G(1,1,1/2)=\langle 0,1,0\rangle$$

Computing the vector cross product:

$$\mathbf{v}=\nabla F\times\nabla G=\begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ 2 & 2 & 2\\ 0 & 1 & 0\end{vmatrix}=-2\mathbf{i}+0\mathbf{j}+2\mathbf{k}=\langle -2,0,2\rangle$$

We can scale this parallel vector to $\langle -1,0,1\rangle$. The tangent line passes through $(1,1,1/2)$ in this direction, yielding the parametric equations:

$$x=1-t,\quad y=1,\quad z=\frac{1}{2}+t\quad \text{for } t\in\mathbb{R}$$

12. Find the local maximum values, local minimum values, and saddle points of the function:

$$f(x,y)=x^2-xy+y^2+2x+2y-4$$

Find the function's value at each of these points.

Solution. We first find the critical points by calculating the first-order partial derivatives and setting them equal to zero:

$$f_x=2x-y+2=0 \implies y=2x+2$$

$$f_y=-x+2y+2=0$$

Substituting $y=2x+2$ into the second equation:

$$-x+2(2x+2)+2=0 \implies 3x+6=0 \implies x=-2$$

Using $x=-2$, we find $y=2(-2)+2=-2$. This gives a single unique critical point at $(-2,-2)$.

To classify this point using the Second Derivatives Test, we compute the second partial derivatives:

$$f_{xx}=2,\quad f_{yy}=2,\quad f_{xy}=-1$$

We assemble these into the discriminant formula:

$$D=f_{xx}f_{yy}-(f_{xy})^2=(2)(2)-(-1)^2=4-1=3$$

Since $D>0$ and $f_{xx}=2>0$, the function has a **local minimum** at $(-2,-2)$. Evaluating the function at this point yields:

$$f(-2,-2)=(-2)^2-(-2)(-2)+(-2)^2+2(-2)+2(-2)-4=4-4+4-4-4-4=-6$$

13. Find the absolute maximum and absolute minimum values of $f(x,y) = x^2 + xy + y^2 - 3x + 3y$ on the closed triangular region in the first quadrant bounded by the lines $x = 0$, $y = 0$, and $x + y = 4$.

Solution. First, we check for critical points in the interior of the triangle:

$$f_x = 2x + y - 3 = 0 \quad \text{and} \quad f_y = x + 2y + 3 = 0$$

Solving this linear system yields $x = 3, y = -3$. However, the point $(3, -3)$ lies outside the first quadrant region $(y \geq 0)$. Thus, there are no valid interior critical points.

Next, we test the three boundary lines of the triangular region:

1 Along the boundary $x = 0$ (for $0 \leq y \leq 4$): The function simplifies to $f(0, y) = y^2 + 3y$. Its derivative with respect to y is $2y + 3 = 0 \implies y = -1.5$ (out of bounds). Evaluating the endpoints: $f(0, 0) = 0$ and $f(0, 4) = 4^2 + 3(4) = 28$.

2 Along the boundary $y = 0$ (for $0 \leq x \leq 4$): The function simplifies to $f(x, 0) = x^2 - 3x$. Its derivative with respect to x is $2x - 3 = 0 \implies x = 1.5$. Evaluating this critical point and the remaining endpoint: $f(1.5, 0) = -2.25$ and $f(4, 0) = 4^2 - 3(4) = 4$.

3 Along the slanted boundary $y = 4 - x$ (for $0 \leq x \leq 4$): Substituting this condition gives:

$$f(x, 4 - x) = x^2 + x(4 - x) + (4 - x)^2 - 3x + 3(4 - x) = x^2 - 11x + 28$$

Its derivative is $2x - 11 = 0 \implies x = 5.5$ (out of bounds).

Comparing all valid candidates, the **absolute maximum value** is 28 at $(0, 4)$, and the **absolute minimum value** is -2.25 at $(1.5, 0)$.

14. Find the second-order partial derivatives of the function $f(x, y) = e^x + y\sin(x)$.

Solution. We begin by computing the first-order partial derivatives of f with respect to x and y:

$$f_x = \frac{\partial}{\partial x}(e^x + y \sin x) = e^x + y \cos x$$

$$f_y = \frac{\partial}{\partial y}(e^x + y \sin x) = \sin x$$

Next, we differentiate these expressions a second time to find the four second-order partial derivatives:

$$f_{xx} = \frac{\partial}{\partial x}(e^x + y \cos x) = e^x - y \sin x$$

$$f_{xy} = \frac{\partial}{\partial y}(e^x + y \cos x) = \cos x$$

$$f_{yx} = \frac{\partial}{\partial x}(\sin x) = \cos x$$

$$f_{yy} = \frac{\partial}{\partial y}(\sin x) = 0$$

Notice that the mixed partial derivatives are equal ($f_{xy} = f_{yx} = \cos x$), as guaranteed by Clairaut's Theorem.

15. Find the absolute max and min values of $f(x, y) = x^3 + y^3 + 3x^2 - 3y^2$ on the square region enclosed by the lines $x = \pm 1$ and $y = \pm 1$.

Solution. First, we find the critical points in the interior of the region by setting the partial derivatives to zero:

$$f_x(x, y) = 3x^2 + 6x = 3x(x + 2) = 0 \implies x = 0,\ x = -2$$
$$f_y(x, y) = 3y^2 - 6y = 3y(y - 2) = 0 \implies y = 0,\ y = 2$$

Since our region is bounded by $[-1, 1] \times [-1, 1]$, the values $x = -2$ and $y = 2$ lie outside the square. The only interior critical point is $(0, 0)$. Evaluating the function here yields:

$$f(0, 0) = 0$$

Next, we check the four boundaries of the square region:

- On the right boundary ($x = 1, -1 \le y \le 1$): $g_1(y) = f(1, y) = y^3 - 3y^2 + 4$. Setting $g_1'(y) = 3y^2 - 6y = 0$ gives $y = 0$. We check the critical point and the edge endpoints:

$$f(1, 0) = 4, \quad f(1, -1) = 0, \quad f(1, 1) = 2$$

- On the left boundary ($x = -1, -1 \le y \le 1$): $g_2(y) = f(-1, y) = y^3 - 3y^2 + 2$. The critical point is $y = 0$. Checking these locations gives:

$$f(-1, 0) = 2, \quad f(-1, -1) = -2, \quad f(-1, 1) = 0$$

- On the top boundary ($y = 1, -1 \le x \le 1$): $h_1(x) = f(x, 1) = x^3 + 3x^2 - 2$. Setting $h_1'(x) = 3x^2 + 6x = 0$ gives $x = 0$. Checking these locations gives:

$$f(0, 1) = -2, \quad f(-1, 1) = 0, \quad f(1, 1) = 2$$

- On the bottom boundary ($y = -1, -1 \le x \le 1$): $h_2(x) = f(x, -1) = x^3 + 3x^2 - 4$. The critical point is $x = 0$. Checking these locations gives:

$$f(0, -1) = -4, \quad f(-1, -1) = -2, \quad f(1, -1) = 0$$

Comparing all computed values, the absolute maximum value is 4 occurring at $(1, 0)$, and the absolute minimum value is -4 occurring at $(0, -1)$.

16. If $w = \ln(x^2 + y^2 + 2z)$, $x = r + s$, $y = r - s$, $z = 2rs$, find w_r and w_s by the Chain Rule. Then check your answer another way.

Solution. Method 1: Chain Rule

First, we find the partial derivatives of w with respect to its intermediate variables x, y, and z:

$$w_x = \frac{2x}{x^2 + y^2 + 2z}, \quad w_y = \frac{2y}{x^2 + y^2 + 2z}, \quad w_z = \frac{2}{x^2 + y^2 + 2z}$$

Now, we calculate the partial derivatives of the inner components with respect to r and s:

$$x_r = 1, \quad y_r = 1, \quad z_r = 2s \qquad \text{and} \qquad x_s = 1, \quad y_s = -1, \quad z_s = 2r$$

Applying the Chain Rule for w_r:

$$\begin{aligned} w_r &= w_x x_r + w_y y_r + w_z z_r \\ &= \frac{2x(1) + 2y(1) + 2(2s)}{x^2 + y^2 + 2z} = \frac{2(r + s) + 2(r - s) + 4s}{(r + s)^2 + (r - s)^2 + 2(2rs)} \\ &= \frac{2r + 2s + 2r - 2s + 4s}{(r^2 + 2rs + s^2) + (r^2 - 2rs + s^2) + 4rs} = \frac{4r + 4s}{2r^2 + 4rs + 2s^2} \\ &= \frac{4(r + s)}{2(r + s)^2} = \frac{2}{r + s} \end{aligned}$$

Applying the Chain Rule for w_s:

$$\begin{aligned}
w_s &= w_x x_s + w_y y_s + w_z z_s \\
&= \frac{2x(1) + 2y(-1) + 2(2r)}{x^2 + y^2 + 2z} = \frac{2(r+s) - 2(r-s) + 4r}{2(r+s)^2} \\
&= \frac{2r + 2s - 2r + 2s + 4r}{2(r+s)^2} = \frac{4r + 4s}{2(r+s)^2} = \frac{2}{r+s}
\end{aligned}$$

Method 2: Direct Substitution

We can check our work by substituting x, y, and z into the expression for w before differentiating:

$$\begin{aligned}
x^2 + y^2 + 2z &= (r+s)^2 + (r-s)^2 + 2(2rs) \\
&= (r^2 + 2rs + s^2) + (r^2 - 2rs + s^2) + 4rs \\
&= 2r^2 + 4rs + 2s^2 = 2(r+s)^2
\end{aligned}$$

Thus, the function can be rewritten directly as:

$$w = \ln(2(r+s)^2) = \ln(2) + 2\ln(r+s)$$

Differentiating directly with respect to r and s yields:

$$w_r = \frac{2}{r+s} \qquad \text{and} \qquad w_s = \frac{2}{r+s}$$

Both methods yield identical results.

Chapter 7

Homework

Chapter 8

Course Documents

MA-105-BL: Survey of Math

Worcester State University, DGCE, Summer 2026

Course Information

Instructor	Dr. Hansun To, Professor of Mathematics, hto1@worcester.edu.
Meeting Information	This is a **Hybrid Course**, combining online and in-person learning to provide a flexible and tailored experience. You'll have the opportunity to learn at your own pace using MyOpenMath, participate in Zoom discussions as needed, and attend in-person sessions for exams. *In-person attendance is required* for the following sessions: • Two midterm exams (1 hour and 30 minutes): ◦ (In person) Thursday, June 4th, from 03:30 PM to 05:00 PM ◦ (Online) Thursday, June 25th. (The exam will be available on Thursday, June 25th, at 08:00 AM. Please submit your work by 06:30 PM, including any necessary show work attachments.) • Final exam (three hours): ◦ (In person) Thursday, July 2nd, from 03:30 PM to 05:30 PM All in-person exams will be held in Sullivan room S-136. Please plan accordingly to ensure your attendance at these mandatory sessions.
Prerequisite	Math placement exam code 3 or higher, or a weighted high school GPA of 2.7 or higher within the past 3 years

Responsibility for Learning. Every student is responsible for their own learning. The instructor will make every effort to support you; however, if you need any help it is your responsibility to reach out with questions

Course Content. In this course we will learn to improve the level of quantitative awareness of students using familiar situations that provide a sense of purpose for studying mathematics. The objective is not to make mathematicians of the students, but to help them deal as comfortably as possible with an environment that increasingly makes use of quantitative reasoning. Topics will include voting and apportionment methods, personal finance, set theory, and probability theory.

Required Materials. A calculator is required for this course. There is no required textbook; notes will be provided in class (or in pdf format) and other materials will be freely available on the MyOpenMath platform.

Calculators. A scientific, non-graphing calculator capable of computing logarithmic and trigonometric values is required, though its use will be limited to only a handful of sections. I recommend the ***TI-30X IIS***, which you can usually find for around $15. Graphing calculators, phones, tablets and any unauthorized devices are never allowed!

MyOpenMath. MyOpenMath (MOM) is a free, open source, online course management system designed for mathematics which we will use extensively in this course. It is available at https://www.myopenmath.com/ . You will need to create an account and register for our course on MyOpenMath to get started; we will do this as a group during the first class meeting. All course documents and materials will be posted on MyOpenMath; and all homework will be completed through MyOpenMath.

Assignments and Grading

Homework. Practice is a primary component of the mathematical learning process. For this reason, homework sets will be assigned on MyOpenMath after most class meetings. Assignments will always be due on Wednesdays by 11:59pm. This means you will often have two homework sets due on Wednesday, one which opened on the previous Monday and the other which opened on the previous Wednesday. While the number of problems in each homework set will vary, your score will be considered as a percentage and each set will be weighted equally when computing your homework grade. The problems on the homework sets are sometimes meant to deepen or extend the understanding you have gained from the class lecture and discussion; you should expect to find some of the exercises difficult.

The online homework is set up so you can try most problems two times, then the correct answer will be shown (there may be some questions in which you have only one try, this will be indicated in the homework set). If you miss a problem and want to improve your score, you can click the ``Try a similar problem'' link, which will give you a new question of the same type. You can keep on working on versions of a question until you get it correct. ***You must earn at least 50% on an assignment for your score to count towards your grade; any set in which you earn less than 50% will be counted as a 0.*** Assignments will remain available in "Practice" mode after their due date to provide additional practice of concepts; new versions of problems can be generated and answered, but scores will not be saved. ***Be aware, opening an online homework assignment in "Practice" mode will disqualify***

you from being able to redeem a LatePass on that assignment! If you are having trouble on a problem, you may use the "Message Instructor" link, which will allow you to send me a message through MyOpenMath; the specific problem you are working on will be automatically included in the message. I can then reply to the message with some guidance. These messages are internal to MyOpenMath and not email, so my reply will go to your messages when you login to MyOpenMath. I will check my messages on MyOpenMath at least once per day. Some problems may have a link to a video solution of a similar problem.

To account for any unexpected emergencies or technical issues that may occur, we will utilize the LatePass feature in MyOpenMath. A LatePass can be redeemed for an automatic 48-hour extension on any online homework set which has NOT been viewed in "Practice" mode. To use a LatePass, click the "Use LatePass" button next to the appropriate assignment on MyOpenMath and the platform will automatically extend the due date for that assignment to 48 hours after the original due date. Each student will be given 8 LatePasses to use as they wish throughout the semester. Note a maximum of two LatePasses can be used on any single assignment. If circumstances require a more substantial extension, contact your instructor as soon as possible to discuss this possibility.

Exams. There will be four exams in this course, one of which will be a cumulative final exam. Tentative dates are June 4, June 25, May 6, and July 2.

Grade Allocation. Your course grade will be determined by the following distribution. ***You must get at least a 50% on the final exam in order to pass the class***.

Assignment	Percentage
MyOpenMath Online Homework	10%
MyOpenMath Online Interactive Activities	10%
MyOpenMath Online Quizzes	10%
Exam 1	20%
Exam 2 (online)	20%
Final Exam	30%

Grading Scale. Letter grades will be assigned as follows. Grades will not be curved.

Letter Grade	Percent Needed
A	93%
A-	90%
B+	87%
B	83%
B-	80%
C+	77%
C	73%
C-	70%
D+	67%
D	63%
D-	60%
E	Below 60%

Course Policies

Email. I try to respond to emails and MyOpenMath messages within 24 hours. You are encouraged to email/message regarding the homework; however, emails sent within 24 hours of a due date may not get a response before the deadline. So do not wait until the last minute to attempt the assignments!

Studying. To succeed in this course, you will need to do more than simply attend class. Mathematics is a subject which is learned by doing, meaning it is important to work problems outside of class to fully understand the concepts we cover. The assigned homework is the minimum amount you should be completing; if you feel like you don't fully understand a concept, ask me to guide you to some additional practice problems. A general rule for studying (in all college courses) is for each hour of class per week you spend a minimum of two hours outside of class on homework, studying, reading the text, etc.

Since this is a hybrid course and it is three credit hours, you should plan to spend at least six hours per week working on course material for a regular semester. Since this is a 7-week summer course, that time is doubled to 12 hours per week.

Academic Integrity and AI Usage. The WSU Mathematics Department does not tolerate academic dishonesty! For an explanation of what is considered "academic dishonesty" at WSU, please see the policy included in the 2025-2026 Undergraduate Catalog. All instructors in the WSU Mathematics Department are required to report all incidents of academic dishonesty observed, detected, or otherwise determined to have occurred to the WSU Academic Central File. WSU requires some penalty be imposed for every reported violation of the academic honesty policy. All incidents of academic dishonesty in this course will result in (at least) a score of 0 on that assignment; additional penalties will be imposed at the instructor's discretion and/or as required by the Mathematics Department Academic Honesty Policy.

Artificial Intelligence (AI) Policy: Artificial intelligence (AI) tools (ChatGPT and similar platforms) may be used for *study support and practice only.* Acceptable uses include reviewing concepts, generating practice problems, checking understanding of definitions, or exploring alternative explanations of course material. AI tools are not permitted on any graded assignments, quizzes, or examinations unless explicitly authorized by the instructor. This includes, but is not limited to, using AI to generate solutions, explanations, written work, computations, graphs, or code that is submitted for credit. All submitted work must reflect the student's own understanding and effort. Reliance on AI-generated responses in place of independent problem-solving undermines the learning objectives of this course and is not allowed. Students are responsible for ensuring that any work they submit complies with this policy. Use of AI in violation of this policy will be treated as a violation of the Academic Honesty Policy and may result in penalties including a score of zero on the assignment, quiz, or exam, and additional disciplinary action as required by university policy.

Instructor Rights. ***As the instructor of this course, I reserve the right to make changes to the above course policies at any time. All changes will be announced in class.***

Student Learning Outcomes

Students will be able to:

1. Students will be able to communicate their mathematical reasoning.
2. Financial Management: Students will be able to calculate simple and compound interest, loan installments and mortgage payments.
3. Probability Theory: Students will be able to determine sample spaces for experiments, calculate probabilities with fundamental counting principles, permutations and combinations.
4. Voting Methods: Students will be able to discuss the pros and cons of several voting methods and how they can be manipulated by a knowledgeable individual.

LASC

This is a Quantitative Reasoning (QR) course in the Liberal Arts and Sciences Curriculum (LASC).

The course addresses the following LASC Quantitative Reasoning objectives:

1. Acquaint students with formal systems, procedures, and sequences of operations.
2. Strengthen students' understanding of variables and functions.
3. Apply mathematical techniques to the analysis and solution of real-life problems.
4. Emphasize the importance of accuracy, including precise language and careful definitions of mathematical concepts.
5. Understand both the underlying principles and practical applications of one or more fields of mathematics.
6. Develop an understanding of and facility with statistical analysis an analysis, including understanding of its applications and limitations.

and the following LASC overarching objectives:

1. Communicate effectively orally and in writing.
2. Understand and employ quantitative and qualitative reasoning.
3. Apply skills in critical thinking.
4. Understand the roles of science and technology in our modern world.
5. Become socially responsible agents in the world.
6. Make connections across courses and disciplines.
7. Develop as healthy individuals--physically, emotionally, socially, ethically, and intellectually.

Course Schedule

Here is a tentative schedule of topics. We will try to stay on schedule as much as possible. If class is canceled for any reason, then the schedule will be adjusted to accommodate this cancelation. This could include a rearrangement of topics, removing a "review" day, or skipping a topic entirely, depending on the timing of the cancelation. Any changes to this schedule will be announced in class and via email.

Table 8.1 Course Schedule and Exam Dates

Unit	Topics	Due Dates
1	Voting methods, apportionment	Fri 5/22
2	Simple interest, compound interest	Wed 5/27
3	Annuities, installment loans ***(Quiz 1)***	Wed 5/27
4	Mortgages, credit cards	Wed 6/3
	In-Person Exam 1 Thu 6/4 (3:30-5:00pm) (S-136)	
5	Set basics, subsets	Wed 6/10
6	Set operations, surveys ***(Quiz 2)***	Wed 6/10
7	Intro to probability, events involving “not”, “or”, “and”	Wed 6/17
8	Counting, permutations, combinations ***(Quiz 3)***	Wed 6/17
	Online Exam 2 (Thu 6/25)	
9	Probability with counting, odds, conditional	Fri 6/26
	In-Person FINAL EXAM Thu 7/2 (3:30-6:30pm) (S-136)	